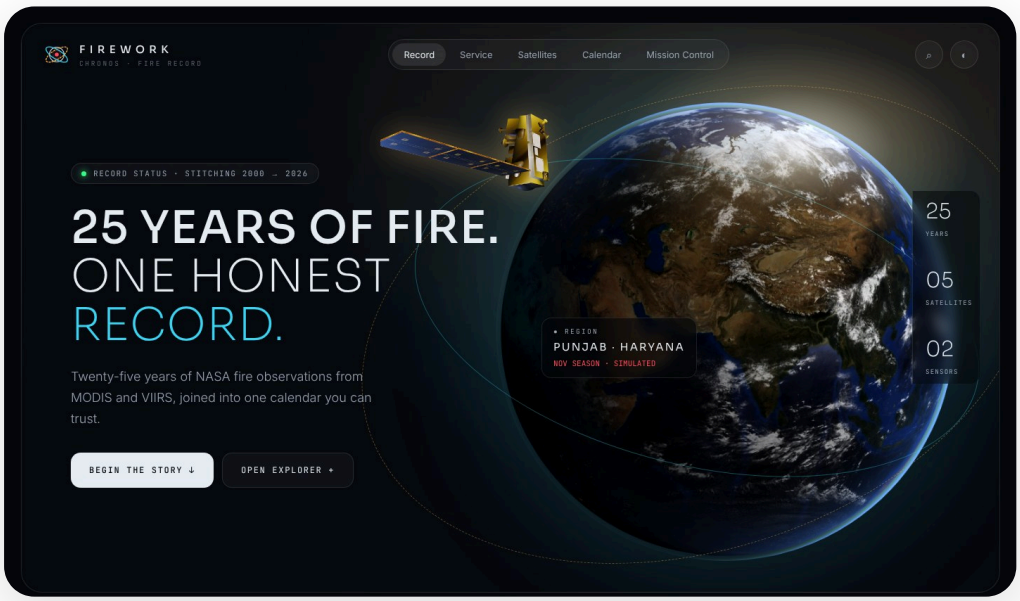
NASA SPACE APPS 2026 · HARMONIZATION OF MODIS AND VIIRS HOT SPOTS

FIREWORK CHRONOS

Version 2 Research Atlas

A research report in one book: the reset of every earlier FIREWORK document, a scientific audit of all 46 formulas with derivations in English and Hinglish, the four-layer model, the 25-year calendar, three design directions with marked page layouts, data and API contracts, a citation-safe RAG knowledge base, automatic PDF briefs and the execution plan.



OFFICIAL DOCUMENTED · SOURCE-DERIVED · RUN / TESTED LOCALLY · SIMULATED DEMO · PROPOSED / UNVALIDATED

Every number carries one of five provenance labels. Simulated numbers are never results.

Dhruv Sharma and team · October 2026 · A4 report edition V2.0 (27 Oct 2026)

Supersedes the 164-page Atlas (20 Oct 2026) as the working source of truth · research decision support, not a life-safety alert system

— MODIS — VIIRS ● fire

BEFORE YOU READ

How this atlas tells the truth

Every earlier FIREWORK document mixed tested code, ideas, demo numbers and real facts on the same page. This edition separates them. Each figure, number and feature carries exactly one of five labels, and the label decides how much you may trust it.

LABEL	WHAT IT MEANS	EXAMPLE IN THIS BOOK
OFFICIAL DOCUMENTED	Matches a first-party NASA / NOAA / ESA / Earth Engine page or a peer-reviewed paper. URL and access date stored in the claim ledger.	FIRMS S-NPP archive starts 20 Jan 2012 (download page, read 27 Oct 2026)
SOURCE-DERIVED	Computed deterministically from an official dataset by code in this project.	Punjab gridded area 50,444 km² from Natural Earth polygons
RUN / TESTED LOCALLY	Code executed while making this atlas. Inputs are stated; if inputs are synthetic the result is a property test, not field accuracy.	Plain floor misassigns 643 of 1,800 latitude edges
SIMULATED DEMO	Synthetic numbers for layout, teaching or method tests. Never a result about real fires.	the calendar heatmaps, k = 2.6 examples
PROPOSED / UNVALIDATED	An idea or method not yet validated on held-out real data.	orbit-drift-aware normaliser (R1)

Seven rules carried through every chapter

1. A hotspot is a pixel, not a fire; a cell-day is not burned area; harmonised cell-days are not hectares or biomass.
2. Missing coverage, failed sources and unprocessed months are **null**, never 0.
3. Official documentation and validated runs outrank project PDFs, whatever their date.
4. Formulas are identified by name + equation + version, never by number alone (old chat numbering differs).
5. Every interval says its level (90% or 95%) and whether it is a confidence or predictive band.
6. AI writes words around computed numbers; it never creates or edits a measurement.
7. FIRMS-based products have limited accuracy; this is situational-awareness research, not a protection-of-life service.

Three reading paths

READER	PATH	PAGES
Judge / official	Ch. 1 Field Card → Ch. 2 jobs → Ch. 14 layers → Ch. 21 calendar → Ch. 31 checklist	≈ 25
Researcher	Part II primer → Ch. 10 unified model → 46 formula spreads → Ch. 12–13 → Ch. 19 new approaches	≈ 120
Builder	Part V design → Part VI contracts → Ch. 30 milestones → appendices	≈ 60

Page anatomy

Dark **Science Midnight** pages carry concepts, derivations and figures. Cream **Laboratory Light** pages carry audits, code and tests. White **Field Card** pages are print-ready summaries for officials. Sensor colours never change meaning: MODIS amber, VIIRS cyan, fire / alert coral, model estimate violet, validated / crop mint, hatched grey = no coverage.

HINGLISH NOTE

Is kitaab mein har number ke saath ek label hai. Label dekho, phir bharosa karo. SIMULATED number kabhi result nahi hota.

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THE ATLAS AT A GLANCE

One problem, seven parts

NASA asks for a comparable burning calendar from two generations of satellites. Each part of this book answers one question on the way there.

I pp. 6–18

Provenance & reset

What we kept, changed, rejected; sources, conflicts, claims

II pp. 19–26

Remote sensing primer

MODIS vs VIIRS, orbits, clocks, masks, burned area

III pp. 27–128

Science & registry

pipeline, unified model, 46 formula spreads, ladder, gates

IV pp. 129–140

Four layers

L1–L4, Village Fire Card, global design, R1–R10

V pp. 141–170

UX & design

IA, calendar, 3 directions, 13 pages, 17 chart forms

VI pp. 171–180

Data, API, RAG, reports

sources, static CSVs, schema, RAG, auto-PDF

VII pp. 181–188

Execution

milestones, demo, checklist, traceability, code

Companion files

Supabase schema and 19 static CSVs (real reference data), fetch scripts for live data, 46 formula YAML files, V2 reference code with audit and simulation, and the six docs/v2 registers.

WHAT CHANGED

V1 → V2 at a glance

TOPIC	V1 (OCT 14–27 DOCUMENTS)	V2 (THIS ATLAS)
unit of comparison	fire cell-days, MODIS count as denominator	exposure rate F/E + VIIRS-eq cell-days with 95% band
missing data	filled with 0	null + status (not_observable, unavailable, pending)
k	one median ratio, 2.6 in examples	per stratum, NB offset model, pooled log k, interval
confidence	VIIRS l/n/h → 0.25/0.6/0.9	sensor-native classes + sensitivity grid
anomaly	3 inconsistent versions	log((F+.5)/(E+1)), target year left out, n ≥ 8
trend	MK without ties	tie-corrected MK + Hamed–Rao + Sen + BH
intervals	'95%' computed as 90%	explicit α; level and type in every legend
blind spot	B = 'real burning ~2×'	B = disagreement flag; scenario clock
validation	in-sample, MAPE	LOYO + region hold-out, placebo seams, predeclared gates
grid	plain floor; India assumptions	rounded floor, antimeridian, real geodesic areas, global
palette	two inverted sensor palettes	one validated semantic palette (pending approval)
truth labels	E / T / P tags	five provenance labels on every number
RAG	LLM + docs	documents for words, SQL for numbers, citation guard
fire data in DB	synthetic seed corpus	real static CSVs now; fire rows only from your FIRMS fetch

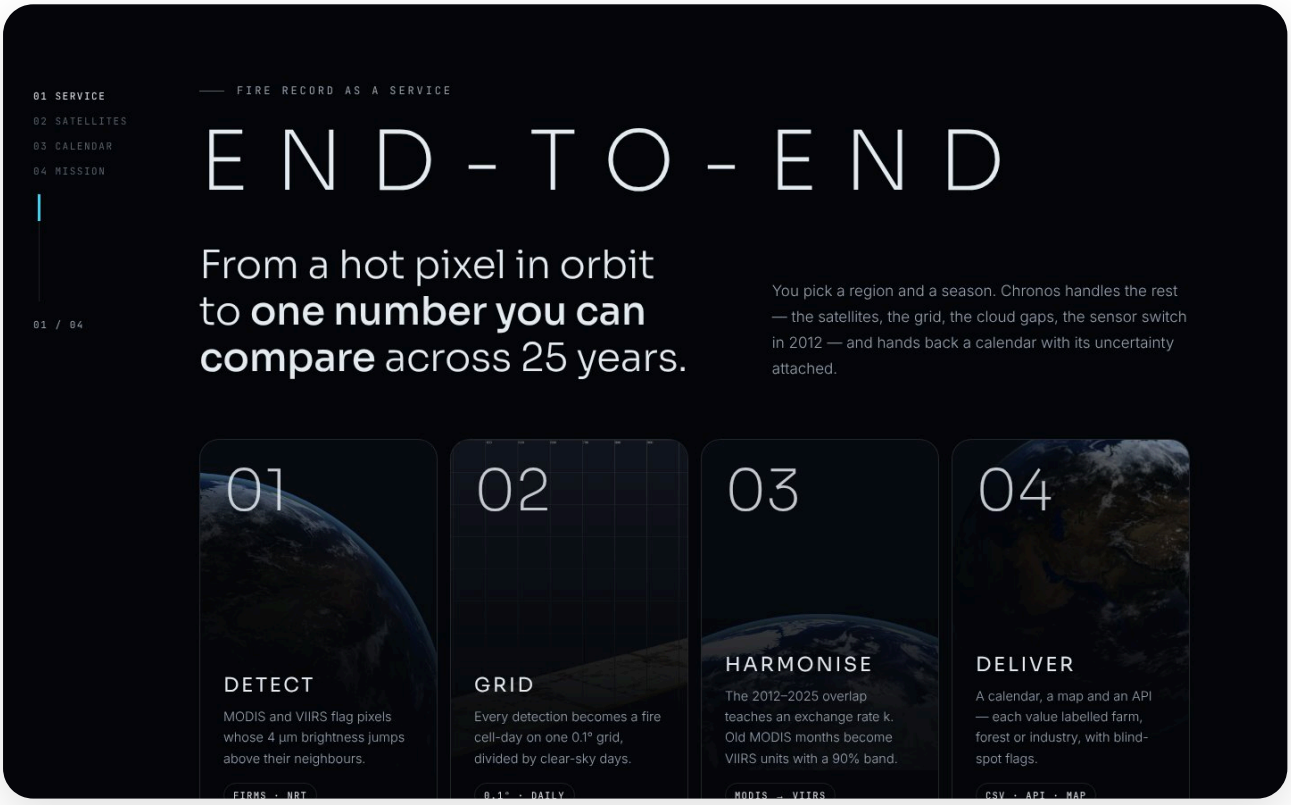
EK LINE MEIN

V1 ne idea diye, V2 ne unhe naapa, theek kiya, aur label lagaya.

PART I

Provenance & research reset

What we kept, what we changed and why — before any new science or design.



CHAPTERS IN THIS PART

Chapter 01 · Executive summary (Field Card)
Chapter 02 · The challenge as testable jobs
Chapter 03 · What went wrong, and the reset
Chapter 04 · Source register
Chapter 05 · Conflicts, idea decisions, open questions
Chapter 06 · Glossary, units and estimands

FIELD CARD · ONE-MINUTE BRIEF

1 · Executive summary

Bottom line: FIREWORK CHRONOS can answer the NASA challenge — a comparable 25-year burning calendar for any area — but the earlier documents over-claimed. This edition keeps the approved product vision, fixes 17 formula or code defects, revises 19 definitions, and labels every number by where it came from.

46
formulas re-audited: 17 FIX, 19 REVISE, 10 OK

36
audit checks executed in code for this atlas

15
conflicts logged; 12 resolved, 3 open for you

0
real fire numbers in this book — archive run is milestone M2

What the product is

A web application that turns MODIS (2000→) and VIIRS (2012→) active-fire hotspots into one month-by-year calendar on a common grid, with an exposure-normalised rate, a harmonised VIIRS-equivalent value with an interval, the sensor split, and data-quality status for every cell. Four explanatory layers sit on top: comparable activity, probable source, what satellites might miss, and whether a month is unusual.

What changed from the 164-page Atlas

- Exposure (valid observations) replaces 'MODIS count' as the denominator everywhere.
- Calendar cells carry **value, coverage, status**; missing is never 0.
- k is learned per stratum (month × day/night × platform pair) with a stated 95% interval, not one 2.6.
- Anomaly z uses a log rate that works at zero fires and leaves the target year out of its own baseline.
- B (burned area ÷ detections) is a warning flag, not a multiplier.
- One accessible sensor palette is proposed for Home and data pages (your decision).

HIGHEST-RISK DEFECTS FOUND

- calendar_matrix fills missing months with 0
- VIIRS l/n/h mapped to fake probabilities
- bootstrap labelled 95% but computes 90%
- Mann–Kendall without tie correction
- grid index floor error at 36–41% of cell edges; antimeridian column 3600

WHAT IS STILL UNPROVEN

No real-data accuracy exists yet. Every k, error, seam ratio and coverage shown here is simulated or a property test. Thresholds (persistence 0.20, night 20%, abstain 0.55) are hypotheses.

WHAT IS SOLID

Archive dates, product definitions and mission plans re-checked on official pages on 27 Oct 2026; cell-area maths validated against real state areas; code defects reproduced and fixed in v2 reference code with tests.

Decisions you need to make

#	DECISION	RECOMMENDED DEFAULT	WHY IT MATTERS
D1	Headline KPI	exposure-normalised rate; cell-days as display unit	rate is what is comparable across clouds and sensors
D2	Sensor palette	MODIS amber / VIIRS cyan on every page	removes the colour inversion between Home and data pages
D3	Explorer confidence control copy	rule picker ('MODIS ≥30 · VIIRS n,h')	the '0.6' label is not a probability
D4	Test regions	California, Mato Grosso, New South Wales	proves the pipeline is global
D5	Interval level	95% everywhere, stated in legends	one convention across API, UI and PDF

Truth status of the main claims

CLAIM	STATUS
S-NPP new-data delivery ends 2 Nov 2026	OFFICIAL DOCUMENTED
Terra end of science Feb 2027 (NASA) · Aqua Sep 2027	OFFICIAL DOCUMENTED
VIIRS 375 m ≠ VNP14A1 1 km	OFFICIAL DOCUMENTED
MCD64A1 uses MODIS active fire → not independent	OFFICIAL DOCUMENTED
0.1° cell = 106 km² in Punjab, 124 at equator	SOURCE-DERIVED
>90% Punjab large fires after 3 pm (2024–25)	SECONDARY REPORT
k ≈ 2.6	SIMULATED DEMO
seam reduced to 2.9% / LOYO 1.6–5% (competitors)	UNREVIEWED
Harmonised calendar beats raw counts	SIMULATED WORLDS ONLY

Next 7 days (before the 14–15 Nov hackathon)

1. Read the full challenge statement released 28 Oct; update Ch. 2 jobs.
2. Request the FIRMS archive (MODIS + S-NPP + NOAA-20) for the India bbox; run `db/fetch_and_load.py` for one overlap year.
3. Run `gee_exposure.py` for the same year; fill cell_month.
4. Run baseline ladder B0–B4 on real data; publish held-out table on /validation.
5. Decide D1–D5; recolour data-page mock-ups if D2 accepted.
6. Re-record Home clips from real pages only after real data lands.

WHERE TO LOOK IN THIS BOOK

Formula fixes: Part III spreads · calendar design: Ch. 21 · data contracts and Supabase CSVs: Ch. 26–27 · milestones: Ch. 30.

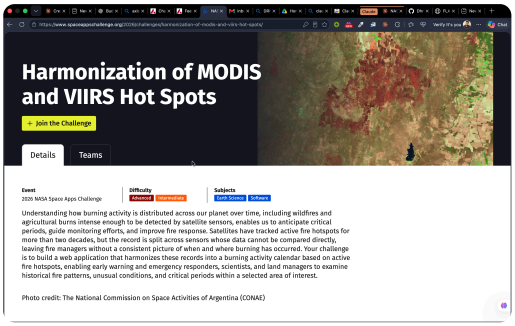
CHAPTER 2

The challenge as testable jobs

NASA asks for a web application that harmonises MODIS and VIIRS hotspot records into a burning-activity calendar for a selected area, so responders, scientists and land managers can see historical patterns, unusual conditions and critical periods.

We turn that sentence into jobs a judge can check. A job is done only when its acceptance test passes on real archive data.

JOB	ACCEPTANCE TEST	SCOPE
J1 Pick any area	polygon/bbox anywhere incl. antimeridian; area shown in km²	MVP
J2 Common grid & time	same 0.1° cells and local days for all platforms	MVP
J3 Transparent harmonisation	k per stratum with interval and method version	MVP
J4 Observation support	exposure E and status per cell-month	MVP
J5 Month × year calendar	25 years, fixed colour scale, null shown hatched	MVP
J6 Raw vs harmonised	toggle changes colours only; seam visible then reduced	MVP
J7 Sensor split	per platform and day/night	MVP
J8 Unusual / critical periods	robust z with sample size; season dates	MVP
J9 Download	CSV/JSON equals API equals chart	MVP
J10 Validation page	held-out errors, seam test, coverage	MVP
J11 Source class	crop/forest/industry/UNKNOWN with abstain	valuable
J12 Missed-burning diagnostic	B flag + overpass clock, scenario labelled	valuable
J13 Briefs, Learn, Studio, RAG	numbers only from API	valuable



Official challenge page (user screenshot, 15 Oct 2026). Full statement and datasets due 28 Oct 2026 — re-check.

Open scientific questions

- What estimand is 'comparable'? (rate vs cell-days)
- Do per-pixel opportunities define a meaningful rate?
- Is k constant locally and over drift years?
- How is daily cloud exposure counted per sensor?
- Can burned area validate activity when it uses active fire as input?
- Can source classes be trusted without labelled sites?

SIMPLE WORDS

NASA ka kaam: purane aur naye satellite ki aag ko ek hi taraazu par tolna, aur har mahine ka calendar banana — kisi bhi jagah ke liye.

Who uses it, and for what decision

PERSONA	QUESTION	PAGE	WHAT THEY MUST NEVER SEE
District officer (Punjab)	Was this November worse than normal, and can I trust that?	Brief, Calendar	a count drop presented as a burning drop
Pollution-board analyst	When does the season open and peak?	Calendar, Blind Spot	season dates without the detection-timing caveat
Fire scientist	How was k estimated, and how wrong is it in held-out years?	Sensor Lab, Validation	in-sample fit called accuracy
Journalist	Is the 2024 'record low' real?	Blind Spot, Brief	B ratio as a multiplier
Student	How do satellites see fire?	Learn, Formulas	teaching numbers without the SIMULATED label
Researcher elsewhere (e.g. California)	Can I run the same method on my region?	Explorer, Studio	India-only assumptions

MVP (must ship)

J1–J10. Punjab–Haryana flagship plus one non-India region rendered by the same code path.

Valuable layers

Source attribution, blind-spot diagnostic, air-quality context, PDF briefs, Learn, Studio, RAG explanations, 3D satellite story.

Not claimed

Burned hectares from hotspots, emissions, fire counts as 'number of fires', emergency alerting, certified accuracy.

Judging criteria and our evidence (equal weight)

CRITERION	EVIDENCE THAT SURVIVES SCRUTINY
Influence	Punjab–Haryana story with honest blind-spot flag; printable briefs
Creativity	exposure-aware calendar, source abstention, overpass clock, placebo-seam validation
Validity	held-out real-data errors, predeclared thresholds, provenance labels
Relevance	J1–J10 complete for any area
Presentation	Concept A v2 Home → calendar 'aha' (seam disappears) in under two minutes

CHAPTER 3

What went wrong, and the reset

Between 14 and 27 October the project produced more than 25 documents. Each one was right for its day, but together they contradicted each other: three versions of the anomaly formula, two sensor palettes, a 90% band called 95%, and teaching numbers that drifted into specifications.

DATE	ARTIFACT	WHAT IT ADDED	WHAT IT GOT WRONG (FOUND NOW)
14–15 Oct	Strategy + Books 1–4	challenge pick, sensors, code, beginner maths	fixed overpass times; 5–95 bootstrap; ln(1+x) z
15 Oct	Accuracy Upgrade (Book 5)	clear-sky rate, day/night k, GLM, EB	rate treated as defined at K=0 only in tests
20 Oct	164-page A4 Atlas	full design + 46 formula pages	mixed status tags; numbers from synthetic data in figures
21 Oct	62-page Build Spec + MD	stack, API, DB, milestones	60 fps claim; 0.6 confidence copy
26 Oct	4 Layers + visual guide	layers, CHANGED formulas	ln(rate) at zero; B = 'many missed'
27 Oct	Final Spec + prompt pack	locked Home/Explorer/Studio, fixtures	palette decision left OPEN
27 Oct	ChatGPT audit (user prompt)	audit flags, truth policy	some claims unverified (e.g. DAY_RANGE)
27 Oct	**This V2 atlas**	verified facts, runnable audit, registry	real-data run still pending

Precedence policy (not 'latest wins')

1. Scientific facts: official product guides and validated runs beat any project document.
2. Home and identity: Concept A v2 + Crossing Orbits logo beat earlier Home ideas.
3. Approved page structures in CLAUDE_CODE_PROMPT.md stay unless a V2 decision replaces them.
4. Everything else: compare old vs new and log the decision (Ch. 5).
5. Nothing is deleted: rejected ideas move to archive/ideas with a reason.

A VERIFIED EXAMPLE OF WHY PRECEDENCE MATTERS

Book 2 and the user's audit say the FIRMS Area API allows 1–5 days per call. The official Area API page read on 27 Oct 2026 says 1–10. The code keeps 5 as a default (valid under both) and reads the limit from configuration. Logged as C-01.

FORMULA-NUMBER COLLISION

Early chat walkthroughs called a calibration idea 'F09' and the bootstrap 'F16'. In the master registry F09 is the exposure rate, F16 the delta-method interval and F24 the bootstrap. This book always prints name + equation + version (alias table in Ch. 5).

CHAPTER 4

Source register

Every artifact read for this edition, with its role after the reset. Hashes are the first 12 hex digits of SHA-256 for files available on disk; duplicates such as "(1)", "(2)" copies were compared by content and listed once.

ARTIFACT	DATE	TYPE	CONTENT USED	ROLE AFTER RESET	SHA256 / STATUS
FIREWORK_CHRONOS_Atlas_A4_Report (4).pdf	20 Oct	PDF 164 pp	design + 46 formulas + pages	archive; superseded by V2 for science	7803fba1ee14
FIREWORK_Chronos_Final_Spec.pdf	27 Oct	PDF 17 pp	locked Home/Explorer/Studio, fixtures	authoritative for page structure	4cf1e929d67e
CLAUDE_CODE_PROMPT.md	27 Oct	MD	prompt pack, sections per page	authoritative for Home/logo/page list	project doc
CLAUDE_BUILD_SPEC (1).md	21 Oct	MD	stack, 46 formula table, API, DB	formula v1 source; science revised	project doc
Final_Design_and_Data_Contract.md	26 Oct	MD	smoothness rules, explorer JSON	keep; validated palette	project doc
Four_Layers_Check.md	26 Oct	MD	layer formulas, findings	keep findings	project doc
FIREWORK_46_Formulas_Visual_Guide.pdf	26 Oct	PDF 12 pp	CHANGED cards F11/F27/F28	partly superseded (F27 zero bug)	project file
Book1 Zero se Hero (Hinglish)	15 Oct	PDF	beginner primer	teaching aid	project file
Book2 Links, APIs & Tested Code	15 Oct	PDF	harmonize.py, calendar_build.py, tests	code re-run here; defects logged	project file
Book3 Formulas, Algorithms & AI	15 Oct	PDF	maths, pseudocode, NN ideas	teaching; 90% bootstrap	project file
Book4 Website Design	15 Oct	PDF	CalendarGrid.jsx, BootSequence.jsx	UI code; see C-13	project file
Accuracy_Upgrade_Summary.md (Book 5)	15 Oct	MD	MUST/SHOULD/COULD	keep, revised	project doc
NASA_Space_Apps_2026_Strategy.md	14 Oct	MD	challenge choice, competitors, Punjab story	keep; numbers re-labelled	project doc

ARTIFACT	DATE	TYPE	CONTENT USED	ROLE AFTER RESET	SHA256 / STATUS
API_Keys_and_Backups.md / Free_Stack	19–20 Oct	MD	keys, free tiers	re-verify tiers before event	project doc
FIREWORK_APIs_Keys_Datasets.pdf	18 Oct	PDF	field-by-field API responses	illustrative samples	project file
FIREWORK_Claude_Code_Build_Spec.pdf	21 Oct	PDF 62 pp	same as MD + images	as MD	project file
ConceptA_v2_full.png	26 Oct	PNG 1440×6661	approved Home	locked	ceb0149c5cad
Logo_v2.png	26 Oct	PNG	Crossing Orbits (recommended)	locked	9670441a6138
FIREWORK_satellite_models_preview.png	26 Oct	PNG	satellite renders	illustrative; licensing to verify	6989bbebcf50
Screenshot 2026-10-15 (challenge)	15 Oct	PNG	official challenge text	evidence of task wording	c03a1d29d416
harmonize.py / calendar_build.py + tests	15 Oct	Python (Book 2)	reconstructed verbatim	tests PASS (synthetic)	code/v1
accuracy_tools.py, failover.py, health_check.py, fetch_firms.py	15–21 Oct	Python	not available in this session	findings carried from user audit, not re-run	MISSING
CalendarGrid.jsx, BootSequence.jsx, mission-control.css, Some-data.txt, .pptx	-	JS/CSS/PPTX	not available in this session	request files	MISSING

FILES NOT AVAILABLE TO THIS SESSION

accuracy_tools.py, failover.py, health_check.py, fetch_firms.py, api_calendar.py (the version with failover), test_accuracy_tools.py, test_failover.py, CalendarGrid.jsx, BootSequence.jsx, mission-control.css, Some-data.txt, Book5 PDF, the .pptx deck. Findings about them (F18, F19, F24 code, F38, F45, failover 'live' label) are taken from the user's 27 Oct audit and marked SOURCE-DERIVED (secondary). Upload them to re-run.

Execution results in this environment

TEST	RESULT	MEANING
test_harmonize.py (v1)	PASS	synthetic invariants only
test_calendar.py (v1)	PASS, z = 6.74	illustrative count table
audit_run.py (v2)	36 checks executed	see Part III spreads
sim_ladder.py	60 worlds × 14 years	method test, not field accuracy
build_db_package.py	20,990 real grid cells	Natural Earth → Supabase CSV

CHAPTER 5

Conflicts, decisions, open questions

ID	TOPIC	OLD POSITION	NEW POSITION	DECISION	STATUS
C-01	FIRMS Area API DAY_RANGE	Book 2 + user audit: 1-5 days; code MAX_DAYS_PER_CALL=5	Area API page fetched 27 Oct 2026 shows 1-10	Keep 5 as default (valid under both), make it config FIRMS_DAY_RANGE	RESOLVED (reversible)
C-02	Sensor colours	Home amber MODIS / cyan VIIRS; data pages blue MODIS / orange VIIRS	One semantic palette proposed: MODIS amber, VIIRS cyan everywhere, dark-surface tints #FFB547/#3DD6F5, light-surface #A86A00/#0B7F99	PROPOSED: unify on amber/cyan; needs user approval	OPEN (user decision)
C-03	F24 interval level	Spec: 95% (2.5-97.5)	Book 3 + code: 5-95 (90%)	Explicit alpha parameter; UI legend states level	RESOLVED
C-04	F27 scale	code: raw counts; atlas: ln(1+x); visual guide: ln(rate)	log((F+0.5)/(E+1)), leave-target-out	Adopt v2	RESOLVED (pending real-data check)
C-05	F14 meaning	described as an independent check of F11	intercept-only offset GLM equals ratio of totals	Rename B3b 'ratio of totals'; GLM proper uses exposure offset	RESOLVED
C-06	Layer 4 name	Atlas: 'Your-data Studio'	Strategy: unusual/trend	L4 = 'Is this month unusual or trending?'; Studio is a tool, not a layer	RESOLVED
C-07	Formula families	Build Spec says 8 families	tables list 7 groups	7 families	RESOLVED
C-08	MODIS end dates	Build Spec: Terra Jan 2027	NASA (Dec 2025): end of science Feb 2027; CAMS (Feb 2026): data collection Jan 2027; Aqua Sep 2027	Show both with source and as-of date	RESOLVED
C-09	VIIRS confidence numbers	UI 'nominal (0.6) same rule both sensors'	categorical sensor-native classes	Remove numeric label	OPEN (touches locked page copy)
C-10	B ratio meaning	'B = 2.4 means real burning ~2x the dots'	B is a disagreement flag	Flag only	RESOLVED
C-11	Calendar zeros	calendar_matrix fill_value=0	null + status	mask-aware cells	RESOLVED
C-12	Grid area claim	'0.1 deg cell ~ 106 km2' used as if constant	area varies 123.6 (equator) to 0.1 km2 (pole)	store area per cell	RESOLVED
C-13	Explorer 60 fps claim	Build Spec: 60 fps at 100k cells	Final contract measured 60 fps on ~hundreds of cells only	Claim only what was measured	OPEN (measure)
C-14	Overpass times	fixed Terra 10:30 / Aqua 13:30	Terra ~10:15 by fall 2022 (NASA)	use observed acq_time per year	RESOLVED
C-15	Formula IDs in chats	chat walkthroughs: 'F09 calibration', 'F16 bootstrap'	registry: F09 exposure rate, F16 delta interval, F24 bootstrap	alias table, IDs never used alone	RESOLVED

Idea decisions register

Every idea from FIREWORK, SIH PyroSentinel and the October documents, with a decision. KEEP = in V2 as is · REVISE = in V2 changed · RESEARCH = test before use · REJECT = archived with a reason.

IDEA	ORIGIN	DECISION	NOTE
Harmonised 25-year calendar (month × year)	Strategy L1, Build Spec C	KEEP	core NASA ask; now mask-aware
VIIRS-equivalent cell-days as common unit	Strategy, Book 3	KEEP	plus exposure rate; never 'hectares'
Clear-sky / exposure normalisation	Book 5	KEEP	platform-specific E, FireMask rules
Single k = 2.6 for everything	visual guide examples	REJECT	k per stratum with interval
Median of monthly ratios as main k	Book 3	REVISE	kept as robustness baseline B3a
Poisson GLM 'confirms' the median	Build Spec F14	REVISE	it is ratio of totals; use exposure offset
Day / night separate k	Book 5	KEEP	with observed acq_time
Empirical Bayes pooling	Book 5	KEEP	R ≥ 3 guard
VIIRS l/n/h = 0.25/0.6/0.9	FIREWORK repo	REJECT	sensor-native categories
Same-confidence slider '0.6 both sensors'	Explorer mock-up	REVISE	rule picker
Block bootstrap bands	Book 3	KEEP	explicit α
Robust z on ln(1+x)	Atlas F27	REVISE	log((F+.5)/(E+1)), LOO
Mann–Kendall + Sen + BH	Strategy L4	KEEP	tie-corrected
Pettitt seam test as proof	Validation page	REVISE	plus placebo seams
Source attribution crop/forest/industry/unknown	SIH, Strategy L2	KEEP	abstain ≥ 'unknown'
FIS = 0.45T+0.30I+0.15WCS+0.10C	SIH deck	REJECT	uncalibrated hand weights
Weather as a class decision input	-	REJECT	context only (FIREWORK rule)
Diurnal fraction seen (F45)	Book 5	REVISE	scenario, union of windows
B ratio as 'real burning × B'	4 Layers PDF	REJECT	flag only
Geostationary hourly FRP for India	Strategy L3	RESEARCH	access unverified (Meteosat IODC, GK-2A, INSAT)
Sentinel-3 SLSTR morning continuity	Strategy L5	RESEARCH	post-Terra morning view
Kalman multi-sensor fusion	Book 5	RESEARCH	experimental tab only
Conformal source sets	Book 5	RESEARCH	needs labelled sites
LightGBM + SHAP classifier	Book 3	RESEARCH	after labels exist
LSTM 14-day nowcast	Book 3	RESEARCH	must beat seasonal naive
Realistic three.js Earth on Home only	Atlas Summary	KEEP	approved
Concept A v2 Home + Crossing Orbits logo	Final Spec	KEEP	locked
Two sensor palettes (Home vs data pages)	Build Spec	REVISE	one palette proposed (C-02)
User Data Studio (upload + compare)	Build Spec E	KEEP	USER label, never mixed
LLM writes words only	FIREWORK rule	KEEP	citation guard + numeric service
Raw FIRMS rows in a vector DB	-	REJECT	numbers come from SQL, not RAG
60 fps at 100k cells claim	Build Spec	RESEARCH	measure before claiming
seed/evaluation_metrics.json F1 values	FIREWORK repo	REJECT	impossible F1 > both P and R
Punjab–Haryana flagship, global pipeline	user decision	KEEP	+ California, Mato Grosso, NSW test regions

Formula aliases (old labels → final IDs)

ALIAS	WHERE IT APPEARED	FINAL ID	NOTE
F09 calibration	early chat walkthrough	F10	calibration ratio concepts now F10-F14
F16 bootstrap	early chat walkthrough	F24	bootstrap is F24; F16 is the delta-method interval
clear-sky normalisation	Book 5 accuracy upgrade	F09	renamed 'exposure rate'
k_shrunk	Strategy doc	F17	teaching rule; production F18
seam test	Strategy / Book 3	F35	plus F33/F34 placebo
FIS = 0.45T+0.30I+0.15WCS+0.10C	SIH deck	REJECT	hand-weighted score not used; see IDEA-17
L_context	CLAUDE.md	F43	sign: + = industry
diurnal fraction	Blind Spot spec	F45	scenario only

Open questions

1. Which comparable estimand is the headline: exposure-normalised rate or VIIRS-equivalent cell-days? (proposal: rate is the KPI; cell-days are the display unit)
2. Is the MODIS→VIIRS relation stationary across 2012–2025 while Terra/Aqua drift? (test R1, R5)
3. Does exposure from the 1 km daily composite represent opportunity for 375 m point detections? (R2)
4. Unify sensor colours across Home and data pages? (C-02, user decision)
5. 2026 Space Apps rules on code written before the event (re-check after 28 Oct release).
6. Final regions beyond Punjab–Haryana (proposed test set: California, Mato Grosso, New South Wales).
7. Geostationary hourly fire data access for India.
8. Labelled industrial / kiln sites for Layer 2 evaluation — who can provide lawful references?

FILES WRITTEN FOR THESE REGISTERS

docs/v2/SOURCE_REGISTER.md · CONFLICT_LOG.md · IDEA_DECISIONS.md · OPEN_QUESTIONS.md · CLAIM_LEDGER.csv · SCIENCE_ERROR_BUDGET.md — and the same tables as Supabase CSVs (conflict_log.csv, claim_ledger.csv, formula_alias.csv).

CHAPTER 6

Glossary, units and estimands

TERM	MEANING IN THIS PROJECT
Active fire / hotspot	a satellite pixel flagged as containing a fire hotter than its background at the overpass moment
Cell-day	one grid cell on one day with ≥1 filtered detection (per platform, day/night)
Exposure E	number of valid observation days for a cell-month and platform (FireMask clear land, water or fire)
Exposure rate r	F / E: fire cell-days per valid observation day
VIIRS-equivalent	a MODIS-era value scaled by k to the VIIRS 375 m detection scale (an estimate)
k	exchange rate VIIRS ÷ MODIS in a stratum during overlap
Stratum	a group with one k: region × month × day/night × platform pair
NRT / SP	near real-time vs standard processing (SP replaces NRT after ~5 months)
LMST	local mean solar time = UTC + longitude/15
Seam	artificial step in a series when a new sensor joins (2012, 2018, 2023)
Placebo seam	a fake change year used to test whether the seam test raises false alarms
LOYO	leave-one-year-out: fit without a year, predict it
Robust z	(value − median)/(1.4826·MAD) with a floor
Burned area (BA)	MCD64A1 monthly 500 m burn scar map — physical area, separate unit
FRP	fire radiative power (MW) at the overpass moment
Abstain / UNKNOWN	classifier refuses a label when evidence is weak
Provenance label	one of the five truth labels on every number
Config hash	hash of all parameters used to produce a published value

Units that must never be mixed

QUANTITY	UNIT	IS NOT
detections	pixels	fires
cell-days	cells × days	burned area, fires
exposure rate	cell-days per valid day	probability of burning
harmonised value	VIIRS-eq cell-days (estimate)	observation
burned area	km² (MCD64A1)	hotspot output
FRP	MW (instantaneous)	energy (MJ) or biomass
anomaly z	unitless	a volume
B ratio	unitless flag	multiplier

The estimand we publish

For region r, month t, platform s and day/night δ:

$$r_{r,t} = \frac{\sum_{g\in r}\sum_{d\in t} c_{s,g,d,\delta}}{\sum_{g\in r}\sum_{d\in t} 1[\text{valid}_{s,g,d,\delta}]}, \qquad \hat{V}^{\text{eq}}_{r,t} = \hat{k}_{r,\sigma}\, M_{r,t}$$

The rate is the comparable activity indicator (Layer 1 KPI). VIIRS-equivalent cell-days are the familiar display unit. Neither is burned area.

YAAD RAKHO

Pixel ≠ aag. Cell-day ≠ jali zameen. Number ke saath unit aur label hamesha.

Claim ledger (extract)

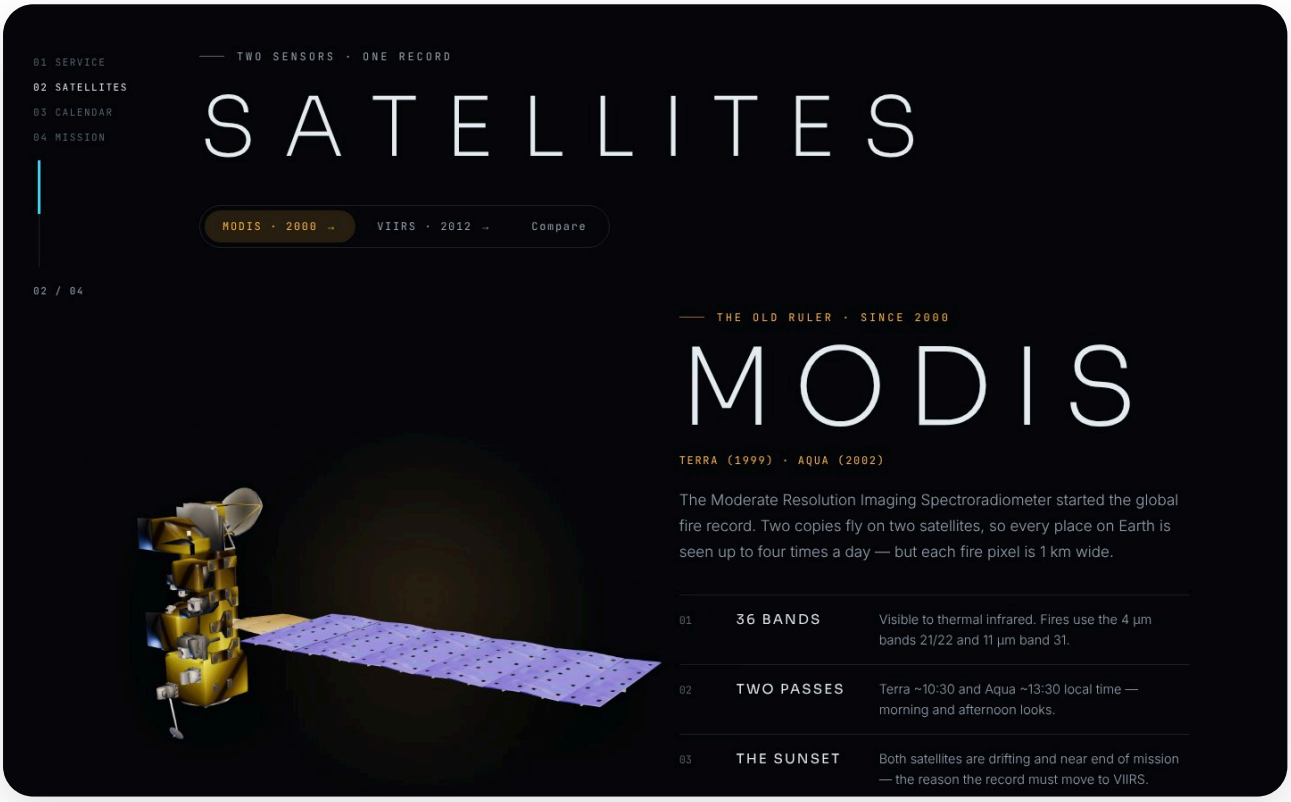
Full ledger: db/csv/claim_ledger.csv (Supabase table claim_ledger). Each claim has value, region, period, unit, URL, verification date, status and test.

ID	CLAIM	VALUE	STATUS	VERIFIED
CL-01	FIRMS MODIS C6.1 archive start	2000-11-01	OFFICIAL DOCUMENTED	2026-10-27
CL-02	FIRMS VIIRS S-NPP archive start	2012-01-20	OFFICIAL DOCUMENTED	2026-10-27
CL-03	FIRMS VIIRS NOAA-20 archive start	2018-04-01	OFFICIAL DOCUMENTED	2026-10-27
CL-04	FIRMS VIIRS NOAA-21 archive start	2023-03-01	OFFICIAL DOCUMENTED	2026-10-27
CL-05	NRT replaced by standard data	~5 months lag	OFFICIAL DOCUMENTED	2026-10-27
CL-06	S-NPP new data delivery ceases	2 Nov 2026 (future as of 27 Oct)	OFFICIAL DOCUMENTED (plan)	2026-10-27
CL-07	Terra end of science	Feb 2027 (NASA); data collection Jan 2027 (CAMS)	OFFICIAL DOCUMENTED (plan)	2026-10-27
CL-08	Aqua end of science	Sep 2027	OFFICIAL DOCUMENTED (plan)	2026-10-27
CL-09	Terra equator crossing by fall 2022	~10:15	OFFICIAL DOCUMENTED	2026-10-27
CL-10	CAMS GFAS added VIIRS	4 Dec 2025	OFFICIAL DOCUMENTED	2026-10-27
CL-11	MOD14A1 FireMask 4 = cloud, 5 = non-fire land, 6 = unknown, 7-9 fire	classes	OFFICIAL DOCUMENTED	2026-10-27
CL-12	VNP14A1 is 1 km daily (not 375 m)	1000 m	OFFICIAL DOCUMENTED	2026-10-27
CL-13	MCD64A1 uses 1 km MODIS active fire as input	yes	OFFICIAL DOCUMENTED	2026-10-27
CL-14	Punjab large farm fires after 3 pm	>90% in 2024-25 (3% in 2021)	SOURCE-DERIVED (secondary report)	2026-10-15
CL-15	Punjab burnt area	~31,447 km2 (2022) → ~20,000 km2 (2025)	SOURCE-DERIVED (secondary report)	2026-10-15
CL-16	Kiln Watch seam reduction 2.9%	self-reported	UNREVIEWED	2026-10-15
CL-17	FireCal LOYO 1.6-5%	self-reported	UNREVIEWED	2026-10-15
CL-18	Gridded area Punjab (Natural Earth)	50,444 km2	SOURCE-DERIVED	2026-10-27
CL-19	VIIRS I4 saturates on ~8% of daytime fire pixels	~8%	OFFICIAL DOCUMENTED (paper, via project notes)	2026-10-15
CL-20	MODIS confidence field is experimental	-	OFFICIAL DOCUMENTED	2026-10-27

PART II

Remote sensing primer

MODIS, VIIRS, orbits, clocks, clouds and products — the physics behind every later formula, in plain words with Hinglish notes.



CHAPTERS IN THIS PART

Chapter 07 · How satellites see fire

Chapter 08 · Orbits, time and the end of MODIS

Chapter 09 · Products, masks and what they do not measure

CHAPTER 7

How satellites see fire

A fire is far hotter than the land around it. Hot objects shine most strongly at short infrared wavelengths, so a pixel containing even a small fire looks much brighter near 4 µm than at 11 µm. Both MODIS and VIIRS look for exactly that contrast.



Satellite renders from the project (illustrative; 3D asset licence to verify; NOAA-21 has no verified public model).

The detection idea in four steps

1. **Measure** brightness temperature in a mid-infrared band (MODIS band 21/22 ≈ 4 µm; VIIRS I-4 ≈ 3.7 µm) and a thermal band (MODIS 31 ≈ 11 µm; VIIRS I-5 ≈ 11.5 µm).
2. **Mask** clouds, water and sun glint; these pixels are not observable for fire.
3. **Compare with neighbours**: a pixel is a candidate if it is much hotter, and hotter at 4 µm than 11 µm, than the valid background around it (contextual test).
4. **Assign confidence**: MODIS gives 0–100 (described by FIRMS as experimental); VIIRS gives low / nominal / high.

Why the size of the pixel matters: a 1 km MODIS pixel must contain a larger or hotter fire to stand out than a 375 m VIIRS pixel, because the fire covers a smaller share of the pixel. This is the physical reason VIIRS detects many more small agricultural fires and why k is usually larger than 1.

$$\lambda_{\max} = \frac{2898\ \mu\text{m K}}{T} \quad \Rightarrow \quad T = 800\,\text{K} : \lambda_{\max} \approx 3.6\,\mu\text{m};\ \ T = 300\,\text{K} : \lambda_{\max} \approx 9.7\,\mu\text{m}$$

HINGLISH

Aag 4 micron par chamakti hai, zameen 10–11 micron par. Dono band ka farak = aag ka sabse bada saboot. Chhota pixel = chhoti aag bhi dikh jaati hai.

	MODIS (TERRA, AQUA)	VIIRS (S-NPP, NOAA-20, NOAA-21)
fire pixel at nadir	~1 km	375 m (I-band)
edge pixel	~2 × 4.8 km	~0.8 × 0.8 km
swath	2,330 km	~3,000 km
mid-IR / thermal	bands 21/22, 31	I-4, I-5
confidence	0–100 %	l / n / h
record starts (FIRMS)	1 Nov 2000 (Terra); Aqua Jul 2002	20 Jan 2012; N20 1 Apr 2018; N21 1 Mar 2023
overpass (LMST)	Terra ~10:30 ↓ (drifted ~10:15), Aqua ~13:30 ↑	~13:30 ↑ (all three)
FRP caveat	1 km mixes fire and background	I-4 saturates on part of daytime fire pixels
planned end	Terra Feb 2027, Aqua Sep 2027 (NASA)	S-NPP data delivery ends 2 Nov 2026

OFFICIAL DOCUMENTED dates read on FIRMS download page, NASA Terra article (29 Dec 2025, upd. 20 Mar 2026) and Earthdata transition page on 27 Oct 2026. Pixel sizes from Giglio et al. 2016 and Schroeder et al. 2014.

Four reasons raw MODIS and VIIRS counts disagree

1 · Pixel size

VIIRS 375 m sees small fires that MODIS 1 km misses. Same field, more dots.

2 · Timing

Terra looks in the morning, Aqua and VIIRS in the early afternoon; drift moves these times; burning after 15:00 is missed by all.

3 · Exposure

Clouds, smoke, swath gaps and bow-tie deletion remove different observations for each sensor.

4 · Processing

Different algorithms, confidence scales, collections (C6.1 vs C2) and NRT vs standard versions.

Cell-day = presence, not size

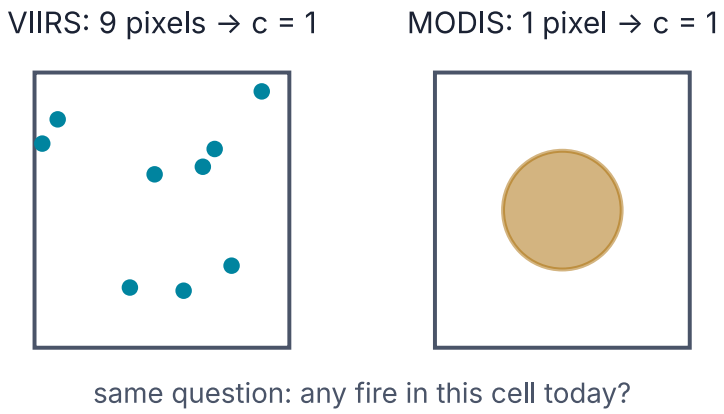


Fig. 7.1. Counting presence per cell-day removes most of the pixel-size effect but not timing or exposure. SIMULATED DEMO

The 2012 seam

When S-NPP VIIRS starts in January 2012, raw counts jump. The jump is an instrument effect. If the calendar simply concatenates MODIS before and VIIRS after, every trend test will 'find' a rise in 2012. Harmonisation exists to remove this artificial step while keeping real change.

Not one seam but three

NOAA-20 (2018) and NOAA-21 (2023) add more afternoon passes. If several VIIRS platforms are merged into one cell-day, exposure rises in 2018 and 2023 too. V2 keeps platforms separate in storage and decides the merge rule explicitly.

COMMON MISTAKE

Calling the 2012 jump a 'fire increase' or calling the post-2021 Punjab drop in counts a 'burning collapse' — both can be measurement effects.

CHAPTER 8

Orbits, time and the end of MODIS

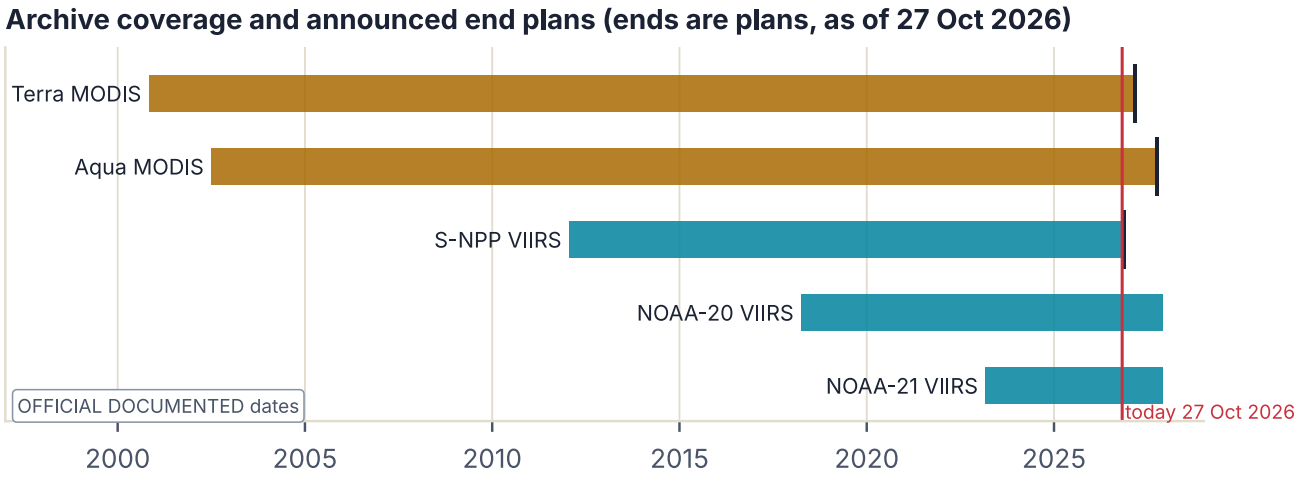


Fig. 8.1. FIRMS archive coverage by platform and the announced end plans (white ticks). S-NPP new-data delivery ends 2 Nov 2026 — six days after this edition; it has not happened yet. OFFICIAL DOCUMENTED

Sun-synchronous orbits and drift

All five platforms are in sun-synchronous polar orbits: each crosses the equator at about the same local solar time every day. Terra (descending, ~10:30) and Aqua (ascending, ~13:30) have stopped orbit-maintenance burns to save fuel, so their crossing times drift. NASA reports Terra crossing at about 10:15 by fall 2022. A fixed 10:30 assumption is therefore wrong for late years; V2 always uses the actual acquisition times stored in FIRMS.

Local solar time is not IST

$$t_{\mathrm{LMST}} = t_{\mathrm{UTC}} + \lambda/15$$

Ludhiana (75.85°E): 08:06 UTC = 13:09 LMST = 13:36 IST. IST is a political clock fixed to 82.5°E. All diurnal analysis uses LMST.

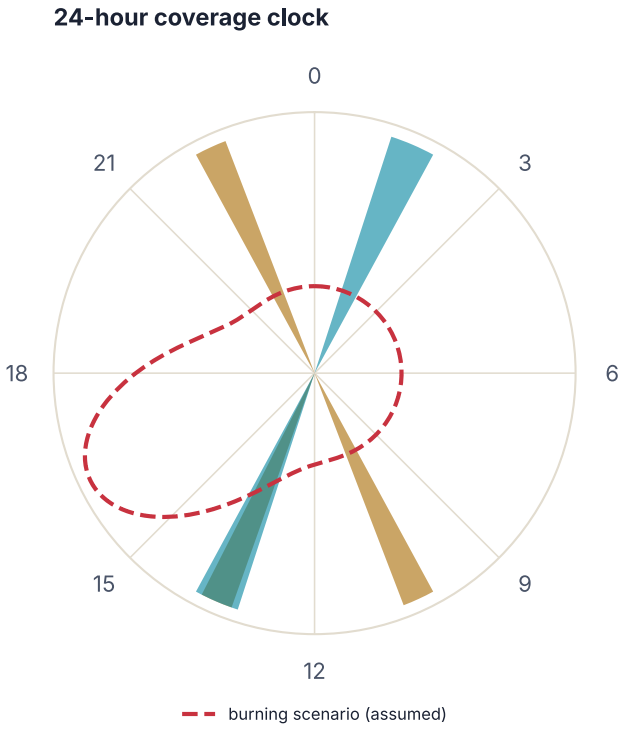


Fig. 8.2. Coverage clock: polar-orbit windows (amber MODIS, cyan VIIRS) vs an assumed late-afternoon burning curve (dashed = scenario, not measurement). SIMULATED DEMO

WHY NOW

With MODIS ending in 2027, the overlap period (2012–2026) is the only time both rulers exist. Calibrating the old record against VIIRS must happen from this overlap; after 2027 it can no longer be extended. CAMS added VIIRS to its global fire assimilation on 4 Dec 2025 for the same reason.

NRT vs standard processing

	NRT	STANDARD (SP)
latency	≈ 3 hours	replaces NRT after ~5 months
use	live map, last-7-days warning	calendar, calibration, validation
FIRMS source	MODIS_NRT, VIIRS_*_NRT	MODIS_SP, VIIRS_SNPP_SP, VIIRS_NOAA20_SP
status label	LIVE NRT	ARCHIVE STANDARD

Rules that follow

- Never mix NRT and SP in one calibration.
- When SP arrives, recompute and version the affected months.
- The UI must show which one a value came from (status chip from meta).
- A fallback to an archive copy is ARCHIVE or CACHED — never 'live' (the v1 failover bug).

Scan geometry

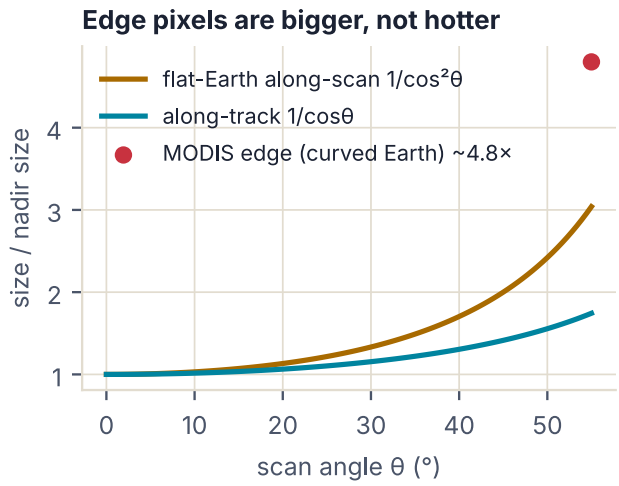


Fig. 8.3. Edge pixels are larger; the footprint is scan × track from FIRMS columns.

Time and date conventions

FIELD	CONVENTION
acq_date, acq_time	UTC, from FIRMS
local day for burning	date in LMST
calendar month	LMST month (state it)
daynight	FIRMS D/N flag (solar zenith based)
weather API	request timezone=UTC (FIREWORK rule)

LEAP YEARS AND MONTH LENGTH

Rates divide by exposure, so a 29-day February is handled automatically; raw counts are not comparable across months of different length.

HINGLISH

NRT jaldi aata hai par kaccha hai; SP 5 mahine baad pakka. Calendar hamesha SP se.

CHAPTER 9

Products, masks and what they do not measure

Four products, four different things. Mixing them silently is the most common source of wrong numbers in fire calendars.

PRODUCT	GEOMETRY	WHAT IT MEASURES	WHAT IT DOES NOT
FIRMS VIIRS 375 m (VNP14IMG / VJ114IMG / VJ214IMG)	points (pixel centres)	where/when a fire was hot at overpass	fire size, burned area, clear-sky days
FIRMS MODIS C6.1 (MCD14ML)	points (1 km pixels)	same, coarser	same
MOD14A1 / MYD14A1 / VNP14A1 daily	1 km raster, daily	FireMask class per pixel-day → exposure	375 m detections (VNP14A1 is 1 km)
MCD64A1 burned area	500 m raster, monthly	burn date of scars	independent check (uses active fire input); small field burns often missed

MOD14A1 / MYD14A1 FireMask

VALUE	MEANING (GEE CATALOGUE, 27 OCT 2026)	V2 EXPOSURE STATE
0	not defined on catalogue page	unknown
1	not processed (obsolete)	unknown
2	not processed (other reason)	unknown
3	non-fire water	observed_no_fire
4	cloud (land or water)	not_observable
5	non-fire land	observed_no_fire
6	unknown (land or water)	unknown
7	fire, low confidence	fire_low
8	fire, nominal confidence	fire_nominal
9	fire, high confidence	fire_high

VNP14A1 FireMask

VALUE	MEANING	V2 EXPOSURE STATE
0	not processed (no/poor data)	unknown
1	not processed (bowtie deletion)	unknown
2	unused	unknown
3	water	observed_no_fire
4	cloud	not_observable
5	land	observed_no_fire
6	unclassified	unknown
7	low-confidence fire	fire_low
8	nominal-confidence fire	fire_nominal
9	high-confidence fire	fire_high

TWO DOCUMENTED TRAPS

Class 6 is **unknown**, not clear land — do not count it as an observation opportunity. Class 0 means different things in the two products. The GEE pages also show inconsistent end dates (e.g. 2025 vs 2026) — check the collection itself before a run.

Exposure: from FireMask to a denominator

$$E_{s,g,m} = \sum_{d \in m} 1[\text{FireMask}_{s,g,d} \in \{3,5,7,8,9\}], \qquad \text{cloud}_{s,g,m} = \sum_{d} 1[\text{FireMask} = 4]$$

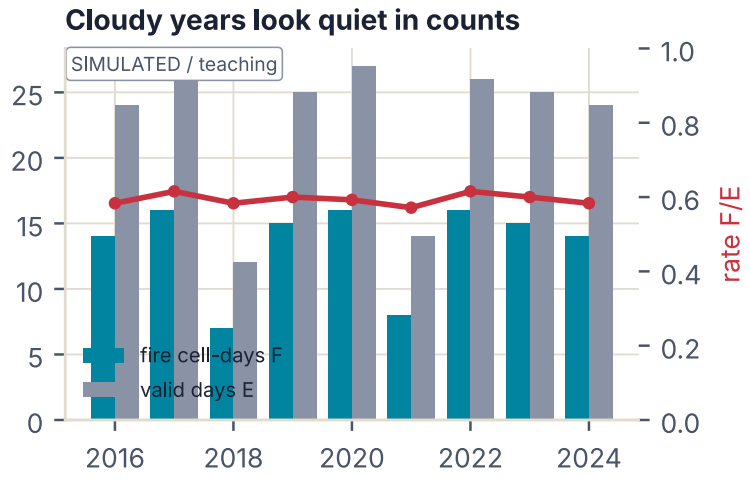


Fig. 9.1. A cloudy month (2018, 2021) looks quiet in counts; the rate removes most of that effect. SIMULATED DEMO

Subtleties the rate must survive

- Thick smoke may be flagged as cloud exactly when burning peaks.
- One clear composite day can hide two or more passes — it is one opportunity, not several (R2 tests this).
- A 0.1° cell holds ~100 MODIS 1 km pixels; exposure is a mean over pixels — store it as a fraction too.
- VNP14A1 is 1 km, so VIIRS point detections and VIIRS exposure come at different resolutions.

Myths to unlearn

"More dots = more fire"

More dots can mean a finer sensor, more passes, or fewer clouds.

"Counts fell 90%, burning fell 90%"

Burning moved after the overpasses (iFOREST 2025, secondary report). Burned area fell much less.

"Burned area validates hotspots"

MCD64A1 uses MODIS active fire as an input; agreement is partly built in.

"Confidence 0.6 is 60%"

VIIRS classes are categories; the numbers in old code were labels.

"Bigger pixel, bigger fire"

Edge pixels are bigger footprints, not bigger fires.

"No detection = no fire"

No detection under cloud or outside a pass is *unknown*.

Burned area and its limits

MCD64A1 maps burn scars at 500 m each month, giving a burn date, an uncertainty in days and QA flags. It measures a different physical quantity (area burned) from hotspots (heat at an instant), so it is shown in its own unit and never added to cell-days.

BAND	MEANING
BurnDate	0 unburned, 1–366 day of year
Uncertainty	days
QA	land/water, valid data, shortened period, relabelled; special codes for unburned cells
FirstDay / LastDay	reliable detection window

- Latency: monthly and lagged → recent months are PENDING, not low.
- Small agricultural burns (< 25 ha) are often missed at 500 m.
- Sentinel-2-based burned area used by iFOREST is another method; compare trends only.

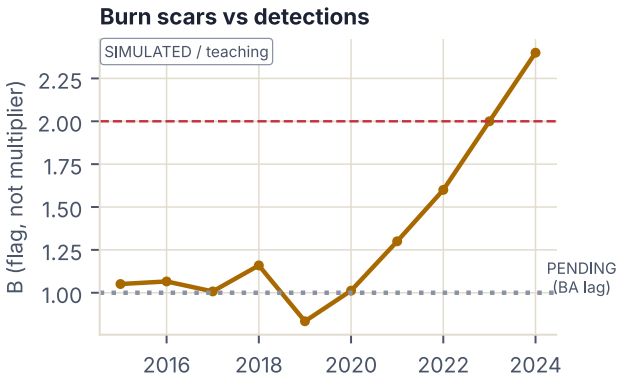


Fig. 9.2. B compares burn scar per detected cell-day with a reference period; the last year is PENDING while BA is lagged. SIMULATED DEMO

HOW V2 USES BURNED AREA

Only as a disagreement diagnostic (Layer 3, F46) and as a separate physical-area view in the calendar. It never rescales hotspot counts.

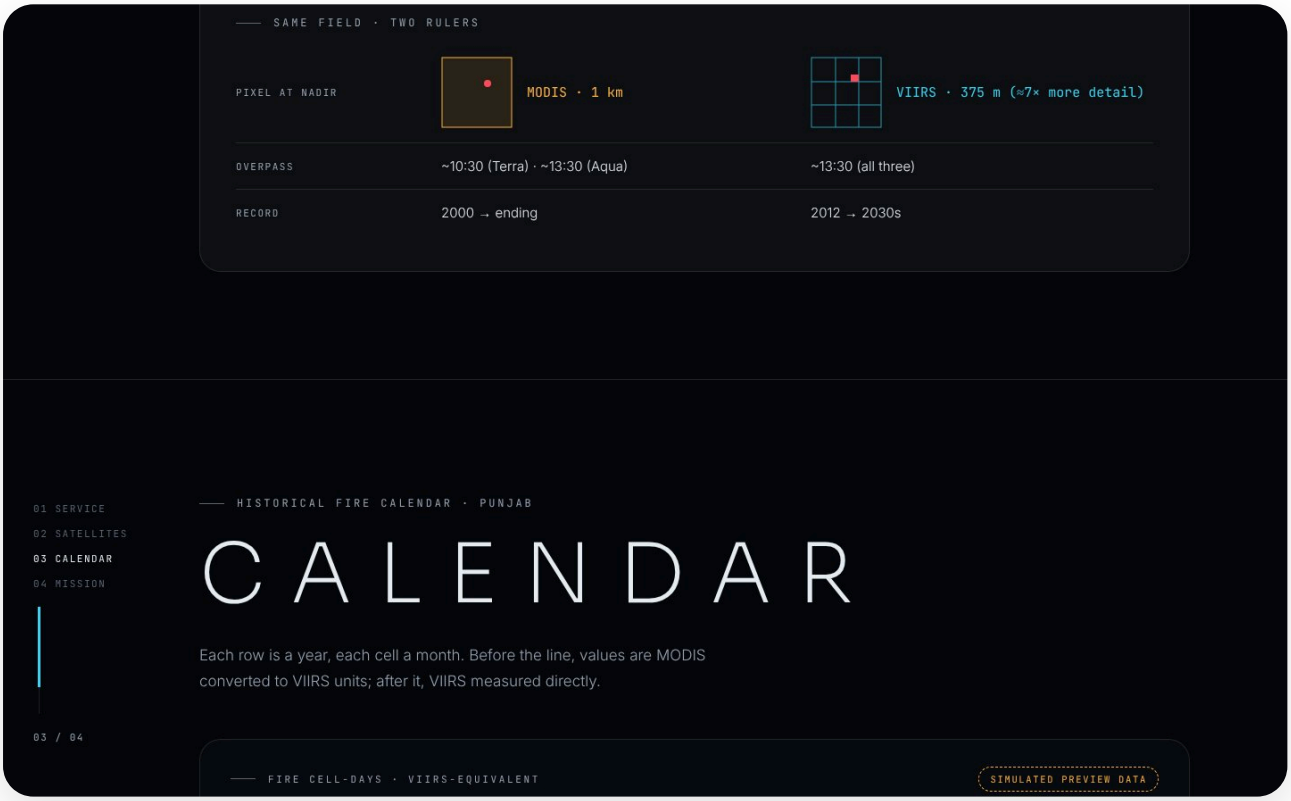
Context data (never fire ground truth)

DATASET	USED FOR	LIMIT
ESA WorldCover 10 m (2021 epoch)	crop / tree / built-up context (L2)	fixed epoch; classification errors
OpenStreetMap (Overpass)	industry, kilns, flares (L2)	India kilns under-mapped → NULL not 0
Open-Meteo / NASA POWER	wind, humidity, rain panels	context only, not a class input
OpenAQ v3	PM2.5 impact panel	sparse stations
CelesTrak GP	overpass prediction	use FIRMS acq_time for history
Natural Earth / geoBoundaries	regions and districts	de facto boundaries; official briefs need official boundaries

PART III

Science & the formula registry

One pipeline, one unified measurement model, and all 46 formulas re-derived, audited and re-tested — each on a two-page spread.



CHAPTERS IN THIS PART

Chapter 10 · End-to-end pipeline and the unified model

Chapter 11 · Registry index and the ten biggest risks

Chapter F01–F46 · Formula spreads

Chapter 12 · Baseline ladder and validation gates

Chapter 13 · How the formulas combine

CHAPTER 10

End-to-end pipeline and the unified model

Raw immutable data → normalised → gridded → exposure-joined → calibrated → uncertainty → validated → published, with a config hash at every arrow. Science is computed once in the backend; charts never recompute it.

01	02	03	04	05	06	07	08	09
Ingest	**Normalise**	**Grid**	**Cell-days**	**Exposure**	**Calibrate**	**Uncertainty**	**Validate**	**Publish**
FIRMS SP archive + NRT, checksum, columns as returned	sensor-native QA, F07 rule grid, type mask	F03 rounded index, F04 area, local day	F05 per platform × D/N	F09 from FireMask {3,5,7,8,9}	F14 NB offset model, F18 pooling	F24 years, F25 log-k, F26 small n	F36 LOYO, F33–35 seams, F23	calendar_value + meta + hash

Unified candidate measurement model

PROPOSED / UNVALIDATED

For platform s, grid cell g and time bin t, let Y be the observed fire cell-days, E the valid exposure, A the latent comparable fire activity and p the detection efficiency of that platform under those conditions:

$$Y_{s,g,t} \sim \mathrm{NegBin}(\mu_{s,g,t}, \theta_s), \qquad \mu_{s,g,t} = E_{s,g,t}\, p_{s,g,t}\, A_{g,t}$$

$$\log p_{s,g,t} = \gamma_s + \beta_{s,\delta} + \beta_s^{\top} x_{g,t} + \eta_s(\mathrm{year}), \qquad \log A_{g,t} = \alpha_g + f_{\mathrm{season}}(t) + u_{g,t}$$

x holds view angle, land cover and local time; η captures orbit drift. Harmonisation is the ratio of efficiencies:

$$k_\sigma = \frac{p_{V,\sigma}}{p_{M,\sigma}} = e^{\gamma_V - \gamma_M + \beta_{V,\delta} - \beta_{M,\delta}}$$

IDENTIFIABILITY WARNING

From hotspots alone, A and p are only identified up to a constant: doubling every p and halving A gives the same data. What we can estimate is the **ratio** of efficiencies between sensors where both observe the same activity (overlap), and changes relative to a reference sensor. So the product reports a comparative index (VIIRS-equivalent), never 'true fire activity'.

Assumptions that make k usable

1. Both sensors see the same A in a stratum and period (paired overlap).
2. k is stable outside the overlap, or its drift is modelled (η).
3. Exposure E is measured consistently for both.
4. Static sources are removed before calibration.

STABLE MVP INSIDE THIS MODEL

Fix A by stratum, drop covariates → exposure-offset Poisson/NB with a sensor term = stratified ratio of exposure rates (baseline B3). More complex variants are used only if they win on held-out data (Ch. 12).

Module boundaries and data layers

MODULE	RESPONSIBILITY	FORMULAS	OUTPUT TABLE
ingest/	source adapters, auth from .env, retries, raw checksum	-	raw files (immutable)
normalize/	columns, sensor-native QA, type mask, local time	F01, F02, F07	fire_detection
grid/	cell index, area, region membership	F03, F04	grid_cell, region_cell
coverage/	exposure from daily FireMask, overpass times	F09, F45	cell_month
calibrate/	overlap pairing, NB offset model, pooling	F10–F19	calibration
uncertainty/	bootstrap, Monte Carlo, exact Poisson	F16, F24–F26	calendar_value.lo/hi
validate/	LOYO, seams, BA agreement, coverage	F23, F33–F37	validation_run
classify/	sites, features, abstaining classifier	F08, F21, F39–F44	source tables
blindspot/	B ratio, overpass scenarios	F45, F46	blindspot
summarize/	calendar, anomalies, trends, briefs	F27–F32	calendar_value, RunReport.json

Data layers

1. **raw**: files exactly as downloaded + sha256
2. **normalised**: Parquet, immutable per version
3. **curated**: cell_day, cell_month aggregates
4. **results**: calibration, calendar_value with method_version + config_hash
5. **served**: cached API payloads {data, meta}

Every published number carries

UNIT	e.g. VIIRS-eq cell-days
INTERVAL	lo, hi, level, type
COVERAGE	valid days / possible
STATUS	ok · not_observable · unavailable · pending · insufficient
PROVENANCE	one of the five labels
VERSION	method_version + config_hash
SOURCE	product + processing (SP/NRT)

CHAPTER 11

Registry index

v1 status from the Build Spec (E existing, T tested on synthetic data, P proposed). Verdict from this audit. Each formula also exists as `formula_registry/Fxx.yaml` and as a row of the Supabase table `formula`.

ID	FORMULA	LAYER	V1	AUDIT VERDICT	V2 STATUS
F01	Local solar time of an overpass	L1 / L3	E	OK + EXTEND	RUN / TESTED LOCALLY
F02	Pixel growth with scan angle	L1	P	REVISE	REVISE
F03	Grid cell index	L1	E	FIX	RUN / TESTED LOCALLY
F04	Area of a grid cell on the sphere	L1	P	OK + VALIDATED	RUN / TESTED LOCALLY
F05	Fire cell-day indicator	L1	E	REVISE	REVISE
F06	Monthly cell-day count	L1	E	FIX (structural)	FIX
F07	Equal-confidence filter	L1	P	FIX	FIX
F08	Persistence (static-source mask)	L2	E	FIX	FIX
F09	Clear-sky (exposure) fire rate	L1 / L4	T	REVISE	REVISE
F10	Monthly overlap ratio	L1	E	FIX	FIX
F11	Median of ratios	L1	ET	REVISE	REVISE
F12	Harmonised MODIS value	L1	E	REVISE	REVISE
F13	Separate day and night factors	L1	P	REVISE	REVISE
F14	Poisson GLM with exposure offset	L1	T	REVISE	REVISE
F15	Negative-binomial variance	L1	P	OK + RULE	RUN / TESTED LOCALLY
F16	Delta-method interval for a ratio	L1	P	REVISE	REVISE
F17	Shrink local factor toward global	L1	E	REVISE	REVISE
F18	Empirical Bayes partial pooling of log k	L1	T	FIX	FIX
F19	Power law vs ratio (model choice)	L1	T	FIX	FIX
F20	Detection-probability curve (FRP50)	L1	T	FIX	FIX
F21	Log transform of FRP	L2	P	OK	OK
F22	Quantile mapping	L1 (FRP only)	T	REVISE	REVISE
F23	Bland–Altman agreement	Validation	T	REVISE	REVISE
F24	Block bootstrap over years	L1 / L4	T	FIX	FIX

ID	FORMULA	LAYER	V1	AUDIT VERDICT	V2 STATUS
F25	Monte Carlo error propagation	L1	T	FIX	FIX
F26	Exact Poisson interval (Garwood)	L1	P	OK	RUN / TESTED LOCALLY
F27	Robust anomaly z-score	L4	ET	FIX	FIX
F28	Month baseline and spread	L4	E	FIX	FIX
F29	Season start, peak and end (10/90%)	L4	E	REVISE	REVISE
F30	Mann–Kendall trend test	L4	T	FIX	FIX
F31	Hamed–Rao autocorrelation correction	L4	T	OK + LIMIT	RUN / TESTED LOCALLY
F32	Sen's slope	L4	E	OK	RUN / TESTED LOCALLY
F33	Pettitt change-point test	Validation	T	REVISE	REVISE
F34	PELT multiple change points	Validation	P	REVISE	REVISE
F35	Seam ratio	Validation	E	REVISE	REVISE
F36	Held-out error (LOYO and blocked CV)	Validation	P	FIX	FIX
F37	Benjamini–Hochberg false discovery rate	L4	T	OK	OK
F38	Kalman fusion (local-level model)	Experimental	T	FIX	FIX
F39	Split conformal prediction sets	L2	T	REVISE	REVISE
F40	Haversine distance	L2	P	OK	RUN / TESTED LOCALLY
F41	DBSCAN fire clustering	L2	P	REVISE	REVISE
F42	Distance-decay exposure (IE / AE)	L2	P	REVISE	REVISE
F43	Context log-ratio L	L2	E	OK	OK
F44	Night fraction	L2	P	REVISE	REVISE
F45	Diurnal fraction seen by overpasses	L3	T	FIX	FIX
F46	Burned-area to detection ratio B	L3	P	FIX	FIX

The ten biggest mathematical risks

#	RISK	WHERE	CONSEQUENCE IF IGNORED
1	missing coverage stored as 0	calendar_matrix, F06, F08	fake quiet months; trends biased
2	MODIS counts used as exposure	F14, F10	zero-month bias, errors-in-variables
3	k assumed stationary through drift	F12, F13	pre-2012 values biased by drift years
4	target year inside its own baseline	F27, F28	anomalies understated
5	log of zero	F19, F27 (ln rate)	crashes or silently dropped months
6	interval level mislabelled	F24	90% band read as 95%
7	independence assumed for paired sensors	F16, F25	wrong band width
8	ties ignored in trend tests	F30	p-values too small
9	scenario shown as measurement	F45, F46	false blind-spot magnitudes
10	floating-point grid edges, antimeridian	F03	detections in the wrong cell

How to read a formula spread

Left page · Science Midnight

- number, family, layer and the formula's neighbours in its family
- one-paragraph purpose
- **V1** equation as registered (grey) and **V2** revised equation (accent)
- symbols with units, assumptions
- a figure generated by code in this atlas
- where the output is used on the website

Right page · Laboratory Light

- audit verdict and the finding
- worked example (numbers re-computed where marked RUN)
- executable pseudocode
- edge conditions and tests
- validation plan, sources, Hinglish explanation
- the audit checks actually executed

The seven families and their layer

FAMILY	FORMULAS	FEEDS
Time, space and counting	F01–F09	Layer 1 inputs
Calibration between sensors	F10–F23	Layer 1 harmonisation
Uncertainty	F24–F26	bands everywhere
Anomalies and seasons	F27–F29	Layer 4
Trends and seam tests	F30–F37	Layer 4 + validation
Fusion and AI	F38–F41	experimental + Layer 2 sites
Source attribution and blind spot	F42–F46	Layers 2 and 3

PADHNE KA TAREEKA

Bayan page: formula kya hai, kyun hai. Dayan page: galti kya thi, theek kaise kiya, code aur test. Har number ke saath label dekho.

FORMULA 01 / 46 · TIME, SPACE AND COUNTING · LAYER L1 / L3

F01 Local solar time of an overpass

F01 F02 F03 F04 F05 F06 F07 F08 F09

Converts the FIRMS UTC acquisition time into the Sun's time at the pixel, so that overpass times, day/night strata and burning-hour comparisons are expressed on one clock everywhere on Earth.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$t_{\mathrm{LST}} = t_{\mathrm{UTC}} + \frac{\lambda}{15^\circ\,\mathrm{h}^{-1}}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$t_{\mathrm{LMST}} = \left(t_{\mathrm{UTC}} + \frac{\lambda}{15}\right) \bmod 24, \qquad t_{\mathrm{LAST}} = t_{\mathrm{LMST}} + \frac{E(d)}{60}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
t_UTC	acquisition time from acq_time (HHMM)	hours
λ	pixel longitude, east positive	degrees
E(d)	equation of time for day-of-year d	minutes, about −14 to +16

DERIVATION, STEP BY STEP

1. Earth turns 360° in 24 h, so the Sun moves 15° of longitude per hour.
2. At longitude λ the Sun is λ/15 hours ahead of Greenwich.
3. Add that offset to UTC and wrap with mod 24.
4. Optionally add the equation of time $E(d)$ for apparent (sundial) time.

ASSUMPTIONS

- Local mean solar time is enough for binning; apparent solar time differs by up to ~16 min.
- acq_time is the granule/scan time, not the moment the fire started.

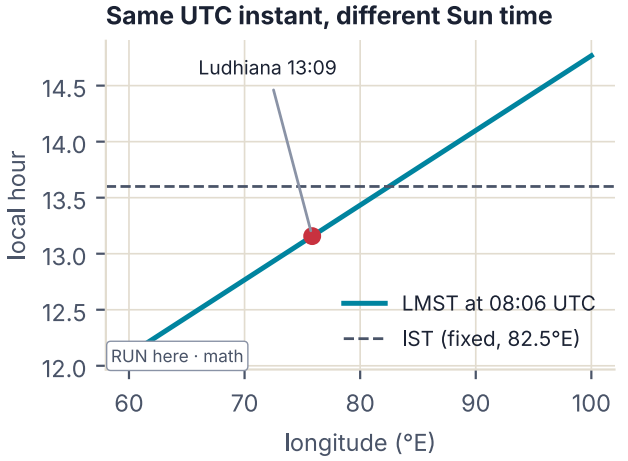


Fig. F01. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

A pass at 13:09 LMST is 'early afternoon' anywhere on Earth; that is why Aqua and VIIRS are comparable.

WHERE THE OUTPUT GOES

Explorer overpass card, Blind Spot clock, Sensor Lab day/night split.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F01 V2**	**Output**	**Shown on**
t_UTC, λ, E(d)	raw FIRMS / FireMask	Local solar time of an overpass	A pass at 13:09 LMST is 'early afternoon' anywhere on Earth; that is why Aqua and VIIRS ar	Explorer overpass card, Blind Spot clock, Sensor Lab day/night split.

V1 → V2 IN ONE LINE

Equation correct.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F01 · AUDIT, EXAMPLE, CODE, TESTS

RUN / TESTED LOCALLY

AUDIT VERDICT · OK + EXTEND

Equation correct. Added modulo-24 and the equation-of-time note. Local solar time is not IST: Ludhiana (75.85°E) solar noon is ~12:06 LMST while IST is fixed to 82.5°E.

WORKED EXAMPLE

08:06 UTC at 75.85°E → 8.10 + 75.85/15 = 8.10 + 5.057 = 13.157 h = 13:09 LMST (run here: 13:09).

EXECUTABLE PSEUDOCODE

```
def lmst(utc_h, lon):
    return (utc_h + lon / 15.0) % 24.0
```

EDGE CONDITIONS

- Longitude −180/180 must give the same answer.
- Times near midnight wrap to the previous/next local day: assign the local date, not the UTC date, when binning by local day.

WHY THIS, NOT THAT

Time zones (IST) are political and constant over a country; burning behaviour and satellite overpasses follow the Sun. Using IST would shift Ludhiana by ~27 min and Kolkata the other way.

COMMON DOUBTS

Q. Why not use the daynight column only?
It is a two-class flag; overpass and burning-hour questions need a continuous clock.

Q. Which date does a 23:00 UTC pass belong to?
Use the LMST date for 'burning day' analysis; record the rule in config.

TESTS (UNIT / PROPERTY)

- lmst(8.1, 75.85) == 13.157 ± 0.001
- lmst(23, 30) == 1.0 (wrap)

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F01 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- FIRMS attribute table (acq_time UTC)
- NOAA solar calculator (equation of time)

HINGLISH MEIN SAMJHO

Satellite ki ghadi UTC hai, aag ki ghadi Sooraj ki. Longitude/15 jodo to Sooraj ka time milta hai — yeh IST nahi hai.

IMPLEMENTATION CONTRACT (V2)

MODULE	`normalize/, grid/, coverage/`
FUNCTION	`local_solar_time_of_an_overpass_v2()`
INPUTS	t_UTC [hours]; λ [degrees]; E(d) [minutes, about −14 to +16]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F02, F03
ACCEPTANCE	lmst(8.1, 75.85) == 13.157 ± 0.001; lmst(23, 30) == 1.0 (wrap)
MAIN FAILURE MODE	Longitude −180/180 must give the same answer.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
08:06 UTC at 75.85E	13:09	13:09	OK

YOU MAY SAY

'X fire cell-days were detected per valid observation day in this cell-month.'

YOU MAY NOT SAY

'X fires burned' or 'X hectares burned'; '0' for an unobserved month.

Chart spec. Explorer cell inspector rows + calendar cell tooltip; unit in every label; hatched null.

FORMULA 02 / 46 · TIME, SPACE AND COUNTING · LAYER L1

F02 Pixel growth with scan angle

F01 F02 F03 F04 F05 F06 F07 F08 F09

Explains why one detection near the swath edge represents a much larger footprint than one at nadir, and gives the per-detection footprint area used for sensitivity checks (equal-effective-swath experiments).

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\frac{s(\theta)}{s_0} \approx \frac{1}{\cos^2\theta}, \qquad \frac{t(\theta)}{t_0} \approx \frac{1}{\cos\theta}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$A_{\text{pix}} = \text{scan} \times \text{track}$ (use FIRMS columns, km^2), flat-Earth $1/\cos^2\theta$ is a teaching approximation

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
θ	view (scan) angle	rad
s, t	along-scan and along-track pixel size	km
scan, track	FIRMS columns giving actual pixel dimensions	km

DERIVATION, STEP BY STEP

1. A scanner sweeps a fixed angular step; the ground distance per step grows with slant range.
2. On a flat Earth, along-scan size grows like $1/\cos^2\theta$ and along-track like $1/\cos\theta$.
3. Earth curvature makes the slant range longer still near the edge, so real MODIS growth is larger.
4. FIRMS reports the actual scan and track sizes, so area = scan × track.

ASSUMPTIONS

- Earth curvature makes real edge growth larger than the flat formula (MODIS ~2×4.8 km at 55°).
- VIIRS aggregation/bow-tie deletion keeps its growth much smaller than MODIS.

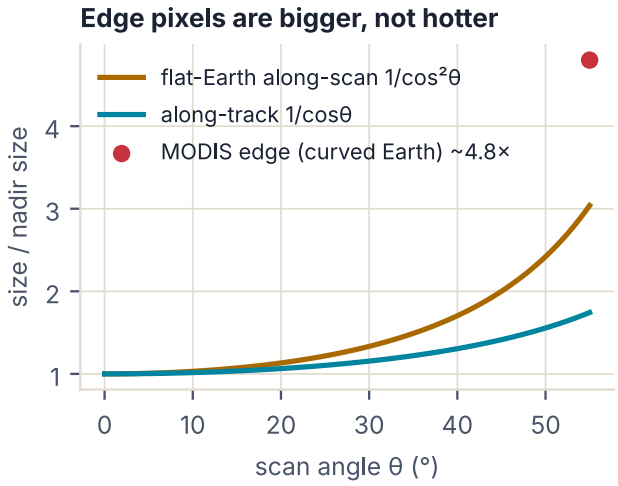


Fig. F02. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Use the footprint to explain detection differences and to test equal-swath sensitivity — not to estimate fire size.

WHERE THE OUTPUT GOES

Sensor Lab: 'why the dots differ' panel; Learn lesson 2.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F02 V2	04 Output	05 Shown on
θ, s, t, scan, track	F01	Pixel growth with scan angle	Use the footprint to explain detection differences and to test equal-swath sensitivity — n	Sensor Lab: 'why the dots differ' panel; Learn lesson 2.

V1 → V2 IN ONE LINE

The $1/\cos^2$ law is a flat-Earth approximation and underestimates MODIS edge growth.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F02 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

The $1/\cos^2$ law is a flat-Earth approximation and underestimates MODIS edge growth. FIRMS already supplies scan and track, so use them. A bigger pixel does not mean a bigger fire.

WORKED EXAMPLE

MODIS nadir 1.0×1.0 km; FIRMS edge rows show scan≈4.8, track≈2.0 → footprint ≈9.6 km². VIIRS 375 m edge ≈0.8×0.8 km.

EXECUTABLE PSEUDOCODE

```
df['pix_km2'] = df.scan * df.track
sensitivity = df[df.scan ⩽ scan_max]
```

EDGE CONDITIONS

- Do not filter scan ≤ 2 km for MODIS without testing: it removes most of the swath and changes revisit frequency.

WHY THIS, NOT THAT

Dropping edge pixels (scan ≤ 2 km) makes pixels more alike but removes many observations and changes revisit, so it is a sensitivity test, not a default.

COMMON DOUBTS

Q. Does a big edge pixel mean a big fire?
No — the same small fire fills a smaller share of a bigger pixel and is harder to detect.

Q. Is VIIRS free of this?
Mostly: its aggregation scheme keeps edge growth to about 2× instead of ~10×.

TESTS (UNIT / PROPERTY)

- pix_km2 ≥ nominal area
- Report % of detections removed by any scan filter.

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F02 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Giglio et al. 2016 (MODIS C6)
- Schroeder et al. 2014 (VIIRS 375 m)

HINGLISH MEIN SAMJHO

Kinare ka pixel bada hota hai, aag nahi. Pixel ka area FIRMS ke scan×track se lo.

IMPLEMENTATION CONTRACT (V2)

MODULE	`normalize/, grid/, coverage/`
FUNCTION	`pixel_growth_with_scan_angle_v2()`
INPUTS	θ [rad]; s, t [km]; scan, track [km]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F03, F04
ACCEPTANCE	pix_km2 ≥ nominal area; Report % of detections removed by any scan filter.
MAIN FAILURE MODE	Do not filter scan ≤ 2 km for MODIS without testing: it removes most of the swath and changes revisit frequency.

YOU MAY SAY

'X fire cell-days were detected per valid observation day in this cell-month.'

YOU MAY NOT SAY

'X fires burned' or 'X hectares burned'; '0' for an unobserved month.

Chart spec. Explorer cell inspector rows + calendar cell tooltip; unit in every label; hatched null.

FORMULA 03 / 46 · TIME, SPACE AND COUNTING · LAYER L1

F03 Grid cell index

F01 F02 F03 F04 F05 F06 F07 F08 F09

Assigns every detection to a fixed, team-chosen comparison grid (0.1°). The grid is an analysis decision, not a NASA-provided boundary.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$i = \left\lfloor \frac{\varphi + 90}{\Delta} \right\rfloor, \quad j = \left\lfloor \frac{\lambda + 180}{\Delta} \right\rfloor$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$i = \min\left(\left\lfloor \mathrm{round}_9 \frac{\varphi + 90}{\Delta} \right\rfloor, \frac{180}{\Delta} - 1 \right), \quad j = \min\left(\left\lfloor \mathrm{round}_9 \frac{\mathrm{wrap}(\lambda) + 180}{\Delta} \right\rfloor, \frac{360}{\Delta} - 1 \right)$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
φ, λ	latitude, longitude	degrees
Δ	cell size	degrees (0.1)
round_9	round to 9 decimals before floor	-

DERIVATION, STEP BY STEP

1. Shift latitude by +90 and longitude by +180 so both start at 0.
2. Divide by the cell size Δ.
3. Round to 9 decimals to remove binary floating-point error.
4. Floor to integers and clamp to the last row/column.

ASSUMPTIONS

- Same grid for every sensor and year.
- Longitude wrapped into [−180, 180).

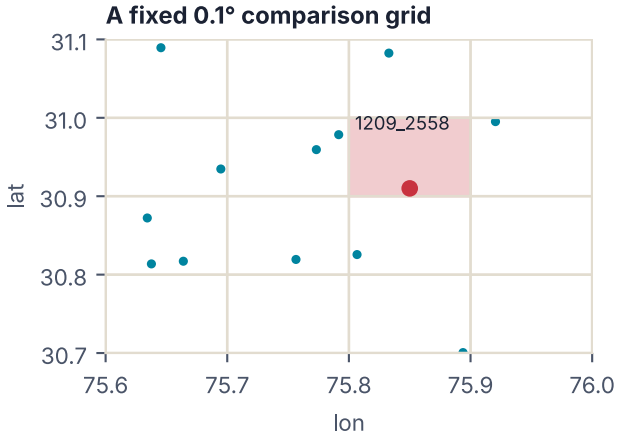


Fig. F03. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Cell 1209_2558 is the same box for every sensor and year; every later count is built on it.

WHERE THE OUTPUT GOES

Explorer cell id chip; Studio grid slider.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F03 V2	04 Output	05 Shown on
φ, λ, Δ, round_9	F01, F02	Grid cell index	Cell 1209_2558 is the same box for every sensor and year; every later count is built on it	Explorer cell id chip; Studio grid slider.

V1 → V2 IN ONE LINE

RUN here: plain floor misassigns 643 of 1,800 latitude edges and 1,287 of 3,600 longitude edges at 0.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F03 · AUDIT, EXAMPLE, CODE, TESTS

RUN / TESTED LOCALLY

AUDIT VERDICT · FIX

RUN here: plain floor misassigns 643 of 1,800 latitude edges and 1,287 of 3,600 longitude edges at 0.1° because of binary floating point (e.g. 28.1°N → row 1180 instead of 1181). Also λ = 180 produces column 3600, outside 0–3599.

WORKED EXAMPLE

(30.91, 75.85) → i = ⌊1209.1⌋ = 1209, j = ⌊2558.5⌋ = 2558 → cell 1209_2558 (matches v1).

EXECUTABLE PSEUDOCODE

```
row = floor(round((lat + 90) / d, 9))
col = floor(round((wrap(lon) + 180) / d, 9))
clamp row, col
```

EDGE CONDITIONS

- Poles: lat = 90 clamps to last row.
- Antimeridian regions: store cell sets, not west<east bboxes.

WHY THIS, NOT THAT

Pixel-to-pixel matching fails because footprints differ and move each orbit. A fixed grid compares 'was there fire in this box today' for both sensors. H3 is useful for display but not equal-area either.

COMMON DOUBTS

Q. Why 0.1°?
≈11 km: larger than both pixels and their geolocation error, small enough for districts. It is our decision — test 0.05° and 0.25° in sensitivity runs.

Q. What about points exactly on an edge?
They go to the upper/right cell consistently once rounding is applied.

TESTS (UNIT / PROPERTY)

- all 0.1° edges map to their own row (1800/1800)
- cell(0,180) == cell(0,-180)

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F03 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Team decision (DEC-03)

HINGLISH MEIN SAMJHO

Poori duniya par graph paper. Computer ka floor kabhi 1180.9999 bana deta hai — pehle round karo.

IMPLEMENTATION CONTRACT (V2)

MODULE	`normalize/, grid/, coverage/`
FUNCTION	`grid_cell_index_v2()`
INPUTS	φ, λ [degrees]; Δ [degrees (0.1)]; round_9 [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F04, F05
ACCEPTANCE	all 0.1° edges map to their own row (1800/1800); cell(0,180) == cell(0,-180)
MAIN FAILURE MODE	Poles: lat = 90 clamps to last row.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
cell of (30.91, 75.85)	1209_2558	1209_2558	OK
boundary 28.1N exact (643 of 1800 lat edges and 1287 of 3600 lon edges misassigned by plain floor)	v1 floor gives row 1180	v2 rounded floor gives row 1181	FIX
longitude 180 and 190	v1 col for 180 = 3600	v2 col = 3599 ; 190E wraps to col 100	FIX

YOU MAY SAY

'X fire cell-days were detected per valid observation day in this cell-month.'

YOU MAY NOT SAY

'X fires burned' or 'X hectares burned'; '0' for an unobserved month.

Chart spec. Explorer cell inspector rows + calendar cell tooltip; unit in every label; hatched null.

FORMULA 04 / 46 · TIME, SPACE AND COUNTING · LAYER L1

F04 Area of a grid cell on the sphere

F01 F02 F03 F04 F05 F06 F07 F08 F09

True cell area for densities, area-weighted roll-ups and fair comparison between latitudes. Region area uses the fraction f_g of each cell inside the polygon.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$A = R^2 \, \Delta\lambda \, (\sin \varphi_2 - \sin \varphi_1)$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$A = R^2 \, \Delta\lambda \, (\sin \varphi_2 - \sin \varphi_1), \quad R = 6371.0088 \text{ km}; \quad A_{\text{region}} = \sum_g f_g A_g$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
R	mean Earth radius	km
Δλ	cell width	rad
φ_1, φ_2	south/north edges	rad
f_g	share of cell inside region	0–1

DERIVATION, STEP BY STEP

1. Area element on a sphere: $dA = R^2 \cos\varphi \, d\varphi \, d\lambda$.
2. Integrate λ over Δλ and φ from φ_1 to φ_2.
3. $\int \cos\varphi \, d\varphi = \sin\varphi$ gives $R^2 \Delta\lambda (\sin\varphi_2 - \sin\varphi_1)$.
4. For a region, sum cell areas times the share of each cell inside the polygon.

ASSUMPTIONS

- Sphere; WGS84 ellipsoid differs by <0.5% at these sizes.

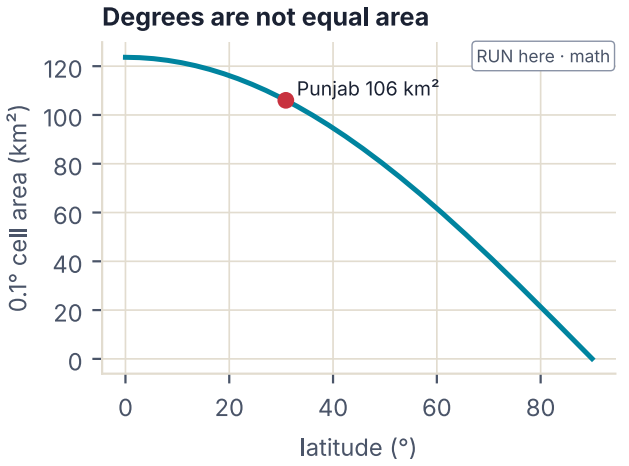


Fig. F04. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Validated on real data: Punjab's gridded area (50,444 km²) matches its published area within 0.2%.

WHERE THE OUTPUT GOES

Explorer inspector 'area', density colour mode.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F04 V2	04 Output	05 Shown on
R, Δλ, φ_1, φ_2, f_g	F02, F03	Area of a grid cell on the sphere	Validated on real data: Punjab's gridded area (50,444 km²) matches its published area with	Explorer inspector 'area', density colour mode.

V1 → V2 IN ONE LINE

RUN here on real Natural Earth polygons: gridded areas Punjab 50,444 km², Haryana 43,997 km², Delhi 1,481 km², Mato Grosso 906,656 km², NSW 802,467 km² — close to published state areas (Punjab ≈50,362, Haryana ≈44,212 km²).

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F04 · AUDIT, EXAMPLE, CODE, TESTS

RUN / TESTED LOCALLY

AUDIT VERDICT · OK + VALIDATED

RUN here on real Natural Earth polygons: gridded areas Punjab 50,444 km², Haryana 43,997 km², Delhi 1,481 km², Mato Grosso 906,656 km², NSW 802,467 km² — close to published state areas (Punjab ≈50,362, Haryana ≈44,212 km²).

WORKED EXAMPLE

0.1° cell at 30.9–31.0°N: 106.0 km²; at the equator 123.6 km²; at 60°N 61.7 km²; at 89.9°N 0.108 km².

EXECUTABLE PSEUDOCODE

```
A = R*R*radians(d)*(sin(radians(lat+d)) -
sin(radians(lat)))
```

EDGE CONDITIONS

- H3 cells are not equal-area either; measure, do not assume.

WHY THIS, NOT THAT

A planar degree-box area (111 km × 111 km cosφ) is close for small cells but fails near the poles and for big boxes; the exact spherical formula costs nothing.

COMMON DOUBTS

Q. Do counts need area?
Cell-day counts no; densities, region comparisons and burned-area ratios yes.

Q. Sphere or ellipsoid?
Sphere error < 0.5% at 0.1°; use an ellipsoid library only for official area statements.

TESTS (UNIT / PROPERTY)

- sum of all cells == $4\pi R^2$ ± 0.01%

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F04 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Spherical area element $R^2 \cos\varphi \, d\varphi \, d\lambda$

HINGLISH MEIN SAMJHO

Upar jaate hue dabba chhota hota hai. Per-km² nikalna ho to asli area se bhaag do.

IMPLEMENTATION CONTRACT (V2)

MODULE	`normalize/, grid/, coverage/`
FUNCTION	`area_of_a_grid_cell_on_the_sphere_v2()`
INPUTS	R [km]; Δλ [rad]; φ_1, φ_2 [rad]; f_g [0–1]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F05, F06
ACCEPTANCE	sum of all cells == $4\pi R^2$ ± 0.01%
MAIN FAILURE MODE	H3 cells are not equal-area either; measure, do not assume.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
0.1 deg cell area	Punjab 106, equator 123 km2	Punjab 106.0, equator 123.6, 60N 61.7, 89.9N 0.108 km2	OK

YOU MAY SAY

'X fire cell-days were detected per valid observation day in this cell-month.'

YOU MAY NOT SAY

'X fires burned' or 'X hectares burned'; '0' for an unobserved month.

Chart spec. Explorer cell inspector rows + calendar cell tooltip; unit in every label; hatched null.

FORMULA 05 / 46 · TIME, SPACE AND COUNTING · LAYER L1

F05 Fire cell-day indicator

F01 F02 F03 F04 **F05** F06 F07 F08 F09

Turns many pixels into one presence flag per cell per day so that a sensor that draws many small pixels and a sensor that draws a few big pixels are compared on the same question: was any fire detected here today?

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$c_{g,d} = 1[\, n_{g,d} \geq 1 \,]$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$c_{s,g,d,\delta} = 1[\, n_{s,g,d,\delta} \geq 1 \,] \quad \text{(per platform } s\text{, local day } d\text{, day/night } \delta\text{)}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
n	number of filtered detections	count
s	platform (Terra, Aqua, S-NPP, N20, N21)	-
δ	D or N	-

DERIVATION, STEP BY STEP

1. Collect all filtered detections of platform s in cell g on local day d, split by day/night δ.
2. If at least one exists, set c = 1, else 0.
3. Keep the count n and max FRP in a side table for diagnostics.

ASSUMPTIONS

- A cell-day is not a fire event and not burned area.
- Several revisits in one day are deduplicated on purpose.

Cell-day = presence, not size

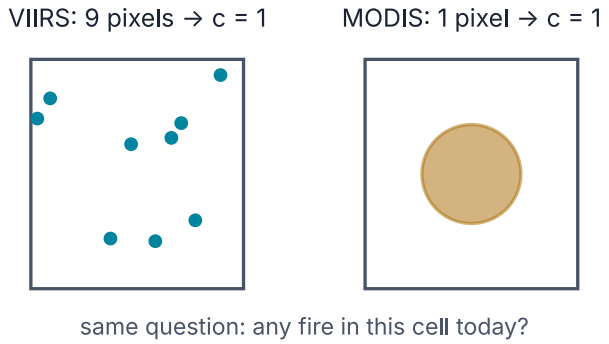


Fig. F05. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

The calendar's basic unit: 'how many cell-days in this region had any detected fire'.

WHERE THE OUTPUT GOES

Calendar metric 'cell-days'; Explorer counts.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F05 V2**	**Output**	**Shown on**
n, s, δ	F03, F04	Fire cell-day indicator	The calendar's basic unit: 'how many cell-days in this region had any detected fire'.	Calendar metric 'cell-days'; Explorer counts.

V1 → V2 IN ONE LINE

v1 deduplicates by (cell, date, sensor) and loses platform and day/night.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F05 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

v1 deduplicates by (cell, date, sensor) and loses platform and day/night. Keep platform and δ so day/night calibration (F13) and coverage (F09) are possible.

WORKED EXAMPLE

3 VIIRS pixels in one cell on 1 Nov → c = 1; MODIS 1 pixel → c = 1.

EXECUTABLE PSEUDOCODE

```
cell_days = det.groupby([cell, local_date, platform,
dn]).size() ⩾ 1
```

EDGE CONDITIONS

- Use local date for burning-day questions; UTC date for pass bookkeeping. Write which one.

WHY THIS, NOT THAT

Counting pixels rewards the finer sensor and multiple revisits; counting cell-days asks a presence question both sensors can answer. It loses intensity, which is why FRP is analysed separately.

COMMON DOUBTS

Q. Is a cell-day a fire?
No: one cell-day can hold many fires, and one large fire can span cells.

Q. Merge NOAA-20 and S-NPP?
Store separately; merging raises exposure after 2018 — decide and document the merge rule.

TESTS (UNIT / PROPERTY)

- v1 test: 2 MODIS pts same cell → 1 cell-day; VIIRS → 2 cells

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F05 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- FireCal / common practice

HINGLISH MEIN SAMJHO

Haazri register: aaj aag thi ya nahi — kitni baar dekhi, farak nahi.

IMPLEMENTATION CONTRACT (V2)

MODULE	`normalize/, grid/, coverage/`
FUNCTION	`fire_cell-day_indicator_v2()`
INPUTS	n [count]; s [-]; δ [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F06, F07
ACCEPTANCE	v1 test: 2 MODIS pts same cell → 1 cell-day; VIIRS → 2 cells
MAIN FAILURE MODE	Use local date for burning-day questions; UTC date for pass bookkeeping. Write which one.

YOU MAY SAY

'X fire cell-days were detected per valid observation day in this cell-month.'

YOU MAY NOT SAY

'X fires burned' or 'X hectares burned'; '0' for an unobserved month.

Chart spec. Explorer cell inspector rows + calendar cell tooltip; unit in every label; hatched null.

FORMULA 06 / 46 · TIME, SPACE AND COUNTING · LAYER L1

F06 Monthly cell-day count

F01 F02 F03 F04 F05 **F06** F07 F08 F09

Region-month total of fire cell-days per platform, the basic count shown in the raw calendar.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$C_{g,m} = \sum_{d \in m} c_{g,d}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$C_{s,r,m} = \sum_{g \in r} \sum_{d \in m} c_{s,g,d}, \quad \text{value} = \text{NULL if } E_{s,r,m} \text{ unknown}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
r	region (set of cells)	-
E	exposure (F09)	valid cell-days

DERIVATION, STEP BY STEP

1. Join cell-days to the region's cell list.
2. Sum c over cells and days in the month.
3. Attach exposure; if exposure is unknown, the month value is null with a status.

ASSUMPTIONS

- Region membership uses region_cell with fractions; for counts use cell centre or fraction ≥0.5 consistently.

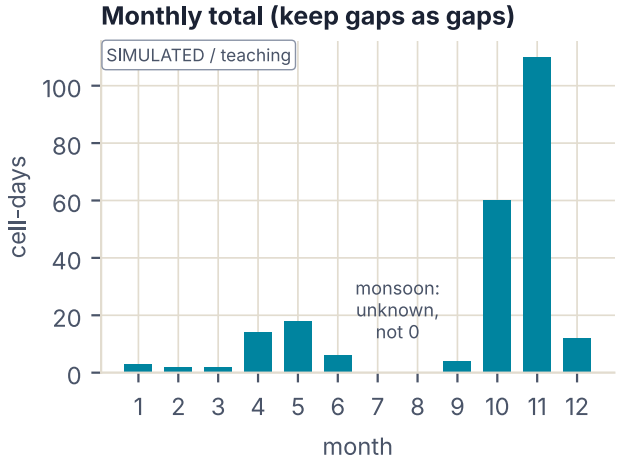


Fig. F06. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

A raw monthly total per platform — comparable only after exposure and calibration.

WHERE THE OUTPUT GOES

Calendar raw mode; Explorer time strip.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F06 V2	04 Output	05 Shown on
r, E	F04, F05	Monthly cell-day count	A raw monthly total per platform — comparable only after exposure and calibration.	Calendar raw mode; Explorer time strip.

V1 → V2 IN ONE LINE

v1 monthly_counts groups by sensor×year×month only, with no region/cell key; the calendar builder then fills absent months with 0.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F06 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX (STRUCTURAL)

v1 monthly_counts groups by sensor×year×month only, with no region/cell key; the calendar builder then fills absent months with 0. Absent = unknown, not zero.

WORKED EXAMPLE

Punjab Nov: S-NPP 14 cell-days in cell 1209_2558; region total = sum over 512 cells.

EXECUTABLE PSEUDOCODE

```
monthly =
cell_days.merge(region_cell).groupby([region,
platform, month]).sum()
```

EDGE CONDITIONS

- Months before a platform's archive start must be NULL, not 0.

WHY THIS, NOT THAT

Summing raw detections double counts revisits; summing cell-days is stable. Area-weighting by coverage fraction is needed when cells straddle the boundary.

COMMON DOUBTS

Q. Why null instead of 0?
0 means 'observed and no fire'; null means 'not observed or not processed'. They lead to different science.

Q. Partial cells?
Use coverage_fraction from region_cell for weighted sums.

TESTS (UNIT / PROPERTY)

- months with no exposure → status 'unavailable'

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F06 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- v1 Book 2 listing 5.1

HINGLISH MEIN SAMJHO

Mahine ki haazri jodo — par jahan dekha hi nahi, wahan 0 mat likho.

IMPLEMENTATION CONTRACT (V2)

MODULE	`normalize/, grid/, coverage/`
FUNCTION	`monthly_cell-day_count_v2()`
INPUTS	r [-]; E [valid cell-days]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F07, F08
ACCEPTANCE	months with no exposure → status 'unavailable'
MAIN FAILURE MODE	Months before a platform's archive start must be NULL, not 0.

YOU MAY SAY

'X fire cell-days were detected per valid observation day in this cell-month.'

YOU MAY NOT SAY

'X fires burned' or 'X hectares burned'; '0' for an unobserved month.

Chart spec. Explorer cell inspector rows + calendar cell tooltip; unit in every label; hatched null.

FORMULA 07 / 46 · TIME, SPACE AND COUNTING · LAYER L1

F07 Equal-confidence filter

F01 F02 F03 F04 F05 F06 F07 F08 F09

Applies a documented, sensor-native quality rule before counting, and tests how much the harmonised answer depends on it.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\text{keep}_i = 1[q_i \geq \tau_s], \quad \tau_{\text{MODIS}} = 30,\ \tau_{\text{VIIRS}} = \text{nominal}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\text{keep}_i = 1[\,\text{class}_i \in \mathcal{Q}_s\,], \quad \mathcal{Q} \text{ chosen from a sensitivity grid } \{\text{MODIS} \geq 0, 30, 50, 80\} \times \{\text{l,n,h};\ \text{n,h};\ \text{h}\}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
q	MODIS confidence 0–100 (experimental per FIRMS)	%
class	VIIRS l / n / h	category

DERIVATION, STEP BY STEP

1. List each sensor's native classes (MODIS 0–100, VIIRS l/n/h).
2. Define candidate filter pairs.
3. For each pair, compute k and held-out error.
4. Choose the pair with the best held-out agreement; report spread across pairs as uncertainty.

ASSUMPTIONS

- MODIS 30 and VIIRS 'nominal' are not proven to be the same strictness.

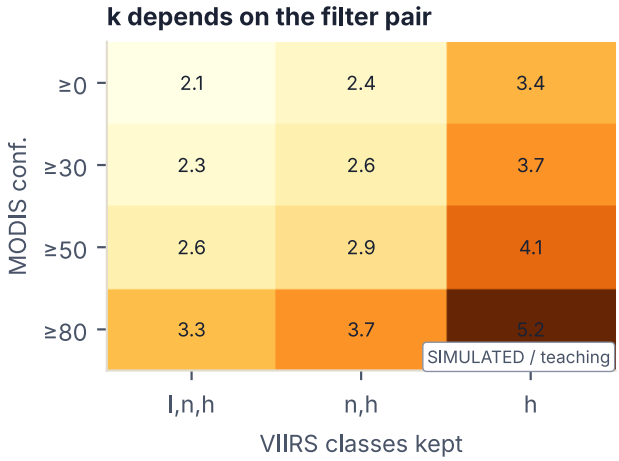


Fig. F07. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

A rule like 'MODIS ≥ 30 · VIIRS n,h' is a configuration, shown on every chart.

WHERE THE OUTPUT GOES

Explorer confidence control becomes a named rule picker.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F07 V2	04 Output	05 Shown on
q, class	F05, F06	Equal-confidence filter	A rule like 'MODIS ≥ 30 · VIIRS n,h' is a configuration, shown on every chart.	Explorer confidence control becomes a named rule picker.

V1 → V2 IN ONE LINE

v1 maps VIIRS l/n/h to 0.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F07 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

v1 maps VIIRS l/n/h to 0.25/0.6/0.9 and UI says 'same confidence 0.6 both sensors'. These numbers are labels, not calibrated probabilities. Keep categories; report results for several filter pairs and pick by held-out agreement.

WORKED EXAMPLE

Grid of 4 MODIS × 3 VIIRS rules → 12 k estimates; show the spread as filter uncertainty.

EXECUTABLE PSEUDOCODE

```
for mt in [0,30,50,80]:
  for vq in [{'l','n','h'},{'n','h'},{'h'}]:
    k[mt,vq] = fit(filter(df, mt, vq))
```

EDGE CONDITIONS

- Low-confidence VIIRS includes many real small fires; dropping them changes k.

WHY THIS, NOT THAT

A fixed mapping to numbers pretends the scales are equal. Letting held-out agreement choose the pair is empirical and auditable.

COMMON DOUBTS

Q. Is MODIS 30 equal to VIIRS nominal?
Unknown; that is exactly what the sensitivity grid tests.

Q. Drop low-confidence VIIRS?
It removes many real small fires; decide with data.

TESTS (UNIT / PROPERTY)

- UI never displays '0.6' as a VIIRS probability

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F07 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- FIRMS notes: MODIS confidence experimental

HINGLISH MEIN SAMJHO

Dono teacher ka passing mark same 'number' nahi ho sakta jab ek marks deta hai aur doosra grade.

IMPLEMENTATION CONTRACT (V2)

MODULE	`normalize/, grid/, coverage/`
FUNCTION	`equal-confidence_filter_v2()`
INPUTS	q [%]; class [category]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F08, F09
ACCEPTANCE	UI never displays '0.6' as a VIIRS probability
MAIN FAILURE MODE	Low-confidence VIIRS includes many real small fires; dropping them changes k.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
VIIRS l/n/h mapping	{'l':.25,'n':.6,'h':.9}	keep categorical; filter rule as sensitivity grid	FIX

YOU MAY SAY

'X fire cell-days were detected per valid observation day in this cell-month.'

YOU MAY NOT SAY

'X fires burned' or 'X hectares burned'; '0' for an unobserved month.

Chart spec. Explorer cell inspector rows + calendar cell tooltip; unit in every label; hatched null.

FORMULA 08 / 46 · TIME, SPACE AND COUNTING · LAYER L2

F08 Persistence (static-source mask)

F01 F02 F03 F04 F05 F06 F07 **F08** F09

Share of observed days a site was hot over several years; high values point to kilns, flares, industry or volcanoes, which should not drive the agricultural calendar or calibration.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$p_g = \frac{\text{active days}_g}{\text{total days}}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$p_g = \frac{\sum_d c_{g,d}}{\sum_d 1[\text{valid}_{g,d}]}, \quad p_g = \text{NULL if denominator} = 0$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
valid	day with usable observation	-

DERIVATION, STEP BY STEP

1. Choose a multi-year window per platform.
2. Count days with detection at the site (cluster or cell).
3. Divide by days the site was observable.
4. Flag as static if p and night share both exceed tuned thresholds.

ASSUMPTIONS

- Threshold 0.20 is a hypothesis to fit on labelled sites.

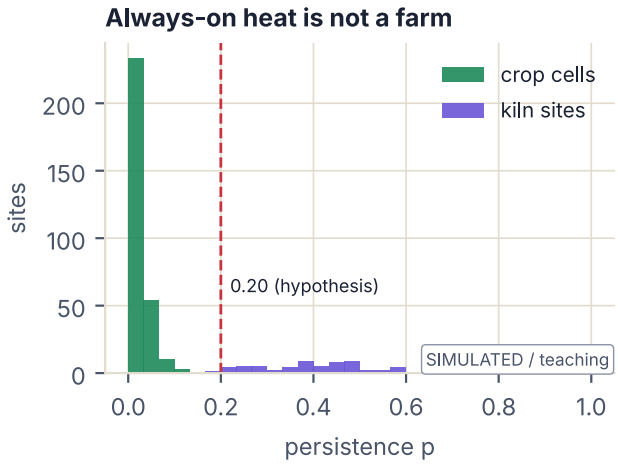


Fig. F08. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Static sources are removed from calibration and shown separately in Layer 2.

WHERE THE OUTPUT GOES

Source page evidence card.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F08 V2	04 Output	05 Shown on
valid	F06, F07	Persistence (static-source mask)	Static sources are removed from calibration and shown separately in Layer 2.	Source page evidence card.

V1 → V2 IN ONE LINE

RUN here: v1 persistence(0, 0) returns 0.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F08 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

RUN here: v1 persistence(0, 0) returns 0.0 (no activity) instead of unknown. Denominator should be observed days, not calendar days.

WORKED EXAMPLE

Kiln: 300 hot of 365 observed → 0.82; paddy cell: 7 of 365 → 0.02.

EXECUTABLE PSEUDOCODE

```
p = active / observed if observed > 0 else None
```

EDGE CONDITIONS

- Multi-year window crossing a sensor change mixes detection sensitivities: compute per sensor.

WHY THIS, NOT THAT

A known-sites list (flares, kilns) is better where it exists; persistence is the fallback that also finds unlisted sites.

COMMON DOUBTS

Q. Why per platform?
VIIRS detects more small hot sources, so the same kiln has higher p in VIIRS.

Q. Does persistence remove crop fires?
Rarely: crop burning is seasonal (p ≈ 0.02).

TESTS (UNIT / PROPERTY)

- persistence(0,0) is None

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F08 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- SIH FIREWORK rules

HINGLISH MEIN SAMJHO

Jo jagah saal bhar garam, woh khet nahi, bhatti hai.

IMPLEMENTATION CONTRACT (V2)

MODULE	`normalize/, grid/, coverage/`
FUNCTION	`persistence_static-source_mask_v2()`
INPUTS	valid [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F09
ACCEPTANCE	persistence(0,0) is None
MAIN FAILURE MODE	Multi-year window crossing a sensor change mixes detection sensitivities: compute per sensor.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
persistence(0,0)	0.0	{'value': None, 'status': 'unavailable', 'note': 'no observed days'}	FIX

YOU MAY SAY

'X fire cell-days were detected per valid observation day in this cell-month.'

YOU MAY NOT SAY

'X fires burned' or 'X hectares burned'; '0' for an unobserved month.

Chart spec. Explorer cell inspector rows + calendar cell tooltip; unit in every label; hatched null.

FORMULA 09 / 46 · TIME, SPACE AND COUNTING · LAYER L1 / L4

F09 Clear-sky (exposure) fire rate

F01 F02 F03 F04 F05 F06 F07 F08 F09

Fire cell-days per valid observation day. Removes the cloud effect so a cloudy November is not mistaken for a quiet November.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$r_{g,m} = \frac{F_{g,m}}{K_{g,m}}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$r_{s,g,m} = \frac{F_{s,g,m}}{E_{s,g,m}}, \quad E = \#\{\text{days with FireMask} \in \{3,5,7,8,9\}\}, \quad r = \text{NULL if } E = 0$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
F	fire cell-days	count
E	valid observable cell-days for the same platform	count
K	old name for E	-

DERIVATION, STEP BY STEP

1. Read daily FireMask for the platform over each cell.
2. Count valid days (classes 3, 5, 7, 8, 9).
3. Divide fire cell-days by valid days.
4. Store both numbers with the rate.

ASSUMPTIONS

- Exposure from the 1 km daily composite (MOD14A1/VNP14A1) approximates opportunity for FIRMS point detections; record the mismatch.
- One valid composite day may contain several passes: it is one opportunity, not several.

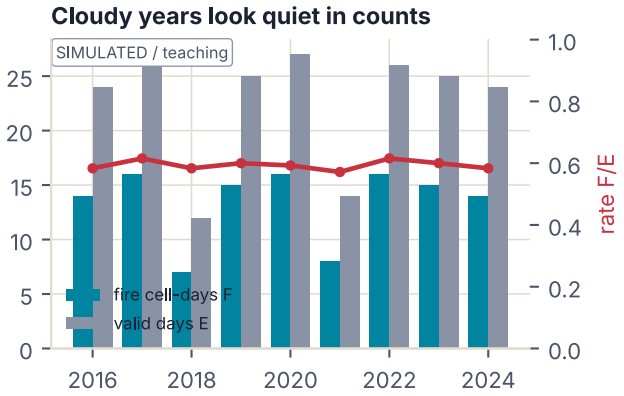


Fig. F09. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

The Layer 1 KPI: comparable across clouds and sensors when combined with k.

WHERE THE OUTPUT GOES

Calendar 'exposure rate' metric; Explorer 'valid days 22/30'.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F09 V2	04 Output	05 Shown on
F, E, K	F07, F08	Clear-sky (exposure) fire rate	The Layer 1 KPI: comparable across clouds and sensors when combined with k.	Calendar 'exposure rate' metric; Explorer 'valid days 22/30'.

V1 → V2 IN ONE LINE

Correct at K=0 → None (v1 test).

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: re-computed in code (RUN). No real-fire result appears here.

F09 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Correct at K=0 → None (v1 test). v2 makes E platform-specific, defines which FireMask classes count (6 'unknown' is not clear), and stores E alongside every rate.

WORKED EXAMPLE

30 fire days / 600 valid cell-days = 0.05 (v1 test, RUN here).

EXECUTABLE PSEUDOCODE

```
E = count(FireMask in {3,5,7,8,9}); r = F/E if E > 0
else None
```

EDGE CONDITIONS

- Thick smoke can be flagged as cloud → exposure falls exactly when burning peaks.

WHY THIS, NOT THAT

Dividing by calendar days treats cloudy days as 'no fire'. Dividing by valid days asks 'how often was fire seen when the ground could be seen'.

COMMON DOUBTS

Q. Why not class 6?
It is 'unknown', not clear.

Q. Smoke as cloud?
Yes, possible; it biases exposure down in peak weeks — monitor with cloud share.

TESTS (UNIT / PROPERTY)

- rate(30,600)=0.05
- rate(5,0)=None

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F09 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- GEE MOD14A1/VNP14A1 catalogue, FireMask classes (verified 27 Oct 2026)

HINGLISH MEIN SAMJHO

Strike rate = runs/balls. Fire rate = aag-din / saaf-din.

IMPLEMENTATION CONTRACT (V2)

MODULE	`normalize/, grid/, coverage/`
FUNCTION	`clear-sky_exposure_fire_rate_v2()`
INPUTS	F [count]; E [count]; K [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	Layer output
ACCEPTANCE	rate(30,600)=0.05; rate(5,0)=None
MAIN FAILURE MODE	Thick smoke can be flagged as cloud → exposure falls exactly when burning peaks.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
rate(30,600)	0.05	0.05	OK
rate(5,0)	None	{'value': None, 'status': 'unavailable', 'note': 'zero valid observations'}	OK

YOU MAY SAY

'X fire cell-days were detected per valid observation day in this cell-month.'

YOU MAY NOT SAY

'X fires burned' or 'X hectares burned'; '0' for an unobserved month.

Chart spec. Explorer cell inspector rows + calendar cell tooltip; unit in every label; hatched null.

FORMULA 10 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F10 Monthly overlap ratio

F10 F11 F12 F13 F14 F15 F16 F17 F18 F19 F20 F21 F22 F23

Exploratory per-stratum ratios of VIIRS to MODIS cell-days during real overlap; inputs to diagnostics, not the final estimator.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$k_m = \frac{V_m}{M_m} \quad (m \in \text{overlap } 2012\text{–}2025)$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$k_{r,\sigma,y} = \frac{V_{r,\sigma,y}}{M_{r,\sigma,y}} \text{ only for strata } \sigma = (\text{month}, \delta, \text{platform pair}) \text{ with both sensors observing}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
V, M	VIIRS and MODIS cell-days on the same cells/days	count
σ	stratum	-

DERIVATION, STEP BY STEP

1. Pair cells and days observed by both platforms in the same day/night window.
2. Sum cell-days for each sensor per stratum and year.
3. Ratio = V/M per stratum-year; keep zero-M strata in a 'dropped' table.

ASSUMPTIONS

- Same cells, same days, same exposure for both sensors.

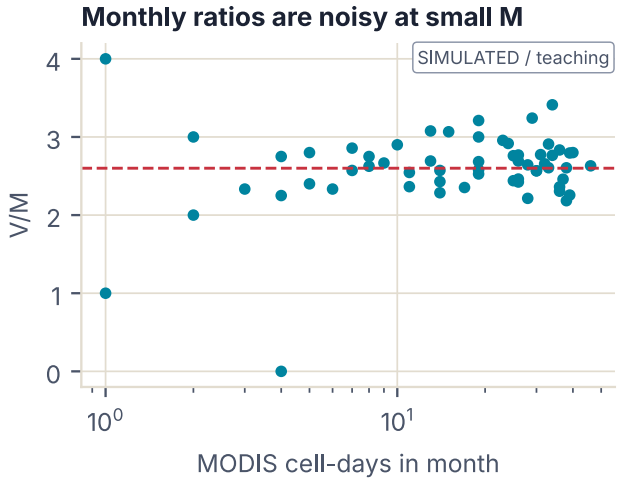


Fig. F10. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Exploratory: shows how stable the sensor ratio is across months and years.

WHERE THE OUTPUT GOES

Sensor Lab bar chart of k by month with n shown.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F10 V2**	**Output**	**Shown on**
V, M, σ	raw FIRMS / FireMask	Monthly overlap ratio	Exploratory: shows how stable the sensor ratio is across months and years.	Sensor Lab bar chart of k by month with n shown.

V1 → V2 IN ONE LINE

v1 silently drops months with M=0 (a VIIRS=9 month vanished in the test) and treats all post-2012 years as one pool.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F10 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

v1 silently drops months with M=0 (a VIIRS=9 month vanished in the test) and treats all post-2012 years as one pool. Pair real overlapping periods and platforms (Aqua↔S-NPP afternoon; Terra has no VIIRS partner).

WORKED EXAMPLE

v1 test: M=[10,20,0,40], V=[25,52,9,100] → ratios 2.5, 2.6, 2.5 → 2.5; the zero month is lost.

EXECUTABLE PSEUDOCODE

```
pairs = join(modis, viirs, on=[cell, date, dn]); ratio per stratum
```

EDGE CONDITIONS

- Ratio of small integers is lumpy: 1→3 gives 3.0, 2→3 gives 1.5.

WHY THIS, NOT THAT

Pooling all post-2012 data mixes Terra (no partner) with Aqua and uses unmatched days. Pairing makes the ratio about sensitivity, not about different weather or passes.

COMMON DOUBTS

Q. Terra has no VIIRS partner — what then?
Calibrate Terra to Aqua in their own overlap (2002→), then chain to VIIRS, with combined uncertainty.

Q. Why keep dropped strata?
Silent removal hides selection bias.

TESTS (UNIT / PROPERTY)

- no stratum with M=0 is silently removed: it is counted in a 'dropped' table

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F10 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Book 2 overlap_ratio

HINGLISH MEIN SAMJHO

Exchange rate tabhi seekho jab dono dukaan ek hi din khuli hon.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`monthly_overlap_ratio_v2()`
INPUTS	V, M [count]; σ [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F11, F12
ACCEPTANCE	no stratum with M=0 is silently removed: it is counted in a 'dropped' table
MAIN FAILURE MODE	Ratio of small integers is lumpy: 1→3 gives 3.0, 2→3 gives 1.5.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
overlap_ratio test	2.5	2.5	OK

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 11 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F11 Median of ratios

F10 F11 F12 F13 F14 F15 F16 F17 F18 F19 F20 F21 F22 F23

Robust exchange rate between sensors that resists one strange month.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING + TESTED (SYNTHETIC)

$$\hat{k} = \text{median}\{k_m\} \quad \text{(v1.1: only } M_m \geq 5\text{)}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$\hat{k}_{\text{robust}} = \text{median}\{k_m : M_m \geq 5\}$ kept as baseline B3a; primary estimator is ratio of totals / GLM (F14

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
k_m	monthly ratios	-

DERIVATION, STEP BY STEP

1. Compute stratum ratios k_m with M_m ≥ 5.
2. Sort and take the middle value.
3. Compare with ratio of totals; large disagreement flags heterogeneity.

ASSUMPTIONS

- Ratios exchangeable across years; no drift.

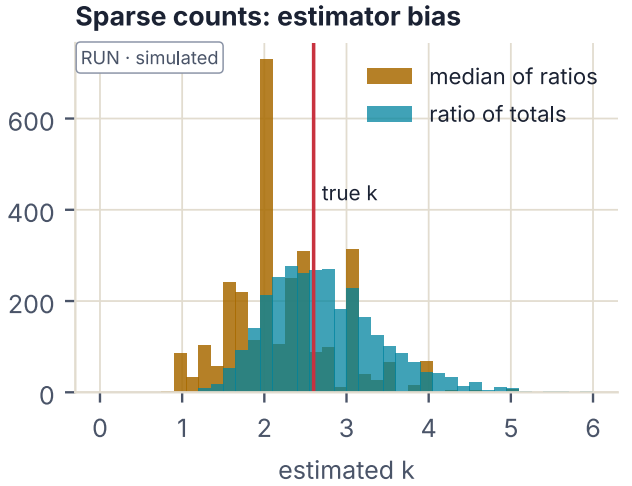


Fig. F11. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

If median and totals disagree by > 20%, report the stratum as heterogeneous.

WHERE THE OUTPUT GOES

Sensor Lab model panel row 'median'.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F11 V2**	**Output**	**Shown on**
k_m	F10	Median of ratios	If median and totals disagree by > 20%, report the stratum as heterogeneous.	Sensor Lab model panel row 'median'.

V1 → V2 IN ONE LINE

RUN here (simulated, true k=2.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F11 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

RUN here (simulated, true k=2.6, MODIS 0–3 per month): median of monthly ratios averaged 2.22 (biased); ratio of totals 2.72. Median is a useful robustness check, not the default estimator for sparse counts. The ≥5 rule from the visual guide helps but discards data.

WORKED EXAMPLE

Ratios {2.45, 2.5, 2.5, 2.6} → median 2.5.

EXECUTABLE PSEUDOCODE

```
k_med = median([v/m for v,m in pairs if m ⩾ 5])
```

EDGE CONDITIONS

- Fewer than 3 months → return status 'insufficient', fall back to pooled k.

WHY THIS, NOT THAT

Median resists outliers but is biased for small integer counts; ratio of totals equals the offset-GLM MLE and uses all data.

COMMON DOUBTS

Q. Why keep the median at all?
As a robustness check (B3a) against one strange year.

Q. Mean of ratios?
Worst option: dominated by small-M months.

TESTS (UNIT / PROPERTY)

- median vs totals compared in the baseline ladder

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F11 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- 46-formula visual guide (CHANGED card)

HINGLISH MEIN SAMJHO

Median ek ajeeb mahine se nahi darta — par jab ginti chhoti ho to median bhi jhoot bolta hai.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`median_of_ratios_v2()`
INPUTS	k_m [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F12, F13
ACCEPTANCE	median vs totals compared in the baseline ladder
MAIN FAILURE MODE	Fewer than 3 months → return status 'insufficient', fall back to pooled k.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
overlap_ratio test	2.5	2.5	OK
sparse-month bias (simulated, true k=2.6)	median of monthly ratios mean=2.22	ratio of totals mean=2.72	FIX

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 12 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F12 Harmonised MODIS value

F10 F11 F12 F13 F14 F15 F16 F17 F18 F19 F20 F21 F22 F23

Expresses MODIS-era months in VIIRS-equivalent cell-days so the 25-year calendar uses one ruler.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$H_{g,m} = \hat{k} \cdot M_{g,m} \quad \text{for } m < 2012$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\hat{V}^{\text{eq}}_{r,\sigma,t} = \hat{k}_{r,\sigma}\, M_{r,\sigma,t} \;\; \text{(estimate, with interval)}; \;\; t \geq 2012 : \; V_{r,t} \text{ observed}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
V^eq	VIIRS-equivalent cell-days (estimate)	cell-days

DERIVATION, STEP BY STEP

1. Take k for the stratum of month t (month × day/night).
2. Multiply MODIS cell-days.
3. Propagate the interval of k and count noise (F25).
4. From 2012, show VIIRS observed values.

ASSUMPTIONS

- k learned in overlap applies before 2012 (stationarity) — the biggest assumption in the project.
- Terra morning passes have no VIIRS analogue; Terra-era scaling is an extrapolation.

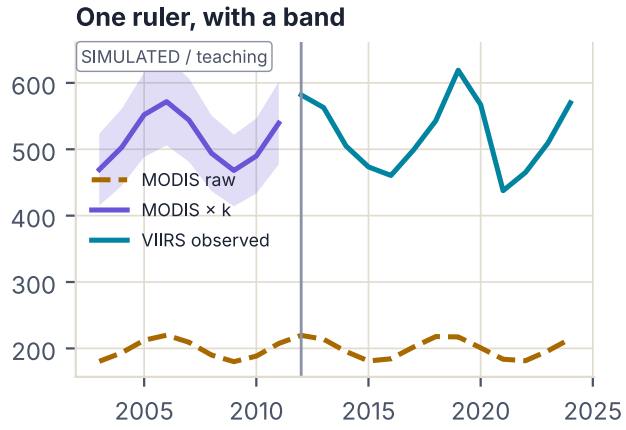


Fig. F12. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Old months on the new ruler — always with a band and the k version.

WHERE THE OUTPUT GOES

Calendar harmonised mode.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F12 V2**	**Output**	**Shown on**
V^eq	F10, F11	Harmonised MODIS value	Old months on the new ruler — always with a band and the k version.	Calendar harmonised mode.

V1 → V2 IN ONE LINE

Fine as arithmetic.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F12 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Fine as arithmetic. Visual guide example '200 → 521 with k=2.6' should be 520. Label output as an estimate with an interval and the k version used.

WORKED EXAMPLE

Nov 2005 MODIS 200 cell-days × k 2.6 = 520 VIIRS-eq cell-days (interval from F24/F25).

EXECUTABLE PSEUDOCODE

```
H = k * M ; H_lo, H_hi from propagated interval
```

EDGE CONDITIONS

- Never multiply NULL by k; NULL stays NULL.

WHY THIS, NOT THAT

Quantile mapping (F22) or regression on log scale are alternatives; a ratio keeps zero at zero and is easiest to audit.

COMMON DOUBTS

Q. Is the harmonised value observed?
No — it is an estimate with a band.

Q. Use MODIS after 2012 too?
Yes, as an overlap check (Bland–Altman, F23).

TESTS (UNIT / PROPERTY)

- harmonise_modis(100, 2.5) == 250

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F12 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Book 3 §2.6

HINGLISH MEIN SAMJHO

Purani 2 MP photo ko naye 48 MP ke scale par laana — andaaza hai, naap nahi.

IMPLEMENTATION CONTRACT (V2)

MODULE	calibrate/
FUNCTION	harmonised_modis_value_v2()
INPUTS	V^eq [cell-days]
RETURNS	{value, status, note, unit, version}
STATUS VALUES	ok · unavailable · insufficient_data · pending
NEXT CONSUMER	F13, F14
ACCEPTANCE	harmonise_modis(100, 2.5) == 250
MAIN FAILURE MODE	Never multiply NULL by k; NULL stays NULL.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
200 × 2.6	visual guide prints 521	520	FIX

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 13 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F13 Separate day and night factors

F10 F11 F12 F13 F14 F15 F16 F17 F18 F19 F20 F21 F22 F23

Day and night detection behaviour differs; mixing them hides shifts in night burning.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\hat{k}_{\text{day}} = \text{median}\,\frac{V_m^D}{M_m^D}, \qquad \hat{k}_{\text{night}} = \text{median}\,\frac{V_m^N}{M_m^N}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$\hat{k}_\delta$ per stratum $\delta \in \{D, N\}$ using matched overpass windows (actual acq_time, not fixed 10:30/13:30)

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
δ	day or night	-

DERIVATION, STEP BY STEP

1. Split pairs by FIRMS daynight flag.
2. Estimate k for each δ (with pooling for sparse night strata).
3. Harmonise day and night counts separately, then add.

ASSUMPTIONS

- Aqua and S-NPP afternoon passes are close in time; Terra's drift (10:30 → ~10:15 by 2022) changes what 'morning' means.

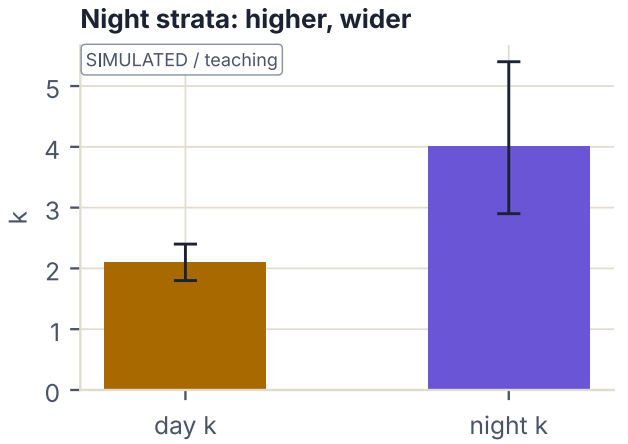


Fig. F13. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Two exchange rates, added up.

WHERE THE OUTPUT GOES

Sensor Lab day/night switch.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F13 V2**	**Output**	**Shown on**
δ	F11, F12	Separate day and night factors	Two exchange rates, added up.	Sensor Lab day/night switch.

V1 → V2 IN ONE LINE

Use observed acquisition times per year; generic Terra 10:30 / Aqua 13:30 is wrong for late years (NASA: Terra crossing ~10:15 by fall 2022).

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F13 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Use observed acquisition times per year; generic Terra 10:30 / Aqua 13:30 is wrong for late years (NASA: Terra crossing ~10:15 by fall 2022).

WORKED EXAMPLE

k_day 2.1, k_night 4.0 (teaching example, SIMULATED).

EXECUTABLE PSEUDOCODE

```
for dn in ['D','N']: k[dn] = fit(pairs[pairs.dn ==
dn])
```

EDGE CONDITIONS

- Night strata are sparse; pool with EB (F18).

WHY THIS, NOT THAT

A single k averages over a changing day/night mix; if night burning grows, one k biases the series.

COMMON DOUBTS

Q. What if night counts are tiny?
Pool with EB (F18) or report 'insufficient'.

Q. Are night detections comparable?
Different thresholds apply; k_night absorbs that.

TESTS (UNIT / PROPERTY)

- night k not computed → '— needs archive run'

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F13 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- NASA Terra end-of-era article (Dec 2025)

HINGLISH MEIN SAMJHO

Din aur raat ki dukaan ka hisaab alag.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`separate_day_and_night_factors_v2()`
INPUTS	δ [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F14, F15
ACCEPTANCE	night k not computed → '— needs archive run'
MAIN FAILURE MODE	Night strata are sparse; pool with EB (F18).

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 14 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F14 Poisson GLM with exposure offset

F10 F11 F12 F13 **F14** F15 F16 F17 F18 F19 F20 F21 F22 F23

Model-based exchange rate with honest uncertainty, using real exposure (valid observations) as the offset instead of MODIS counts.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$\log \mathbb{E}[V_m] = \log M_m + \beta, \qquad \hat{k} = e^{\hat{\beta}}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\log \mu_{s,r,t} = \log E_{s,r,t} + \alpha_r + \gamma_s + x^\top \beta, \;\; Y \sim \text{NegBin}(\mu, \theta); \;\; \hat{k} = e^{\hat{\gamma}_V - \hat{\gamma}_M}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
E	exposure for that sensor	valid cell-days
γ_s	sensor effect	log scale
θ	NB dispersion	-

DERIVATION, STEP BY STEP

1. Write expected counts as exposure × efficiency × activity.
2. Take logs: $\log \mu = \log E + \alpha + \gamma_s$.
3. Fit with Poisson or NB likelihood; read $\gamma_V - \gamma_M$.
4. Exponentiate for k with a profile or bootstrap interval.

ASSUMPTIONS

- Both sensors see the same latent activity in a stratum (paired design).
- Covariates x (day/night, season, view geometry) explain sensitivity differences.

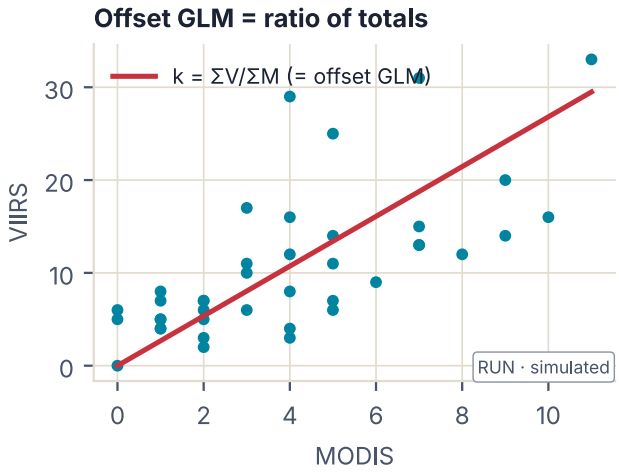


Fig. F14. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

k with an honest interval, from one auditable model.

WHERE THE OUTPUT GOES

Sensor Lab model panel.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F14 V2**	**Output**	**Shown on**
E, γ_s, θ	F12, F13	Poisson GLM with exposure offset	k with an honest interval, from one auditable model.	Sensor Lab model panel.

V1 → V2 IN ONE LINE

RUN here: intercept-only Poisson GLM with offset log M gives exactly ΣV/ΣM (2.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F14 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

RUN here: intercept-only Poisson GLM with offset log M gives exactly ΣV/ΣM (2.6000 = 2.6000). It is the ratio of totals, not a different 'confirmation' of F11. Using MODIS counts as exposure is errors-in-variables and breaks at M=0. v2 uses exposure offsets and a sensor term.

WORKED EXAMPLE

V=[25,52,9,100,61], M=[10,20,3,40,22] → k = 247/95 = 2.600.

EXECUTABLE PSEUDOCODE

```
glm(Y ~ C(sensor) + C(region) + C(dn), offset=log(E),
family=NB)
```

EDGE CONDITIONS

- Check Pearson dispersion; if >1.5 switch to NB or quasi-Poisson.

WHY THIS, NOT THAT

Ratio estimators ignore exposure and give no model checks; the GLM gives dispersion diagnostics, covariates and held-out deviance.

COMMON DOUBTS

Q. Why not offset log(M)?
M is a noisy count and zero in some months; exposure E is the right offset.

Q. Poisson or NB?
Decide by dispersion (F15) and held-out deviance.

TESTS (UNIT / PROPERTY)

- intercept-only offset GLM == ratio of totals (RUN)

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F14 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- McCullagh & Nelder, GLM

HINGLISH MEIN SAMJHO

Badi trip par mileage par zyada bharosa — GLM wahi karta hai, par 'trip ki lambai' MODIS count nahi, saaf din hone chahiye.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`poisson_glm_with_exposure_offset_v2()`
INPUTS	E [valid cell-days]; γ_s [log scale]; θ [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F15, F16
ACCEPTANCE	intercept-only offset GLM == ratio of totals (RUN)
MAIN FAILURE MODE	Check Pearson dispersion; if >1.5 switch to NB or quasi-Poisson.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Poisson offset GLM vs ratio of totals	k_glm=2.6000	sumV/sumM=2.6000	NOTE

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 15 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F15 Negative-binomial variance

F10 F11 F12 F13 F14 **F15** F16 F17 F18 F19 F20 F21 F22 F23

Fire counts arrive in bursts (spatial clustering, weather), so intervals from a Poisson model are too narrow.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\mathrm{Var}(V) = \mu + \frac{\mu^2}{\theta}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\mathrm{Var}(Y) = \mu + \frac{\mu^2}{\theta}, \quad \hat{\phi}_{\mathrm{Pearson}} = \frac{1}{n-p}\sum \frac{(y-\hat{\mu})^2}{\hat{\mu}} \text{ decides Poisson vs NB}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
θ	dispersion (smaller = more bursty)	-
φ	Pearson dispersion	-

DERIVATION, STEP BY STEP

1. Gamma–Poisson mixture: λ ~ Gamma(θ, μ/θ), Y | λ ~ Poisson(λ).
2. Mean μ; variance $\mu + \mu^2/\theta$.
3. Estimate θ by ML; check Pearson dispersion.

ASSUMPTIONS

- Overdispersion is constant within a stratum.

Bursty counts widen bands

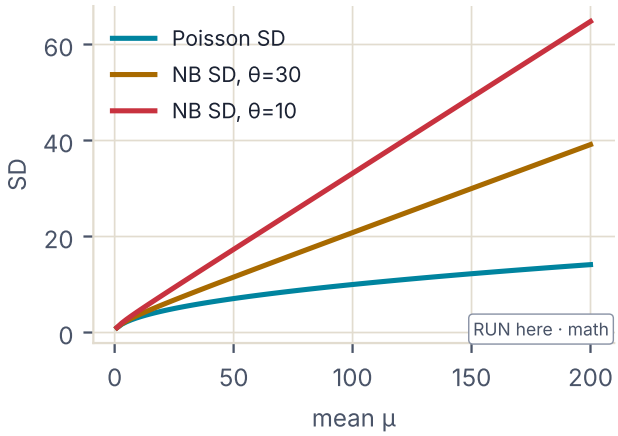


Fig. F15. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Bands from Poisson alone are usually too narrow.

WHERE THE OUTPUT GOES

Validation page 'dispersion' tile.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F15 V2	04 Output	05 Shown on
θ, φ	F13, F14	Negative-binomial variance	Bands from Poisson alone are usually too narrow.	Validation page 'dispersion' tile.

V1 → V2 IN ONE LINE

Correct.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F15 · AUDIT, EXAMPLE, CODE, TESTS

RUN / TESTED LOCALLY

AUDIT VERDICT · OK + RULE

Correct. Add the decision rule: compute $\hat{\varphi}$; NB if $\hat{\varphi} > 1.5$. RUN here on the small example: $\hat{\varphi} = 0.16$ (under-dispersed toy data — real data must decide).

WORKED EXAMPLE

μ=100, θ=10 → Var = 100 + 1000 = 1100, SD 33 (Poisson SD 10).

EXECUTABLE PSEUDOCODE

```
phi = sum((y-mu)**2/mu)/(n-p)
```

EDGE CONDITIONS

- θ → ∞ returns Poisson.

WHY THIS, NOT THAT

Quasi-Poisson only scales the variance; NB models how it grows with μ. Both are fine; choose by fit.

COMMON DOUBTS

Q. Why are fire counts overdispersed?
Weather and burning practices cluster fires in time and space.

Q. Can data be under-dispersed?
Rarely; the toy example is — real data decides.

TESTS (UNIT / PROPERTY)

- NB SD ≥ Poisson SD

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F15 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Hilbe, Negative Binomial Regression

HINGLISH MEIN SAMJHO

Ice-cream ki bikri jhatkon mein — variance mean se zyada.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`negative-binomial_variance_v2()`
INPUTS	θ [-]; φ [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F16, F17
ACCEPTANCE	NB SD ≥ Poisson SD
MAIN FAILURE MODE	θ → ∞ returns Poisson.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Pearson dispersion on example	-	0.16	NOTE

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 16 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F16 Delta-method interval for a ratio

F10 F11 F12 F13 F14 F15 **F16** F17 F18 F19 F20 F21 F22 F23

Quick interval for k; v2 shows that paired sensors are positively correlated, so the independent formula is not exact.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\operatorname{Var}\left(\log \tfrac{V}{M}\right) \approx \frac{1}{V} + \frac{1}{M}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\operatorname{Var}(\log \hat{k}) \approx \frac{1}{V} + \frac{1}{M} - \frac{2\,\operatorname{Cov}(V, M)}{V M}; \quad \text{default: paired block bootstrap (F24)}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
V, M	totals	count

DERIVATION, STEP BY STEP

1. log k = log V − log M.
2. Var(log V) ≈ 1/V, Var(log M) ≈ 1/M (Poisson).
3. Add, minus 2Cov/(VM) if dependent.
4. Interval = k·exp(±1.96 se).

ASSUMPTIONS

- Independent Poisson counts — false for the same fires seen by two sensors.

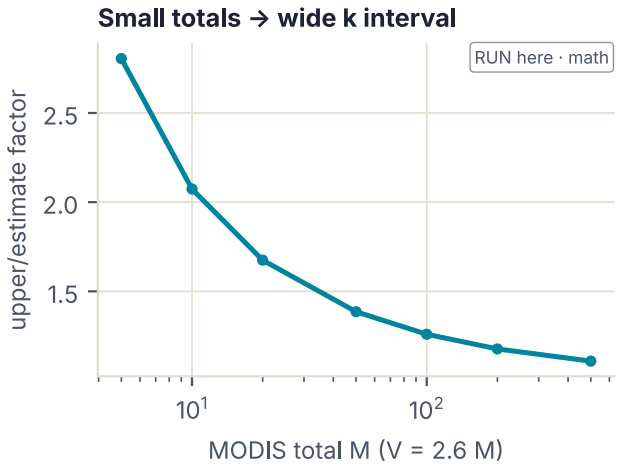


Fig. F16. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

A quick sanity band, not the published one.

WHERE THE OUTPUT GOES

Sensor Lab k ± band (labelled 'approx.').

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F16 V2**	**Output**	**Shown on**
V, M	F14, F15	Delta-method interval for a ratio	A quick sanity band, not the published one.	Sensor Lab k ± band (labelled 'approx.').

V1 → V2 IN ONE LINE

RUN here: totals 247/95 → k 2.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F16 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

RUN here: totals 247/95 → k 2.60, 95% [2.05, 3.29] under independence. Correlation narrows the true interval, overdispersion widens it; bootstrap over years captures both. Undefined for V or M = 0.

WORKED EXAMPLE

k = 2.60, se(log k) = √(1/247 + 1/95) = 0.121 → [2.05, 3.29].

EXECUTABLE PSEUDOCODE

```
se = sqrt(1/V + 1/M) if V>0 and M>0 else None
```

EDGE CONDITIONS

- Zero counts → no interval; use Garwood bounds (F26) on each count.

WHY THIS, NOT THAT

Fast and closed-form, but assumes independence; the paired bootstrap handles correlation and overdispersion.

COMMON DOUBTS

Q. When is it OK?
Quick UI band for large totals, labelled 'approx.'.

Q. Zero counts?
Undefined; show Garwood bounds.

TESTS (UNIT / PROPERTY)

- CI contains k; returns None at zero

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F16 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Delta method

HINGLISH MEIN SAMJHO

10 logon ka survey ka error bada, 1000 ka chhota.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`delta-method_interval_for_a_ratio_v2()`
INPUTS	V, M [count]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F17, F18
ACCEPTANCE	CI contains k; returns None at zero
MAIN FAILURE MODE	Zero counts → no interval; use Garwood bounds (F26) on each count.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Delta CI for k (example totals)	-	2.60 [2.05, 3.29]	NOTE

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 17 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F17 Shrink local factor toward global

F10 F11 F12 F13 F14 F15 F16 **F17** F18 F19 F20 F21 F22 F23

Stops a small region with two noisy months from getting an extreme k.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$\tilde{k} = \frac{n\,k_{\text{loc}} + n_0\,k_{\text{glob}}}{n + n_0}$$

V2 · REVISED DEFINITION USED FROM NOW ON

kept only as a teaching / fallback rule; production uses F18 on $\log k$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
n_0	prior strength	years

DERIVATION, STEP BY STEP

1. Treat global k as a prior with weight n_0 'pseudo-years'.
2. Weighted average of local and global.

ASSUMPTIONS

- n_0 hand-picked (6 in docs, 4 in tests).

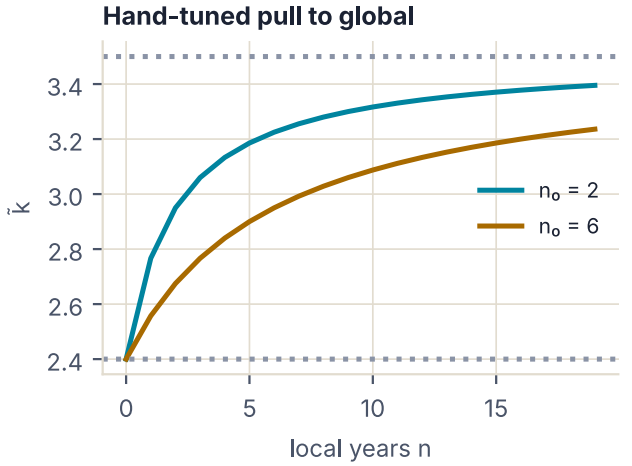


Fig. F17. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Little local data → borrow from the group.

WHERE THE OUTPUT GOES

Sensor Lab 'small area' note.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F17 V2	04 Output	05 Shown on
n_0	F15, F16	Shrink local factor toward global	Little local data → borrow from the group.	Sensor Lab 'small area' note.

V1 → V2 IN ONE LINE

Arithmetic is right (3,4,2,4 → 2.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F17 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Arithmetic is right (3,4,2,4 → 2.5, RUN). Shrinking on the ratio scale is not the same as partial pooling on log k (F18); do not present both as one method.

WORKED EXAMPLE

k_loc 3.0 (4 yrs), k_glob 2.0, n_0 4 → 2.5.

EXECUTABLE PSEUDOCODE

```
(n*k + n0*kg)/(n + n0)
```

EDGE CONDITIONS

- k_loc None → k_glob

WHY THIS, NOT THAT

Simple and explainable, but n_0 is arbitrary and the average is on the ratio scale.

COMMON DOUBTS

Q. Why keep it?
Teaching and fallback when EB cannot run.

Q. Typical n_0?
Docs used 4 and 6 — exactly the problem.

TESTS (UNIT / PROPERTY)

- v1 tests pass

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F17 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Book 3 §2.5

HINGLISH MEIN SAMJHO

Naya batsman — team ke average ki taraf kheencho.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`shrink_local_factor_toward_global_v2()`
INPUTS	n_0 [years]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F18, F19
ACCEPTANCE	v1 tests pass
MAIN FAILURE MODE	k_loc None → k_glob

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
shrink_ratio(3,4,2,4)	2.5	2.5	OK

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 18 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F18 Empirical Bayes partial pooling of log k

F10 F11 F12 F13 F14 F15 F16 F17 F18 F19 F20 F21 F22 F23

Lets the data decide how much each region borrows from the group, with weights from real precision.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$\widehat{\log k}_r = w_r \overline{\log k}_r + (1 - w_r)\,\mu, \qquad w_r = \frac{\tau^2}{\tau^2 + s_r^2/n_r}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$y_r = \log \hat{k}_r,\ \ s_r^2 = \widehat{\mathrm{Var}}(y_r);\ \ \hat{\tau}^2_{\mathrm{DL}} = \max\left(0, \frac{Q - (R-1)}{c}\right);\ \ w_r = \frac{\hat{\tau}^2}{\hat{\tau}^2 + s_r^2};\ \ R \geq 3$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
τ^2	between-region variance	$\log^2$
Q	Cochran heterogeneity statistic	-
R	number of regions	-

DERIVATION, STEP BY STEP

1. Estimate y_r = log k_r and its variance s_r² per region.
2. Estimate between-region variance τ² (DerSimonian–Laird).
3. Weight w_r = τ²/(τ² + s_r²); shrink toward the weighted mean μ.
4. Return None if fewer than 3 regions.

ASSUMPTIONS

- Regions exchangeable; s_r² estimated well.

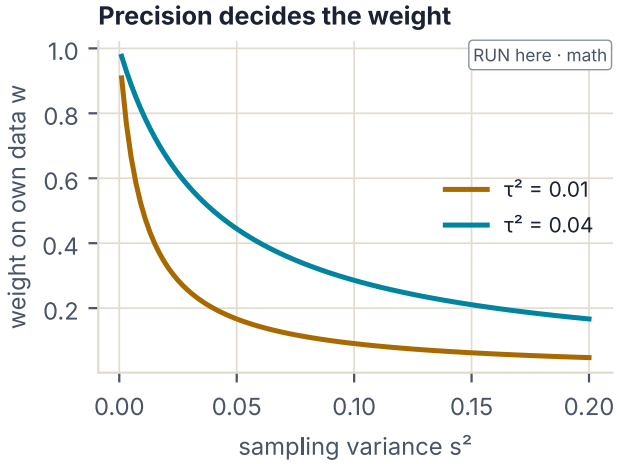


Fig. F18. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Small regions get a stable k; large ones keep their own.

WHERE THE OUTPUT GOES

Validation 'pooling weight' column.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F18 V2**	**Output**	**Shown on**
τ², Q, R	F16, F17	Empirical Bayes partial pooling of log k	Small regions get a stable k; large ones keep their own.	Validation 'pooling weight' column.

V1 → V2 IN ONE LINE

v1 divides by (R−1): undefined for one region; empty groups not guarded.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F18 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

v1 divides by (R−1): undefined for one region; empty groups not guarded. v2 uses DerSimonian–Laird τ^2 with $R \geq 3$ guard and returns None otherwise (RUN here).

WORKED EXAMPLE

Regions with 2 months get the lowest weight (v1 must-pass test).

EXECUTABLE PSEUDOCODE

```
if R < 3: return None
tau2 = max(0,(Q-(R-1))/c)
```

EDGE CONDITIONS

- $\tau^2 = 0 \rightarrow$ full pooling; report it.

WHY THIS, NOT THAT

Data-driven pooling replaces hand-picked n_0; log scale makes ratios symmetric.

COMMON DOUBTS

Q. What if $\tau^2 = 0$?
Full pooling: all regions share μ — report it.

Q. Why log?
k = 2 and k = 0.5 are symmetric on log scale.

TESTS (UNIT / PROPERTY)

- EB(1 region) is None
- weight increases with n

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F18 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- DerSimonian & Laird 1986

HINGLISH MEIN SAMJHO

Kitna bharosa apne data par — yeh data hi bataye.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`empirical_bayes_partial_pooling_of_v2()`
INPUTS	τ^2 [log²]; Q [-]; R [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F19, F20
ACCEPTANCE	EB(1 region) is None; weight increases with n
MAIN FAILURE MODE	$\tau^2 = 0 \rightarrow$ full pooling; report it.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
EB with 1 region	v1: divides by len-1 = 0 (per user audit)	None	FIX

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 19 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F19 Power law vs ratio (model choice)

F10 F11 F12 F13 F14 F15 F16 F17 F18 **F19** F20 F21 F22 F23

Tests whether the relation is proportional or curved (e.g. MODIS saturating in big fire months).

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$V = aM^b \text{ vs } V = kM, \qquad \text{AIC} = n \ln \frac{\text{RSS}}{n} + 2p$$

V2 · REVISED DEFINITION USED FROM NOW ON

compare $\log \mu = \log E + \beta_0 + b \log(M + 0.5)$ vs $b = 1$ by held-out Poisson/NB deviance (blocked CV

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
b	elasticity	-

DERIVATION, STEP BY STEP

1. Fit V ∝ M and V ∝ M^b on the count scale (offset model).
2. Score each on held-out years (blocked CV).
3. Prefer the simpler model unless deviance improves materially.

ASSUMPTIONS

- Same likelihood family for both models.

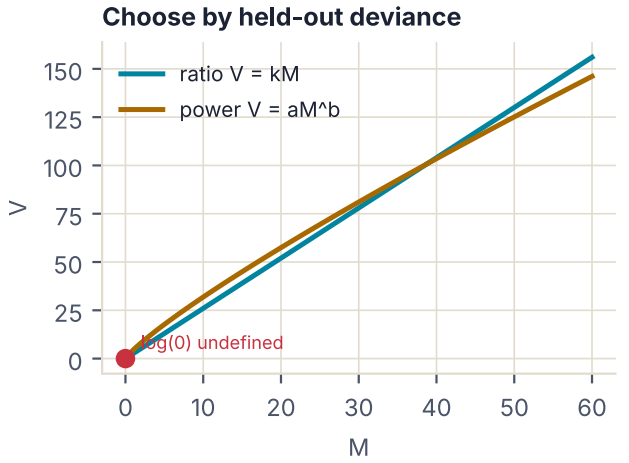


Fig. F19. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Straight or curved — the hidden years decide.

WHERE THE OUTPUT GOES

Sensor Lab 'model shape' chip.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F19 V2	04 Output	05 Shown on
b	F17, F18	Power law vs ratio (model choice)	Straight or curved — the hidden years decide.	Sensor Lab 'model shape' chip.

V1 → V2 IN ONE LINE

log(M) is −∞ at zero months; Gaussian RSS on logs is not the count likelihood; in-sample AIC is not predictive accuracy.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F19 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

log(M) is −∞ at zero months; Gaussian RSS on logs is not the count likelihood; in-sample AIC is not predictive accuracy.

WORKED EXAMPLE

If held-out deviance improves < 2%, keep the simpler ratio.

EXECUTABLE PSEUDOCODE

```
for fold in year_blocks: fit both; score deviance on fold
```

EDGE CONDITIONS

- Small n: prefer simpler model.

WHY THIS, NOT THAT

In-sample AIC on log-log RSS is neither the count likelihood nor predictive accuracy.

COMMON DOUBTS

Q. When would b ≠ 1?
Saturation in big months, or MODIS missing more small fires in light seasons.

Q. Small n?
Keep b = 1.

TESTS (UNIT / PROPERTY)

- no log(0) in any fit

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F19 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Burnham & Anderson

HINGLISH MEIN SAMJHO

Seedhi line ya curve — data aur hidden saal decide karein.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`power_law_vs_ratio_model_choice_v2()`
INPUTS	b [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F20, F21
ACCEPTANCE	no log(0) in any fit
MAIN FAILURE MODE	Small n: prefer simpler model.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
log(M) with zero month	-inf in fit	use Poisson/NB likelihood with held-out deviance	FIX

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 20 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1

F20 Detection-probability curve (FRP50)

F10 F11 F12 F13 F14 F15 F16 F17 F18 F19 F20 F21 F22 F23

Explains why k > 1: MODIS misses small fires that VIIRS sees. Uses near-simultaneous matched pairs (Aqua vs S-NPP within ~15 min).

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$P(\text{MODIS detects} \mid \text{FRP}) = \frac{1}{1 + e^{-(\text{FRP}-\text{FRP}_{50})/s}}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\text{logit}\, P(D_M = 1 \mid F_V) = \beta_0 + \beta_1 \ln F_V + \beta_2\, \theta + \ldots, \quad \text{FRP}_{50} = e^{-\beta_0/\beta_1} \ (\text{at } \theta = 0)$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
F_V	VIIRS FRP of the matched fire	MW
θ	view angle	deg

DERIVATION, STEP BY STEP

1. Match VIIRS fires with near-simultaneous MODIS passes (≤ 15 min, Aqua–S-NPP).
2. Label each VIIRS fire detected/not by MODIS.
3. Fit logistic regression on log FRP and view angle.
4. FRP50 = exp(−β₀/β₁).

ASSUMPTIONS

- VIIRS detection used as the reference label (it also misses fires: ascertainment bias).

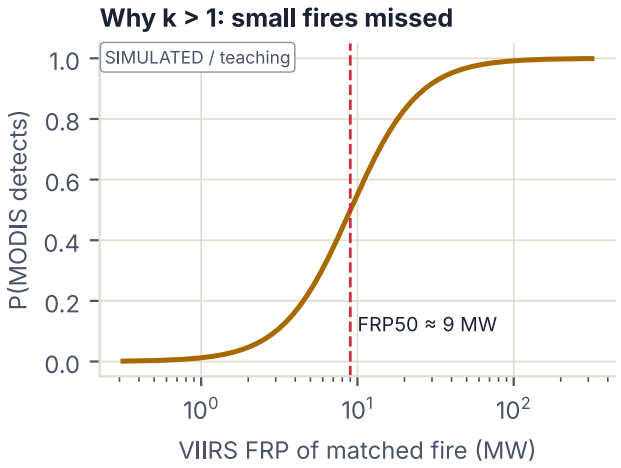


Fig. F20. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Below ~FRP50 MODIS sees fewer than half of the fires VIIRS sees.

WHERE THE OUTPUT GOES

Sensor Lab detection curve.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F20 V2	04 Output	05 Shown on
F_V, θ	F18, F19	Detection-probability curve (FRP50)	Below ~FRP50 MODIS sees fewer than half of the fires VIIRS sees.	Sensor Lab detection curve.

V1 → V2 IN ONE LINE

Printed equation is logistic in FRP; uploaded code fits log(FRP).

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F20 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

Printed equation is logistic in FRP; uploaded code fits log(FRP). Pick one (log form, because FRP is skewed) and publish FRP50 = exp(−β₀/β₁).

WORKED EXAMPLE

$\beta_0 = -4.4$, $\beta_1 = 2.0$ → FRP50 = e^{2.2} = 9.0 MW (teaching, SIMULATED).

EXECUTABLE PSEUDOCODE

```
LogisticRegression on [log(frp_v), view_angle]
```

EDGE CONDITIONS

- VIIRS I4 saturation (~8% of day fire pixels) biases high FRP.

WHY THIS, NOT THAT

Explains k physically; without matched pairs, k is a black-box ratio.

COMMON DOUBTS

Q. Is VIIRS the truth?
No — fires missed by both are invisible (ascertainment bias).

Q. Why log FRP?
FRP is heavily skewed.

TESTS (UNIT / PROPERTY)

- FRP50 recovered on simulated data within 10%

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F20 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Schroeder et al. 2014

HINGLISH MEIN SAMJHO

Eye-test chart: kis size ki aag aadhi baar dikhti hai.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`detection-probability_curve_frp50_v2()`
INPUTS	F_V [MW]; θ [deg]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F21, F22
ACCEPTANCE	FRP50 recovered on simulated data within 10%
MAIN FAILURE MODE	VIIRS I4 saturation (~8% of day fire pixels) biases high FRP.

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 21 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L2

F21 Log transform of FRP

F10 F11 F12 F13 F14 F15 F16 F17 F18 F19 F20 **F21** F22 F23

Tames the long right tail of fire radiative power before z-scores and quantile maps.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$y = \ln(1 + \mathrm{FRP})$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$y = \ln(1 + \mathrm{FRP}/1\,\mathrm{MW})$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
FRP	fire radiative power	MW

DERIVATION, STEP BY STEP

1. Add 1 MW so zero stays zero.
2. Take the natural log.

ASSUMPTIONS

- FRP is comparable only within a sensor; MODIS and VIIRS FRP differ by pixel size and saturation.

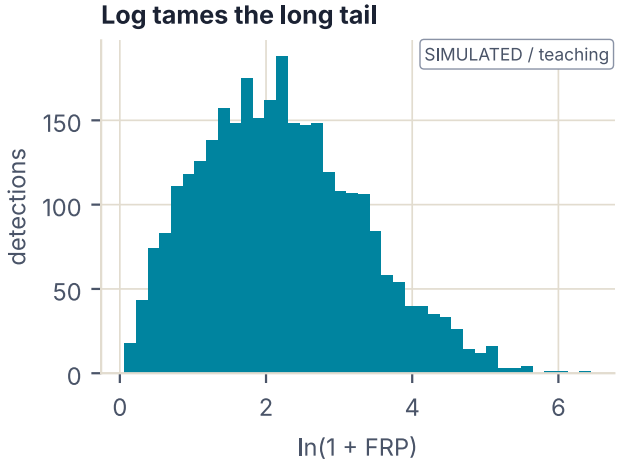


Fig. F21. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

A 1000 MW fire becomes 6.9 instead of dominating the scale.

WHERE THE OUTPUT GOES

Source page FRP distribution (ECDF).

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F21 V2**	**Output**	**Shown on**
FRP	F19, F20	Log transform of FRP	A 1000 MW fire becomes 6.9 instead of dominating the scale.	Source page FRP distribution (ECDF).

V1 → V2 IN ONE LINE

Correct; units made explicit.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: unchanged. Example values: teaching / simulated. No real-fire result appears here.

F21 · AUDIT, EXAMPLE, CODE, TESTS

OK AS WRITTEN

AUDIT VERDICT · OK

Correct; units made explicit.

WORKED EXAMPLE

1000 MW → 6.909.

EXECUTABLE PSEUDOCODE

```
np.log1p(frp)
```

EDGE CONDITIONS

- Negative/missing FRP → NULL.

WHY THIS, NOT THAT

Square-root or Box-Cox also work; log1p is standard and interpretable.

COMMON DOUBTS

Q. Compare MODIS and VIIRS FRP directly?
Not without matching: pixel size and saturation differ.

Q. Units?
Keep MW in storage; transform only for statistics.

TESTS (UNIT / PROPERTY)

- log1p(0)=0

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F21 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- -

HINGLISH MEIN SAMJHO

Ek arabpati average bigaad deta hai — log usko shaant karta hai.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`log_transform_of_frp_v2()`
INPUTS	FRP [MW]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F22, F23
ACCEPTANCE	log1p(0)=0
MAIN FAILURE MODE	Negative/missing FRP → NULL.

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 22 / 46 · CALIBRATION BETWEEN SENSORS · LAYER L1 (FRP ONLY)

F22 Quantile mapping

F10 F11 F12 F13 F14 F15 F16 F17 F18 F19 F20 F21 **F22** F23

Maps MODIS FRP distribution onto VIIRS's for distribution-level comparisons.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$x' = F_V^{-1}(F_M(x))$$

V2 · REVISED DEFINITION USED FROM NOW ON

$x' = F_V^{-1}(F_M(x))$ fitted per stratum on matched overlap; extrapolation beyond training range flagged

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
F	empirical CDF	-

DERIVATION, STEP BY STEP

1. Build empirical CDFs of MODIS and VIIRS FRP in matched overlap.
2. For a MODIS value, read its percentile.
3. Return the VIIRS value at that percentile.

ASSUMPTIONS

- Distributions stable over time.

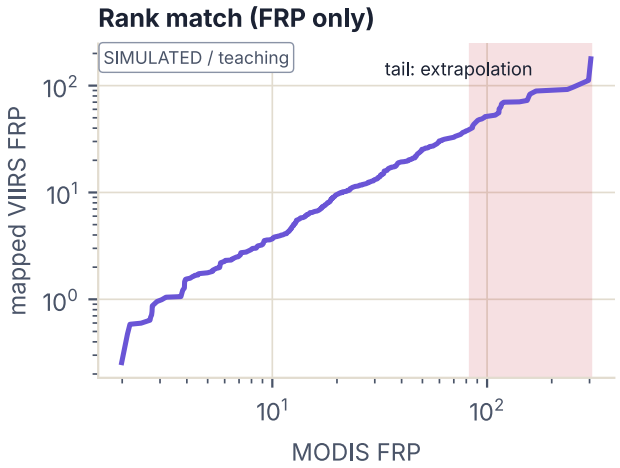


Fig. F22. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Same rank, same value — for FRP views only.

WHERE THE OUTPUT GOES

Sensor Lab advanced tab.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F22 V2**	**Output**	**Shown on**
F	F20, F21	Quantile mapping	Same rank, same value — for FRP views only.	Sensor Lab advanced tab.

V1 → V2 IN ONE LINE

Quantile mapping forces the old era's distribution to look like the new one and can erase real extremes or trends in the tails; never apply to cell-day counts, and flag values outside the training range.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F22 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Quantile mapping forces the old era's distribution to look like the new one and can erase real extremes or trends in the tails; never apply to cell-day counts, and flag values outside the training range.

WORKED EXAMPLE

MODIS median 25 MW → VIIRS median 9 MW.

EXECUTABLE PSEUDOCODE

```
np.interp(F_M(x), qV_probs, qV_values)
```

EDGE CONDITIONS

- Ties at FRP floor.

WHY THIS, NOT THAT

Useful for distribution plots; dangerous for counts or trends because it imposes the overlap distribution on the past.

COMMON DOUBTS

Q. Why not for counts?
It would erase real changes in extremes.

Q. Outside training range?
Flag as extrapolation.

TESTS (UNIT / PROPERTY)

- median maps to median

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F22 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Climate bias-correction literature

HINGLISH MEIN SAMJHO

JEE percentile wala trick — par record toda hua saal bhi 'normal' lagne lagta hai.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`quantile_mapping_v2()`
INPUTS	F [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F23
ACCEPTANCE	median maps to median
MAIN FAILURE MODE	Ties at FRP floor.

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 23 / 46 · CALIBRATION BETWEEN SENSORS · LAYER VALIDATION

F23 Bland–Altman agreement

F10 F11 F12 F13 F14 F15 F16 F17 F18 F19 F20 F21 F22 **F23**

Shows bias and agreement limits between harmonised MODIS and observed VIIRS in held-out overlap years.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$\bar{d} = \overline{a - b}, \qquad \text{LoA} = \bar{d} \pm 1.96\,\text{SD}(a - b)$$

V2 · REVISED DEFINITION USED FROM NOW ON

$d_t = \log(a_t + 0.5) - \log(b_t + 0.5)$, a = harmonised MODIS, b = observed VIIRS; $\bar{d} \pm 1.96\,\text{SD}(d)$ on h

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
a, b	two measurements of the same stratum	same units

DERIVATION, STEP BY STEP

1. Compute differences a − b (or log differences).
2. Mean = bias; ±1.96 SD = limits of agreement.
3. Plot differences against means; check trend in the cloud.

ASSUMPTIONS

- Differences roughly constant across magnitude; otherwise use log scale.

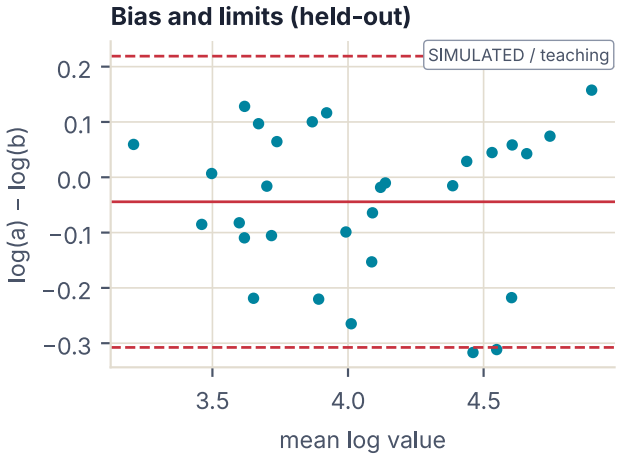


Fig. F23. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

How far apart two rulers are, and whether one is always higher.

WHERE THE OUTPUT GOES

Validation Bland–Altman plot.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F23 V2**	**Output**	**Shown on**
a, b	F21, F22	Bland–Altman agreement	How far apart two rulers are, and whether one is always higher.	Validation Bland–Altman plot.

V1 → V2 IN ONE LINE

Fix the sign convention in code, docs and axis label (a − b).

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: re-computed in code (RUN). No real-fire result appears here.

F23 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Fix the sign convention in code, docs and axis label (a − b). Counts are heteroscedastic, so use log scale. Must be done on held-out data, not the fit years.

WORKED EXAMPLE

a=[10,12,15], b=[8,13,12] → bias +1.33 (RUN).

EXECUTABLE PSEUDOCODE

```
d = log(a+.5)-log(b+.5); bias=d.mean()
```

EDGE CONDITIONS

- n < 8 → do not draw limits.

WHY THIS, NOT THAT

Correlation measures association, not agreement; two series can correlate at 0.99 and differ by 40%.

COMMON DOUBTS

Q. Which way is the sign?
a − b with a = harmonised MODIS, b = VIIRS — printed on the axis.

Q. In-sample?
No: held-out years only.

TESTS (UNIT / PROPERTY)

- sign test: a>b everywhere → bias > 0

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F23 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Bland & Altman 1986

HINGLISH MEIN SAMJHO

Do taraazu — ek hamesha 50 g zyada dikhaye to bias.

IMPLEMENTATION CONTRACT (V2)

MODULE	`calibrate/`
FUNCTION	`bland_altman_agreement_v2()`
INPUTS	a, b [same units]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	Layer output
ACCEPTANCE	sign test: a>b everywhere → bias > 0
MAIN FAILURE MODE	n < 8 → do not draw limits.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Bland-Altman sign	-	bias(a-b)=1.33	NOTE

YOU MAY SAY

'MODIS-era values are expressed on the VIIRS scale with k = a [lo, hi] for this stratum.'

YOU MAY NOT SAY

'k is a physical constant' or 'harmonised values are observations'.

Chart spec. Sensor Lab: k bars by stratum with 95% whiskers and n; paired scatter with ratio line.

FORMULA 24 / 46 · UNCERTAINTY · LAYER L1 / L4

F24 Block bootstrap over years

F24 F25 F26

Interval for k and for any summary, respecting that months inside a year are correlated.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$\hat{\theta}^{*(b)} = T(\text{years resampled}),\ \ \text{CI} = [q_{2.5}, q_{97.5}]\ \ (\text{code: } q_5, q_{95})$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\text{CI}_{1-\alpha} = \left[q_{\alpha/2}(\hat{\theta}^*),\ q_{1-\alpha/2}(\hat{\theta}^*)\right],\ \alpha \text{ explicit},\ \text{resample whole years (or seasons)}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
α	1 − coverage	-
B	bootstrap draws (≥2000)	-

DERIVATION, STEP BY STEP

1. Group data into blocks (years or seasons).
2. Resample blocks with replacement, B ≥ 2000 times.
3. Recompute the statistic each time (ratio of totals).
4. Take the α/2 and 1−α/2 quantiles.

ASSUMPTIONS

- Years are exchangeable blocks; ≥8 blocks for a usable interval.

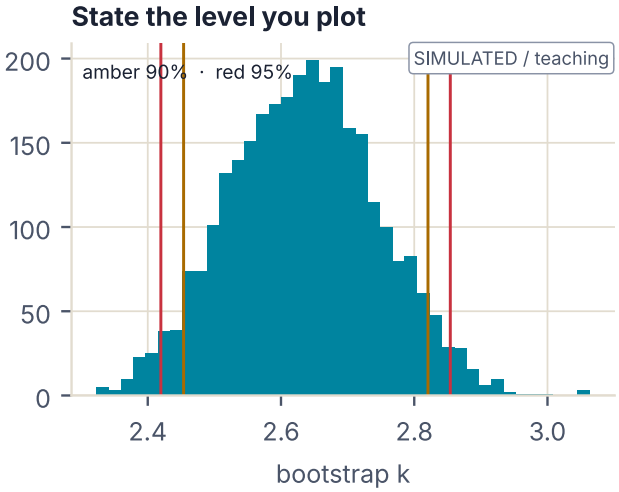


Fig. F24. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

The published 95% band for k.

WHERE THE OUTPUT GOES

Every band shows its level (90% or 95%) in the legend.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F24 V2**	**Output**	**Shown on**
α, B	raw FIRMS / FireMask	Block bootstrap over years	The published 95% band for k.	Every band shows its level (90% or 95%) in the legend.

V1 → V2 IN ONE LINE

Docs say 95% (2.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F24 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

Docs say 95% (2.5–97.5) but Book 3 and the code use 5–95 = 90%. RUN here with {2.45, 2.5, 2.5, 2.6}: 90% [2.450, 2.550] vs 95% [2.450, 2.600]. With 4 years the bootstrap distribution has only a handful of values: report 'too few years'.

WORKED EXAMPLE

ratios {2.45,2.5,2.5,2.6}, B=2000, seed 42 → 95% [2.45, 2.60].

EXECUTABLE PSEUDOCODE

```
def boot(v, m, alpha=0.05): ... quantile(alpha/2), quantile(1-alpha/2)
```

EDGE CONDITIONS

- Resample totals, not ratios, to avoid zero-denominator draws.

WHY THIS, NOT THAT

Resampling months breaks within-year correlation and gives bands that are too narrow.

COMMON DOUBTS

Q. How many years are enough?
≥ 8; with 4 the bootstrap has a handful of distinct values.

Q. Percentile or BCa?
Percentile for MVP; BCa if skewed.

TESTS (UNIT / PROPERTY)

- coverage test on simulated data ≈ nominal (see Ch. 14)

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F24 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Efron & Tibshirani; Künsch 1989

HINGLISH MEIN SAMJHO

Saalon ki parchiyan dabbe mein — baar baar nikaalo, kitna hilta hai dekho.

IMPLEMENTATION CONTRACT (V2)

MODULE	`uncertainty/`
FUNCTION	`block_bootstrap_over_years_v2()`
INPUTS	α [-]; B [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F25, F26
ACCEPTANCE	coverage test on simulated data ≈ nominal (see Ch. 14)
MAIN FAILURE MODE	Resample totals, not ratios, to avoid zero-denominator draws.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
bootstrap ratios {2.45,2.5,2.5,2.6}	q5-q95 = [2.450, 2.550] (90%)	q2.5-q97.5 = [2.450, 2.600] (95%)	FIX

YOU MAY SAY

'The 95% band from year-block bootstrap is [lo, hi].'

YOU MAY NOT SAY

a band without its level or type; a band computed from fewer than 8 years without a warning.

Chart spec. Ribbon / whisker with legend stating level and type (confidence vs predictive).

FORMULA 25 / 46 · UNCERTAINTY · LAYER L1

F25 Monte Carlo error propagation

F24 F25 F26

Band on each calendar cell combining calibration uncertainty and count noise.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$H^{(s)} = k^{(s)} M^{(s)}, \quad k^{(s)} \sim \mathcal{N}(\hat{k}, \sigma_k^2), \; M^{(s)} \sim \text{Poisson}(M)$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\log k^{(s)} \sim \mathcal{N}(\log \hat{k}, \; \text{se}^2_{\log k}), \quad H^{(s)} = k^{(s)} \, M^{(s)}, \quad M^{(s)} \sim \text{NB}(M, \hat{\theta}) \text{ (predictive)}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
se	standard error of log k	-

DERIVATION, STEP BY STEP

1. Draw log k from its sampling distribution.
2. Draw counts from NB(M, θ) for a predictive band.
3. Multiply and take quantiles.

ASSUMPTIONS

- Calibration and count noise independent; do not add the same noise twice (k's bootstrap already contains count noise in overlap years).

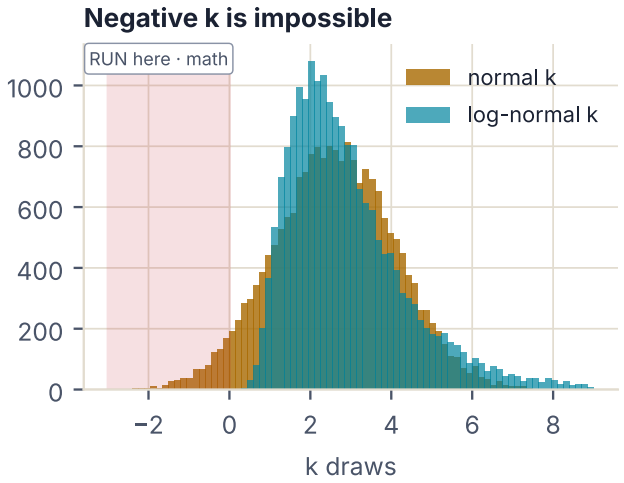


Fig. F25. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Each calendar cell's band combines calibration and counting uncertainty.

WHERE THE OUTPUT GOES

Calendar inspector band.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F25 V2**	**Output**	**Shown on**
se	F24	Monte Carlo error propagation	Each calendar cell's band combines calibration and counting uncertainty.	Calendar inspector band.

V1 → V2 IN ONE LINE

RUN here: with k=2.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F25 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

RUN here: with k=2.6, σ=1.5 the normal draws are negative 4.2% of the time. Use log scale. Say whether the band is a confidence interval (for the expected value) or a predictive interval (for a new count).

WORKED EXAMPLE

M=200, k=2.6 (se log 0.08) → 95% ≈ [430, 620] predictive.

EXECUTABLE PSEUDOCODE

```
k_s = exp(normal(log k, se, S)); M_s = poisson(M, S);
H_s = k_s*M_s
```

EDGE CONDITIONS

- M = 0 still has uncertainty: Garwood upper bound.

WHY THIS, NOT THAT

Analytic propagation is possible for products but breaks with zeros and skew; simulation is simple and honest.

COMMON DOUBTS

Q. Confidence or predictive?
Predictive includes count noise; confidence does not — label which.

Q. Double counting?
Do not add count noise twice if k's bootstrap already includes it for overlap years.

TESTS (UNIT / PROPERTY)

- all draws positive

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F25 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- JCGM 101 (GUM supplement 1)

HINGLISH MEIN SAMJHO

10,000 alag subah socho — reach-time ka range.

IMPLEMENTATION CONTRACT (V2)

MODULE	`uncertainty/`
FUNCTION	`monte_carlo_error_propagation_v2()`
INPUTS	se [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F26
ACCEPTANCE	all draws positive
MAIN FAILURE MODE	M = 0 still has uncertainty: Garwood upper bound.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Normal draws of k (k=2.6, sd=1.5)	P(k<0)=0.042	simulate log k ~ N(log k, se^2)	FIX

YOU MAY SAY

'The 95% band from year-block bootstrap is [lo, hi].'

YOU MAY NOT SAY

a band without its level or type; a band computed from fewer than 8 years without a warning.

Chart spec. Ribbon / whisker with legend stating level and type (confidence vs predictive).

FORMULA 26 / 46 · UNCERTAINTY · LAYER L1

F26 Exact Poisson interval (Garwood)

F24 F25 F26

Correct interval for tiny counts (0–10) where normal approximations fail.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\left[\tfrac{1}{2}\chi^2_{2n,\,\alpha/2},\ \tfrac{1}{2}\chi^2_{2(n+1),\,1-\alpha/2}\right]$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\left[\tfrac{1}{2}\chi^2_{2n,\,\alpha/2},\ \tfrac{1}{2}\chi^2_{2(n+1),\,1-\alpha/2}\right],\quad \text{lower} = 0 \text{ when } n = 0$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
n	observed count	-

DERIVATION, STEP BY STEP

1. Poisson tail probabilities relate to χ^2 quantiles.
2. Lower bound ½χ^2(2n, α/2); upper ½χ^2(2n+2, 1−α/2).
3. For n = 0 the lower bound is 0.

ASSUMPTIONS

- Count is Poisson (no overdispersion).

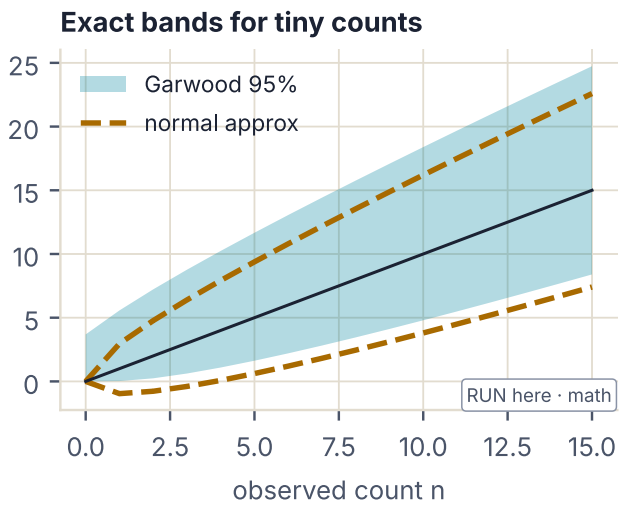


Fig. F26. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Three detections is consistent with a true rate from 0.6 to 8.8.

WHERE THE OUTPUT GOES

Explorer inspector for small cells.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F26 V2	04 Output	05 Shown on
n	F24, F25	Exact Poisson interval (Garwood)	Three detections is consistent with a true rate from 0.6 to 8.8.	Explorer inspector for small cells.

V1 → V2 IN ONE LINE

RUN here: n=3 → [0.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F26 · AUDIT, EXAMPLE, CODE, TESTS

RUN / TESTED LOCALLY

AUDIT VERDICT · OK

RUN here: n=3 → [0.62, 8.77] (doc 0.6–8.8). Add n=0 rule.

WORKED EXAMPLE

n=0 → [0, 3.69]; n=3 → [0.62, 8.77].

EXECUTABLE PSEUDOCODE

```
lo = chi2.ppf(a/2, 2n)/2 if n>0 else 0; hi =
chi2.ppf(1-a/2, 2n+2)/2
```

EDGE CONDITIONS

- Conservative (coverage ≥ nominal).

WHY THIS, NOT THAT

Normal approximations give negative lower bounds for small n.

COMMON DOUBTS

Q. Conservative?
Yes, coverage ≥ nominal.

Q. Overdispersion?
Garwood assumes Poisson; widen with NB if needed.

TESTS (UNIT / PROPERTY)

- garwood(3) == (0.619, 8.767)

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F26 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Garwood 1936

HINGLISH MEIN SAMJHO

3 accident — asli rate 1 bhi ho sakta hai, 8 bhi.

IMPLEMENTATION CONTRACT (V2)

MODULE	`uncertainty/`
FUNCTION	`exact_poisson_interval_garwood_v2()`
INPUTS	n [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	Layer output
ACCEPTANCE	garwood(3) == (0.619, 8.767)
MAIN FAILURE MODE	Conservative (coverage ≥ nominal).

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Garwood n=3	0.6 to 8.8	0.62 to 8.77	OK

YOU MAY SAY

'The 95% band from year-block bootstrap is [lo, hi].'

YOU MAY NOT SAY

a band without its level or type; a band computed from fewer than 8 years without a warning.

Chart spec. Ribbon / whisker with legend stating level and type (confidence vs predictive).

FORMULA 27 / 46 · ANOMALIES AND SEASONS · LAYER L4

F27 Robust anomaly z-score

F27 F28 F29

How unusual is this month compared with the same month in other years, on the exposure-normalised scale, with the target year left out of its own baseline.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING + TESTED (SYNTHETIC)

$$z = \frac{\ln(1+x) - \text{med}}{\max(1.4826\,\text{MAD},\ \sigma_{\min})} \quad (\text{v1.1: } \ln r)$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$y_t = \log\frac{F_t + 0.5}{E_t + 1}, \quad z_t = \frac{y_t - \text{med}(y_{\mathcal{B}\setminus t})}{\max(1.4826\,\text{MAD}(y_{\mathcal{B}\setminus t}),\ 0.15)}, \quad |\mathcal{B}| \geq 8$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
F, E	fire cell-days and exposure	count
$\mathcal{B}$	reference years for that calendar month	-

DERIVATION, STEP BY STEP

1. Transform the target month: y = log((F+0.5)/(E+1)).
2. Take the same calendar month in reference years, excluding the target year.
3. Median and 1.4826·MAD with a 0.15 floor.
4. z = (y − median)/spread; require ≥ 8 reference years.

ASSUMPTIONS

- Reference years are on the same harmonised scale; their own uncertainty is ignored by z (show it separately).

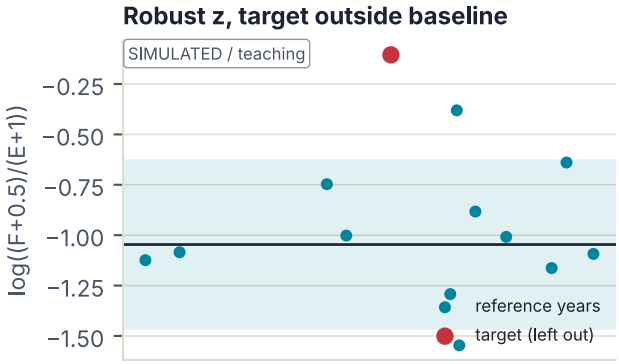


Fig. F27. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

|z| > 2 = unusual for that month, shown with its sample size.

WHERE THE OUTPUT GOES

Calendar 'anomaly z' metric with sample-size badge.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F27 V2**	**Output**	**Shown on**
F, E, $\mathcal{B}$	raw FIRMS / FireMask	Robust anomaly z-score	|z| > 2 = unusual for that month, shown with its sample size.	Calendar 'anomaly z' metric with sample-size badge.

V1 → V2 IN ONE LINE

Three versions exist: code uses raw counts with the target year inside the baseline; Atlas uses ln(1+x); visual guide uses ln(rate), which is undefined when rate = 0 (RUN: math domain error).

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: re-computed in code (RUN). No real-fire result appears here.

F27 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

Three versions exist: code uses raw counts with the target year inside the baseline; Atlas uses ln(1+x); visual guide uses ln(rate), which is undefined when rate = 0 (RUN: math domain error). v2 uses the Anscombe-style log((F+0.5)/(E+1)), leave-target-out, ≥8 reference years.

WORKED EXAMPLE

Nov history 300, 280, 320, 310; target 410 → in-sample z 6.74 vs leave-out z 7.08 (RUN; teaching numbers).

EXECUTABLE PSEUDOCODE

```
ref = [y for yr,y in hist if yr ≠ target]
if len(ref) < 8: return insufficient
```

EDGE CONDITIONS

- MAD = 0 → floor; report 'floor used'.
- Missing months excluded, not zero.

WHY THIS, NOT THAT

Classic z (mean/SD) is pulled by the outlier it should detect; ln(rate) fails at zero; ln(1+x) squeezes small rates.

COMMON DOUBTS

Q. Why +0.5 and +1?
Anscombe-style continuity: finite at F = 0, little effect for large counts.

Q. Why leave the target out?
Otherwise the anomaly inflates its own baseline.

TESTS (UNIT / PROPERTY)

- z(rate=0) finite
- target not in its baseline

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F27 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Visual guide CHANGED card; Book 3 §3.3

HINGLISH MEIN SAMJHO

7-foot ka bachcha — median wala z usko sach mein ajeeb dikhata hai.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/anomaly.py`
FUNCTION	`robust_anomaly_z-score_v2()`
INPUTS	F, E [count]; $\mathscr{B}$ [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F28, F29
ACCEPTANCE	z(rate=0) finite; target not in its baseline
MAIN FAILURE MODE	MAD = 0 → floor; report 'floor used'.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Nov 2024 z	in-sample z=6.74 (target year inside baseline)	leave-target-out z=7.08	FIX
ln(rate) when rate=0	math domain error	log((F+0.5)/(K+1)) = -3.738	FIX

YOU MAY SAY

'November 2024 is +2.3 robust SD above its 2012–2023 normal (n = 12 years).'

YOU MAY NOT SAY

'record burning' when the anomaly is in detections only; season dates as ignition dates.

Chart spec. Calendar diverging overlay on z (grey midpoint) with sample-size badge; season band.

FORMULA 28 / 46 · ANOMALIES AND SEASONS · LAYER L4

F28 Month baseline and spread

F27 F28 F29

The 'normal' November to compare against.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$\text{base}_m = \text{median}_y\, H_{y,m}, \quad \text{spread}_m = 1.4826\,\text{MAD}_y H_{y,m}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\text{base}_{m,-t} = \text{med}_{y \in \mathcal{B} \setminus t}\; y_{y,m}, \quad \text{spread}_{m,-t} = \max(1.4826\,\text{MAD}, 0.15) \text{ on the F27 scale}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
y	log exposure rate (F27)	-

DERIVATION, STEP BY STEP

1. For each calendar month m, gather reference-year values on the F27 scale.
2. Median = normal; 1.4826·MAD = spread.
3. Store per month with n.

ASSUMPTIONS

- Same scale as F27.

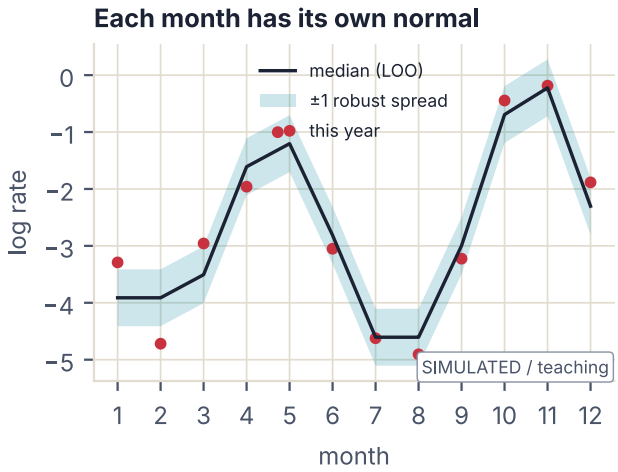


Fig. F28. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

November is compared with past Novembers only.

WHERE THE OUTPUT GOES

Calendar inspector 'normal for November'.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F28 V2	04 Output	05 Shown on
y	F27	Month baseline and spread	November is compared with past Novembers only.	Calendar inspector 'normal for November'.

V1 → V2 IN ONE LINE

calendar_build.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F28 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

calendar_build.month_baseline uses raw counts with floor 1.0 and includes the target year. Must use the same scale, floor and leave-out rule as F27.

WORKED EXAMPLE

Nov normal 480 ± 90 (teaching).

EXECUTABLE PSEUDOCODE

```
see F27
```

EDGE CONDITIONS

- Small history (<8) → 'indicative'.

WHY THIS, NOT THAT

One baseline for all months hides seasonality.

COMMON DOUBTS

Q. Floor 1.0 (v1)?
It belongs to raw counts; on log scale use 0.15.

Q. Few years?
Mark 'indicative'.

TESTS (UNIT / PROPERTY)

- same function used for API and calendar

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F28 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- calendar_build.py

HINGLISH MEIN SAMJHO

December ka bill pichle December se milao, June se nahi.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/anomaly.py`
FUNCTION	`month_baseline_and_spread_v2()`
INPUTS	y [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F29
ACCEPTANCE	same function used for API and calendar
MAIN FAILURE MODE	Small history (<8) → 'indicative'.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Nov 2024 z	in-sample z=6.74 (target year inside baseline)	leave-target-out z=7.08	FIX

YOU MAY SAY

'November 2024 is +2.3 robust SD above its 2012–2023 normal (n = 12 years).'

YOU MAY NOT SAY

'record burning' when the anomaly is in detections only; season dates as ignition dates.

Chart spec. Calendar diverging overlay on z (grey midpoint) with sample-size badge; season band.

FORMULA 29 / 46 · ANOMALIES AND SEASONS · LAYER L4

F29 Season start, peak and end (10/90%)

F27 F28 F29

Critical-period dates for the problem statement: when does the burning season open, peak and close?

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$t_{\text{start}} = \min\{t : S(t) \geq 0.10\, S_{\text{tot}}\}, \quad t_{\text{end}} = \min\{t : S(t) \geq 0.90\, S_{\text{tot}}\}$$

V2 · REVISED DEFINITION USED FROM NOW ON

same, on a declared season window $[t_a, t_b]$ (e.g. 15 Sep–31 Dec), weekly bins; NULL if any bin unobserv

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
S(t)	cumulative detected activity	cell-days

DERIVATION, STEP BY STEP

1. Choose the season window (e.g. 15 Sep–31 Dec).
2. Weekly totals; cumulative share.
3. Start = first week ≥ 10%, end = first ≥ 90%, peak = max week.

ASSUMPTIONS

- Detected activity timing ≈ burning timing (false when burning shifts after overpasses).

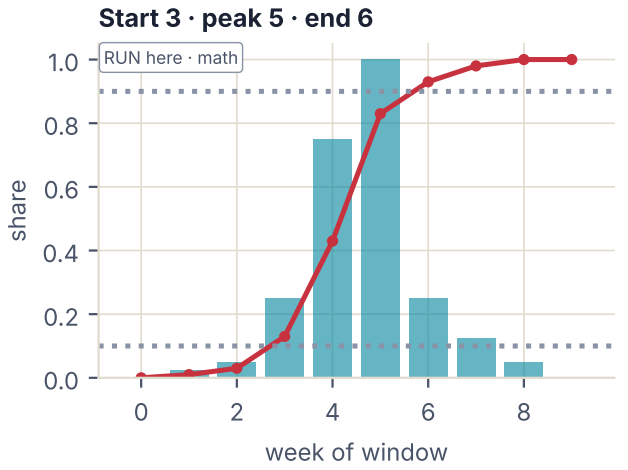


Fig. F29. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Critical period for officials, with the timing caveat.

WHERE THE OUTPUT GOES

Calendar season band.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F29 V2**	**Output**	**Shown on**
S(t)	F27, F28	Season start, peak and end (10/90%)	Critical period for officials, with the timing caveat.	Calendar season band.

V1 → V2 IN ONE LINE

Dates depend on the window and on detection timing; they are dates of detected activity, not ignition dates.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: re-computed in code (RUN). No real-fire result appears here.

F29 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Dates depend on the window and on detection timing; they are dates of detected activity, not ignition dates. Declare the window; null when gaps.

WORKED EXAMPLE

weekly [0,1,2,10,30,40,10,5,2,0] → start 3, peak 5, end 6 (RUN).

EXECUTABLE PSEUDOCODE

```
c = cumsum(w); start = argmax(c ⩾ .1*total)
```

EDGE CONDITIONS

- Two seasons (rabi/kharif) need two windows.

WHY THIS, NOT THAT

First/last detection dates are noisy; share-based dates are robust.

COMMON DOUBTS

Q. Is this the ignition date?
No — detected-activity timing.

Q. Two seasons?
Two windows.

TESTS (UNIT / PROPERTY)

- v1 test passes; gaps → None

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F29 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Book 3 §3.4

HINGLISH MEIN SAMJHO

Festival sale ka season 10% bikri se shuru, 90% par khatam.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/anomaly.py`
FUNCTION	`season_start,_peak_and_end_10_90%_v2()`
INPUTS	S(t) [cell-days]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	Layer output
ACCEPTANCE	v1 test passes; gaps → None
MAIN FAILURE MODE	Two seasons (rabi/kharif) need two windows.

YOU MAY SAY

'November 2024 is +2.3 robust SD above its 2012–2023 normal (n = 12 years).'

YOU MAY NOT SAY

'record burning' when the anomaly is in detections only; season dates as ignition dates.

Chart spec. Calendar diverging overlay on z (grey midpoint) with sample-size badge; season band.

FORMULA 30 / 46 · TRENDS AND SEAM TESTS · LAYER L4

F30 Mann–Kendall trend test

F30 F31 F32 F33 F34 F35 F36 F37

Is the seasonal total rising or falling across years, without assuming a straight line or normal errors?

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$S = \sum_{i<j} \operatorname{sgn}(x_j - x_i), \quad \operatorname{Var}(S) = \frac{n(n-1)(2n+5)}{18}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\operatorname{Var}(S) = \frac{n(n-1)(2n+5) - \sum_p t_p(t_p-1)(2t_p+5)}{18} \quad \text{(tie groups } t_p\text{)}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
t_p	size of each tie group	-

DERIVATION, STEP BY STEP

1. Count concordant minus discordant pairs: S.
2. Variance with tie correction.
3. Continuity-corrected Z and two-sided p.

ASSUMPTIONS

- Independent years (see F31).

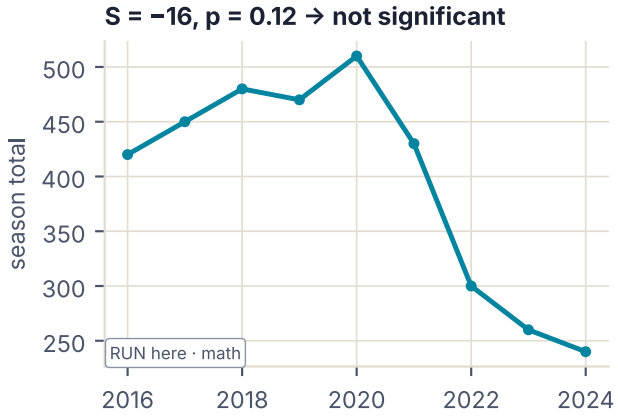


Fig. F30. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Direction of change with an honest significance test.

WHERE THE OUTPUT GOES

Calendar trend line with p and effect size.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F30 V2**	**Output**	**Shown on**
t_p	raw FIRMS / FireMask	Mann–Kendall trend test	Direction of change with an honest significance test.	Calendar trend line with p and effect size.

V1 → V2 IN ONE LINE

RUN here: tie correction missing in v1.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: re-computed in code (RUN). No real-fire result appears here.

F30 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

RUN here: tie correction missing in v1. Series [5,3,6,2,5,4,5,3]: v1 var 65.33 vs corrected 60.67 (pymannkendall agrees, p = 0.700). Counts often tie (zeros), so this matters.

WORKED EXAMPLE

Book 3 series (b) 9 years: S = −16, Z = −1.56, p = 0.12 → not significant (RUN, matches).

EXECUTABLE PSEUDOCODE

```
var = (n(n-1)(2n+5) - sum(t(t-1)(2t+5)))/18
```

EDGE CONDITIONS

- n < 8: report 'too short'.

WHY THIS, NOT THAT

Linear regression assumes normal residuals and is outlier-sensitive; MK only uses order.

COMMON DOUBTS

Q. Is p > 0.05 'no change'?
No — 'not shown with this data'.

Q. Effect size?
Report Sen slope (F32) with it.

TESTS (UNIT / PROPERTY)

- matches pymannkendall.original_test

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F30 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Kendall 1975

HINGLISH MEIN SAMJHO

Har baad wala saal pehle se upar ya neeche — sirf order ginte hain.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/trend.py, validate/`
FUNCTION	`mann_kendall_trend_test_v2()`
INPUTS	t_p [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F31, F32
ACCEPTANCE	matches pymannkendall.original_test
MAIN FAILURE MODE	n < 8: report 'too short'.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
MK with ties [5,3,6,2,5,4,5,3]	var=65.33 z=-0.371 p=0.711	var=60.67 z=-0.385 p=0.700 (pymannkendall p=0.700)	FIX
Book 3 series (b)	S=-16 Z=-1.56 p=0.12	S=-16 Z=-1.56 p=0.12	OK

YOU MAY SAY

'No significant trend detected (p = 0.12); Sen slope −26 cell-days/yr.'

YOU MAY NOT SAY

'no change' from p > 0.05; 'seam removed' from a single non-significant test.

Chart spec. Trend line with Sen slope label and p; event-study chart with placebo seams.

FORMULA 31 / 46 · TRENDS AND SEAM TESTS · LAYER L4

F31 Hamed–Rao autocorrelation correction

F30 F31 F32 F33 F34 F35 F36 F37

Neighbouring years are similar, so the effective sample is smaller; this widens the test.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$\mathrm{Var}^*(S) = \mathrm{Var}(S) \cdot \frac{n}{n_s^*}, \quad \frac{n}{n_s^*} = 1 + \frac{2}{n(n-1)(n-2)} \sum_k (n-k)(n-k-1)(n-k-2)\rho_k$$

V2 · REVISED DEFINITION USED FROM NOW ON

same, with ρ_k of detrended ranks, only significant lags; seasonal Kendall for monthly series

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
ρ_k	rank autocorrelation at lag k	-

DERIVATION, STEP BY STEP

1. Detrend with Sen slope; compute rank autocorrelations ρ_k.
2. Keep significant lags; compute variance inflation n/n*.
3. Recompute Z with the inflated variance.

ASSUMPTIONS

- Enough years to estimate ρ_k (≥15).

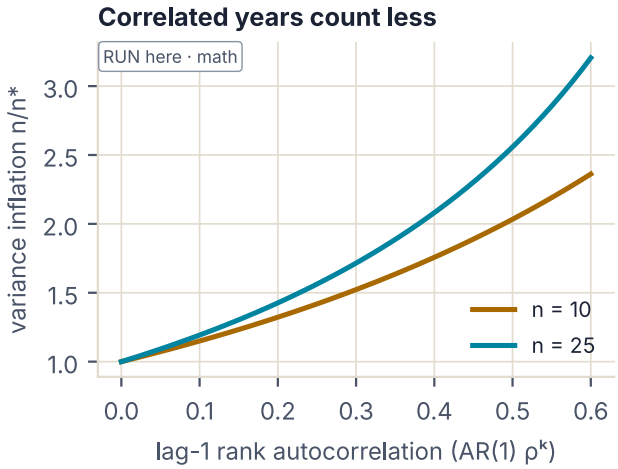


Fig. F31. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Fairer p-value when neighbouring years resemble each other.

WHERE THE OUTPUT GOES

Trend tooltip 'autocorrelation-adjusted'.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F31 V2	04 Output	05 Shown on
ρ_k	F30	Hamed–Rao autocorrelation correction	Fairer p-value when neighbouring years resemble each other.	Trend tooltip 'autocorrelation-adjusted'.

V1 → V2 IN ONE LINE

RUN here on 9 years: z = −1.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F31 · AUDIT, EXAMPLE, CODE, TESTS

RUN / TESTED LOCALLY

AUDIT VERDICT · OK + LIMIT

RUN here on 9 years: z = −1.56, p = 0.118 (no correction applied — lags not significant). With 25 years ρ_k is noisy; report both raw and corrected p.

WORKED EXAMPLE

Book 3 series (b): unchanged p = 0.118.

EXECUTABLE PSEUDOCODE

```
pymannkendall.hamed_rao_modification_test(x)
```

EDGE CONDITIONS

- Do not mix with BH across cells without stating it.

WHY THIS, NOT THAT

Ignoring autocorrelation makes trends look more significant than they are.

COMMON DOUBTS

Q. With 25 years?
ρ estimates are noisy; show both p-values.

Q. Seasonal data?
Use Seasonal Kendall across months.

TESTS (UNIT / PROPERTY)

- equals pymannkendall

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F31 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Hamed & Rao 1998

HINGLISH MEIN SAMJHO

Aaj ka mood kal se juda — saat din saat alag gawah nahi.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/trend.py, validate/`
FUNCTION	`hamed_rao_autocorrelation_correcti_v2()`
INPUTS	ρ_k [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F32, F33
ACCEPTANCE	equals pymannkendall
MAIN FAILURE MODE	Do not mix with BH across cells without stating it.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Hamed-Rao on series (b)	-	z=-1.56 p=0.118	NOTE

YOU MAY SAY

'No significant trend detected (p = 0.12); Sen slope −26 cell-days/yr.'

YOU MAY NOT SAY

'no change' from p > 0.05; 'seam removed' from a single non-significant test.

Chart spec. Trend line with Sen slope label and p; event-study chart with placebo seams.

FORMULA 32 / 46 · TRENDS AND SEAM TESTS · LAYER L4

F32 Sen's slope

F30 F31 F32 F33 F34 F35 F36 F37

How fast the series changes per year, robust to outliers.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$\beta = \text{median}_{i<j} \frac{x_j - x_i}{j - i}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\beta = \text{median}_{i<j} \frac{x_j - x_i}{t_j - t_i}$$ (use real year gaps; report % of baseline and CI)

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
t	year	yr

DERIVATION, STEP BY STEP

1. All pairwise slopes (x_j − x_i)/(t_j − t_i).
2. Take the median.

ASSUMPTIONS

- Monotonic trend.

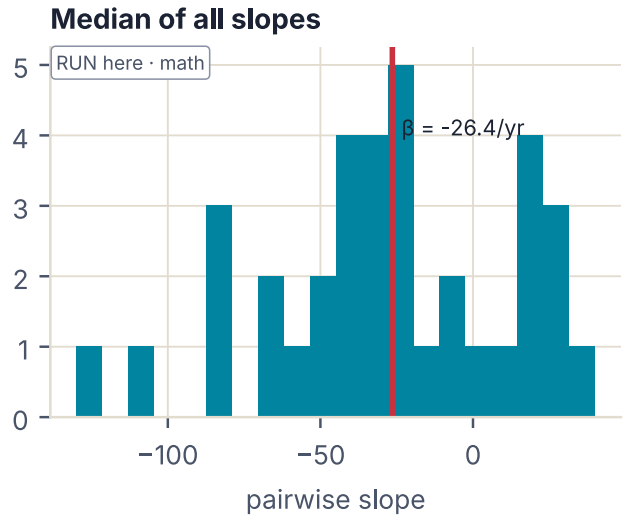


Fig. F32. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

How fast the season changes per year.

WHERE THE OUTPUT GOES

Calendar trend label 'β per year'.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F32 V2**	**Output**	**Shown on**
t	F30, F31	Sen's slope	How fast the season changes per year.	Calendar trend label 'β per year'.

V1 → V2 IN ONE LINE

RUN here: series (b) → −26.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F32 · AUDIT, EXAMPLE, CODE, TESTS

RUN / TESTED LOCALLY

AUDIT VERDICT · OK

RUN here: series (b) → −26.4 per year (matches). Use t gaps when years are missing.

WORKED EXAMPLE

[1,3,5,7] → 2.0.

EXECUTABLE PSEUDOCODE

```
median of pairwise slopes
```

EDGE CONDITIONS

- Missing years: skip pairs with NULL.

WHY THIS, NOT THAT

OLS slope is pulled by one extreme year.

COMMON DOUBTS

Q. Units?
cell-days per year (or % of baseline per year).

Q. Interval?
Rank-based CI from the slope distribution.

TESTS (UNIT / PROPERTY)

- v1 test passes

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F32 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Sen 1968

HINGLISH MEIN SAMJHO

Har do saal ke beech ki dhalaan — beech wali.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/trend.py, validate/`
FUNCTION	`sens_slope_v2()`
INPUTS	t [yr]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F33, F34
ACCEPTANCE	v1 test passes
MAIN FAILURE MODE	Missing years: skip pairs with NULL.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Sen slope series (b)	-26.4/yr	-26.4/yr	CHECK

YOU MAY SAY

'No significant trend detected (p = 0.12); Sen slope −26 cell-days/yr.'

YOU MAY NOT SAY

'no change' from p > 0.05; 'seam removed' from a single non-significant test.

Chart spec. Trend line with Sen slope label and p; event-study chart with placebo seams.

FORMULA 33 / 46 · TRENDS AND SEAM TESTS · LAYER VALIDATION

F33 Pettitt change-point test

F30 F31 F32 **F33** F34 F35 F36 F37

Finds the single most likely jump year; used to show the 2012 seam before harmonisation and its absence after.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$U_t = \sum_{i \le t} \sum_{j > t} \operatorname{sgn}(x_j - x_i), \quad K = \max_t |U_t|, \quad p \approx 2e^{-6K^2/(n^3+n^2)}$$

V2 · REVISED DEFINITION USED FROM NOW ON

same, plus placebo distribution: K at random non-sensor years and at 2018/2023 platform additions

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
K	max absolute U	-

DERIVATION, STEP BY STEP

1. For each split t, U_t counts how much later values exceed earlier ones.
2. K = max |U_t|.
3. Approximate p from K and n.

ASSUMPTIONS

- One change point; p approximate.

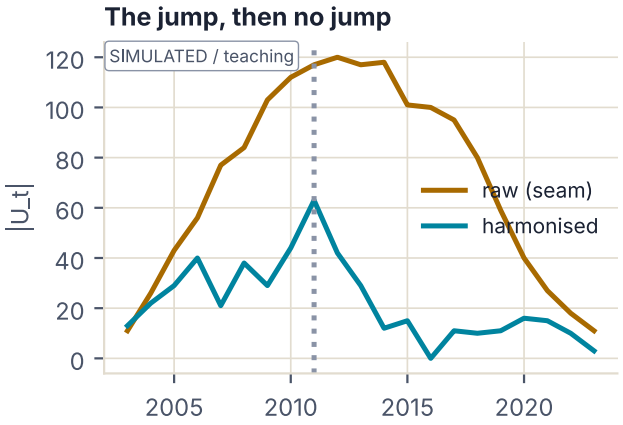


Fig. F33. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Seam test: the jump should be visible in raw, not in harmonised.

WHERE THE OUTPUT GOES

Validation seam tile.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F33 V2**	**Output**	**Shown on**
K	F31, F32	Pettitt change-point test	Seam test: the jump should be visible in raw, not in harmonised.	Validation seam tile.

V1 → V2 IN ONE LINE

A non-significant Pettitt after harmonisation does not prove the seam is gone (low power with 25 points).

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F33 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

A non-significant Pettitt after harmonisation does not prove the seam is gone (low power with 25 points). Combine with placebo years and the seam ratio interval.

WORKED EXAMPLE

Synthetic break at year 10 found before, not after (v1 must-pass test).

EXECUTABLE PSEUDOCODE

```
for t: U_t = ...; K = max |U|
```

EDGE CONDITIONS

- Real changes around 2012 (policy, weather) confound.

WHY THIS, NOT THAT

Detects one shift without assuming its size; weak with few points.

COMMON DOUBTS

Q. No break after harmonisation proves success?
No — test power is low; use placebo seams too.

Q. Several breaks?
Use PELT (F34).

TESTS (UNIT / PROPERTY)

- detects planted break; placebo false-positive rate ≈ α

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F33 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Pettitt 1979

HINGLISH MEIN SAMJHO

Dukaan ne supplier kab badla — sabse bada farak wala din.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/trend.py, validate/`
FUNCTION	`pettitt_change-point_test_v2()`
INPUTS	K [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F34, F35
ACCEPTANCE	detects planted break; placebo false-positive rate ≈ α
MAIN FAILURE MODE	Real changes around 2012 (policy, weather) confound.

YOU MAY SAY

'No significant trend detected (p = 0.12); Sen slope −26 cell-days/yr.'

YOU MAY NOT SAY

'no change' from p > 0.05; 'seam removed' from a single non-significant test.

Chart spec. Trend line with Sen slope label and p; event-study chart with placebo seams.

FORMULA 34 / 46 · TRENDS AND SEAM TESTS · LAYER VALIDATION

F34 PELT multiple change points

F30 F31 F32 F33 **F34** F35 F36 F37

Find all jump points (2012 S-NPP, 2018 NOAA-20, 2023 NOAA-21, processing changes).

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\min_{\tau} \sum_{i} \mathcal{C}\left(x_{(\tau_{i-1}+1):\tau_i}\right) + \beta\,|\tau|$$

V2 · REVISED DEFINITION USED FROM NOW ON

Poisson/NB segment cost on monthly exposure rates; penalty $\beta = \mathrm{BIC} \cdot \log n$ declared before running

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
β	penalty	-

DERIVATION, STEP BY STEP

1. Define a segment cost (Poisson/NB or L2 on deseasoned log rates).
2. Add penalty β per break.
3. Dynamic programming with pruning finds the optimal break set.

ASSUMPTIONS

- Penalty fixed in advance.

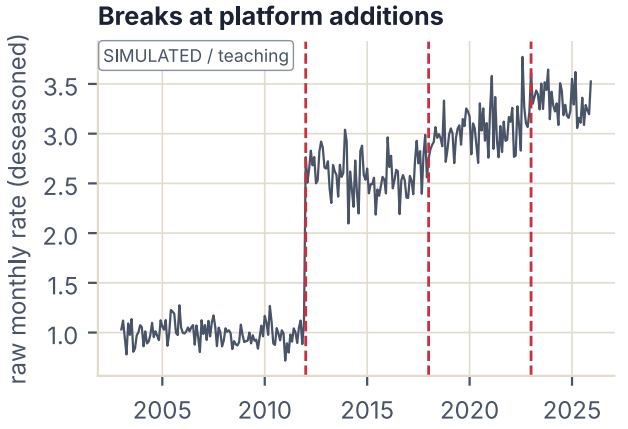


Fig. F34. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

All jump years, including 2018 and 2023 platform additions.

WHERE THE OUTPUT GOES

Validation event study.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F34 V2	04 Output	05 Shown on
β	F32, F33	PELT multiple change points	All jump years, including 2018 and 2023 platform additions.	Validation event study.

V1 → V2 IN ONE LINE

Choose cost for counts (not rbf by default); fix penalty before seeing results to avoid tuning to a story.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F34 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Choose cost for counts (not rbf by default); fix penalty before seeing results to avoid tuning to a story.

WORKED EXAMPLE

Expect breaks near platform additions in raw counts, none after harmonisation.

EXECUTABLE PSEUDOCODE

```
ruptures.Pelt(model='l2').fit(y).predict(pen=pen)
```

EDGE CONDITIONS

- Seasonality must be removed first.

WHY THIS, NOT THAT

Binary segmentation is greedy; PELT is exact under the cost.

COMMON DOUBTS

Q. Which penalty?
Declare BIC-type penalty in advance.

Q. Seasonality?
Remove first.

TESTS (UNIT / PROPERTY)

- synthetic two-break series recovered

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F34 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Killick et al. 2012

HINGLISH MEIN SAMJHO

Gaane ko verse/chorus mein todna — penalty zyada tukde rokti hai.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/trend.py, validate/`
FUNCTION	`pelt_multiple_change_points_v2()`
INPUTS	β [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F35, F36
ACCEPTANCE	synthetic two-break series recovered
MAIN FAILURE MODE	Seasonality must be removed first.

YOU MAY SAY

'No significant trend detected (p = 0.12); Sen slope −26 cell-days/yr.'

YOU MAY NOT SAY

'no change' from p > 0.05; 'seam removed' from a single non-significant test.

Chart spec. Trend line with Sen slope label and p; event-study chart with placebo seams.

FORMULA 35 / 46 · TRENDS AND SEAM TESTS · LAYER VALIDATION

F35 Seam ratio

F30 F31 F32 F33 F34 **F35** F36 F37

How much of the instrument step remains after harmonisation.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$R_{\text{seam}} = \frac{|H_{\text{after}} - H_{\text{before}}|}{|V_{\text{after,raw}} - M_{\text{before}}|}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$R_{\text{seam}} = \frac{|\overline{\log H}_{2012\text{–}14} - \overline{\log H}_{2009\text{–}11}|}{|\overline{\log V}_{2012\text{–}14} - \overline{\log M}_{2009\text{–}11}|}$$ with bootstrap CI and placebo seams

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
H, V, M	harmonised, VIIRS raw, MODIS raw	cell-days

DERIVATION, STEP BY STEP

1. Mean log value 3 years after minus 3 years before, harmonised.
2. Same for raw (VIIRS after vs MODIS before).
3. Ratio of absolute steps, with bootstrap CI.

ASSUMPTIONS

- No real change exactly at the seam.

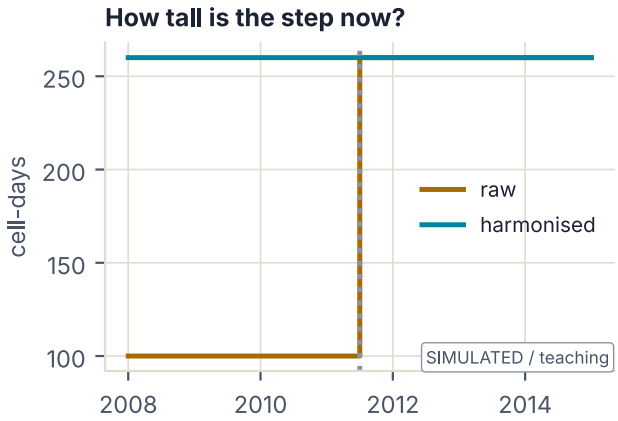


Fig. F35. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

How much of the instrument step is left.

WHERE THE OUTPUT GOES

Home stat + Validation tile.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F35 V2**	**Output**	**Shown on**
H, V, M	F33, F34	Seam ratio	How much of the instrument step is left.	Home stat + Validation tile.

V1 → V2 IN ONE LINE

One-year-to-one-year comparison mixes real interannual change with the instrument step.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: re-computed in code (RUN). No real-fire result appears here.

F35 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

One-year-to-one-year comparison mixes real interannual change with the instrument step. Use 3-year windows, log scale, interval, and compare against placebo seams.

WORKED EXAMPLE

seam_ratio(100, 260, 104.68) = 0.029 (RUN).

EXECUTABLE PSEUDOCODE

```
see v2
```

EDGE CONDITIONS

- Denominator 0 → undefined (v1 returns 0.0 — fix to None).

WHY THIS, NOT THAT

One-year steps are dominated by weather; windows and logs make it a step-size estimate.

COMMON DOUBTS

Q. Can it be 0 by accident?
Yes if a real change cancels the step — check placebo seams.

Q. Target?
Upper CI ≤ 0.25 (predeclared).

TESTS (UNIT / PROPERTY)

- placebo seams distribution reported

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F35 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Kiln Watch README (self-reported 2.9%)

HINGLISH MEIN SAMJHO

Seedhi ki step kitni bachi.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/trend.py, validate/`
FUNCTION	`seam_ratio_v2()`
INPUTS	H, V, M [cell-days]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F36, F37
ACCEPTANCE	placebo seams distribution reported
MAIN FAILURE MODE	Denominator 0 → undefined (v1 returns 0.0 — fix to None).

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
seam_ratio(100,260,104.68)	0.02925	0.02925	NOTE

YOU MAY SAY

'No significant trend detected (p = 0.12); Sen slope −26 cell-days/yr.'

YOU MAY NOT SAY

'no change' from p > 0.05; 'seam removed' from a single non-significant test.

Chart spec. Trend line with Sen slope label and p; event-study chart with placebo seams.

FORMULA 36 / 46 · TRENDS AND SEAM TESTS · LAYER VALIDATION

F36 Held-out error (LOYO and blocked CV)

F30 F31 F32 F33 F34 F35 F36 F37

Honest out-of-sample error: hide a year (or region), fit on the rest, predict the hidden one.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\mathrm{MAPE}_{\mathrm{LOYO}} = \frac{1}{Y}\sum_{y}\frac{|\hat{k}_{-y}M_y - V_y|}{V_y}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\mathrm{MAE}_y = |\hat{k}_{-y}M_y - V_y|,\quad D_y = 2\left[V_y \ln\frac{V_y}{\hat{\mu}_y} - (V_y - \hat{\mu}_y)\right],\quad \text{sMAPE with } V_y + \hat{\mu}_y > 0$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
D	Poisson deviance	-

DERIVATION, STEP BY STEP

1. For each year y in overlap: fit without y.
2. Predict VIIRS from MODIS in y.
3. Score MAE, Poisson deviance, guarded sMAPE.

ASSUMPTIONS

- Hidden years independent of training years (block by season).

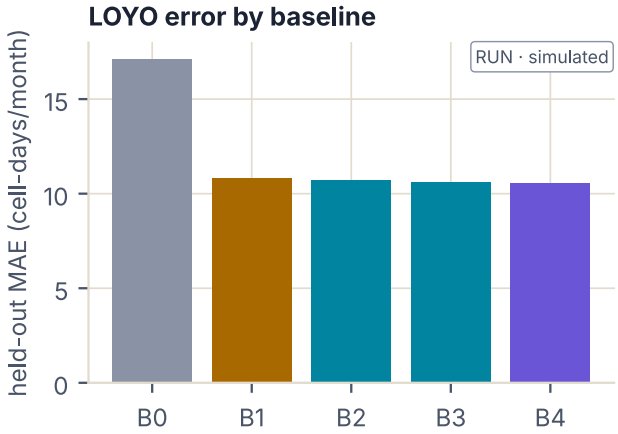


Fig. F36. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

The honest error the Validation page shows.

WHERE THE OUTPUT GOES

Validation LOYO chart.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F36 V2**	**Output**	**Shown on**
D	F34, F35	Held-out error (LOYO and blocked CV)	The honest error the Validation page shows.	Validation LOYO chart.

V1 → V2 IN ONE LINE

MAPE is infinite when V = 0 and favours under-prediction.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F36 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

MAPE is infinite when V = 0 and favours under-prediction. Use MAE + Poisson deviance + guarded sMAPE; also hold out regions.

WORKED EXAMPLE

Book 3 LOYO: hidden errors 0, 0, 3.85, 2.04% → median 1.0% (teaching).

EXECUTABLE PSEUDOCODE

```
for y in years: k = fit(exclude y); score(y)
```

EDGE CONDITIONS

- Thresholds must be declared before running.

WHY THIS, NOT THAT

In-sample fit always looks good; LOYO measures generalisation.

COMMON DOUBTS

Q. Why not MAPE?
Infinite at zero, biased to under-prediction.

Q. Regions?
Also leave-one-region-out.

TESTS (UNIT / PROPERTY)

- no division by zero

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F36 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- FireCal README (self-reported 1.6–5%)

HINGLISH MEIN SAMJHO

Ek purana paper chhupao, baaki se padho, chhupe wale par test do.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/trend.py, validate/`
FUNCTION	`held-out_error_loyo_and_blocked_cv_v2()`
INPUTS	D [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F37
ACCEPTANCE	no division by zero
MAIN FAILURE MODE	Thresholds must be declared before running.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
MAPE when VIIRS=0	division by zero / inf	use MAE, Poisson deviance, sMAPE with guard	FIX

YOU MAY SAY

'No significant trend detected (p = 0.12); Sen slope −26 cell-days/yr.'

YOU MAY NOT SAY

'no change' from p > 0.05; 'seam removed' from a single non-significant test.

Chart spec. Trend line with Sen slope label and p; event-study chart with placebo seams.

FORMULA 37 / 46 · TRENDS AND SEAM TESTS · LAYER L4

F37 Benjamini–Hochberg false discovery rate

F30 F31 F32 F33 F34 F35 F36 F37

When thousands of cells are tested for trends, keeps the share of false 'significant' cells near α.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$\text{reject } H_{(i)} \text{ for } i \le \max\{i : p_{(i)} \le \tfrac{i}{m}\alpha\}$$

V2 · REVISED DEFINITION USED FROM NOW ON

same; m = number of cells actually tested (with enough data); BY variant if spatially dependent

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
m	tests	-

DERIVATION, STEP BY STEP

1. Sort p-values.
2. Find the largest i with p_(i) ≤ iα/m.
3. Reject all tests up to i.

ASSUMPTIONS

- Independent or positively dependent tests.

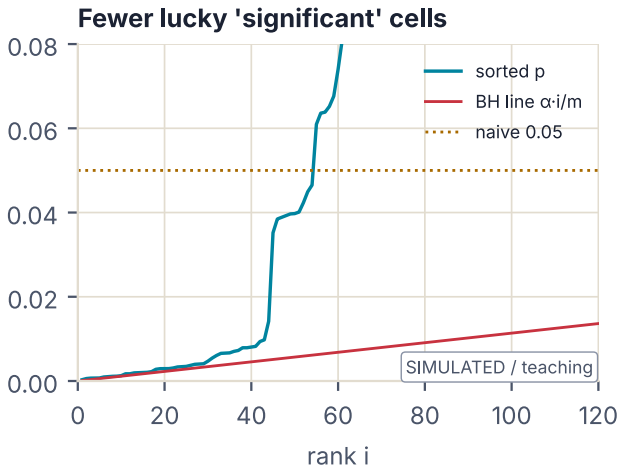


Fig. F37. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

A trend map with ~5% false 'significant' cells.

WHERE THE OUTPUT GOES

Explorer trend map 'FDR 5%'.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F37 V2**	**Output**	**Shown on**
m	F35, F36	Benjamini–Hochberg false discovery rate	A trend map with ~5% false 'significant' cells.	Explorer trend map 'FDR 5%'.

V1 → V2 IN ONE LINE

Correct.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: unchanged. Example values: teaching / simulated. No real-fire result appears here.

F37 · AUDIT, EXAMPLE, CODE, TESTS

OK AS WRITTEN

AUDIT VERDICT · OK

Correct. Define m honestly (cells with data).

WORKED EXAMPLE

94 nominal → 53 after BH (v1 synthetic test).

EXECUTABLE PSEUDOCODE

```
see chronos_core.benjamini_hochberg
```

EDGE CONDITIONS

- Spatial correlation: use BY or cluster-level tests.

WHY THIS, NOT THAT

Bonferroni is too strict for thousands of cells; BH controls the false discovery share.

COMMON DOUBTS

Q. Spatial dependence?
BY variant or cluster-level testing.

Q. m?
Only cells actually tested.

TESTS (UNIT / PROPERTY)

- monotone rejection set

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F37 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Benjamini & Hochberg 1995

HINGLISH MEIN SAMJHO

1000 paase phenko — kuch luck se 'special' lagenge.

IMPLEMENTATION CONTRACT (V2)

MODULE	`summarize/trend.py, validate/`
FUNCTION	`benjamini_hochberg_false_discovery_v2()`
INPUTS	m [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	Layer output
ACCEPTANCE	monotone rejection set
MAIN FAILURE MODE	Spatial correlation: use BY or cluster-level tests.

YOU MAY SAY

'No significant trend detected (p = 0.12); Sen slope −26 cell-days/yr.'

YOU MAY NOT SAY

'no change' from p > 0.05; 'seam removed' from a single non-significant test.

Chart spec. Trend line with Sen slope label and p; event-study chart with placebo seams.

FORMULA 38 / 46 · FUSION AND AI · LAYER EXPERIMENTAL

F38 Kalman fusion (local-level model)

F38 F39 F40 F41

One smoothed activity series from several sensors, each weighted by its noise.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$\hat{x} = \hat{x}^{-} + K(z - \hat{x}^{-}), \qquad K = \frac{P^{-}}{P^{-} + R}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$x_t = x_{t-1} + w_t,\ z_{s,t} = x_t + v_{s,t};\ \ P_t^{-} = P_{t-1} + Q\Delta t,\ K = \frac{P^{-}}{P^{-} + R_s};\ \text{skip update when } z \text{ missing; init } c$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
Q	process noise	-
R_s	measurement noise per sensor	-

DERIVATION, STEP BY STEP

1. Predict: $x^{-} = x$, $P^{-} = P + Q\Delta t$.
2. If a measurement exists: $K = P^{-}/(P^{-} + R)$, update x and P.
3. If not: keep the prediction.

ASSUMPTIONS

- Constant Q, R; Gaussian; one state.

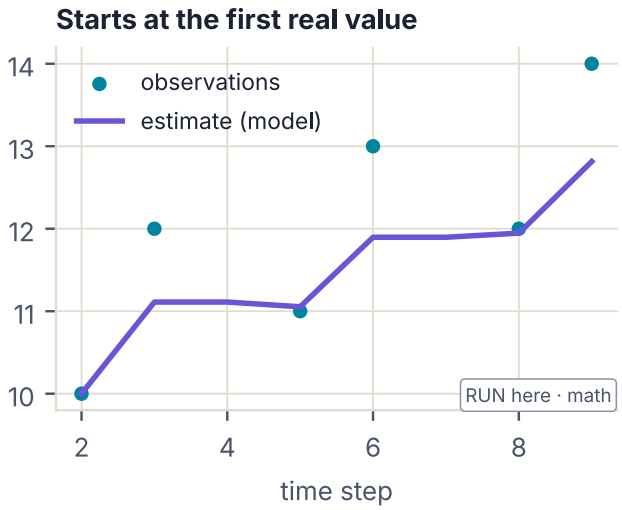


Fig. F38. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

A smoothed experimental series — never the headline number.

WHERE THE OUTPUT GOES

Sensor Lab experimental tab (PROPOSED).

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F38 V2**	**Output**	**Shown on**
Q, R_s	raw FIRMS / FireMask	Kalman fusion (local-level model)	A smoothed experimental series — never the headline number.	Sensor Lab experimental tab (PROPOSED).

V1 → V2 IN ONE LINE

v1 can start with level None when the first values are missing.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F38 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

v1 can start with level None when the first values are missing. RUN here with [None, None, 10, 12, None, 11] → [None, None, 10, 11.11, 11.11, 11.05]. Output is a model estimate, never 'observed'.

WORKED EXAMPLE

see audit run

EXECUTABLE PSEUDOCODE

```
if x is None: init on first valid z
```

EDGE CONDITIONS

- Irregular time steps scale Q by Δt.

WHY THIS, NOT THAT

Simple averaging ignores noise differences; Kalman weights by trust and handles gaps.

COMMON DOUBTS

Q. Is the output observed?
No — it is a model state.

Q. Q and R?
Estimate from overlap residuals.

TESTS (UNIT / PROPERTY)

- no arithmetic on None

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F38 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Kalman 1960; Durbin & Koopman

HINGLISH MEIN SAMJHO

Google Maps ka blue dot — GPS aur sensor mila ke.

IMPLEMENTATION CONTRACT (V2)

MODULE	`classify/, experimental/`
FUNCTION	`kalman_fusion_local-level_model_v2()`
INPUTS	Q [-]; R_s [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F39, F40
ACCEPTANCE	no arithmetic on None
MAIN FAILURE MODE	Irregular time steps scale Q by Δt.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Kalman with leading missing values	level=None then arithmetic error (per user audit)	[None, None, 10.0, 11.11, 11.11, 11.05]	FIX

YOU MAY SAY

'Model estimate (experimental)'; 'classifier set {crop, kiln} at 90% marginal coverage.'

YOU MAY NOT SAY

a fused value labelled as observed; a class without the abstain option.

Chart spec. Experimental tab with violet 'model' style; set chips for classes.

FORMULA 39 / 46 · FUSION AND AI · LAYER L2

F39 Split conformal prediction sets

F38 F39 F40 F41

Source class sets with a marginal coverage guarantee: {crop} or {crop, kiln}.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$\hat{q} = \text{Quantile}_{\lceil (n+1)(1-\alpha) \rceil / n}\{1 - \hat{p}_{y_i}(x_i)\}, \quad C(x) = \{y : 1 - \hat{p}_y(x) \le \hat{q}\}$$

V2 · REVISED DEFINITION USED FROM NOW ON

same; calibration set = held-out sites (GroupKFold by site/region); report coverage by region; abstain

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
α	miscoverage	-

DERIVATION, STEP BY STEP

1. Train a probabilistic classifier on training sites.
2. On held-out calibration sites, scores 1 − p(true class).
3. Threshold = adjusted quantile; prediction set = classes under it.

ASSUMPTIONS

- Exchangeability between calibration and new sites.

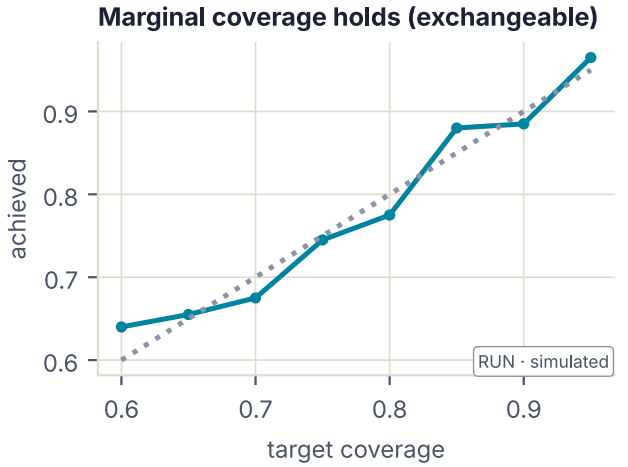


Fig. F39. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

A set like {crop, kiln} instead of a forced single label.

WHERE THE OUTPUT GOES

Source page set chips.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F39 V2**	**Output**	**Shown on**
α	F38	Split conformal prediction sets	A set like {crop, kiln} instead of a forced single label.	Source page set chips.

V1 → V2 IN ONE LINE

Guarantee is marginal, not per district.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F39 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Guarantee is marginal, not per district. Needs real labelled sites; none exist yet → feature remains PROPOSED.

WORKED EXAMPLE

n=100 calibration, α=0.1 → q̂ is the 91st smallest score.

EXECUTABLE PSEUDOCODE

```
q = quantile(scores, ceil((n+1)(1-a))/n, 'higher')
```

EDGE CONDITIONS

- Shift in region → coverage breaks; monitor.

WHY THIS, NOT THAT

Raw probabilities are often miscalibrated; conformal sets give a coverage guarantee under exchangeability.

COMMON DOUBTS

Q. Per-district guarantee?
No — marginal only.

Q. Without labels?
Not possible; feature stays PROPOSED.

TESTS (UNIT / PROPERTY)

- simulated coverage ≈ 0.9

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F39 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Vovk et al.; Angelopoulos & Bates 2021

HINGLISH MEIN SAMJHO

Mausam app 'baarish ya baadal' — shak ko imaandari se dikhana.

IMPLEMENTATION CONTRACT (V2)

MODULE	`classify/, experimental/`
FUNCTION	`split_conformal_prediction_sets_v2()`
INPUTS	α [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F40, F41
ACCEPTANCE	simulated coverage ≈ 0.9
MAIN FAILURE MODE	Shift in region → coverage breaks; monitor.

YOU MAY SAY

'Model estimate (experimental)'; 'classifier set {crop, kiln} at 90% marginal coverage.'

YOU MAY NOT SAY

a fused value labelled as observed; a class without the abstain option.

Chart spec. Experimental tab with violet 'model' style; set chips for classes.

FORMULA 40 / 46 · FUSION AND AI · LAYER L2

F40 Haversine distance

F38 F39 F40 F41

Distances for clustering and industrial/agricultural context.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$d = 2R \arcsin \sqrt{\sin^2 \frac{\Delta\varphi}{2} + \cos\varphi_1 \cos\varphi_2 \sin^2 \frac{\Delta\lambda}{2}}$$

V2 · REVISED DEFINITION USED FROM NOW ON

same, $R = 6371.0088$ km; for < 1 km precision use geodesic (Karney)

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
R	Earth radius	km

DERIVATION, STEP BY STEP

1. Convert to radians.
2. Haversine of central angle from Δφ, Δλ.
3. d = 2R·asin(√a).

ASSUMPTIONS

- Sphere (0.3% error).

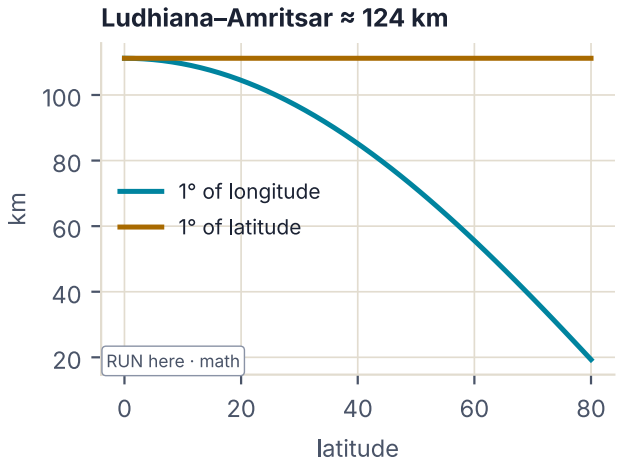


Fig. F40. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Kilometres on the round Earth.

WHERE THE OUTPUT GOES

Source evidence 'nearest kiln 0.8 km'.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F40 V2**	**Output**	**Shown on**
R	F38, F39	Haversine distance	Kilometres on the round Earth.	Source evidence 'nearest kiln 0.8 km'.

V1 → V2 IN ONE LINE

RUN here: Ludhiana–Amritsar ≈ 124 km (doc 125).

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F40 · AUDIT, EXAMPLE, CODE, TESTS

RUN / TESTED LOCALLY

AUDIT VERDICT · OK

RUN here: Ludhiana–Amritsar ≈ 124 km (doc 125).

WORKED EXAMPLE

(30.901,75.857) → (31.634,74.872): 124 km.

EXECUTABLE PSEUDOCODE

```
chronos_core.haversine_km
```

EDGE CONDITIONS

- Antipodal points numerically fine with asin form.

WHY THIS, NOT THAT

Planar distance distorts with latitude; haversine is exact on a sphere and stable for small distances.

COMMON DOUBTS

Q. Precision?
~0.3% from the sphere assumption.

Q. Many points?
Use BallTree with haversine metric.

TESTS (UNIT / PROPERTY)

- d(p,p)=0, symmetric

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F40 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Sinnott 1984

HINGLISH MEIN SAMJHO

Delhi–Mumbai flight ki doori gol dharti par.

IMPLEMENTATION CONTRACT (V2)

MODULE	`classify/, experimental/`
FUNCTION	`haversine_distance_v2()`
INPUTS	R [km]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F41
ACCEPTANCE	d(p,p)=0, symmetric
MAIN FAILURE MODE	Antipodal points numerically fine with asin form.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Ludhiana-Amritsar	125 km	124 km	OK

YOU MAY SAY

'Model estimate (experimental)'; 'classifier set {crop, kiln} at 90% marginal coverage.'

YOU MAY NOT SAY

a fused value labelled as observed; a class without the abstain option.

Chart spec. Experimental tab with violet 'model' style; set chips for classes.

FORMULA 41 / 46 · FUSION AND AI · LAYER L2

F41 DBSCAN fire clustering

F38 F39 F40 F41

Groups detections into events/sites; feeds persistence and night fraction per site.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\text{core}(p) \iff |N_\varepsilon(p)| \geq \text{minPts}, \quad N_\varepsilon(p) = \{q : d(p,q) \leq \varepsilon\}$$

V2 · REVISED DEFINITION USED FROM NOW ON

ST-DBSCAN on (space, time) with $\varepsilon_s, \varepsilon_t$ per sensor resolution; sites from multi-year persistent cluster

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
ε	radius	km
minPts	min neighbours	-

DERIVATION, STEP BY STEP

1. Choose ε (space) and ε_t (time) and minPts.
2. Core points have ≥ minPts neighbours.
3. Expand clusters through core points; others are noise.

ASSUMPTIONS

- ε larger than geolocation error (MODIS ~1 km, VIIRS ~375 m).

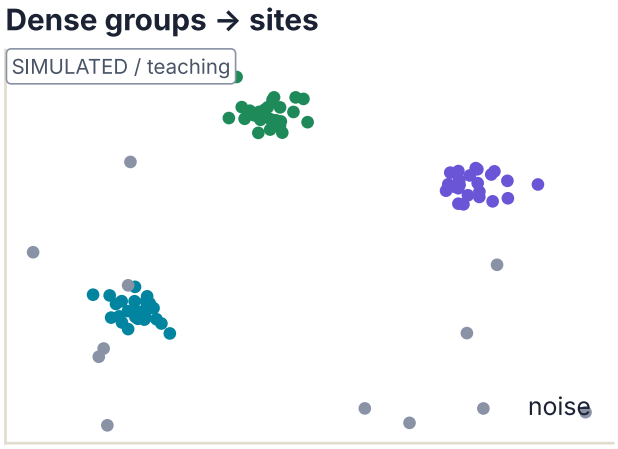


Fig. F41. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Sites for persistence and night-share features.

WHERE THE OUTPUT GOES

Source map site outlines.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F41 V2**	**Output**	**Shown on**
ε, minPts	F39, F40	DBSCAN fire clustering	Sites for persistence and night-share features.	Source map site outlines.

V1 → V2 IN ONE LINE

Spatial-only clustering merges unrelated fires on different days; add time.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F41 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Spatial-only clustering merges unrelated fires on different days; add time. Clusters are not fires 'counted'.

WORKED EXAMPLE

ε=1.5 km, minPts=3 on a year of VIIRS → kiln sites.

EXECUTABLE PSEUDOCODE

```
DBSCAN(eps=eps/R, metric='haversine')
```

EDGE CONDITIONS

- Mixed sensors need sensor-specific ε.

WHY THIS, NOT THAT

k-means needs k and round clusters; DBSCAN finds arbitrary shapes and noise.

COMMON DOUBTS

Q. Is a cluster a fire?
Not necessarily — a site or an event.

Q. Mixed sensors?
Sensor-specific ε.

TESTS (UNIT / PROPERTY)

- synthetic sites recovered

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F41 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Ester et al. 1996

HINGLISH MEIN SAMJHO

Party mein paas khade log ek group.

IMPLEMENTATION CONTRACT (V2)

MODULE	`classify/, experimental/`
FUNCTION	`dbscan_fire_clustering_v2()`
INPUTS	ε [km]; minPts [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	Layer output
ACCEPTANCE	synthetic sites recovered
MAIN FAILURE MODE	Mixed sensors need sensor-specific ε.

YOU MAY SAY

'Model estimate (experimental)'; 'classifier set {crop, kiln} at 90% marginal coverage.'

YOU MAY NOT SAY

a fused value labelled as observed; a class without the abstain option.

Chart spec. Experimental tab with violet 'model' style; set chips for classes.

FORMULA 42 / 46 · SOURCE ATTRIBUTION AND BLIND SPOT · LAYER L2

F42 Distance-decay exposure (IE / AE)

F42 F43 F44 F45 F46

How much industrial vs agricultural context surrounds a detection.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\mathrm{IE}_g = \sum_{f \in \text{industry}} e^{-d_{g,f}/L}, \qquad \mathrm{AE}_g = \sum_{c \in \text{crop}} e^{-d_{g,c}/L}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\mathrm{IE}_g = \sum_{f} S_f \, e^{-d_{g,f}/L}, \quad L \text{ fitted on labelled sites; NULL if OSM coverage unknown}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
L	decay length	km
S_f	feature weight	-

DERIVATION, STEP BY STEP

1. For each detection, find mapped features within a search radius.
2. Weight each by S·exp(−d/L).
3. Sum separately for industrial (IE) and agricultural (AE) features.

ASSUMPTIONS

- OSM completeness; in India kilns are under-mapped.

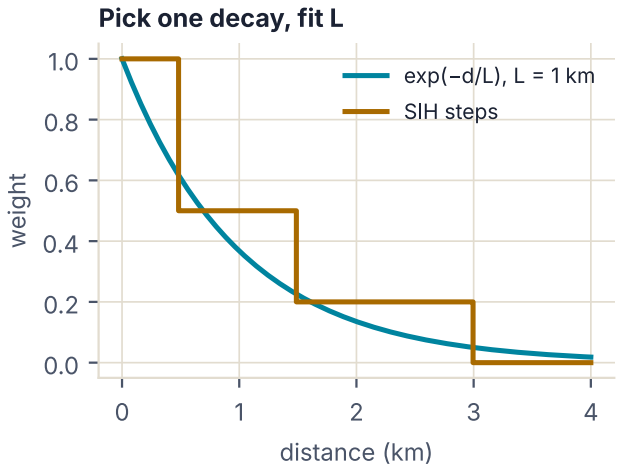


Fig. F42. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

How much industrial vs farm context surrounds a fire.

WHERE THE OUTPUT GOES

Source evidence card.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F42 V2**	**Output**	**Shown on**
L, S_f	raw FIRMS / FireMask	Distance-decay exposure (IE / AE)	How much industrial vs farm context surrounds a fire.	Source evidence card.

V1 → V2 IN ONE LINE

Two versions exist (smooth exp vs SIH steps 1/0.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F42 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Two versions exist (smooth exp vs SIH steps 1/0.5/0.2). Pick one; L and weights are hypotheses. Missing OSM must be NULL, not 0.

WORKED EXAMPLE

Refinery S=5 at 300 m, factory S=3 at 1 km (L=1 km) → IE = 5e^{-0.3}+3e^{-1} = 4.81.

EXECUTABLE PSEUDOCODE

```
sum(S*exp(-d/L))
```

EDGE CONDITIONS

- Area features vs points: use distance to polygon edge.

WHY THIS, NOT THAT

Nearest-distance only uses one feature; decay sums capture density.

COMMON DOUBTS

Q. Missing OSM?
NULL, never 0.

Q. Choose L?
Fit on labelled sites.

TESTS (UNIT / PROPERTY)

- monotone in distance

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F42 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- SIH framework PDF

HINGLISH MEIN SAMJHO

Bakery ki khushboo darwaze par tez, door jaate kam.

IMPLEMENTATION CONTRACT (V2)

MODULE	`classify/, blindspot/`
FUNCTION	`distance-decay_exposure_ie___ae_v2()`
INPUTS	L [km]; S_f [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F43, F44
ACCEPTANCE	monotone in distance
MAIN FAILURE MODE	Area features vs points: use distance to polygon edge.

YOU MAY SAY

'Context suggests industrial heat (L = +0.8, night share 45%).' 'Burned-area / detection disagreement flag B = 2.4.'

YOU MAY NOT SAY

'this is a kiln' without labelled validation; 'real burning is B × the dots'.

Chart spec. Evidence card for/against; stacked source area with UNKNOWN; overpass clock with dashed scenario.

FORMULA 43 / 46 · SOURCE ATTRIBUTION AND BLIND SPOT · LAYER L2

F43 Context log-ratio L

F42 F43 F44 F45 F46

One signed context number for the classifier.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS EXISTING

$$L_g = \ln \frac{\mathrm{IE}_g + 1}{\mathrm{AE}_g + 1}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$L_g = \ln \frac{\mathrm{IE}_g + 1}{\mathrm{AE}_g + 1} \; (+ = \text{industry}); \quad \text{a feature, not a label}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
L	log ratio	-

DERIVATION, STEP BY STEP

1. Add 1 to both exposures.
2. Log of their ratio.

ASSUMPTIONS

- +1 pseudo-count choice changes small values.

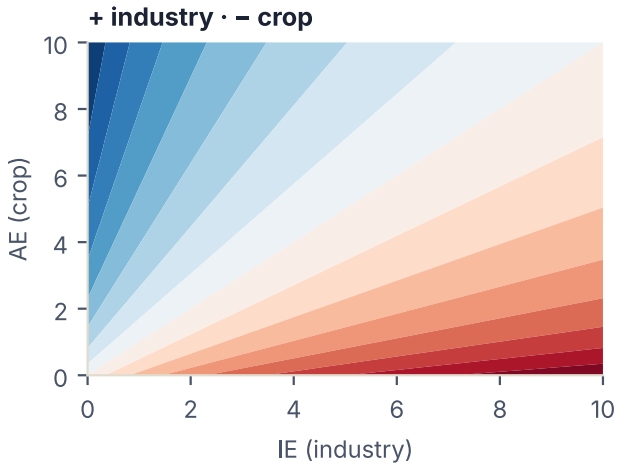


Fig. F43. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

One signed context number.

WHERE THE OUTPUT GOES

Source evidence 'L'.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F43 V2**	**Output**	**Shown on**
L	F42	Context log-ratio L	One signed context number.	Source evidence 'L'.

V1 → V2 IN ONE LINE

Sign convention fixed (old deck had it inverted and a wrong 1.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: unchanged. Example values: teaching / simulated. No real-fire result appears here.

F43 · AUDIT, EXAMPLE, CODE, TESTS

OK AS WRITTEN

AUDIT VERDICT · OK

Sign convention fixed (old deck had it inverted and a wrong 1.46 example; correct ln(6/4) = 0.405, RUN).

WORKED EXAMPLE

IE 6.5, AE 2.5 → ln(7.5/3.5) = 0.762.

EXECUTABLE PSEUDOCODE

```
log((ie+1)/(ae+1))
```

EDGE CONDITIONS

- Both missing → NULL.

WHY THIS, NOT THAT

A difference IE − AE depends on scale; a log ratio is symmetric.

COMMON DOUBTS

Q. Sign?
+ industry, − crop.

Q. A label?
No — a feature for the classifier.

TESTS (UNIT / PROPERTY)

- l_context(5,3)=ln1.5

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F43 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- CLAUDE.md fix

HINGLISH MEIN SAMJHO

Ghar ke aas-paas factory zyada ya khet — ek number.

IMPLEMENTATION CONTRACT (V2)

MODULE	`classify/, blindspot/`
FUNCTION	`context_log-ratio_l_v2()`
INPUTS	L [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F44, F45
ACCEPTANCE	l_context(5,3)=ln1.5
MAIN FAILURE MODE	Both missing → NULL.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
L sign	deck: ln((AE+1)/(IE+1))	ln((IE+1)/(AE+1)) = 0.405 for IE=5, AE=3	OK

YOU MAY SAY

'Context suggests industrial heat (L = +0.8, night share 45%).' 'Burned-area / detection disagreement flag B = 2.4.'

YOU MAY NOT SAY

'this is a kiln' without labelled validation; 'real burning is B × the dots'.

Chart spec. Evidence card for/against; stacked source area with UNKNOWN; overpass clock with dashed scenario.

FORMULA 44 / 46 · SOURCE ATTRIBUTION AND BLIND SPOT · LAYER L2

F44 Night fraction

F42 F43 F44 F45 F46

Industrial sources burn at night; crop fires mostly in the afternoon.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$\nu_g = \frac{N_g^{\text{night}}}{N_g^{\text{day}} + N_g^{\text{night}}}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$\nu_g = \frac{N_g^N / E_g^N}{N_g^D / E_g^D + N_g^N / E_g^N}$$ (exposure-adjusted; night passes are fewer for MODIS)

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
E^D, E^N	day and night opportunities	-

DERIVATION, STEP BY STEP

1. Count night and day detections at a site.
2. Divide each by its exposure (night passes are fewer for MODIS).
3. Share of the night rate.

ASSUMPTIONS

- Night detection sensitivity differs (cooler background, different thresholds).

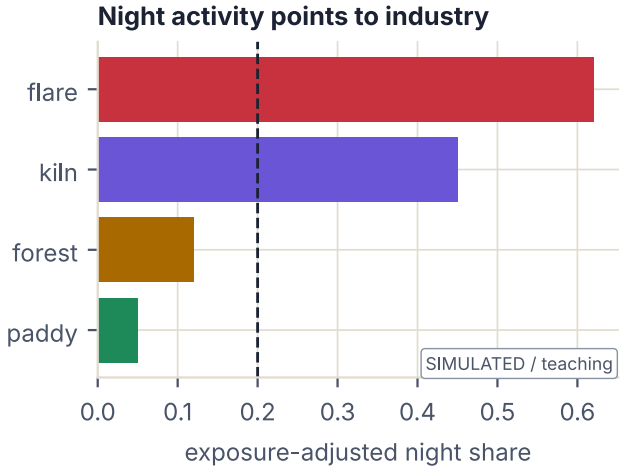


Fig. F44. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

Night-active sites point to industry.

WHERE THE OUTPUT GOES

Source evidence 'night share'.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F44 V2**	**Output**	**Shown on**
E^D, E^N	F42, F43	Night fraction	Night-active sites point to industry.	Source evidence 'night share'.

V1 → V2 IN ONE LINE

Raw share is biased by unequal day/night opportunities.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F44 · AUDIT, EXAMPLE, CODE, TESTS

REVISED IN V2

AUDIT VERDICT · REVISE

Raw share is biased by unequal day/night opportunities. Threshold 20% is a hypothesis.

WORKED EXAMPLE

Kiln 0.45, paddy 0.05 (teaching).

EXECUTABLE PSEUDOCODE

```
exposure-adjusted share
```

EDGE CONDITIONS

- Few detections (<10) → NULL.

WHY THIS, NOT THAT

Raw night share is biased by unequal pass numbers.

COMMON DOUBTS

Q. Threshold?
20% is a hypothesis.

Q. Few detections?
NULL below 10.

TESTS (UNIT / PROPERTY)

- equal rates → 0.5

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F44 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- Visual guide CUT-OFF card

HINGLISH MEIN SAMJHO

Raat 2 baje bhi light on — 24-ghante wali dukaan.

IMPLEMENTATION CONTRACT (V2)

MODULE	`classify/, blindspot/`
FUNCTION	`night_fraction_v2()`
INPUTS	E^D, E^N [-]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F45, F46
ACCEPTANCE	equal rates → 0.5
MAIN FAILURE MODE	Few detections (<10) → NULL.

YOU MAY SAY

'Context suggests industrial heat (L = +0.8, night share 45%).' 'Burned-area / detection disagreement flag B = 2.4.'

YOU MAY NOT SAY

'this is a kiln' without labelled validation; 'real burning is B × the dots'.

Chart spec. Evidence card for/against; stacked source area with UNKNOWN; overpass clock with dashed scenario.

FORMULA 45 / 46 · SOURCE ATTRIBUTION AND BLIND SPOT · LAYER L3

F45 Diurnal fraction seen by overpasses

F42 F43 F44 F45 F46

Scenario tool: if burning follows a diurnal curve g, what share happens inside the times the polar satellites look?

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS TESTED (SYNTHETIC)

$$f_{\text{seen}} = \sum_{o} \int_{t_o - w}^{t_o + w} \frac{1}{\sigma\sqrt{2\pi}} e^{-(t-\mu)^2/2\sigma^2}\, dt$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$f_{\text{seen}} = \int_{\bigcup_o [t_o - w,\, t_o + w]} g(t)\, dt, \quad g \text{ from geostationary/hourly evidence; otherwise labelled SCENARIO}$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
t_o	observed overpass times (per year)	h LMST
w	half-window	h
g	diurnal burning density	1/h

DERIVATION, STEP BY STEP

1. List actual overpass windows for the place and year.
2. Merge overlapping windows.
3. Integrate the assumed diurnal density over the union.

ASSUMPTIONS

- g is an assumption unless fitted to hourly data (e.g. GEO-KOMPSAT-2A / Meteosat IODC; access unverified).

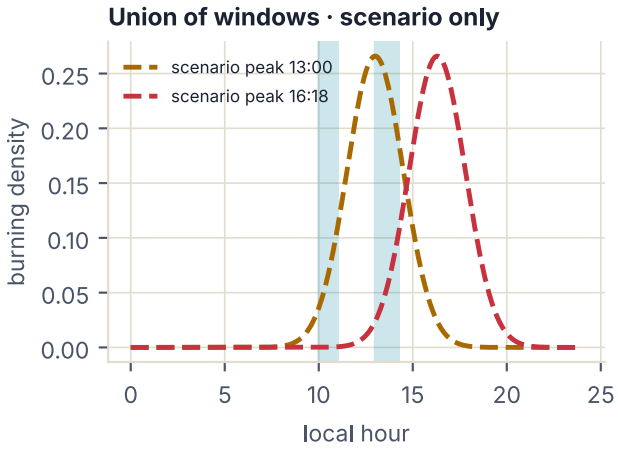


Fig. F45. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

If burning peaks at 16:18, polar passes see only a small share — a scenario to explain missed burning.

WHERE THE OUTPUT GOES

Blind Spot clock — dashed 'scenario' style.

LINEAGE OF THIS FORMULA

01 Inputs	02 Upstream	03 F45 V2	04 Output	05 Shown on
t_o, w, g	F43, F44	Diurnal fraction seen by overpasses	If burning peaks at 16:18, polar passes see only a small share — a scenario to explain mis	Blind Spot clock — dashed 'scenario' style.

V1 → V2 IN ONE LINE

RUN here: Aqua and NOAA-20 windows overlap near 13:30; summing integrals gives 54.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F45 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

RUN here: Aqua and NOAA-20 windows overlap near 13:30; summing integrals gives 54.7% instead of the union 36.6% (μ=13, σ=1.5 h). Result is a scenario, not a measured share of fires.

WORKED EXAMPLE

Peak 13:00 vs peak 16:18 scenarios (v1 must-pass 31.6% vs 3.3%) — teaching.

EXECUTABLE PSEUDOCODE

```
merge windows; integrate once
```

EDGE CONDITIONS

- Windows crossing midnight split in two.

WHY THIS, NOT THAT

Summing per-satellite integrals double counts overlapping passes.

COMMON DOUBTS

Q. Measured share?
No — a scenario unless g comes from hourly data.

Q. Where could g come from?
Geostationary FRP (access unverified for India).

TESTS (UNIT / PROPERTY)

- union ≤ 1; union ≤ naive sum

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F45 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- iFOREST 2025 (reported shift after 15:00)

HINGLISH MEIN SAMJHO

CCTV sirf 10:30 aur 13:30 par — 4 baje wala chor chhoot jaata hai.

IMPLEMENTATION CONTRACT (V2)

MODULE	`classify/, blindspot/`
FUNCTION	`diurnal_fraction_seen_by_overpasse_v2()`
INPUTS	t_o [h LMST]; w [h]; g [1/h]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	F46
ACCEPTANCE	union ≤ 1; union ≤ naive sum
MAIN FAILURE MODE	Windows crossing midnight split in two.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
Overlapping windows (Aqua + NOAA-20 near 13:30)	sum = 54.7%	union = 36.6%	FIX

YOU MAY SAY

'Context suggests industrial heat (L = +0.8, night share 45%).' 'Burned-area / detection disagreement flag B = 2.4.'

YOU MAY NOT SAY

'this is a kiln' without labelled validation; 'real burning is B × the dots'.

Chart spec. Evidence card for/against; stacked source area with UNKNOWN; overpass clock with dashed scenario.

FORMULA 46 / 46 · SOURCE ATTRIBUTION AND BLIND SPOT · LAYER L3

F46 Burned-area to detection ratio B

F42 F43 F44 F45 **F46**

Warning flag: burn scars stay high while detections fall → detections may be missing burning.

V1 · AS WRITTEN IN THE 46-FORMULA REGISTRY · STATUS PROPOSED

$$B_{g,y} = \frac{\mathrm{BA}_{g,y}/C_{g,y}}{\mathrm{median}_{y'<2021}\left(\mathrm{BA}_{g,y'}/C_{g,y'}\right)}$$

V2 · REVISED DEFINITION USED FROM NOW ON

$$B_{r,y} = \frac{\mathrm{BA}_{r,y}/C_{r,y}}{\mathrm{med}_{y'\in\mathcal{R}}(\mathrm{BA}_{r,y'}/C_{r,y'})},\quad |\mathcal{R}| \geq 3;\ \text{PENDING if BA lagged; NULL if } C = 0$$

SYMBOLS, INPUTS AND UNITS

SYMBOL	MEANING	UNIT
BA	MCD64A1 burned area	km²
C	detected cell-days	count

DERIVATION, STEP BY STEP

1. Compute BA per detected cell-day each year.
2. Median over ≥ 3 reference years.
3. B = current ratio / reference median; pending if BA lagged.

ASSUMPTIONS

- MCD64A1 itself uses MODIS active fire → not independent.
- BA omission of small agricultural burns.

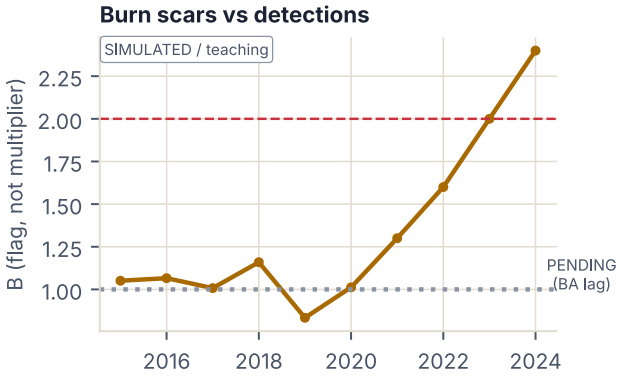


Fig. F46. Visual intuition generated by code in this atlas.

HOW TO READ THE OUTPUT

A 'check this' warning, shown next to the calendar value.

WHERE THE OUTPUT GOES

Blind Spot B timeline; Brief flag.

LINEAGE OF THIS FORMULA

01	02	03	04	05
Inputs	**Upstream**	**F46 V2**	**Output**	**Shown on**
BA, C	F44, F45	Burned-area to detection ratio B	A 'check this' warning, shown next to the calendar value.	Blind Spot B timeline; Brief flag.

V1 → V2 IN ONE LINE

B is a disagreement flag, never a multiplier ('real burning is 2.

PROVENANCE OF NUMBERS ON THIS SPREAD

Equations: revised in this audit. Example values: teaching / simulated. No real-fire result appears here.

F46 · AUDIT, EXAMPLE, CODE, TESTS

FIX REQUIRED → V2

AUDIT VERDICT · FIX

B is a disagreement flag, never a multiplier ('real burning is 2.4× the dots' is unsupported). Guard zeros, lag (PENDING) and reference-year choice.

WORKED EXAMPLE

Reference median 1.2 km²/cell-day; 2024: 2.9 → B = 2.4 → 'check' flag (teaching).

EXECUTABLE PSEUDOCODE

```
chronos_core.blindspot_ratio
```

EDGE CONDITIONS

- Sentinel-2 based BA (iFOREST) differs in method; compare trends only.

WHY THIS, NOT THAT

A regression of BA on detections would treat BA as truth; B is a transparent disagreement flag.

COMMON DOUBTS

Q. B = 2.4 means 2.4× more burning?
No — it means the two products disagree more than before.

Q. Independent?
No: MCD64A1 uses MODIS active fire input.

TESTS (UNIT / PROPERTY)

- C=0 → unavailable; BA None → pending

REAL-DATA VALIDATION PLAN

Run on the Punjab–Haryana FIRMS archive (M2) with exposure from MOD14A1/VNP14A1; report the result with its interval and the config hash. Until then every displayed value for F46 stays SIMULATED DEMO or PROPOSED / UNVALIDATED.

SOURCES

- GEE MCD64A1 catalogue (inputs include active fire)

HINGLISH MEIN SAMJHO

Raakh zyada, lakdi ki ginti kam — kisi ne ginna band kiya.

IMPLEMENTATION CONTRACT (V2)

MODULE	`classify/, blindspot/`
FUNCTION	`burned-area_to_detection_ratio_b_v2()`
INPUTS	BA [km²]; C [count]
RETURNS	`{value, status, note, unit, version}`
STATUS VALUES	`ok · unavailable · insufficient_data · pending`
NEXT CONSUMER	Layer output
ACCEPTANCE	C=0 → unavailable; BA None → pending
MAIN FAILURE MODE	Sentinel-2 based BA (iFOREST) differs in method; compare trends only.

AUDIT CHECKS EXECUTED FOR THIS ATLAS

CHECK	V1 / DOCUMENTED	V2 / RE-RUN	VERDICT
B with C=0 or BA pending	division error / normal	pending, unavailable	FIX

YOU MAY SAY

'Context suggests industrial heat (L = +0.8, night share 45%).' 'Burned-area / detection disagreement flag B = 2.4.'

YOU MAY NOT SAY

'this is a kiln' without labelled validation; 'real burning is B × the dots'.

Chart spec. Evidence card for/against; stacked source area with UNKNOWN; overpass clock with dashed scenario.

CHAPTER 12

Baseline ladder and validation gates

Select the simplest method that wins on held-out data. A more complex model is not automatically more accurate.

RUNG	METHOD	WHAT IT ADDS	RISK
B0	raw counts concatenated	nothing (reference for 'how bad')	2012 seam
B1	median of monthly cell-day ratios	sensor scale	sparse-count bias, clouds
B2	ratio of exposure rates (pooled)	cloud/coverage correction	day/night mix
B3	B2 stratified by day/night (and month)	night-burning shifts	sparse night strata
B4	B3 + drift term on log k after 2016	orbit drift	overfitting with few years
B5	NB model with view geometry + drift (R1, R4)	physics covariates	identifiability
B6	fusion / detection models (R3, F38)	multi-sensor	complexity

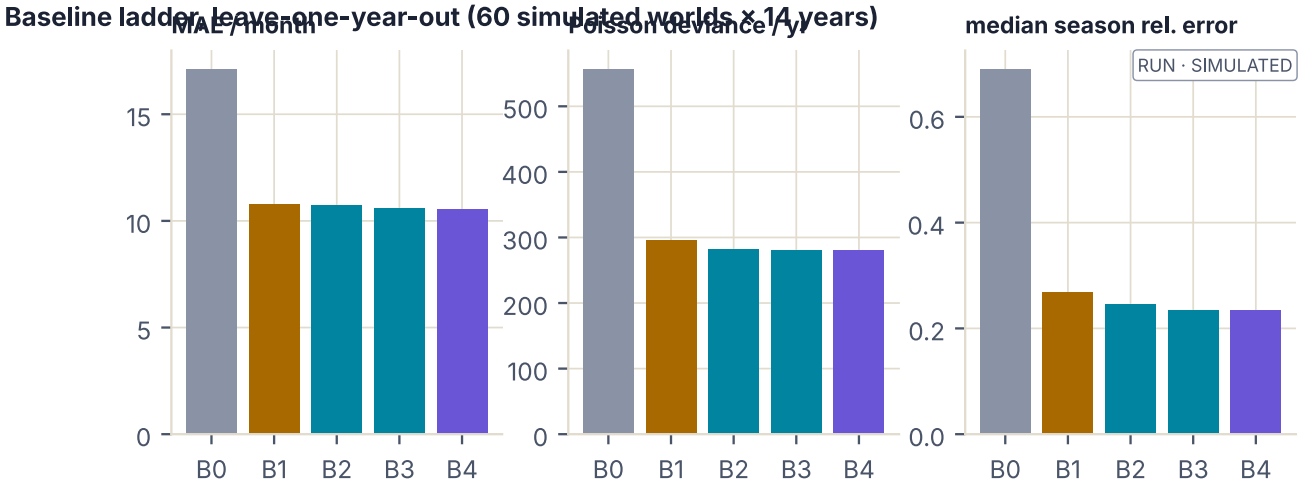


Fig. 12.1. Leave-one-year-out on 60 simulated worlds with known truth (overlap 2012–2025, MODIS drift 2.5%/yr after 2016, NB noise θ = 8). Exposure normalisation (B2) gives most of the gain; stratification and drift add little in this simulation. Real-data ranking may differ. RUN / TESTED LOCALLY

RUNG	MAE / MONTH	DEVIANCE	SEASON REL. ERR.
B0	17.13	557	69.1%
B1	10.79	295	26.9%
B2	10.72	283	24.5%
B3	10.58	281	23.4%
B4	10.54	280	23.4%

READ THIS HONESTLY

The 23% median season error comes from Negative-Binomial noise in small MODIS counts that no k can remove. It is a property of the simulated world, not a forecast for Punjab. Interval coverage of the B3 predictive band was 94.5% against a nominal 95%.

Validation gates (thresholds declared before any real run)

GATE	TEST	PASS RULE (PREDECLARED 27 OCT 2026)	FORMULA
G1 Held-out years	LOYO over 2012–2025, Oct–Nov totals	median relative error ≤ 15% and better than B1	F36
G2 Held-out geography	train on Punjab, test Haryana; and reverse	error ≤ 1.5 × in-region error	F36
G3 Seam	3-yr windows, log scale, with bootstrap CI	seam ratio upper CI ≤ 0.25	F35
G4 Placebo seams	Pettitt/PELT at 20 random non-sensor years	false-positive rate ≤ 10%	F33, F34
G5 Platform seams	2018 and 2023 additions	no break after harmonisation at α = 0.05	F34
G6 Coverage	95% predictive bands on held-out years	achieved 88–99%	F24, F25
G7 Agreement	Bland–Altman log scale, held-out	|bias| ≤ 0.1 (≈10%)	F23
G8 Zero / empty	empty months, K = 0, one region	no crash; status not 0	all
G9 Negatives	water and urban cells	harmonised activity ≈ 0 with status	F07, F08
G10 Persistent heat	known flare/kiln list	excluded from calibration	F08
G11 Classifier	held-out sites (GroupKFold by site and region)	macro-F1 reported with abstain rate; else feature stays PROPOSED	F39
G12 API provenance	every response	meta.data_source, status, version present	contract

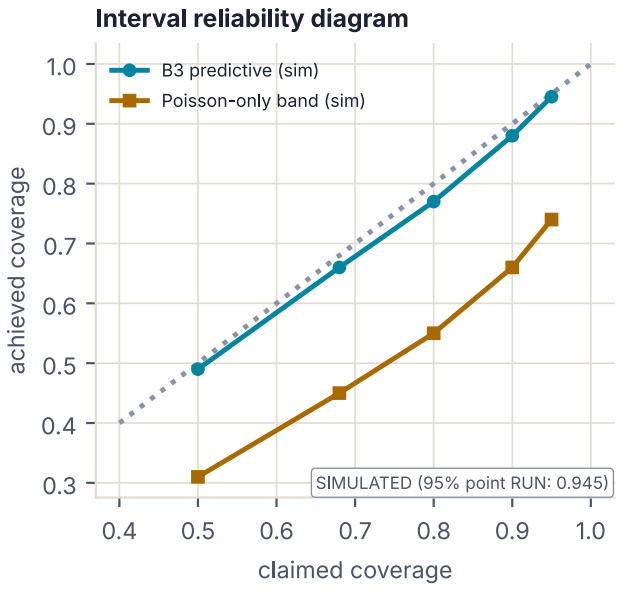


Fig. 12.2. Reliability diagram target: achieved vs claimed coverage.

SIMULATED DEMO

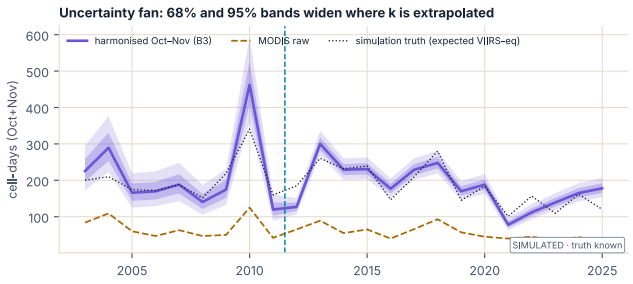


Fig. 12.3. Bands widen where k is extrapolated (before 2012).

SIMULATED DEMO

LEAKAGE CONTROLS

Block by year (seasons are correlated), by region for G2, by site for G11. Thresholds above may only be changed with a dated note in the conflict log — never after seeing results.

CHAPTER 13

How the formulas combine

The 46 formulas are not 46 separate tools. They form one chain per layer. This is the combined algorithm in the order the backend runs it.

```
for each region r and platform s:
  D  = normalise(FIRMS_SP rows)                  # F01 local time, F02 footprint, F07 rule
  D  = D[type not in {1,3}] ; mask static sites   # F08 persistence, F41 sites
  cd = cell_days(D, grid=0.1, by=[cell, local_day, s, dn])    # F03, F05
  E  = exposure(FireMask_s, cells, month)          # F09 (null if E = 0)
  Y  = monthly(cd) ; rate = Y / E                  # F06, F09
overlap = pairs(Aqua, S-NPP, same cells, days, dn) # F10
model   = NB_offset(Y ~ sensor + dn + month, offset=log E)  # F14, F15
k[σ]    = exp(γ_V - γ_M) per stratum ; pool log k  # F18 (R ≥ 3)
band    = block_bootstrap(years, alpha=0.05)       # F24 ; F25 log-k propagate
H[t]    = k[σ(t)] · M[t] (t < 2012) else V[t]      # F12, F13
valid   = LOYO, seams, placebo, Bland-Altman       # F23, F33-F36
z[t]    = robust_z_loo(log((F+.5)/(E+1)))          # F27, F28
trend   = MK_ties + Hamed-Rao + Sen ; BH over cells # F30-F32, F37
season  = 10/50/90% dates in declared window       # F29
L2      = features(F08, F21, F42-F44) → abstaining classifier (F39)
L3      = overpass union (F45, scenario) ; B flag (F46, pending/null)
publish calendar_value{value, lo, hi, coverage, status, provenance, version, hash}
```

Dependency rules

- Every downstream formula receives the status of its inputs; null propagates.
- Uncertainty is computed once at the calibration step and propagated, never re-estimated in the UI.
- Layer 4 uses the same rate scale as Layer 1 — never raw counts.
- Layer 2 and 3 outputs never modify Layer 1 values; they annotate them.

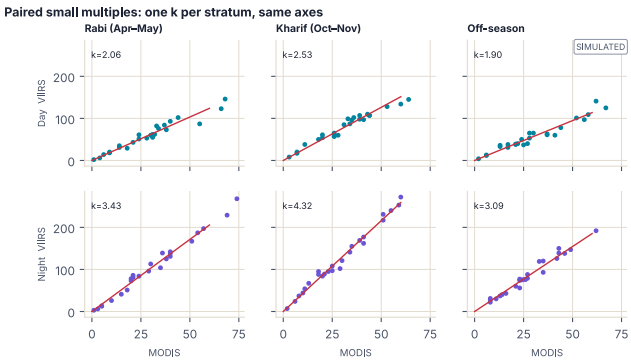


Fig. 13.1. One k per stratum, identical axes. SIMULATED DEMO

Science error budget PROPOSED / UNVALIDATED

Where the uncertainty of a harmonised November value comes from, and how each source is handled. Sizes are to be measured in M3; the order of magnitude column is an expectation to test, not a result.

SOURCE	MECHANISM	HANDLED BY	EXPECTED SIZE	MEASURED?
counting noise	NB variation of cell-days	F15, F25	large at small counts	simulated
calibration	uncertainty of k in overlap	F24, F18	±10–20% of k	pending
non-stationarity	drift, processing changes	R1, R5, R8	unknown	pending
exposure mismatch	1 km composite vs 375 m points	R2	unknown	pending
filter choice	confidence rule pair	F07 grid	±15% of k (sim example)	simulated
static sources	kilns/flares in calibration	F08 mask	region-dependent	pending
timing	burning outside overpasses	L3 flag only	can dominate (Punjab)	not correctable from hotspots
attribution	source class errors	abstain, F39	unknown without labels	pending

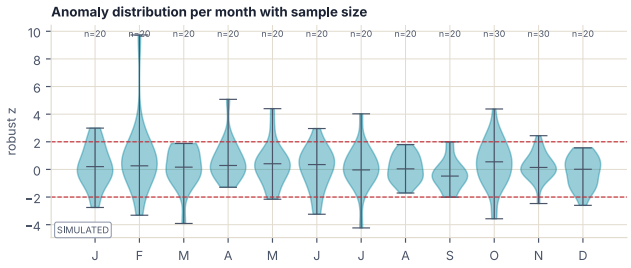


Fig. 13.2. Anomaly z by month with sample size. SIMULATED DEMO

WHAT THE BAND DOES NOT INCLUDE

Timing bias and attribution error are not in the calendar interval. They are shown as separate flags (Layer 3, Layer 2) because no hotspot-only model can estimate them.

SAAF BAAT

Band sirf woh galti dikhata hai jo hum naap sakte hain. Jo satellite dekh hi nahi sakta, uske liye alag warning.

PART IV

Four layers & the evidence behind them

Four separate questions, four views. No hand-weighted 'fire truth score'. Then ten new research approaches, each with a way to fail.

CHAPTERS IN THIS PART

Chapter 14 · Four layers, four questions
Chapter 15 · Layer 2 — what is burning?
Chapter 16 · Layer 3 — what might the satellite miss?
Chapter 17 · Layer 4 — unusual or trending?
Chapter 18 · Global by design
Chapter 19 · New approaches R1–R10

CHAPTER 14

Four layers, four questions

Each layer answers a different question with its own unit. Layers annotate each other; they never multiply each other into a single score.

L1

How much activity was detected on a common scale?

Output. exposure rate; VIIRS-eq cell-days with 95% band; sensor split; coverage

Formulas. F01–F07, F09–F16, F18, F24–F26

When unsure. status ≠ ok → hatched; never 0

L2

What kind of heat source is this?

Output. crop / forest / industrial / UNKNOWN with calibrated probability and evidence for/against

Formulas. F08, F21, F39–F44

When unsure. abstain when max p < threshold or set > 1 class

L3

What might the satellite miss?

Output. overpass clock (scenario), observation opportunity, B disagreement flag

Formulas. F01, F09, F45, F46

When unsure. PENDING while BA lagged; scenario style

L4

Is this month unusual or trending?

Output. robust z with n, season dates, MK + Sen + BH

Formulas. F27–F32, F37

When unsure. insufficient history → 'indicative'

REJECTED: ONE COMBINED SCORE

The SIH deck combined evidence as FIS = 0.45T + 0.30I + 0.15WCS + 0.10C. The weights were never fitted and the result had no unit. V2 shows four calibrated or clearly-labelled views instead.

How the layers connect

01 **Data** FIRMS SP + exposure

02 **L1 activity** rate, VIIRS-eq ± band

03 **L2 annotates** source class per cell

04 **L3 annotates** opportunity + B flag

05 **L4 compares** z, trend, season on L1 scale

06 **Outputs** calendar, inspector, brief

LAYER	PAGE	API	MAY CHANGE L1 VALUES?
L1	Explorer, Calendar, Sensor Lab	/api/calendar, /api/calibration	is L1
L2	Sources	/api/source	no — filters and splits only
L3	Blind Spot	/api/blindspot	no — flags only
L4	Calendar, Validation	/api/calendar?metric=z	no — reads L1

SIMPLE WORDS

Chaar sawaal: kitni aag dikhi? kya jal raha hai? satellite ne kya miss kiya? kya yeh mahina ajeeb hai? Har sawaal ka jawab alag, aur 'pata nahi' bhi ek jawab hai.

Village Fire Card — the four layers on one cell-month

FIELD CARD · CELL 1209_2558 · NEAR LUDHIANA · NOVEMBER · SIMULATED VALUES

How much was detected?

14 VIIRS fire cell-days out of 22 valid days (rate 0.64). Harmonised long-term November normal: 0.41 [0.33, 0.52].

What is burning?

Crop residue (p = 0.81). Evidence for: cropland 92%, seasonal, low persistence 0.03. Against: one kiln 2.1 km away. Set at 90%: {crop}.

What might be missed?

Detections cluster 13:00–14:00 LMST; reported burning after 15:00 would be largely unseen. B flag = 2.4 (check). Scenario only.

Is it unusual?

z = +2.3 vs 12 reference Novembers → unusual. 10-year trend: not significant (p = 0.21).

Not for life-safety decisions. Detection data have limited accuracy.

Why a card per cell

Officials think in villages and districts, not in rasters. The card is the inspector of the Explorer and the unit of the PDF brief: four short answers, each with its own unit, uncertainty and label.

Rules for the card

- Numbers are inserted from the API JSON; the template never writes numbers.
- Each answer has a fixed slot for 'unknown'.
- Colour only supports the text; the card prints in black and white.
- The life-safety caveat is always present.

Layer contract

LAYER	API ROUTE	UNIT	MAY BE NULL WHEN	NEVER
L1	/api/calendar/:region, /api/cells/:cell	rate; VIIRS-eq cell-days	no exposure, archive not processed	0 for unobserved
L2	/api/source/:cell	class probabilities, set	weak evidence, OSM missing	forced label
L3	/api/blindspot/:region	flag; scenario share	BA pending; no hourly data	multiplier
L4	/api/calendar?metric=z	robust z, p, slope	< 8 years	p-only claims

CHAPTER 15

Layer 2 — what is burning?

Decision flow (rule-based v1, classifier v2)

01 MODIS type
1 volcano / 3 offshore → exclude

02 Persistence + night
p ≥ 0.20 and ν ≥ 0.20 → static (hypothesis)

03 Context L
L > 0.5 → industrial context

04 Land + season
cropland in crop window → crop

05 Forest/shrub
→ forest / wildland

06 Else
UNKNOWN

The rule flow is explainable and fast; the v2 classifier (gradient boosting with calibrated probabilities and conformal sets) replaces it only after it beats the rules on held-out labelled sites grouped by site and region.

FEATURE	FORMULA	FOR INDUSTRY	FOR CROP	PITFALL
persistence p	F08	high, all year	low, seasonal	sensor-dependent
night share ν	F44	high	low	unequal day/night passes
context L	F42–F43	positive	negative	OSM gaps
FRP (log)	F21	steady	variable	VIIRS saturation
ΔT (T4 − T11)	-	small hot source	large cool fire	per sensor only
land cover	WorldCover	built-up	cropland	2021 epoch

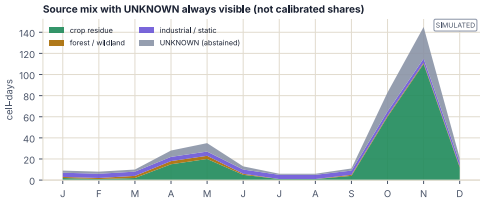


Fig. 15.1. Calendar split by source with UNKNOWN always shown. SIMULATED DEMO

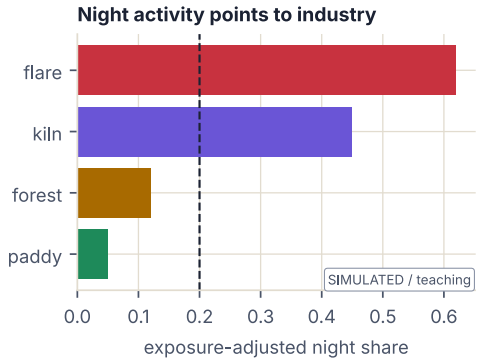


Fig. 15.2. Night share by site type. SIMULATED DEMO

NOT YET VALIDATED

There is no labelled set of Indian kilns/flares in the project. Precision, recall and abstain rate must be measured before any class share is called a 'fraction'.

CHAPTER 16

Layer 3 — what might the satellite miss?

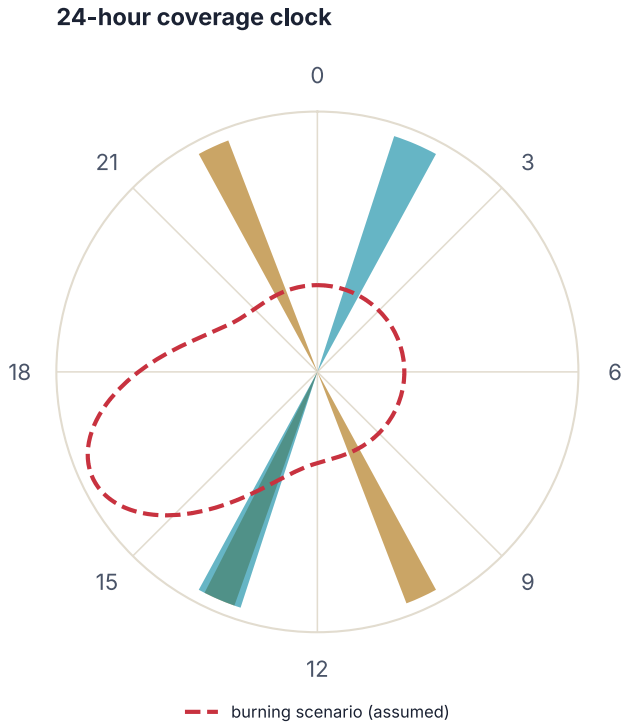


Fig. 16.1. Coverage clock. Windows from actual acquisition times per year; burning curve dashed = scenario. SIMULATED DEMO

Three honest outputs

1. **Observation opportunity**: how many valid passes covered the cell, at which local times.
2. **Scenario share** (F45): if burning followed curve g, what share falls in the windows — labelled scenario.
3. **Disagreement flag** (F46): burned area per detection vs its own past; PENDING while BA is lagged.

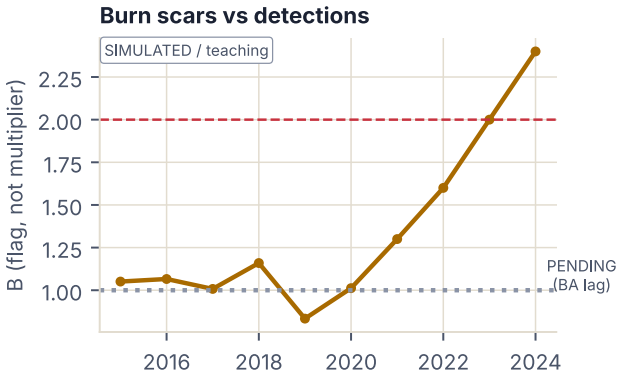


Fig. 16.2. B flag timeline. SIMULATED DEMO

The Punjab story, told carefully

Secondary reporting of iFOREST (2025) says more than 90% of large Punjab farm fires in 2024–25 happened after 3 pm, compared with about 3% in 2021, while reported burnt area fell much less than fire counts. If true, a count-only calendar would say burning collapsed. Layer 3 exists to stop that conclusion — but it cannot tell how much burning was missed. SECONDARY REPORT

NEVER

B × detections = 'true burning'. MCD64A1 uses MODIS active fire as an input, misses many small field burns, and is lagged.

CHAPTER 17

Layer 4 — unusual or trending?

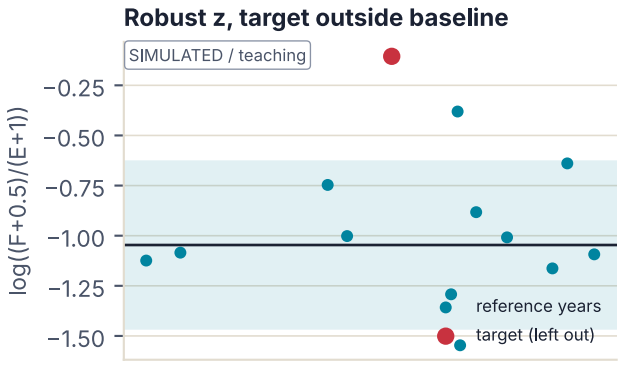


Fig. 17.1. Robust z with the target year left out. SIMULATED DEMO

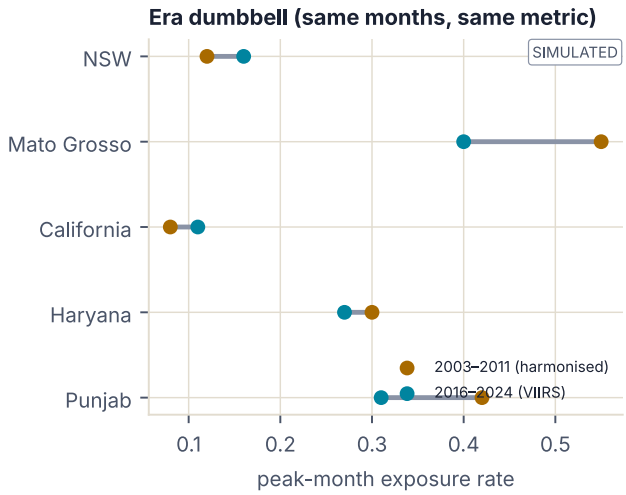


Fig. 17.2. Era comparison on the same metric and months.
SIMULATED DEMO

Rules

- Anomalies on the exposure-rate log scale (F27), never raw counts.
- Report n reference years; < 8 → 'indicative'.
- Trends: tie-corrected Mann–Kendall + Hamed–Rao, Sen slope as effect size, BH across cells.
- p > 0.05 = 'not shown with this data', never 'no change'.
- Season dates are dates of detected activity.

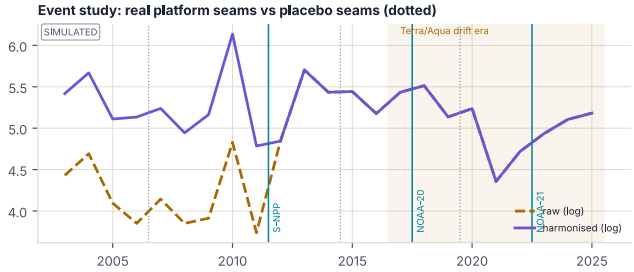


Fig. 17.3. Event study: sensor seams vs placebo seams.
SIMULATED DEMO

YAAD RAKHO

Ajeeb mahina = apne hi purane mahino se compare. Trend = sirf p-value nahi, slope bhi.

CHAPTER 18

Global by design

Punjab–Haryana is the flagship story, not a hard-coded assumption. The same code builds grids for any polygon on Earth.

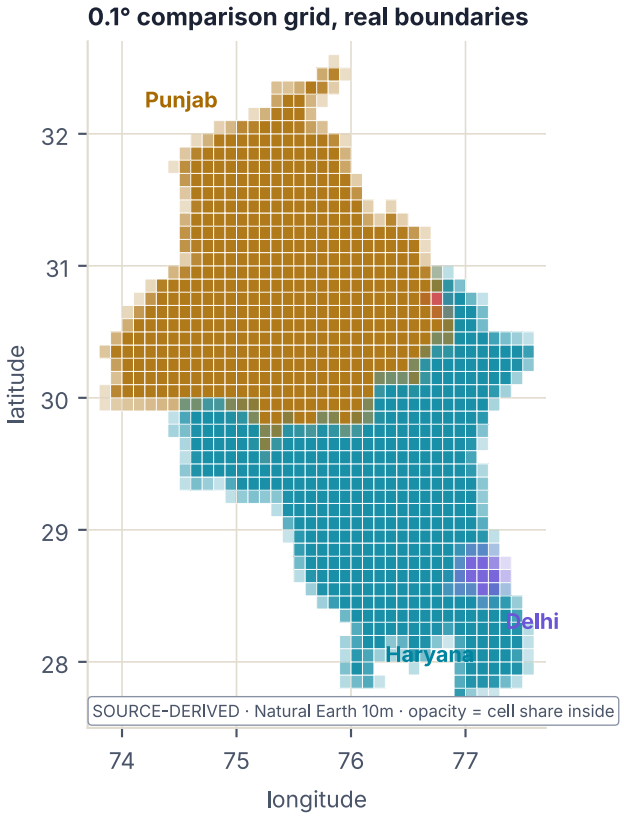


Fig. 18.1. Real 0.1° grid over Punjab, Haryana, Chandigarh and Delhi (opacity = share of cell inside). SOURCE-DERIVED

REGION	ROLE	BBOX (W,S,E,N)	GRIDDED KM²
California	global_test	-124.4092, 32.5317, -114.1191, 41.9995	409,603
Mato Grosso	global_test	-61.6113, -18.0316, -50.2357, -7.3391	906,656
Punjab	flagship	73.8423, 29.5598, 76.9104, 32.5167	50,444
New South Wales	global_test	141.0001, -37.5103, 153.6306, -28.1493	802,466
Delhi	air_quality_context	76.8337, 28.4326, 77.3375, 28.8821	1,481
Chandigarh	flagship	76.6937, 30.6723, 76.8273, 30.799	120
Haryana	flagship	74.4594, 27.6704, 77.588, 30.9343	43,997
Punjab + Haryana + Chandigarh (flagship)	flagship	73.8423, 27.6704, 77.588, 32.5167	94,561

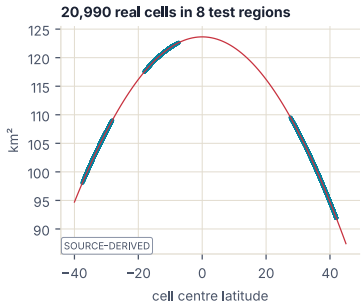


Fig. 18.2. Real cell areas in the eight test regions follow the spherical formula. SOURCE-DERIVED

Global checklist

ISSUE	V2 RULE
antimeridian (e.g. Fiji, Chukotka)	regions stored as cell sets; bbox may have west > east
poles	cells clamp; area tiny but exact
other fire regimes	same estimand; strata by month and day/night are local
different crops / seasons	season window per region in config
local time	LMST from longitude, never a national clock
southern hemisphere seasons	month strata, not 'winter/summer' labels

CHAPTER 19

New approaches R1–R10

Each idea is a hypothesis with a baseline, a test and a way to fail. None is claimed to work. PROPOSED / UNVALIDATED

R1 · Orbit-drift-aware observation normaliser

PROPOSED / UNVALIDATED

PROBLEM	Terra and Aqua crossing times drift after 2020; k learned in early overlap may not apply late.
HYPOTHESIS	k varies smoothly with effective local overpass time; modelling it lowers held-out error in 2017–2025.
INPUTS / COST	FIRMS acq_time per year (free)
BASELINE	B3 stratified k

$$\log k_{\sigma,y} = \kappa_\sigma + \beta\,(\bar{t}^{\,\mathrm{LMST}}_{M,y} - \bar{t}^{\,\mathrm{LMST}}_{V,y})$$

TEST DESIGN	LOYO on 2017–2025 only; compare B3 vs B3+drift
FALSIFIED IF	no improvement or coefficient sign unstable across regions
COMPUTE	medium
DECISION	**adopt if ≥ 5% deviance gain**

R2 · Opportunity-based denominator

PROPOSED / UNVALIDATED

PROBLEM	A clear daily composite can hide several passes; exposure per pass may be the right opportunity unit.
HYPOTHESIS	Pass-level exposure (swath ∩ cell, valid) makes rates more stable across 2018/2023 platform additions.
INPUTS / COST	VNP14IMG / MOD14 swath granules via LAADS (large)
BASELINE	daily FireMask exposure

$$E^{\mathrm{pass}}_{s,g,m} = \sum_{o \in \mathrm{passes}} 1[g \in \mathrm{swath}_o]\, 1[\mathrm{valid}_{g,o}]$$

TEST DESIGN	seam ratio at 2018 and 2023 with each denominator
FALSIFIED IF	no difference within CI
COMPUTE	high (granules)
DECISION	**postpone to after MVP**

R3 · Paired-overlap probabilistic detection model

PROPOSED / UNVALIDATED

PROBLEM	k is a black box; matched pairs reveal detection curves by size, land cover and day/night.
HYPOTHESIS	MODIS detection probability given VIIRS FRP, angle and land cover predicts k in held-out strata.
INPUTS / COST	Aqua–S-NPP matches ≤ 15 min
BASELINE	empirical k

$$\operatorname{logit} P(D_M = 1 \mid F_V, \theta, \ell) = \beta_0 + \beta_1 \ln F_V + \beta_2 \theta + \beta_\ell$$

TEST DESIGN	predict stratum k from the curve; compare to observed k on held-out years
FALSIFIED IF	curve-implied k outside observed CI
COMPUTE	medium
DECISION	**adopt as explanation; use for k only if it wins**

R4 · Spatio-temporal hierarchical NB

PROPOSED / UNVALIDATED

PROBLEM	Regions and seasons differ; independent k per stratum is noisy.
HYPOTHESIS	Partial pooling of log k across regions and months reduces held-out error for small regions.
INPUTS / COST	existing tables
BASELINE	F18 EB

$$\log k_{r,m} = \mu + a_r + b_m + (ab)_{r,m},\ a_r \sim \mathcal{N}(0, \tau_a^2)$$

TEST DESIGN	leave-one-region-out
FALSIFIED IF	no gain vs EB
COMPUTE	low
DECISION	**test in M3**

R5 · Change-point placebo battery

PROPOSED / UNVALIDATED

PROBLEM	A single seam test can miss or invent breaks.
HYPOTHESIS	Harmonised series show no more breaks at sensor years than at random placebo years.
INPUTS / COST	existing
BASELINE	Pettitt at 2012 only

$$\hat{\pi} = \frac{1}{P}\sum_{p=1}^{P} 1[\text{break detected at placebo } p]$$

TEST DESIGN	20 placebo years + 2012/2018/2023
FALSIFIED IF	sensor-year break rate > placebo rate
COMPUTE	low
DECISION	**adopt (validation gate G4)**

R6 · Latency-aware burned-area disagreement

PROPOSED / UNVALIDATED

PROBLEM	Recent BA is incomplete; naive B jumps or drops artificially.
HYPOTHESIS	A model of BA availability and expected BA/detection ratio gives calibrated flags.
INPUTS / COST	MCD64A1 QA + dates
BASELINE	F46 median ratio

$$B^{*}_{r,y} = \frac{\mathrm{BA}_{r,y}/C_{r,y}}{\widehat{\mathbb{E}}[\mathrm{BA}/C \mid r, \text{season}, \text{lag}]}$$

TEST DESIGN	back-test flags on past years with simulated lag
FALSIFIED IF	flags on known normal years > 10%
COMPUTE	low
DECISION	**test**

R7 · Ground-truth triangulation

PROPOSED / UNVALIDATED

PROBLEM	No independent truth exists for detections.
HYPOTHESIS	Independent official burn records and hand-labelled high-resolution imagery reveal missed burning by hour.
INPUTS / COST	Sentinel-2 / Landsat labelling, official records (permission)
BASELINE	none

$$\text{recall}_h = \frac{\#\{\text{labelled burns at hour } h \text{ detected}\}}{\#\{\text{labelled burns at hour } h\}}$$

TEST DESIGN	stratified sample of 200 fields
FALSIFIED IF	labels inconsistent between reviewers (κ < 0.6)
COMPUTE	high (people)
DECISION	**research partnership**

R8 · Distribution-shift monitor

PROPOSED / UNVALIDATED

PROBLEM	Processing versions, new sensors, land-use change can silently invalidate k.
HYPOTHESIS	A monitor on residuals flags 'needs re-fit' before users see biased values.
INPUTS / COST	API outputs
BASELINE	none

$$\text{CUSUM}_t = \max\left(0,\ \text{CUSUM}_{t-1} + \frac{d_t - \bar{d}}{s_d} - \kappa\right)$$

TEST DESIGN	inject synthetic shifts; measure detection delay
FALSIFIED IF	false alarms > 1 per year
COMPUTE	low
DECISION	**adopt (status 'needs re-fit')**

R9 · Uncertainty decomposition

PROPOSED / UNVALIDATED

PROBLEM	One band hides where uncertainty comes from.
HYPOTHESIS	Showing sampling, calibration, structural and attribution parts helps users decide.
INPUTS / COST	existing
BASELINE	single band

$$\mathrm{Var}(H) \approx \underbrace{k^2\,\mathrm{Var}(M)}_{\text{counting}} + \underbrace{M^2\,\mathrm{Var}(k)}_{\text{calibration}} + \sigma^2_{\text{model}}$$

TEST DESIGN	user test with 5 officials: correct decision rate
FALSIFIED IF	no change in decisions
COMPUTE	low
DECISION	**adopt in inspector**

R10 · Active learning for source classes

PROPOSED / UNVALIDATED

PROBLEM	Labels are scarce and expensive.
HYPOTHESIS	Querying the most uncertain sites gives better classifier per label.
INPUTS / COST	lawful public references only
BASELINE	random labelling

$$x^* = \arg\max_x \left(1 - \max_y \hat{p}(y \mid x)\right)$$

TEST DESIGN	precision/recall industry vs crop on held-out sites
FALSIFIED IF	no gain vs random
COMPUTE	medium
DECISION	**postpone until labels exist**

PART V

UX, website & three design directions

A calendar-centred product: one shared analytical state, 13 routes, three coordinated visual directions and seventeen chart forms chosen by the question they answer.



CHAPTERS IN THIS PART

Chapter 20 · Information architecture
Chapter 21 · The calendar as the centre
Chapter 22 · Three design directions
Chapter 23 · The thirteen pages
Chapter 24 · Visual analytics catalogue
Chapter 25 · Palette, accessibility, motion

CHAPTER 20

Information architecture

Primary journey (the NASA task)

01 **Home** — why + trust
02 **Choose area** — search · draw · upload
03 **Calendar** — month × year
04 **Explain difference** — raw ↔ harmonised
05 **Source / Blind spot** — L2, L3
06 **Validation** — how sure
07 **Export** — CSV · JSON · PDF brief

Researcher path

01 **Explorer** — map + inspector
02 **Data Studio** — own data
03 **Formula Explorer** — F01–F46
04 **Model Lab** — B0–B6 ladder
05 **Method report** — RunReport + hash

Learning path

01 **Home**
02 **Satellites** — dossiers
03 **Learn** — 6 lessons EN/Hinglish
04 **Methods** — formula cards

One shared analytical state

```
state = {
  region: 'punjab_haryana',   bbox: null,
  period: ['2003-01', '2025-12'], grid: 0.1,
  metric: 'rate' | 'cell_days' | 'harmonised' | 'z' |
'burned_area',
  sensor: 'all' | 'terra' | 'aqua' | 'snpp' | 'noaa20'
| 'noaa21',
  dn: 'both' | 'D' | 'N',   source: 'all' | 'crop' |
...,
  qa_rule: 'modis30_viirs_nh',
  show: {ci: true, coverage: true, provenance: true},
  compare: {baseline_years: [2012, 2021]}
}
```

Every control writes to this one object; one requestAnimationFrame redraws all linked views; the URL mirrors it so any view can be shared or re-run.

ROUTE	PAGE
/	A Home
/explore	B Explorer
/calendar/:region	C Calendar
/sensors	D Sensor Lab
/studio	E Data Studio
/formulas/:id	F Formula Explorer
/validation	G Validation
/sources	H Sources
/blind-spot	I Blind Spot
/learn	J Learn
/me	K Workspace
/brief/:id	L Research Brief
/sources/health	M Source Health

User journeys with success measures

#	JOURNEY	STEPS	SUCCESS IF	FAILURE WE DESIGN AGAINST
J1	Officer checks this November	Home → Calendar (district) → inspector → Brief	answer + caveat in < 90 s	reads count drop as burning drop
J2	Scientist audits k	Sensor Lab → Validation → Formula F14	finds method, interval, held-out error	in-sample fit shown as accuracy
J3	Journalist on '2024 record low'	Calendar → Blind Spot → Brief	quotes the flag and its limit	B used as multiplier
J4	Student learns	Learn → Formulas → Explorer	passes 3-question quiz	teaching number taken as fact
J5	NGO compares field survey	Studio → compare → export	match table with lineage	user data mixed into public calendar
J6	California researcher	Explorer search → Calendar	same features, correct LMST	India-only assumptions
J7	Developer checks sources	Health → lineage	sees fallback and processing	archive labelled 'live'

Empty

"No detections in this window" + nearest active month. Distinguish observed-zero from unobserved.

Loading / failed

Skeletons; on failure show CACHED with observation date, or UNAVAILABLE. Never zeros, never demo.

Demo

meta.demo = true → amber SIMULATED badge on every number and in the export file name.

CHAPTER 21

The calendar as the centre of the product

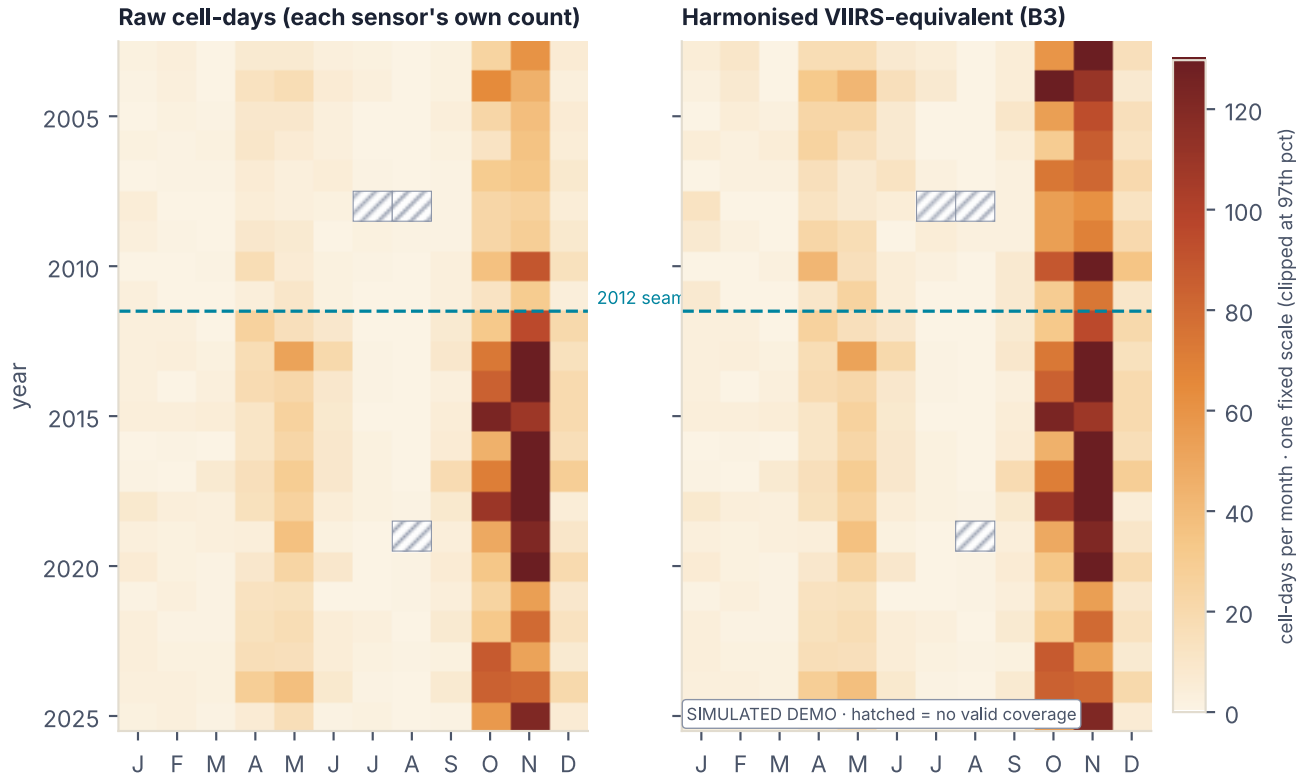


Fig. 21.1. Raw vs harmonised year × month heatmaps on one fixed colour scale; hatched cells have no valid coverage; dashed line = 2012 seam. The 'aha': the step before 2012 fades in harmonised mode while seasonality stays. SIMULATED DEMO

Independent, synchronised toggles

CONTROL	OPTIONS
granularity	year × month · week × year · daily strip
metric	raw detections · cell-days · exposure rate · harmonised VIIRS-eq · anomaly z · burned area (km², own scale)
sensor	all · per platform · day / night
source	all · crop · forest · industrial · UNKNOWN
region	hierarchy: country → state → district → cell; custom polygon
grid	0.05° · 0.1° · 0.25° (sensitivity)
compare	baseline years; era A vs era B
overlays	95% band · coverage · provenance · seams

Encoding rules

- Colour domain fixed per metric across all years and regions in view (never max of one year).
- Sequential single-hue ramp for magnitudes; diverging (grey midpoint) only for z.
- Null = hatched grey; observed zero = the lightest ramp step with a 0 label on hover.
- Low coverage = reduced opacity + dotted border, with E in the tooltip.
- Sensor of origin marked by a 2 px top tick: amber MODIS-derived, cyan VIIRS.

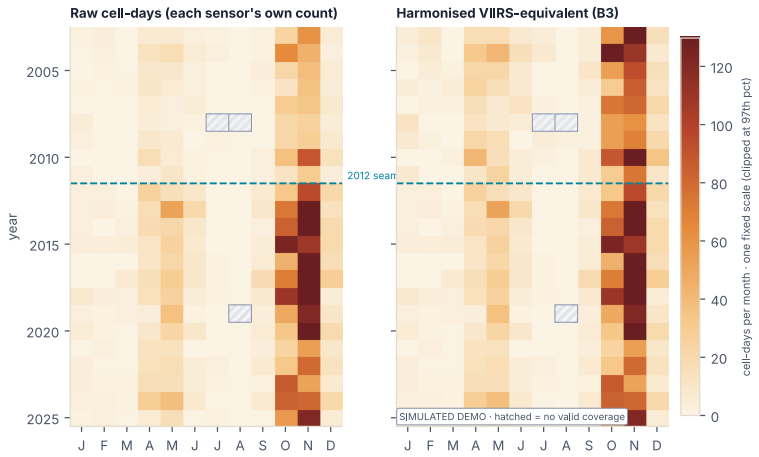


Fig. 21.2. Print / light version, same data and scale. SIMULATED DEMO

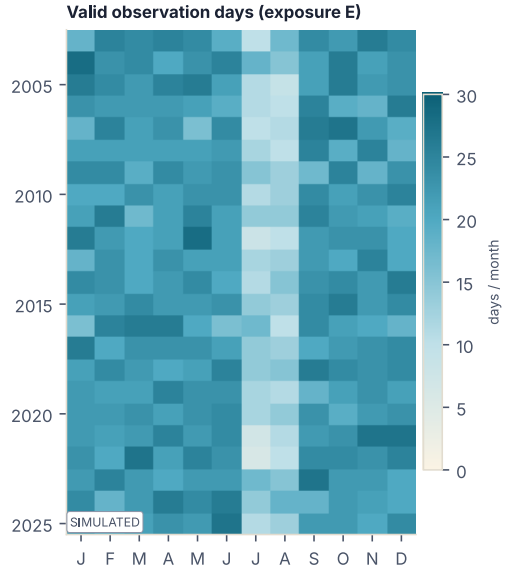


Fig. 21.3. Exposure matrix shown beside the fire heatmap. SIMULATED DEMO

Cell inspector (click a month)

VALUE	harmonised 412 VIIRS-eq cell-days
95% BAND	[356, 478] predictive · year-block bootstrap
EXPOSURE	E = 22 of 30 days · cloud 8
RATE	0.64 fire cell-days per valid day
ANOMALY	z = +2.3 (n = 12 Novembers, target left out)
CONVERSION	k_day 2.41 [2.1, 2.8] · k_night 3.9 [2.6, 5.8] · calib v0.4
10-YR REF.	median 0.41 · IQR 0.33–0.52
QUALITY	SP processing · 0 fallback · 3 % detections removed by QA
LINEAGE	F05 → F09 → F14 → F12 → F25 · hash 7c1e…

SIMULATED DEMO example values.

Keyboard and accessibility

- Arrow keys move between cells; Enter opens the inspector; Esc closes.
- Every cell has an aria-label with value, unit, band and status.
- 'Table view' renders the same data as an accessible table and CSV.
- Pattern/texture for colour-blind and print; contrast ≥ 3:1 for marks.

Export

- CSV and JSON equal the API payload (same numbers, same nulls).
- PNG/SVG of the view with legend, provenance and run hash.
- One-click PDF brief from the current state (Ch. 29).

Beyond the line and bar chart: calendar companions

Standard dashboards stop at lines and bars. The calendar is surrounded by forms that answer questions lines cannot: when the satellites could see, how sure each value is, and whether a change is real.

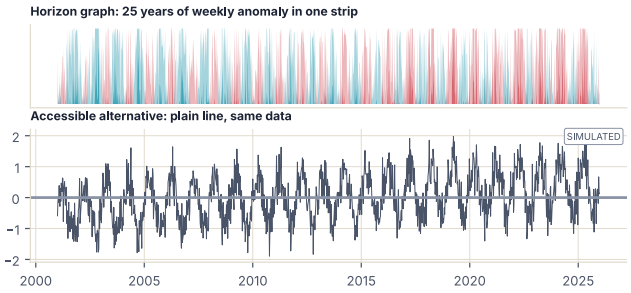


Fig. 21.4. Horizon graph of 25 years of weekly anomaly, with the plain-line alternative for accessibility. SIMULATED DEMO

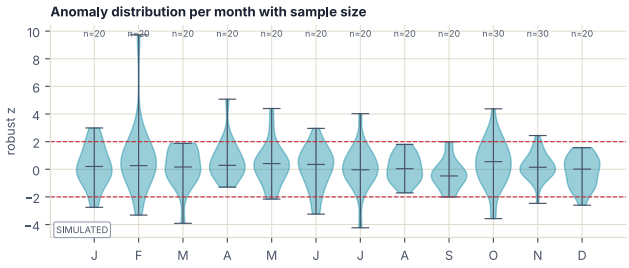


Fig. 21.5. Anomaly distribution per month with n. SIMULATED DEMO

Week × year strip

52 columns × 25 rows; shows the season sliding later (Punjab) at a glance. Same colour domain as the monthly view.

Daily ribbon

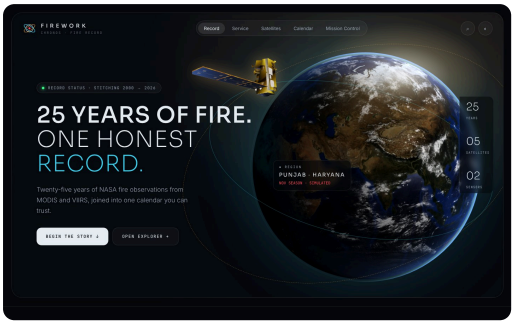
One year as 365 cells with pass markers; good for NRT last-7-days vs history (early warning).

Compare mode

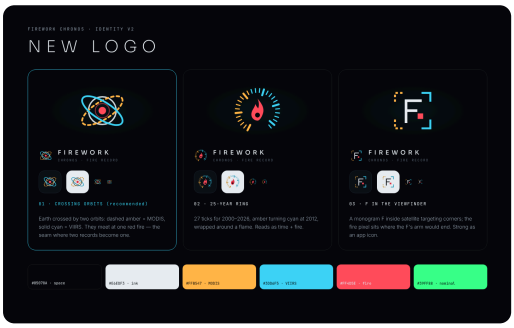
Two regions or two eras side by side with a difference row on a diverging scale.

CHAPTER 22 · DIRECTION A

Orbit Editorial — the story



Approved Concept A v2 hero (locked, user's design).



Crossing Orbits identity (recommended concept 01).

Used for Home and storytelling sections. Preserves the approved Concept A v2: deep black, split-letter Sora headings, cinematic Earth with two orbits (amber dashed MODIS, cyan solid VIIRS), the 4-step End-to-End, satellite dossiers, a calendar teaser and five control tiles.

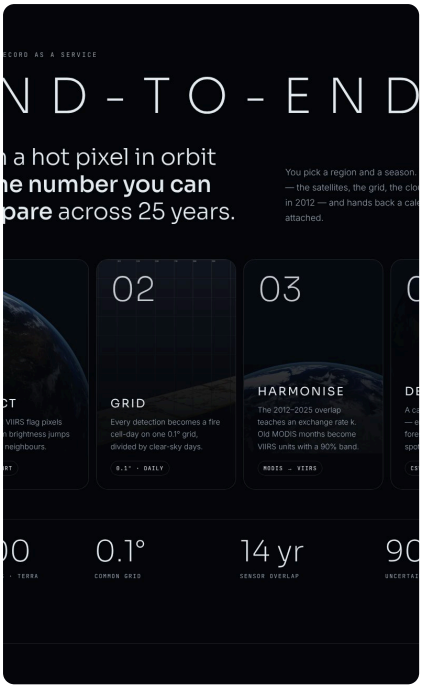
TOKEN	VALUE
bg	`#05070A`
ink	`#E6EDF3`
MODIS	`#FFB547 (brand) · #C98418 (data marks)`
VIIRS	`#3DD6F5 (brand) · #1A9DBA (data marks)`
fire	`#FF4D5E`
nominal	`#39FF88 (status only)`
display	`Sora 200, tracking 0.3 em, split letters`
body / data	`Inter 300 · JetBrains Mono`

Components

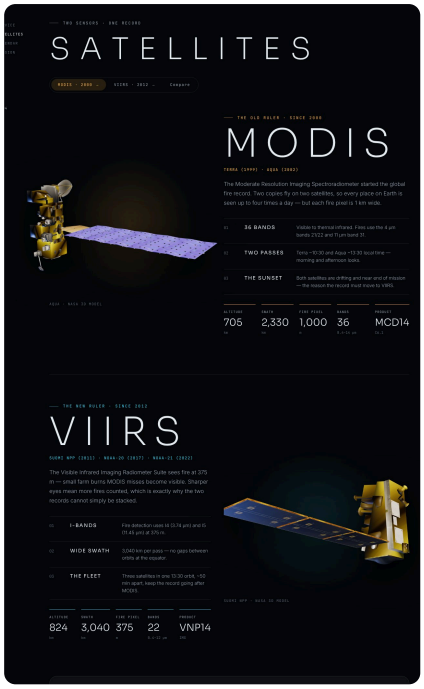
frame card · nav pill · status chip · split heading · step card with progress · dossier spec list · stat counter · teaser heatmap · control tile · side progress nav 01–05.

GUARDRAILS

Story motion (Lenis, GSAP scrubs, letter reveals) stops before analysis sections. NASA logos are not used; no implied endorsement. Satellite models are illustrative until licences are verified.



End-to-end section (Concept A v2).



Satellite dossiers.



Calendar teaser + control tiles.

Breakpoints

WIDTH	LAYOUT
≥ 1440	full cinematic, side nav
768–1439	hero stacks, dossiers 1 column
390	headline 2 lines, clips swipe, tiles 2 × 3

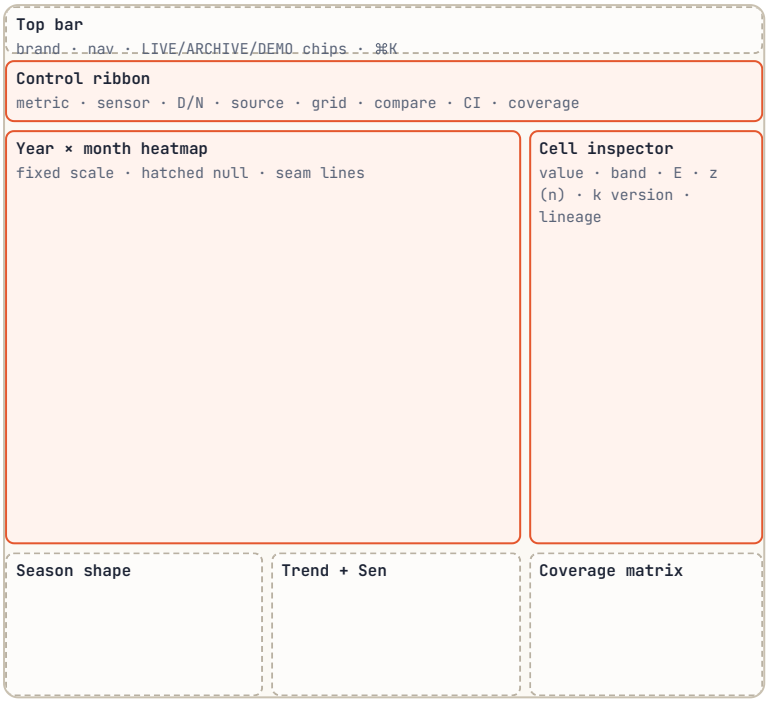
Storyboards

1. Scroll: satellite drifts along orbit; Earth rotates 14°; letters rise; 'Begin the story' → How it works clip 1 autoplays muted in view.
2. Dossiers: Aqua flies in from left with amber 1 km pixels catching 1 of 3 fires; S-NPP from right with 375 m pixels catching 3 of 3 (teaching animation, labelled).

CRITERION	A ORBIT EDITORIAL	B MISSION CONTROL	C CIVIC REPORT
science readability	3	5	4
accessibility	3	4	5
wow / story	5	3	2
feasibility (hackathon)	4	4	5
load time	3	4	5
best for	Home, video, pitch	data pages B-I, M	Brief, Learn, print, officials

CHAPTER 22 · DIRECTION B

Analytical Mission Control — the work



Calendar page in Direction B (1600 × 1000). Hot outlines = primary task zones.

Dense but calm panels for data pages: filter ribbon, map/heatmap, source inspector, monospaced metadata, precise units and provenance in every panel header. No scroll-pinned scenes; values tween 280–320 ms.

TOKEN	VALUE
surface	#0B1220 / panel #111A2E
line	#1E2A42
text	#E6EDF3 / dim #9AA6BD
MODIS mark	#C98418
VIIRS mark	#1A9DBA
model estimate	#8C7CF0
alert	#E04857
type	Inter 13/20 · JetBrains Mono 11 for numbers

PALETTE DECISION (C-02, PROPOSED)

Replace atlas.css MODIS blue / VIIRS orange with the amber / cyan semantics of Home, using validated mark tints. Validator (dataviz skill) on #0B1220: lightness band, chroma, CVD ΔE ≥ 15, contrast ≥ 3:1 — all PASS.

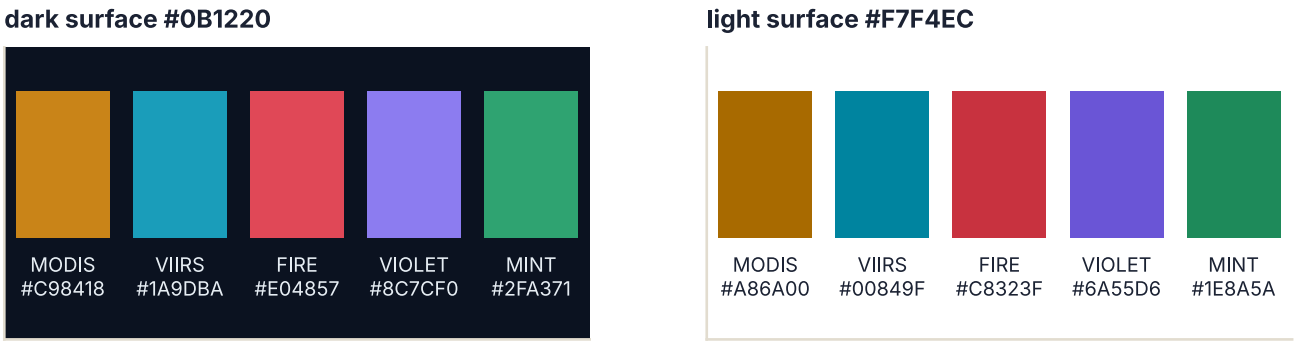
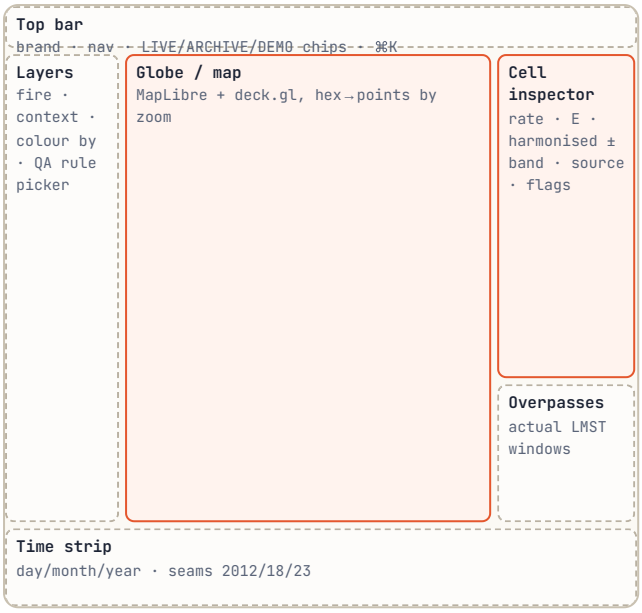


Fig. 22.1. One semantic palette, dark and light steps, validated separately for each surface. RUN / TESTED LOCALLY



Explorer in Direction B (desktop).



Sensor Lab in Direction B.

Component inventory

COMPONENT	STATES
StatusChip	LIVE · ARCHIVE · CACHED · SIMULATED · UNAVAILABLE · USER
FilterRibbon	default · changed · invalid
ValueWithBand	ok · wide · null · pending
CellInspector	loading · ok · not_observable · error
ProvenanceBadge	5 labels
FormulaChip	links to /formulas/:id
EmptyState	unobserved vs observed-zero
DataTable	a11y twin of every chart

Interaction storyboards

1. Click a calendar cell → inspector slides in (240 ms) → map flies to region (1,200 ms) → overpass card updates; all from one state change.
2. Switch metric to 'anomaly z' → colour scale swaps to diverging with grey midpoint; legend and table update together; cells with n < 8 show 'i' badge.

Reduced motion

No camera flights; instant state changes; tweens off; focus ring always visible (2 px cyan outline).

CHAPTER 22 · DIRECTION C

Civic Evidence Report — the decision

PUNJAB · KHARIF SEASON 2024 · FIELD CARD SIMULATED

Detected farm-fire activity was above normal in November, but counts may miss late-afternoon burning.

+2.3	0.64	CHECK
robust z vs 12 Novembers	fire cell-days per valid day	blind-spot flag B = 2.4

What it means. More cells than usual had detected fire on clear days. Because burning after 15:00 is mostly unseen, the true change is uncertain. Not for life-safety decisions.

An accessible off-white / indigo / orange direction for officials, printouts and mobile. Plain-language finding first, three numbers, a coverage warning, and sources at the bottom. It is an independent alternative for Brief and Learn — not a replacement for the approved Home.

TOKEN	VALUE
paper	#FAFAF7
ink	#1B1B1F
primary	indigo #1A237E / #3949AB
warning	orange #E65100
MODIS / VIIRS	#A86A00 / #00849F (light steps)
type	Sora 15/20 headings · Inter 11/16 body
contrast	≥ 7:1 body text



Brief page in Direction C (A4 portrait).



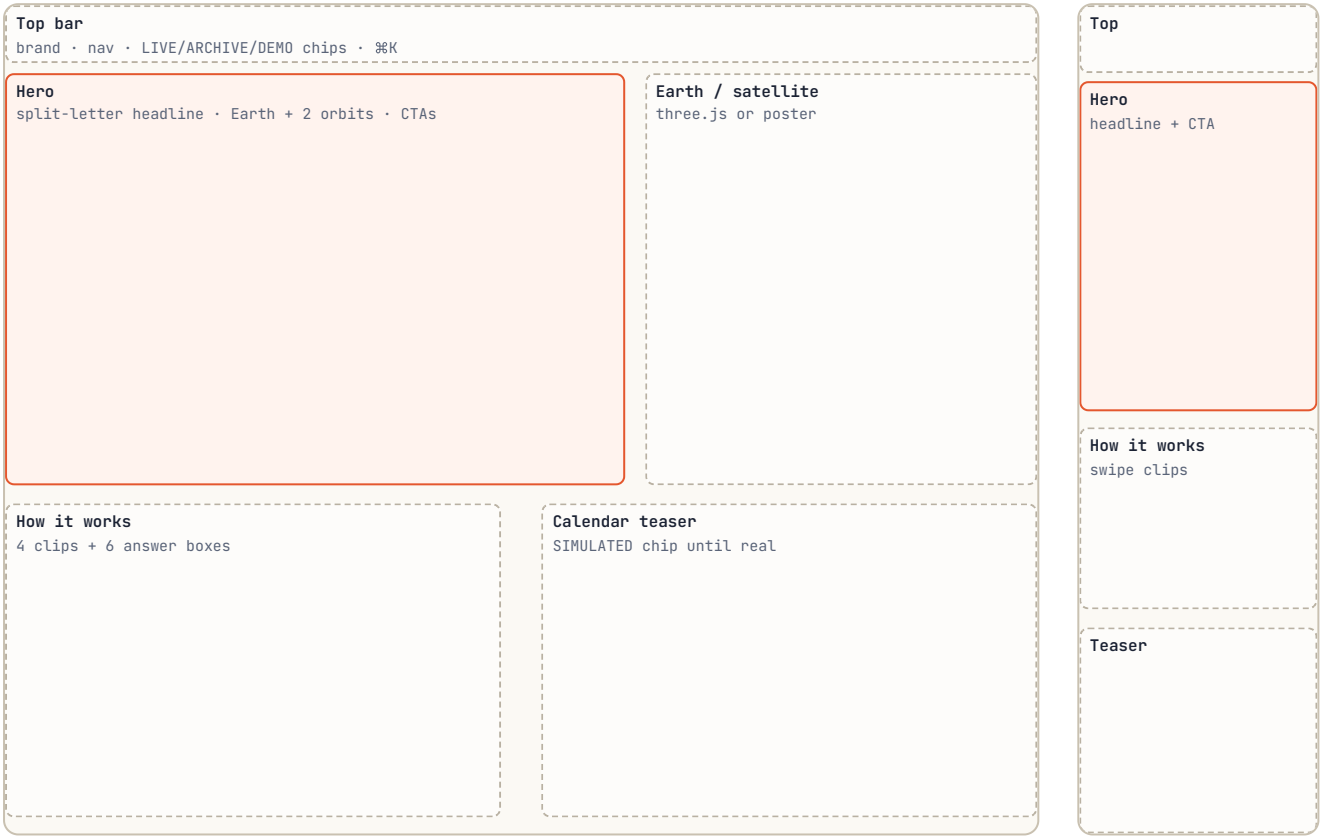
Brief at 390 px.

CHAPTER 23
PAGE A · /

Home

Direction A

Key question: *What is this and why should I trust it?*



Desktop 1600 × 1000 wireframe

Mobile 390 × 844

PERSONA	Everyone (judges, students, officials)
DATA / API	home_content, satellites (static); /api/health for chips
CONTROLS	CTA: Begin story · Open explorer
STATES	poster on load; reduced-motion static; live feed down → CACHED chip
PERF / RULE	Concept A v2 locked; LCP < 2.5 s

PRIMARY VIEW	split-letter hero, orbit illustration, 4-step clip player
ALTERNATIVES	static poster for low-end devices; text-only summary for screen readers
DONE WHEN	screenshot matches A_home_locked.png; LCP < 2.5 s; chips equal /api/health
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE B · /explore

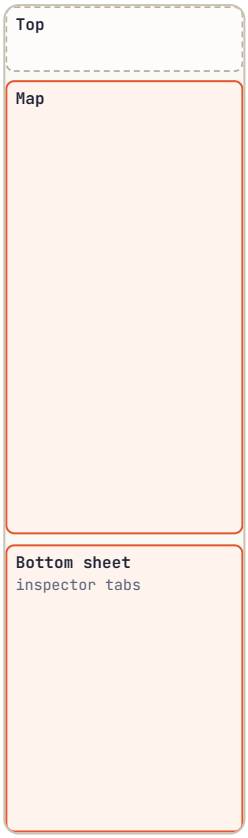
Explorer

Direction B

Key question: *Where and when was burning detected, and how sure is it?*



Desktop 1600 × 1000 wireframe



Mobile 390 × 844

PERSONA	Researchers, officials
DATA / API	/api/cells?bbox&date&res · /api/cells/:cell
CONTROLS	bbox search, date, layers, QA rule
STATES	tiles loading shimmer; no detections → nearest active month; too many points → hex
PERF / RULE	60 fps claim only after measurement

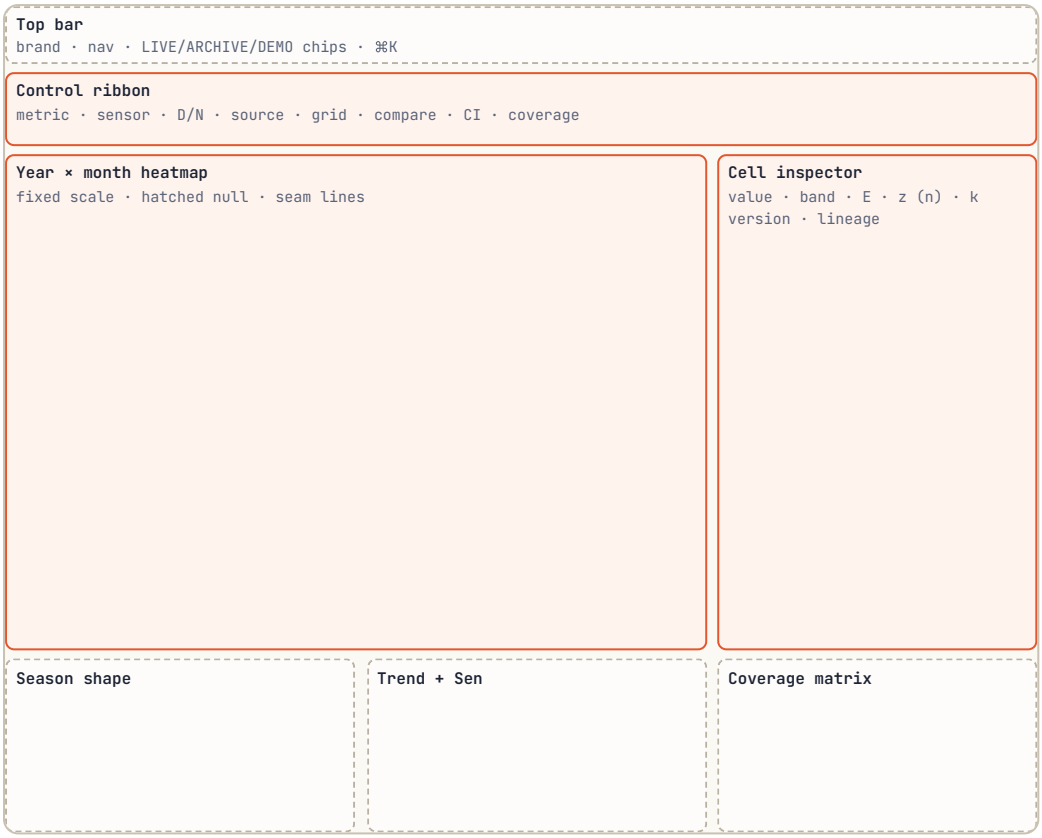
PRIMARY VIEW	map with hex density → points by zoom; inspector key–value list
ALTERNATIVES	small-multiple maps by year; hexbin + uncertainty overlay
DONE WHEN	inspector value equals calendar value for the same cell-month
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE C · /calendar/:region

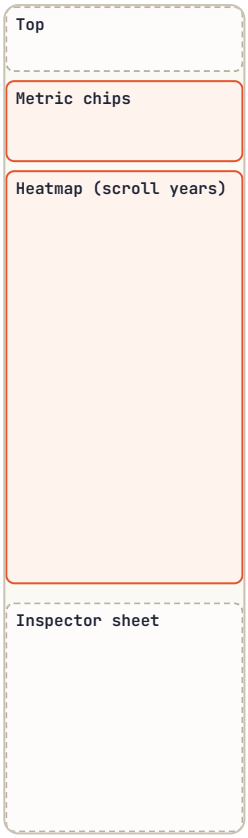
Historical Calendar

Direction B

Key question: *Which months and years were unusual, on one honest scale?*



Desktop 1600 × 1000 wireframe



Mobile 390 × 844

PERSONA	Officials, scientists, journalists
DATA / API	/api/calendar/:region?metric=&sensor=&dn=
CONTROLS	see Ch. 21
STATES	archive not processed → locked with ETA; few fires → 'indicative'
PERF / RULE	25-year render measured on a mid laptop

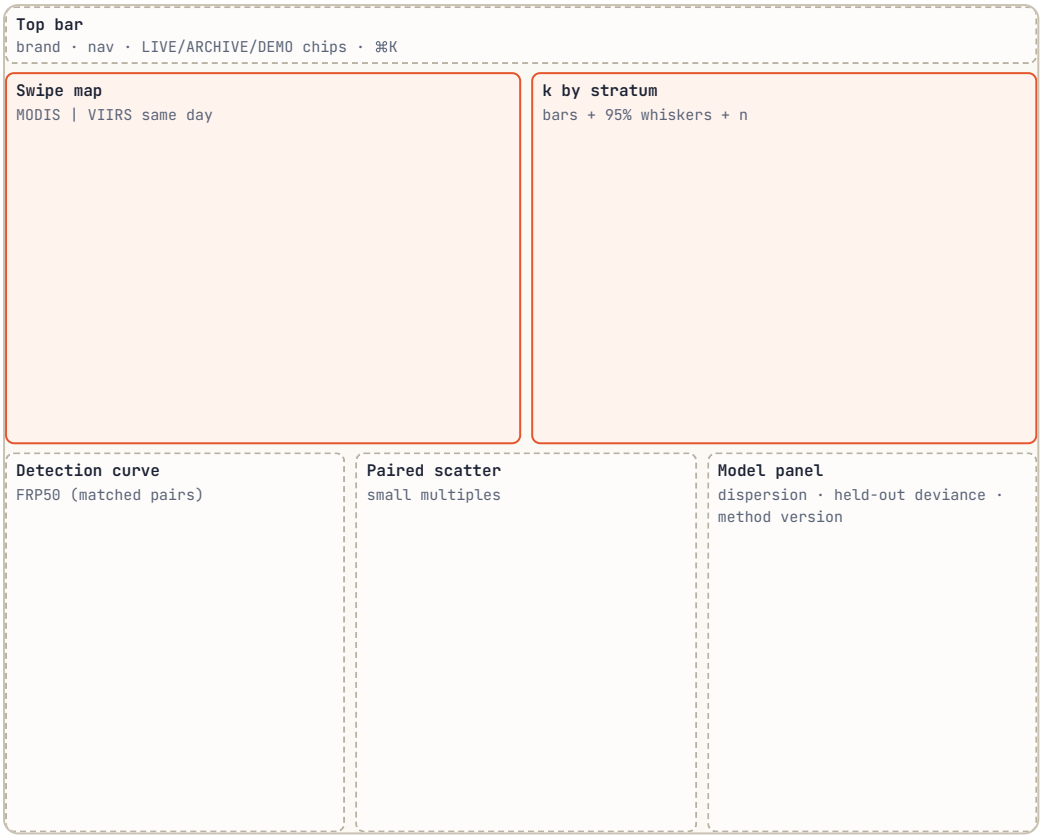
PRIMARY VIEW	year × month heatmap on fixed scale with hatched nulls
ALTERNATIVES	week × year strip, horizon graph, exposure matrix, compare mode
DONE WHEN	toggle changes colours only; CSV = API = chart; 25 years render measured
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE D · /sensors

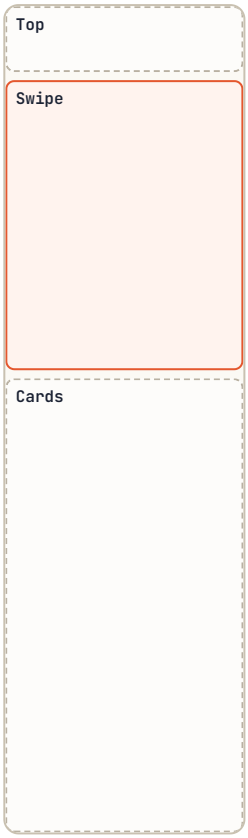
Sensor Lab

Direction B

Key question: *How was k learned and how stable is it?*



Desktop 1600 × 1000 wireframe



Mobile 390 × 844

PERSONA	Researchers, reviewers
DATA / API	/api/calibration/:region · /api/satellite/:platform
CONTROLS	D/N switch, filter pair, stratum
STATES	night k null → 'needs archive run'; no pairs → curve hidden
PERF / RULE	-

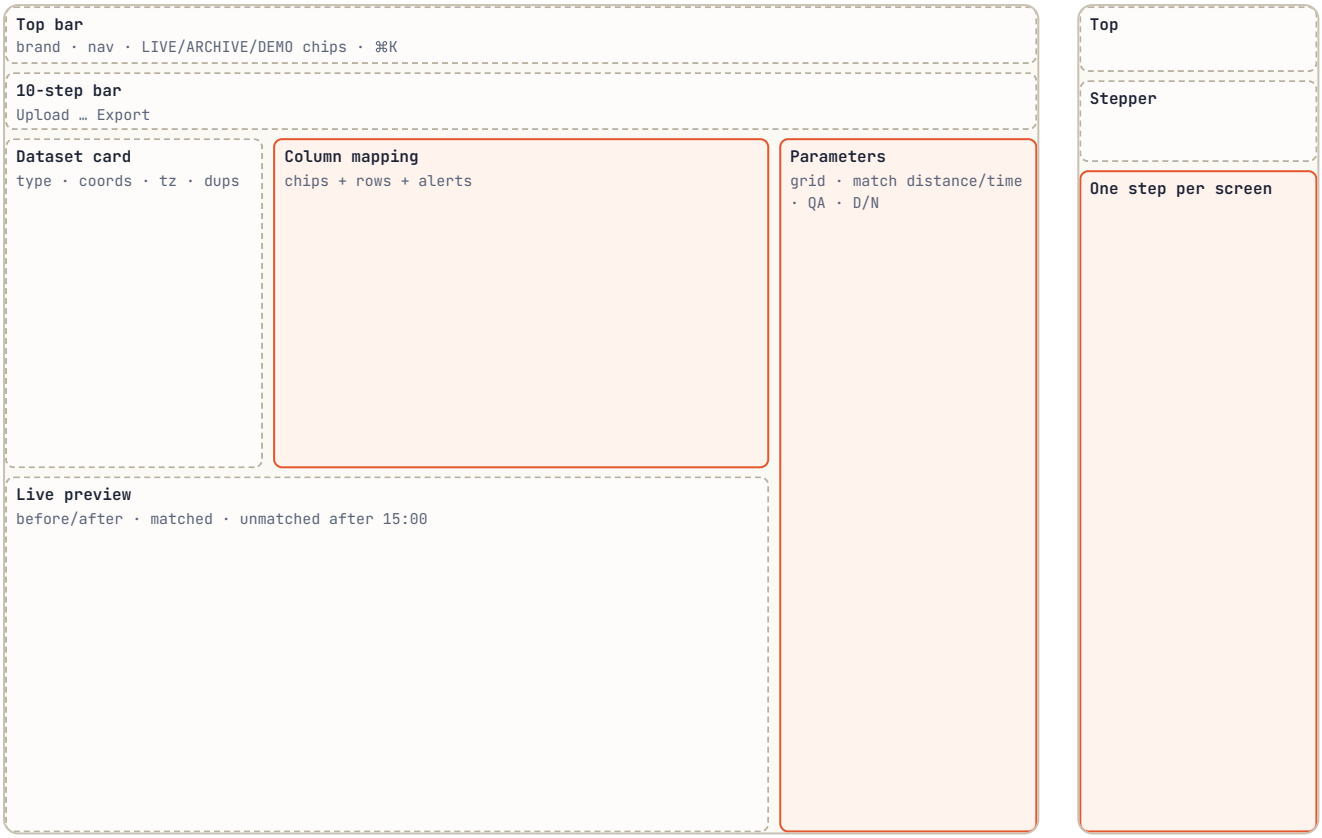
PRIMARY VIEW	k by stratum with 95% whiskers and n
ALTERNATIVES	paired small multiples, detection curve, Bland–Altman
DONE WHEN	k never shown without interval; night k null shows reason
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE E · /studio

Data Studio

Direction B

Key question: *Does my own data agree with the satellite record?*



Desktop 1600 × 1000 wireframe

Mobile 390 × 844

PERSONA	Researchers, NGOs, students
DATA / API	POST /api/studio/upload · /config · /compare
CONTROLS	mapping, parameters
STATES	unsupported file; schema unknown → manual; > 25 MB → refuse
PERF / RULE	uploads isolated, TTL, never sent to LLM

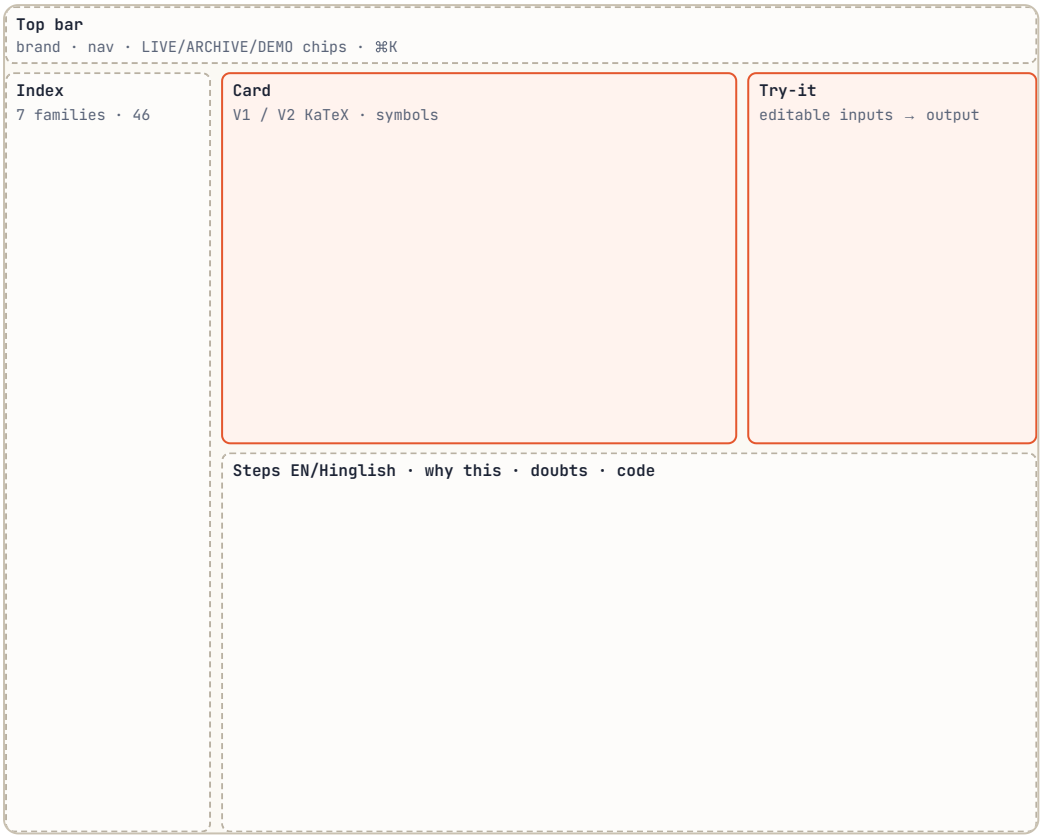
PRIMARY VIEW	editable column-mapping grid + validation list
ALTERNATIVES	before/after histogram, match map
DONE WHEN	impossible values always caught; same config → same result
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE F · /formulas/:id

Formula Explorer

Direction B

Key question: *What does this formula do and can I try it?*



Desktop 1600 × 1000 wireframe



Mobile 390 × 844

PERSONA	Students, reviewers
DATA / API	/api/formulas/:id (static JSON from registry YAML)
CONTROLS	inputs, language
STATES	invalid input → explain rule (e.g. E = 0)
PERF / RULE	-

PRIMARY VIEW	KaTeX card V1/V2 + try-it table
ALTERNATIVES	derivation stepper, figure from code
DONE WHEN	every card: problem, intuition, V2 definition, example, limits, code link
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE G · /validation

Validation

Direction B

Key question: *How far can I trust each result?*



Desktop 1600 × 1000 wireframe

Mobile 390 × 844

PERSONA	Judges, scientists
DATA / API	/api/validation/:region
CONTROLS	region select
STATES	real run pending → '—' + SYNTHETIC banner
PERF / RULE	thresholds immutable once published
PRIMARY VIEW	predeclared pass-rule panel + scorecard tiles
ALTERNATIVES	LOYO chart, reliability diagram, placebo event study
DONE WHEN	SYNTHETIC banner whenever any value is simulated; tiles link to formulas
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE H · /sources

Sources

Direction B

Key question: *What is burning here, and how sure?*



Desktop 1600 × 1000 wireframe

Mobile 390 × 844

PERSONA	Officials, researchers
DATA / API	/api/source/:cell · /api/sites?bbox
CONTROLS	class filter
STATES	abstain → UNKNOWN + top-2; OSM missing → 'no data'
PERF / RULE	-

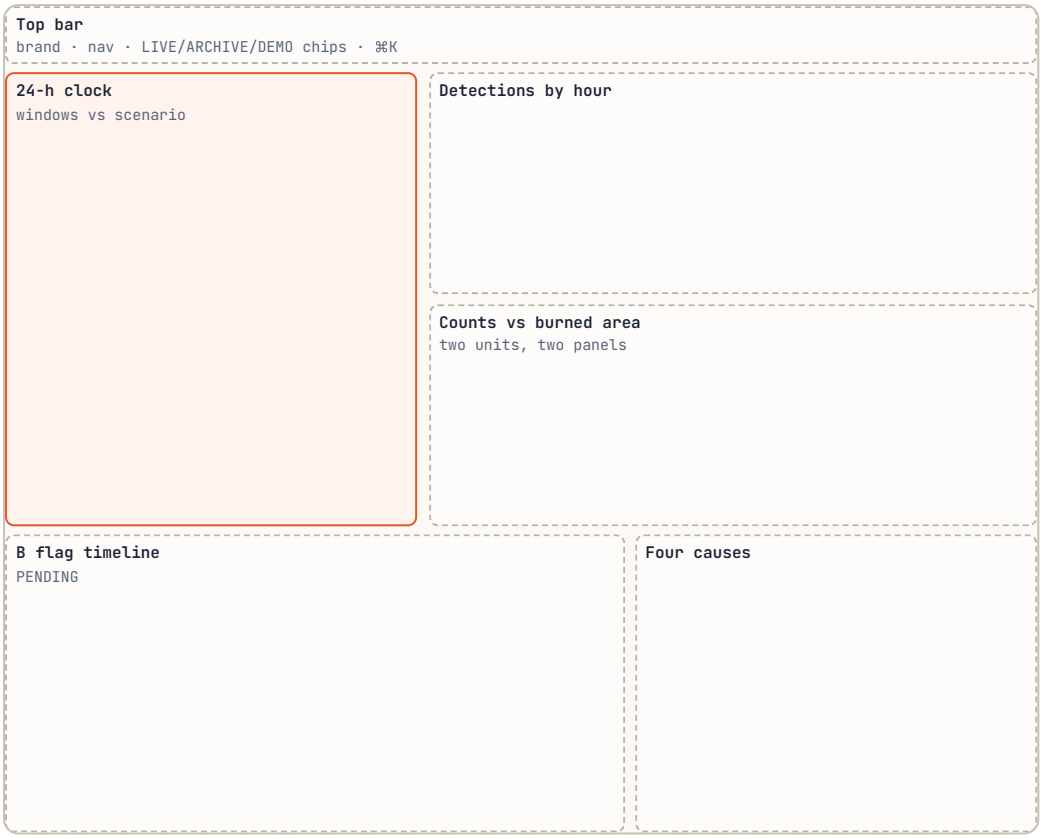
PRIMARY VIEW	map by class with UNKNOWN + evidence card
ALTERNATIVES	source mix stacked area, set chips
DONE WHEN	every class shows evidence for and against; UNKNOWN never hidden
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE I · /blind-spot

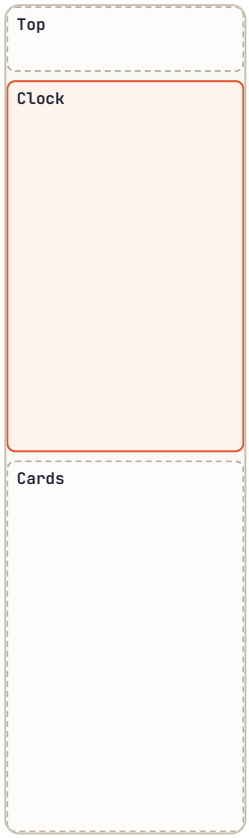
Blind Spot

Direction B

Key question: *When might the satellites miss burning?*



Desktop 1600 × 1000 wireframe



Mobile 390 × 844

PERSONA	Officials, journalists
DATA / API	/api/blindspot/:region
CONTROLS	year, scenario curve
STATES	no hourly data → 'scenario, not measurement'
PERF / RULE	-

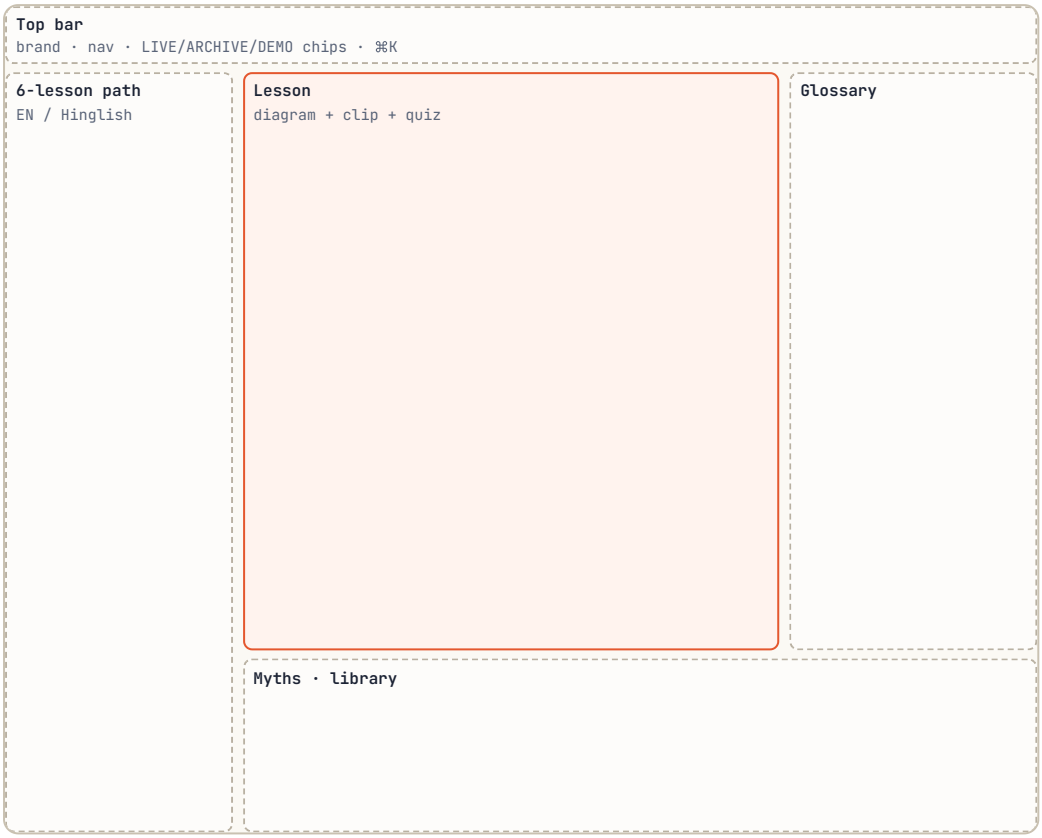
PRIMARY VIEW	24-hour coverage clock (measured solid, scenario dashed)
ALTERNATIVES	detections by hour histogram, B timeline with PENDING
DONE WHEN	scenario visibly different from measured; reported values carry source and year
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE J · /learn

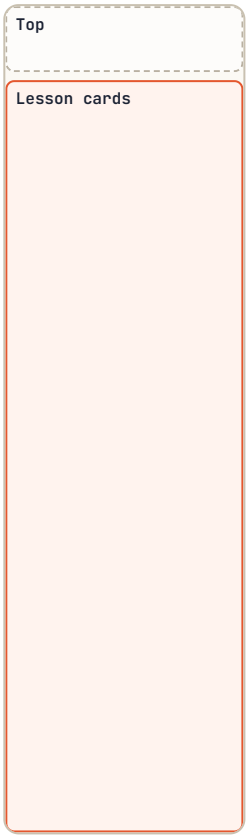
Learn

Direction C

Key question: *How do satellites see fire?*



Desktop 1600 × 1000 wireframe



Mobile 390 × 844

PERSONA	Students, public
DATA / API	static lessons; Home clips
CONTROLS	language
STATES	offline cached
PERF / RULE	teaching numbers labelled

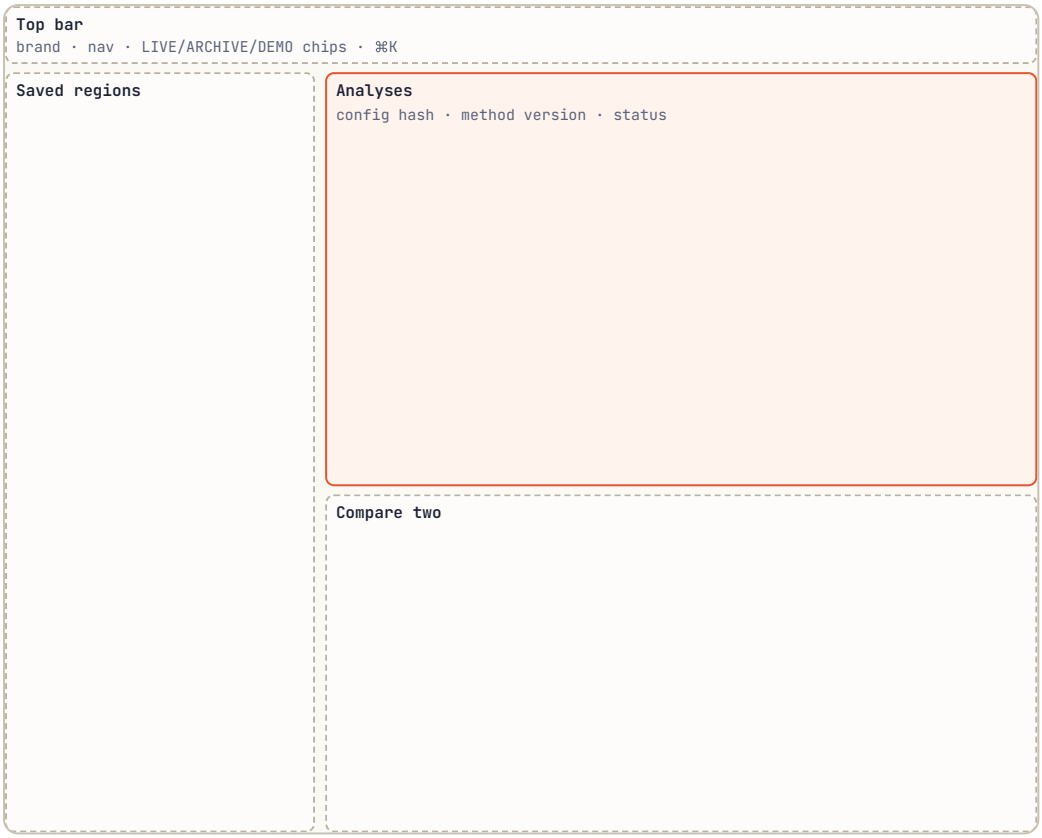
PRIMARY VIEW	lesson cards with diagram + clip + quiz
ALTERNATIVES	myth cards, glossary
DONE WHEN	every lesson cites at least one source; EN/Hinglish switch keeps position
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE K · /me

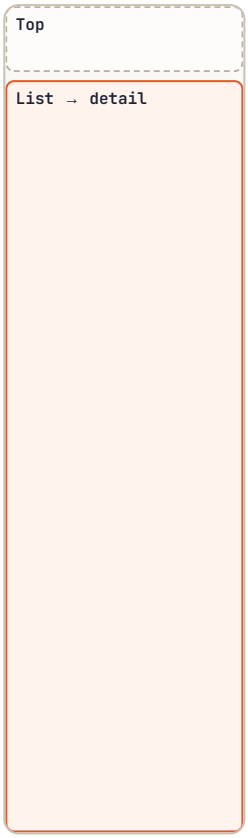
Workspace

Direction B

Key question: *Can I reproduce last week's analysis?*



Desktop 1600 × 1000 wireframe



Mobile 390 × 844

PERSONA	Returning users
DATA / API	/api/me/regions · /api/me/jobs
CONTROLS	select two to compare
STATES	algorithm changed → 're-run' badge
PERF / RULE	private by default

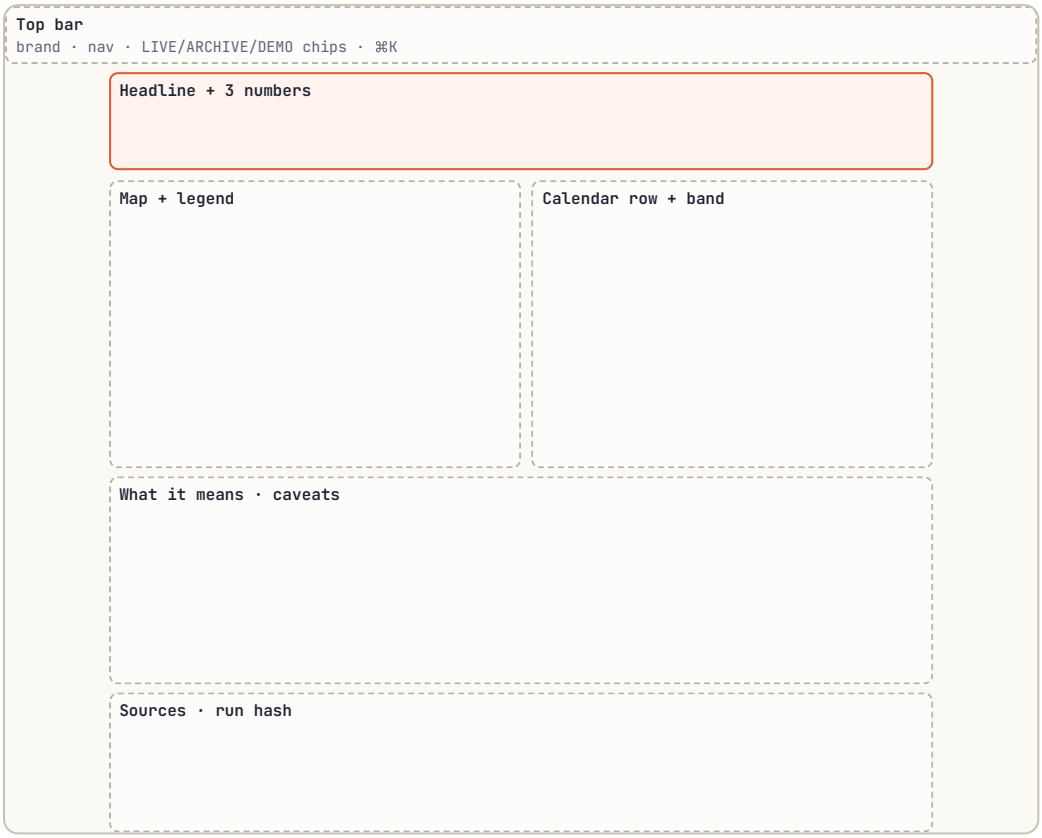
PRIMARY VIEW	analyses table with config hash and status
ALTERNATIVES	side-by-side compare
DONE WHEN	any saved result re-runs to the same numbers
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE L · /brief/:id

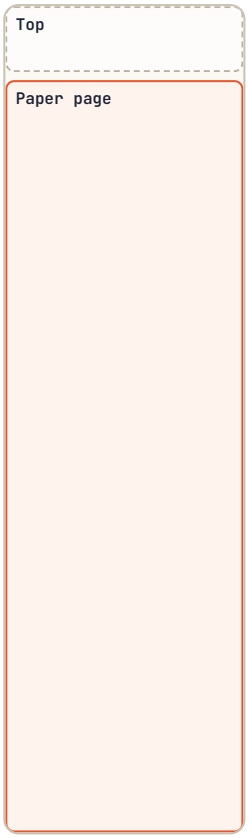
Research Brief

Direction C

Key question: *What happened this season, in one page?*



Desktop 1600 × 1000 wireframe



Mobile 390 × 844

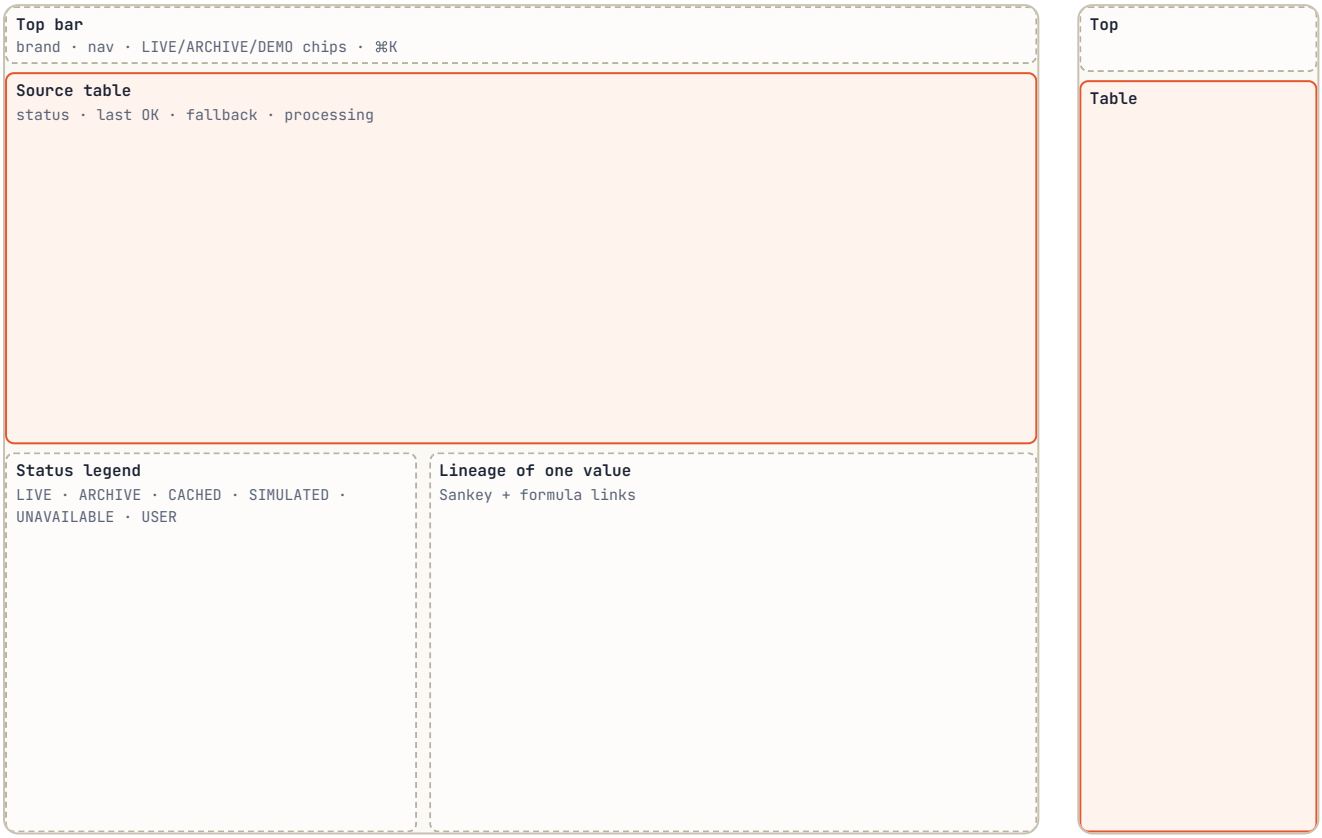
PERSONA	District officials, journalists
DATA / API	/api/brief/:region?season= (RunReport.json)
CONTROLS	audience EN/Hinglish
STATES	LLM down → template text, same numbers
PERF / RULE	every number in text exists in payload
PRIMARY VIEW	paper page: headline, 3 numbers, map, calendar row
ALTERNATIVES	light print template
DONE WHEN	every number in text exists in RunReport.json; caveat always present
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

PAGE M · /sources/health

Source Health

Direction B

Key question: *Where does every number come from, right now?*



Desktop 1600 × 1000 wireframe

Mobile 390 × 844

PERSONA	Developers, judges
DATA / API	/api/health · /api/lineage/:id
CONTROLS	-
STATES	all failing → UNAVAILABLE, never demo
PERF / RULE	health ≠ correctness

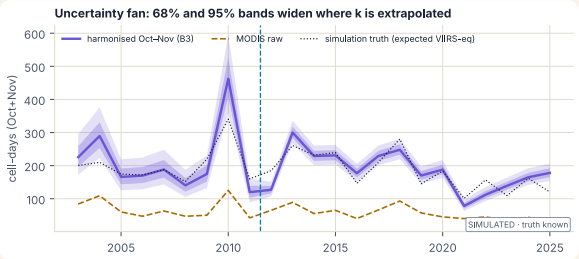
PRIMARY VIEW	source status table + lineage Sankey
ALTERNATIVES	timeline of fallbacks
DONE WHEN	every API response has meta.data_source; fallback events logged
KEYBOARD	Tab order: ribbon → main view → inspector; arrows inside grids; Esc closes panels
WCAG	text ≥ 4.5:1, marks ≥ 3:1, focus ring 2 px, no colour-only meaning
MOTION	≤ 320 ms value tweens; none under reduced motion
TOOLTIP / EXPORT	value + unit + band + status + provenance; CSV equals API
SCREENSHOT TEST	Playwright 1600 × 1000 and 390 px vs reference; only data may differ

CHAPTER 24

Visual analytics catalogue

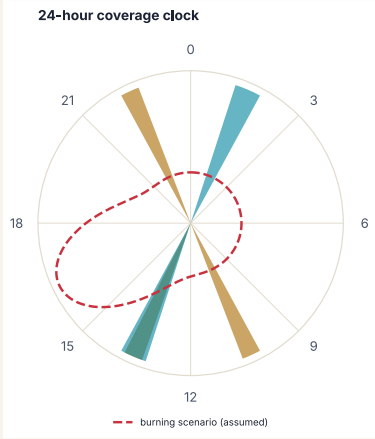
Seventeen forms usually skipped by fire dashboards. Each answers one question. Pick by question, and never more than three on one screen. All figures here are SIMULATED DEMO unless marked.

Uncertainty fan



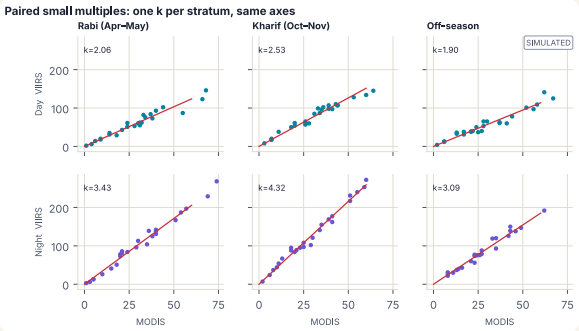
Question. How sure is each year of the long series?
Data. harmonised value, 68/95% bands
Interaction. line + ribbons; hover year → inspector
Validity warning. bands must state level and type

Day × hour coverage clock



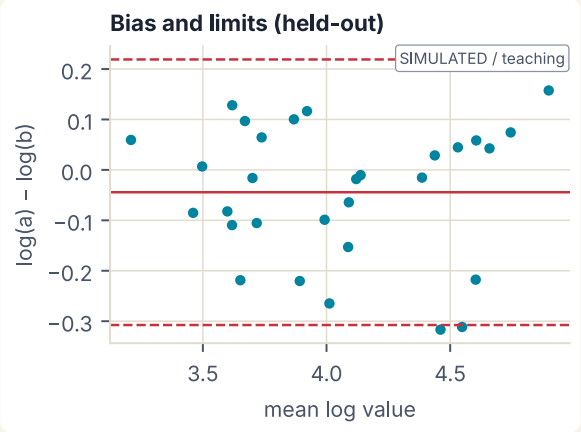
Question. When could satellites see this place?
Data. actual pass windows vs scenario curve
Interaction. scrub hours; two styles: measured solid, scenario dashed
Validity warning. polar distorts area; label hours

Paired small multiples



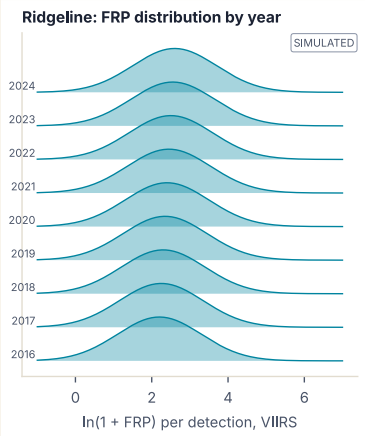
Question. Is k the same in every stratum?
Data. V vs M per stratum with k line
Interaction. shared axes; click stratum → Sensor Lab
Validity warning. few points → hide line

Bland–Altman



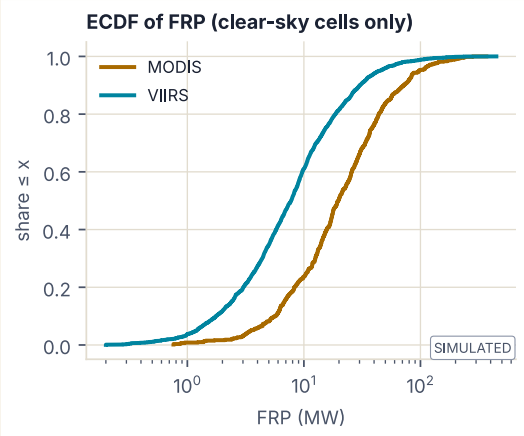
Question. Do harmonised MODIS and VIIRS agree?
Data. log difference vs mean, held-out
Interaction. hover point → year
Validity warning. never in-sample

Ridgeline



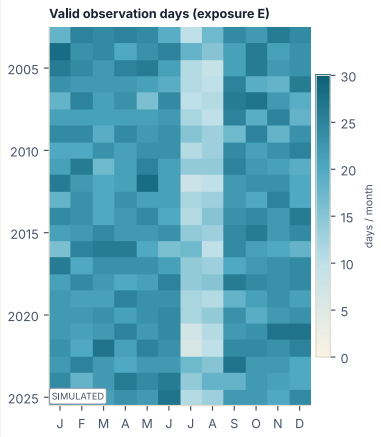
Question. Is the FRP distribution shifting?
Data. density of log FRP by year
Interaction. hover ridge → year stats
Validity warning. overlap hides tails

ECDF of FRP



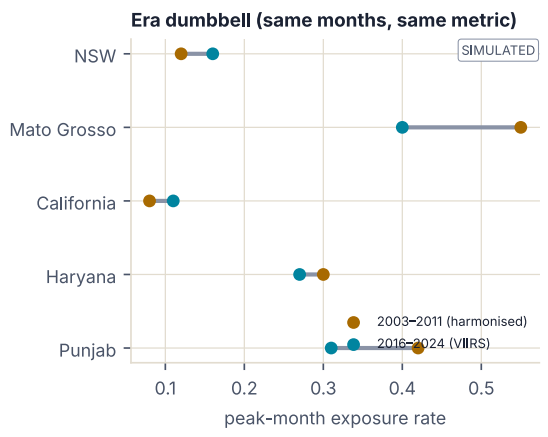
Question. Do the sensors see different fire sizes?
Data. share ≤ FRP by sensor
Interaction. toggle clear-sky only
Validity warning. FRP not comparable across sensors raw

Exposure matrix



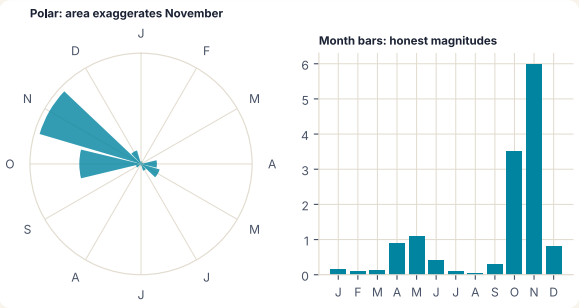
Question. Where is the calendar blind?
Data. valid days per month
Interaction. sits beside heatmap; linked hover
Validity warning. low E → widen band

Era dumbbell



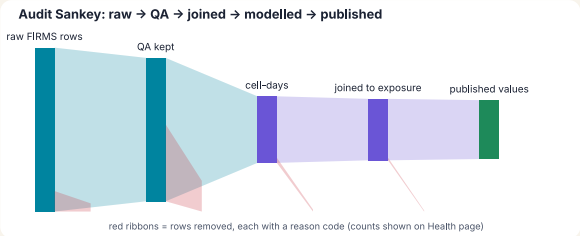
Question. How did each region change between eras?
Data. same metric, same months
Interaction. sort by change
Validity warning. era lengths must match

Polar vs month bars



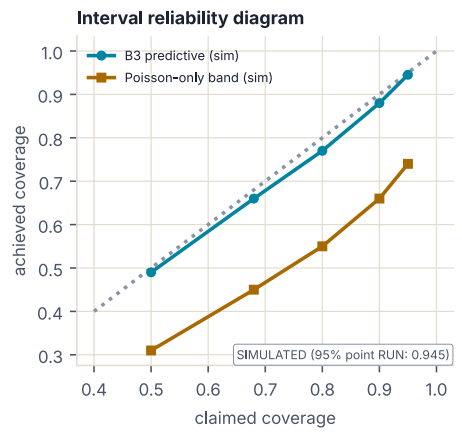
Question. When is the season?
Data. monthly totals
Interaction. use bars by default
Validity warning. radial area exaggerates peaks

Sankey audit



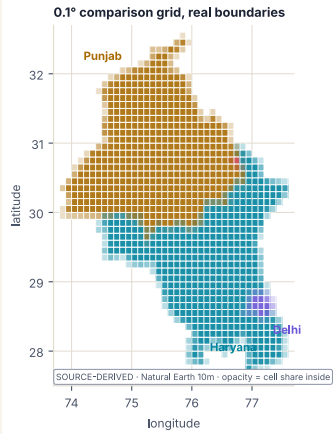
Question. Where did rows go?
Data. row counts per pipeline stage
Interaction. click ribbon → reason codes
Validity warning. same denominator everywhere

Reliability diagram



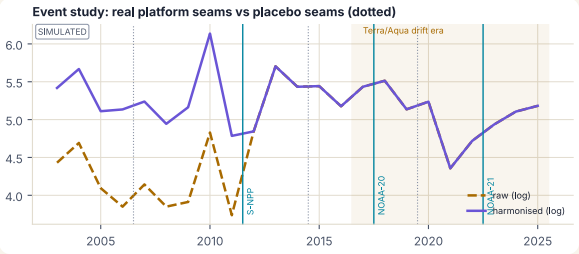
Question. Do 95% bands contain 95%?
Data. achieved vs claimed coverage
Interaction. by region
Validity warning. needs held-out truth

Real grid map



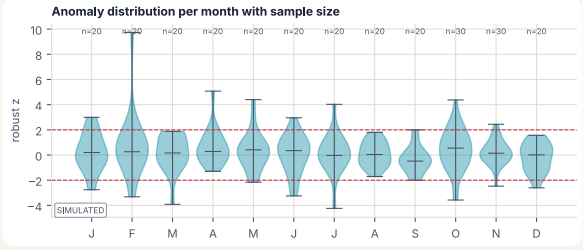
Question. Which cells are in my region?
Data. cell share inside polygon
Interaction. click cell → inspector
Validity warning. boundary source stated

Event study



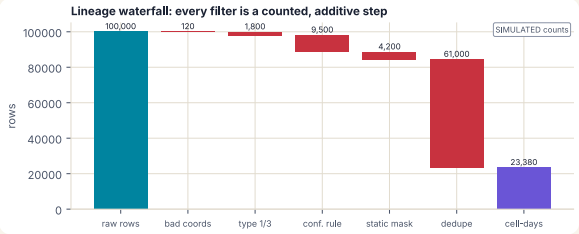
Question. Are sensor seams gone?
Data. log series with seams and placebos
Interaction. toggle placebo years
Validity warning. real change can coincide with seams

Anomaly violins



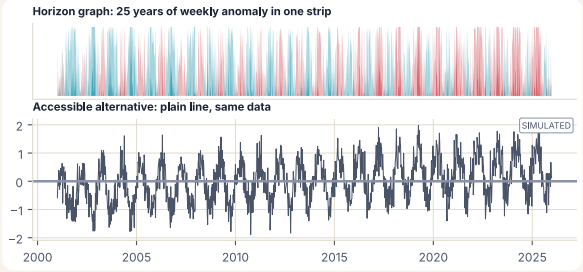
Question. How variable is each month?
Data. robust z by month with n
Interaction. hover n
Validity warning. small n → hide violin

Lineage waterfall



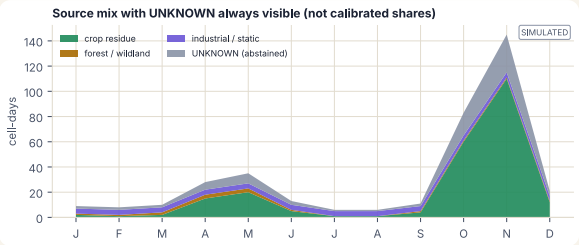
Question. How did filters shrink the data?
Data. rows removed per step
Interaction. click step → code
Validity warning. only additive steps

Horizon graph



Question. 25 years of weekly anomaly at a glance
Data. banded anomaly
Interaction. plus plain-line alternative
Validity warning. hard for new users

Source mix with UNKNOWN



Question. What is burning each month?
Data. cell-days by class
Interaction. toggle classes
Validity warning. not calibrated fractions

MVP vs stretch for chart forms

TIER	FORMS	REASON
MVP	calendar heatmap, exposure matrix, uncertainty fan, coverage clock, Bland–Altman, reliability diagram, source mix with UNKNOWN	each answers a judge's 'how do you know?'
Stretch	ridgeline, ECDF, dumbbell, horizon, violins, event study, Sankey, waterfall	research depth; add when real data exists
Avoid by default	polar seasonality, 3D charts, dual axes	distort magnitude or compare two scales

CHAPTER 25

Palette, accessibility and motion

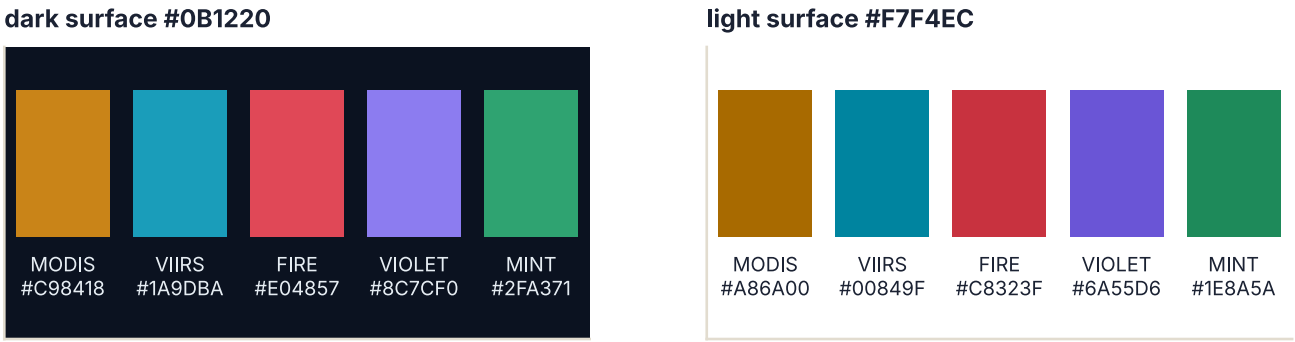


Fig. 25.1. Sensor semantics never change between pages: amber = MODIS, cyan = VIIRS, coral = fire/alert, violet = model estimate, mint = validated/crop. RUN / TESTED LOCALLY

Validator results (run 27 Oct 2026)

CHECK	DARK #0B1220	LIGHT #F7F4EC
lightness band	PASS	PASS
chroma floor	PASS	PASS
CVD adjacent ΔE	PASS (15.2 worst)	PASS (15.4)
normal-vision floor	PASS (24.0)	PASS (22.0)
contrast ≥ 3:1	PASS	PASS

Brand tints (#FFB547, #3DD6F5) are too light for data marks on dark surfaces (lightness FAIL) — keep them for identity, text accents and lines; use #C98418 / #1A9DBA for filled marks.

Motion system

EVENT	DURATION
hover	120 ms
panel slide	240 ms
value tween	280–320 ms
timeline step	600 ms
camera flight	1,200 ms
reduced motion	0 (instant)

Accessibility checklist

- Never colour alone: labels, shapes, hatch.
- Legend for ≥ 2 series; direct labels when ≤ 4.
- Data table twin for every chart.
- Hindi / Hinglish labels with fonts that include Devanagari (Noto Sans Devanagari) — test rendering.
- Keyboard-only completion of journeys J1–J7.

PART VI

Data, API, RAG & automatic reports

Where every input comes from, the static data you can load into Supabase today, the API contract, an own knowledge base that cites instead of inventing, and a deterministic PDF brief builder.

CHAPTERS IN THIS PART

Chapter 26 · Data sources, keys and backups
Chapter 27 · Database, static CSVs and API contract
Chapter 28 · Your own RAG knowledge base
Chapter 29 · Automatic PDF briefs

CHAPTER 26

Data sources, keys and backups

SOURCE	INPUT	ROLE	ENDPOINT / ID	AUTH (ENV VAR)	VERIFIED
firms_area_api	x_fire_points	primary (recent / small areas)	`https://firms.modaps.eosdis.nasa.gov/api/area/csv/{MAP_KEY}/{SOURCE}/{W,S,E,N}/{DAY_RANGE}/{YYYY-MM-DD}`	FIRMS_MAP_KEY	2026-10-27
firms_archive	x_fire_points	primary (history 2000→)	`https://firms.modaps.eosdis.nasa.gov/download/`	Earthdata login or email code	2026-10-27
laads_earthaccess	x_fire_points	backup	`https://ladsweb.modaps.eosdis.nasa.gov/`	EARTHDATA_USERNAME/PASSWORD (~/.netrc)	—
gee_mod14a1	x_clear_days	primary exposure	`ee.ImageCollection('MODIS/061/MOD14A1')`	EE_PROJECT (+ earthengine authenticate)	2026-10-27
gee_vnp14a1	x_clear_days	primary exposure (VIIRS)	`ee.ImageCollection('NASA/VIIRS/002/VNP14A1')`	EE_PROJECT	2026-10-27
planetary_computer_14a1	x_clear_days	backup	`https://planetarycomputer.microsoft.com/`	none for STAC read	—
gee_mcd64a1	x_burned_area	primary	`ee.ImageCollection('MODIS/061/MCD64A1')`	EE_PROJECT	2026-10-27
esa_worldcover	x_land_cover	primary	`ee.ImageCollection('ESA/WorldCover/v200')`	EE_PROJECT	—
osm_overpass	x_industry	primary	`https://overpass-api.de/api/interpreter`	none	—
open_meteo_archive	x_weather	context only	`https://archive-api.open-meteo.com/v1/archive`	none	—
nasa_power	x_weather	backup	`https://power.larc.nasa.gov/api/temporal/daily/point`	none	—
openaq_v3	x_air_quality	impact panel	`https://api.openaq.org/v3/`	OPENAQ_API_KEY	—
celestrak_gp	x_orbits	primary	`https://celestrak.org/NORAD/elements/gp.php?CATNR={id}&FORMAT=json`	none	—
natural_earth	x_boundaries	STATIC CSV (included)	`https://github.com/nvkelso/natural-earth-vector`	none	2026-10-27
geoboundaries	x_boundaries	backup (districts)	`https://www.geoboundaries.org/api/current/gbOpen/IND/ADM2/`	none	—
cmr_search	x_granule_index	health/archive	`https://cmr.earthdata.nasa.gov/search/`	none	—

Static data vs dynamic data

Two kinds of data feed the system. **Static reference data** does not change between requests and has no API you must call at run time — it ships as CSV files you load once into Supabase. **Dynamic observation data** changes daily and must be fetched with a key by sending requests; it is never pre-filled with dummy numbers.

Static CSVs included (real, no dummy)

FILE	ROWS	WHAT IT IS
region.csv	8	regions with real Natural Earth polygons + bbox + gridded area
grid_cell.csv	20,990	0.1° cells with geodesic area and polygon
region_cell.csv	22,060	cell ↔ region with share inside
satellite.csv	5	platforms, NORAD ids, archive starts, end plans
product.csv	5	FIRMS/GEE products, fields, limits
firemask_class.csv	20	FireMask classes + exposure rule
modis_type_code.csv	4	MODIS type field
viirs_confidence_class.csv	3	l / n / h
worldcover_class.csv	11	land-cover legend
firms_column.csv	16	column dictionary
data_source.csv	16	this registry
formula.csv · formula_alias.csv	46 · 8	registry
claim_ledger · conflict_log · audit_check	20 · 15 · 36	evidence
provenance_label · data_status · milestone	5 · 6 · 10	enums

Dynamic data (fetch with keys)

TABLE	SCRIPT	NEEDS
fire_detection	db/fetch_and_load.py	FIRMS_MAP_KEY
cell_day	same (aggregated)	—
cell_month (exposure)	db/gee_exposure.py	EE_PROJECT + earthengine authenticate
burned area	GEE MCD64A1 (extend gee_exposure.py)	EE_PROJECT
weather context	Open-Meteo archive	none
air quality	OpenAQ v3	OPENAQ_API_KEY
orbits	CelesTrak GP	none

WHY NO FIRE ROWS ARE PRE-FILLED

This build environment blocks firms.modaps.eosdis.nasa.gov, Open-Meteo, CelesTrak and NASA POWER (proxy policy). Inventing rows would break the truth policy, so fire tables stay empty until you run the scripts on your machine. Natural Earth came from GitHub, which is reachable.

Fallback, keys and request anatomy

01 **Try primary** — e.g. FIRMS Area API SP

02 **Quality gate** — schema, row count, date range

03 **On failure** — skip source 10 min, try backup

04 **Cache** — store with observation date

05 **Label** — LIVE · ARCHIVE · CACHED · UNAVAILABLE

06 **Return** — {data, meta.data_source, fallback_used, tried}

FIRMS Area API request

```
GET https://firms.modaps.eosdis.nasa.gov/api/area/csv/

{FIRMS_MAP_KEY}/VIIRS_SNPP_SP/73.8,27.6,77.6,32.6/5/2024-
11-01
# order: west,south,east,north · DAY_RANGE 1–10 per page
(27 Oct 2026)
# default 5 in code (FIRMS_DAY_RANGE) · date = YYYY-MM-DD
# sources: MODIS_NRT MODIS_SP VIIRS_SNPP_NRT
VIIRS_SNPP_SP
#          VIIRS_NOAA20_NRT VIIRS_NOAA20_SP
VIIRS_NOAA21_NRT LANDSAT_NRT
```

.env (names only — never values in docs)

```
FIRMS_MAP_KEY=
FIRMS_DAY_RANGE=5
EARTHDATA_USERNAME=
EARTHDATA_PASSWORD=
EE_PROJECT=
OPENAQ_API_KEY=
SUPABASE_URL=
SUPABASE_SERVICE_KEY=
DATABASE_URL=
GROQ_API_KEY=
OPENROUTER_API_KEY=
GEMINI_API_KEY=
```

Data contract template (every input)

PRODUCT / VERSION	e.g. VIIRS_SNPP_SP, collection 2
CADENCE / LATENCY	daily; SP after ~5 months
TIME RANGE	start, end of what was fetched
ACCESS	endpoint + auth env var name
BACKUP	next source in order
GEOMETRY / UNITS	points, MW, K, km
COLUMNS AS RETURNED	recorded from the file, not assumed
MISSING SEMANTICS	empty response ≠ zero fires
QA RULES	F07 rule id
LICENCE / CITATION	NASA open data; cite FIRMS
SIZE / COST	rows, MB, transactions
STATUS	LIVE / ARCHIVE / CACHED / UNAVAILABLE

RE-VERIFY BEFORE THE EVENT

Free tiers (Groq, OpenRouter, Gemini, Supabase, Earth Engine community quota) change often. Check each one in the week before 14 Nov and update data_source.csv.verified_on.

CHAPTER 27

Database, static CSVs and API contract

region

```
region_id PK
name, country, iso_3166_2
role, bbox ×4
area_km2_gridded
boundary_source, url
geom MultiPolygon
```

grid_cell

```
cell_id PK (row_col)
row_idx, col_idx
center_lat/lon
area_km2, size_deg
geom Polygon
```

region_cell

```
region_id FK
cell_id FK
coverage_fraction
area_in_region_km2
```

satellite · product · firemask_class

```
reference enums
(static CSV)
```

fire_detection

```
platform_id, firms_source
processing NRT|SP
lat, lon, acq_date, acq_time_utc
daynight, confidence_raw, frp
scan, track, bright_hi/lo
firms_type, version, cell_id
```

cell_day

```
cell_id, acq_date
platform_id, daynight
n_detections, n_nominal_or_high
max_frp
```

cell_month

```
cell_id, month, platform, dn
fire_cell_days
valid_obs_days, cloud_days
status, exposure_product
```

calibration

```
region, stratum, method
k, k_lo, k_hi, n_years
dispersion, heldout_deviance
status, method_version, hash
```

calendar_value

```
region_id, month, metric
value NULLABLE, lo, hi, level
coverage, status, provenance
method_version, config_hash
```

validation_run · lineage

```
test, metric, value, threshold
passed, predeclared_on
inputs, code_sha, params
```

doc_chunk (RAG)

```
doc_id, source_url, tier
page, heading_path, checksum
body, embedding vector(384)
```

Supabase import order

1. Run `db/schema_v2.sql` in the SQL editor.
2. Import enums: provenance_label, data_status, satellite, product, firemask_class, modis_type_code, viirs_confidence_class, worldcover_class, firms_column.
3. Import region, grid_cell, region_cell; then run the commented UPDATE lines to fill geometry from WKT.
4. Import data_source, formula, formula_alias, claim_ledger, conflict_log, audit_check, milestone.
5. Later: run fetch_and_load.py and gee_exposure.py locally; import csv_real/*.csv into fire_detection, cell_day, cell_month.

API contract


```
{
  "data": { ... },
  "meta": {
    "data_source": {"fire_points": "FIRMS
VIIRS_SNPP_SP",
                    "exposure": "GEE
NASA/VIIRS/002/VNP14A1"},
    "provider": "NASA FIRMS", "product": "VNP14IMG",
"version": "2",
    "processing": "SP", "status": "ARCHIVE_STANDARD",
    "generated_at_utc": "2026-11-14T06:00:00Z",
    "observation_period": ["2024-11-01", "2024-11-
30"],
    "region": "punjab_haryana", "unit": "VIIRS-eq
cell-days",
    "method_version": "calib-0.4", "config_hash":
"7c1e…",
    "coverage": 0.73, "uncertainty": {"level": 0.95,
"type": "predictive"},
    "provenance": "RUN_TESTED_LOCALLY", "demo": false,
    "fallback_used": false, "warnings": [],
"citation_urls": ["…"]
  }
}
```


ROUTE	RETURNS
`GET /api/regions`	region list + bbox + area
`GET /api/cells?bbox&date&res`	aggregates for map
`GET /api/cells/:cell?month`	inspector payload
`GET /api/calendar/:region?metric&sensor&dn`	matrix with value/lo/hi/coverage/status
`GET /api/calibration/:region`	k per stratum + diagnostics
`GET /api/validation/:region`	gates + results
`GET /api/source/:cell`	features, probabilities, set
`GET /api/blindspot/:region`	windows, scenario, B
`GET /api/formulas/:id`	registry YAML as JSON
`GET /api/satellite/:platform`	static facts
`GET /api/health`	status per source
`POST /api/studio/upload | config | compare`	user data flow
`GET /api/brief/:region?season`	RunReport.json + PDF link
`POST /api/explain`	retrieval + citations only

Errors

422 with field list for invalid bbox, dates, metric; 404 unknown region; 503 with status UNAVAILABLE when all sources fail. Shape stays {data: null, meta: {error}}.

Security

Keys server-side only; rate limit per IP; bbox ≤ 25° × 25° per request; uploads ≤ 25 MB, type allow-list, isolated, TTL, never executed or sent to an LLM.

Performance

Aggregates in Parquet/DuckDB; cached payloads; p95 target 300 ms cached — to be measured, not claimed.

CHAPTER 28

Your own RAG knowledge base

A RAG system retrieves passages and cites them. It must never compute fire statistics. Three separate tools keep words and numbers apart.

1 · Document RAG

Indexed NASA guides, papers, this atlas, the formula registry, changelog, API docs, product metadata. Chunk by section with page/URL/version. Returns cited passages.

2 · Numeric query service

Allow-listed, parameterised SQL/DuckDB endpoints built from parsed intent (region, period, metric). Returns numbers with interval and status. The LLM never edits them.

3 · Explanation layer

Templates with slots for computed numbers; LLM may translate to English/Hinglish and explain provenance, never assert new facts.

Directory

```
rag/
  sources/raw_docs/          # NASA/NOAA PDFs, papers
(licence noted)
  sources/verified_web/      # url + accessed_at
snapshots
  sources/formula_registry/  # F01..F46 YAML (from
this atlas)
  sources/method_runs/       # RunReport.json, model
cards, metrics
  indexes/document_manifest.json
  indexes/chunk_catalog.jsonl
  retrieval/chunking.py      # heading-aware, 300-800
tokens
  retrieval/search.py        # hybrid BM25 +
embeddings
  retrieval/citation_guard.py# every sentence ↔ chunk
id
  sql_tools/safe_query.py    # allow-list of query
templates
  evaluations/questions_gold.jsonl
  evaluations/hallucination_tests.py
```

Chunk metadata

```
{doc_id, title, source_url, source_kind,
 provenance_tier:
official|peer_review|project|teaching,
 page, heading_path, product_version, document_date,
 ingested_at, checksum, section_key,
 allowed_use: cite|explain_only, license}
```

Trust tiers and conflict rule

1. official product pages and ATBDs
2. peer-reviewed papers
3. audited project docs (this atlas, registry)
4. teaching books, chats (explain only)

CONFLICT RULE

If retrieved passages disagree (e.g. DAY_RANGE 1–5 vs 1–10), show both with dates and tiers. Never average or silently pick.

Answer flow

01	02	03	04	05	06	07
Question	**Intent parse**	**Numeric → SQL**	**Text → retrieve**	**Compose**	**Citation guard**	**Reply**
EN / Hinglish	numeric? explanatory?	allow-listed template + params	hybrid top-k with tiers	template slots + cited passages	drop uncited sentences, numbers not in result	answer + sources + conflicts

```
def citation_guard(answer_sentences, chunks, numeric_result):
    kept = []
    for s in answer_sentences:
        if s.chunk_id is None and not s.is_template:
            continue
        for number in numbers_in(s.text):
            if number not in numeric_result.values:
                raise RefuseNumber("number not computed: " + number)
        kept.append(s)
    return kept
```

What to feed it, and how to build it step by step

FEED THIS	WHY	TIER
MODIS C6.1 active fire user guide / ATBD	algorithm, confidence, type	official
VIIRS 375 m user guide (Schroeder et al.)	detection, saturation	official / peer
FIRMS FAQ, Area API page, download page	limits, sources, dates	official
GEE catalogue pages (MOD14A1, VNP14A1, MCD64A1)	bands, classes	official
NASA/NOAA mission notices	end dates (with as-of)	official
This V2 atlas + formula YAML	methods, audit, rules	project
RunReport.json of each pipeline run	what was computed, hash	project
Books 1–4 (Hinglish)	teaching phrasing	teaching

Do not feed

- millions of raw FIRMS rows (use SQL)
- API keys, .env, personal data
- unverified competitor numbers as facts
- superseded specs without a 'superseded' tag

Build steps

1. Collect PDFs/HTML into sources/ with licence and accessed_at.
2. Extract text with page numbers; split by headings; drop boilerplate; dedupe by checksum.
3. Embed with a local 384-d sentence-transformer; store in Supabase pgvector (doc_chunk).
4. Hybrid search: BM25 for exact terms (FireMask, DAY_RANGE) + vectors for meaning.
5. Answer with retrieved chunks only; citation_guard rejects sentences without a chunk id.
6. Numeric questions route to safe_query; if no computed result exists → refuse with 'not computed yet'.
7. Evaluate weekly on questions_gold.jsonl.

Gold evaluation questions

QUESTION	MUST ANSWER
What is the denominator of F09?	valid FireMask days {3,5,7,8,9}, per platform
Is VNP14A1 the same as FIRMS VIIRS?	no: 1 km daily vs 375 m points
What changed in F24?	explicit α; 90% vs 95% fix
How many fires burned in Punjab in Nov 2024?	refuse: hotspots are not fires; offer computed cell-days if available
Is B a multiplier?	no: disagreement flag

CHAPTER 29

Automatic PDF briefs from data

The user picks area + period + metric + audience. The server computes first, writes RunReport.json, then renders a PDF only from that file. The same JSON can be re-rendered years later to the same document.

01	02	03	04	05	06	07
Request	**Compute**	**RunReport.json**	**Plots**	**Template**	**Render**	**Check**
region, period, metric, audience EN/Hinglish	SQL + model; result hashes	numbers, intervals, masks, versions	from arrays in the JSON	dark online · light print	HTML → PDF (Chromium)	snapshot test + numbers-in-text test

RunReport.json (schema v1)

```
{
  "schema": "runreport/1", "run_id": "…",
"config_hash": "…",
  "request": {"region": "punjab", "period": ["2024-
10", "2024-11"],
              "metric": "rate", "audience": "officer",
"lang": "en"},
  "results": {"calendar": [...], "z": [...], "trend":
{...},
              "source_mix": {...}, "blindspot":
{...}},
  "quality": {"coverage": 0.73, "fallbacks": 0,
"pending": ["BA 2024-11"]},
  "provenance": {"products": [...], "formulas":
["F05","F09","F14",...],
                 "code_sha": "…", "citations": [...]},
  "text_slots": {"headline": null, "meaning": null}
}
```

Pages of a brief

1. Cover: key question + 3 plain-language findings
2. Map with legend and provenance
3. Observed vs harmonised trend with band
4. Calendar with hatched gaps
5. Source mix with UNKNOWN share
6. Missed-observation warning (BA lag, gaps)
7. Same-month anomaly and trend with honest significance
8. Data-quality table
9. Audit / lineage page: formula IDs, sources, run command
10. Limitations + 'not a life-safety warning'

TESTS

PDF snapshot test per template; every number in text must appear in RunReport.json (regex check); fonts embed Devanagari for Hindi labels; bookmarks and alt text present.

How this atlas itself was produced (reproducible)

```
# 1 audit and simulations
python3 code/v1/test_harmonize.py
python3 code/v1/test_calendar.py
python3 code/v2/audit_run.py        # →
out/audit_results.json
python3 code/v2/sim_ladder.py       # →
out/sim_ladder.json
# 2 static database package
python3 db/build_db_package.py      # Natural Earth →
csv/
python3 db/export_registers.py      # registers +
formula YAML
# 3 figures
python3 build/figs_formula.py ; python3
build/figs_main.py
# 4 book
python3 build/build.py              # HTML → Chromium
PDF + outline
```

INPUT	TYPE
content/formulas.py, formulas_extra.py	46-formula registry text
content/registers.py	sources, conflicts, claims
code/v2/chronos_core.py	reference V2 functions
db/csv/*.csv	static Supabase data
img/*	user's approved designs

SAME PATTERN FOR USER BRIEFS

This book is built exactly like the proposed brief pipeline: data and text in files, figures from code, one renderer, a page-overflow check and a bookmark outline.

PART VII

Execution, checks & traceability

Milestones with quality gates, the submission checklist, what was kept from the old atlas page by page, and the code behind this book.

CHAPTERS IN THIS PART

Chapter 30 · Milestones and quality gates
Chapter 31 · Demo, risks and acceptance checklist
Chapter 32 · Traceability: old atlas → V2
Chapter A · References
Chapter B · Code appendix

CHAPTER 30

Milestones and quality gates

ID	MILESTONE	WORK	DONE WHEN
M0	Preservation & inventory	copy originals, hash, source register, run all tests in pinned env, log skipped tests	register complete; tests reported honestly (pymannkendall dependency installed)
M1	Science audit & data contracts	F01–F46 YAML, estimand decision D1, data contracts, error budget	46 YAML + decision logged
M2	Real small-area extraction	1 district × 1 overlap year (e.g. Ludhiana 2019) from FIRMS SP + GEE exposure	cell_day + cell_month rows; columns recorded as returned
M3	Model comparison	B0–B4 on real overlap years, bootstrap coverage, seam + placebo	held-out table on /validation; gates G1–G6 evaluated
M4	UX decision	palette D2, three boards approved, locked vs revisable surfaces	user approval recorded
M5	Core website	Home, Explorer, Calendar, Sensor Lab, Validation on real API + fixtures labelled	Playwright screenshots; no orphan filters
M6	Optional layers	L2 abstaining rules, L3 pending states, Studio, Formula Explorer, Learn	UNKNOWN visible; scenario styling
M7	RAG & reports	doc index, citation guard, safe_query, RunReport + PDF builder	gold questions pass; numbers-in-text test
M8	Master docs	this atlas updated with real results, traceability, bilingual summaries	no SIMULATED where real exists
M9	Release checks	offline backup demo, a11y, cross-browser, reproducibility script, video	fresh-browser test; references complete

RULES CARRIED FROM THE TEAM WORKFLOW

Beginner-readable code (useState/useEffect/useParams/useNavigate, props, map/filter/find, if/else before return; no ternaries for branching, no custom hooks). Full files, not snippets. No new packages without asking. One branch per milestone; never push to main without approval.

REPORT AFTER EVERY MILESTONE

changed files · data lineage · formulas touched · tests run (and skipped) · screenshots/visual diff · measured performance · known limitations · next milestone. Never say 'all tests pass' when some were skipped.

CHAPTER 31

Demo, risks and acceptance checklist

Two-minute demo storyboard

TIME	SCREEN	LINE
0:00	Home hero	'Satellites are losing the fire record — and burning moved to hours they don't see.'
0:15	Calendar raw	the 2012 step
0:30	Calendar harmonised	the step fades, seasons stay; band on every cell
0:50	Inspector	E, rate, k version, z with n
1:05	Blind Spot	clock + B flag — 'counts may miss late burning'
1:25	Validation	held-out errors, placebo seams
1:40	Brief PDF	one page for an officer, with caveat
1:55	Health	every number has a source

Risk register

RISK	LIKELIHOOD	MITIGATION
archive request delayed	medium	request now; Area API SP fallback
GEE quota	medium	small region first; Planetary Computer backup
k unstable in real data	medium	report per stratum; mark 'indicative'
rules restrict pre-built code	unknown	re-read 2026 rules after 28 Oct
demo data shown as real	low	meta.demo badge + tests
free-tier changes	high	template fallback for text

Final acceptance checklist

- [x] all primary source files inventoried (missing ones listed)
- [x] 46 formulas with dimensions, assumptions, status
- [x] no false accuracy claims; real metrics marked pending
- [x] FIREWORK, SIH ideas and four layers reconciled
- [x] 13 routes specified with states
- [x] three design directions + one validated palette (approval pending)
- [ ] calendar supports 25 years with real data (needs M2–M3)
- [x] chart forms chosen by question
- [x] data sources, keys, versions, fallbacks documented
- [x] RAG separated from numeric computation (design)
- [ ] dynamic PDF generation implemented (design done, code M7)
- [x] 160+ page atlas with bookmarks, formulas, badges, code appendix, traceability
- [x] rendering checked for overflow and broken images
- [ ] challenge page re-verified after 28 Oct release
- [x] states research decision support, not life-safety

CHAPTER 32

Traceability: old atlas → V2

Nothing vanishes silently. Every section of the 164-page Atlas (20 Oct 2026) maps to a V2 chapter or an archive decision.

OLD ATLAS SECTION	V2 LOCATION	DECISION	NOTE
Part I Vision (ch. 1–4)	Ch. 1–3	KEEP / REVISE	story kept; claims re-labelled
2.2 Four reasons counts disagree	Ch. 7	KEEP	extended with processing
2.3 The 2012 seam	Ch. 7, F33–F35	REVISE	+2018/2023 seams, placebo
3.1 FIREWORK today	Ch. 4 register	KEEP	repo fixes listed
4.1 Competitor scan	Ch. 1 claims, CL-16/17	REVISE	marked UNREVIEWED
Part II Design research (ch. 5–7)	Ch. 22	REVISE	3 directions re-scoped A/B/C
6.2 Scorecard (Paper Atlas, Ember Night, Aurora)	Ch. 22 scorecard	REVISE	new criteria incl. accessibility
7.1 Logo concepts (Chrono-Orbit)	Ch. 22 A	REPLACED	Crossing Orbits (Final Spec)
7.2 Tokens (atlas.css)	Ch. 22 B, Ch. 25	REVISE	palette C-02 proposed
8 Information architecture	Ch. 20	KEEP	+ shared state object
9.1–9.13 Thirteen pages	Ch. 23	KEEP / REVISE	states, a11y, provenance added
10 Responsive	Ch. 23, 25	KEEP	
11 How satellites see fire	Ch. 7	KEEP	physics added
12 Dataset encyclopedia	Ch. 9, 26	REVISE	products split, FireMask verified
13 APIs, keys, backups	Ch. 26	REVISE	DAY_RANGE conflict C-01
14 Pipeline and lineage	Ch. 10, 13	REVISE	exposure-centred
15 Validation framework	Ch. 12	REVISE	predeclared gates G1–G12
16 A–O formula template	Ch. 11 spread guide	REPLACED	2-page spread template

OLD ATLAS SECTION	V2 LOCATION	DECISION	NOTE
17 Provenance of 46 formulas	Ch. 11 index	REVISE	audit verdicts
18.1–18.7 The 46 formulas	F01–F46 spreads	REVISE	17 FIX, 19 REVISE, 10 OK
19 Misunderstood concepts, AI/ML	Ch. 9 myths, R1–R10	REVISE	AI only where rules fail
20–24 Architecture	Ch. 10, 27	REVISE	module boundaries, API meta
23 Upload pipeline (Studio)	Ch. 23 E, Ch. 27	KEEP	security limits
25 Workstreams and roadmap	Ch. 30	REPLACED	M0–M9 quality gates
26 MVP vs stretch, demo	Ch. 2, 31	REVISE	jobs J1–J13
27 Seven user journeys	Ch. 20	KEEP	success measures
28 Risk register, checklist	Ch. 31	REVISE	
A Parameter dictionary	Ch. 6, formula symbols	MERGED	
B Glossary	Ch. 6	REVISE	units that must not mix
C Code-to-feature map	Ch. 10 modules	REVISE	
D Evidence register	claim_ledger.csv	REPLACED	ledger with URLs
E Bibliography	Appendix A	REVISE	official sources first

Deletion / retention report

ITEM	STATUS
synthetic seed metrics (impossible F1)	removed from UI; archived
k = 2.6 as a universal constant	archived as teaching example
old Chrono-Orbit logo	archived (exploration)
Paper Atlas / Ember Night / Aurora directions	archived; elements reused in A/B/C
FIS weighted score	rejected, archived with reason
old A–O formula pages	content merged into spreads

APPENDIX A

References

Official sources first. Access dates are when this edition read the page. Re-check close to submission.

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APPENDIX B

Code appendix — V2 reference functions

From `code/v2/chronos_core.py`. Plain functions, if/else, every edge case returns a status instead of 0. RUN / TESTED LOCALLY

```
def cell_index(lat, lon, size):
    if lat < -90 or lat > 90:
        return None
    lon_wrapped = ((lon + 180.0) % 360.0) - 180.0
    if lon == 180.0:
        lon_wrapped = 180.0 - 1e-9
    row = math.floor(round((lat + 90.0) / size, 9))
    col = math.floor(round((lon_wrapped + 180.0) / size, 9))
    max_row = int(round(180.0 / size)) - 1
    max_col = int(round(360.0 / size)) - 1
    if row > max_row:
        row = max_row
    if col > max_col:
        col = max_col
    return row, col

def clear_sky_rate(fire_days, valid_days):
    if valid_days is None:
        return result(None, STATUS_UNAVAILABLE, "no coverage
information")
    if valid_days <= 0:
        return result(None, STATUS_UNAVAILABLE, "zero valid
observations")
    if fire_days > valid_days:
        return result(None, "invalid", "fire days exceed valid
days")
    return result(fire_days / valid_days, STATUS_OK)

def log_rate(fire_days, valid_days):
    if valid_days is None or valid_days <= 0:
        return None
    return math.log((fire_days + 0.5) / (valid_days + 1.0))

def ratio_ci_delta(v_total, m_total, alpha=0.05):
    se = log_ratio_se(v_total, m_total)
    if se is None:
        return None
    zq = stats.norm.ppf(1 - alpha / 2)
    k = v_total / m_total
    return k * math.exp(-zq * se), k, k * math.exp(zq * se)

def empirical_bayes_log_k(log_k_means, se2_list):
    y = np.asarray(log_k_means, dtype=float)
    s2 = np.asarray(se2_list, dtype=float)
    good = np.isfinite(y) & np.isfinite(s2) & (s2 > 0)
    if good.sum() < 3:
        return None
    y = y[good]
    s2 = s2[good]
    w0 = 1.0 / s2
    mu0 = float(np.sum(w0 * y) / np.sum(w0))
    q = float(np.sum(w0 * (y - mu0) ** 2))
    c = float(np.sum(w0) - np.sum(w0 ** 2) / np.sum(w0))
    tau2 = max(0.0, (q - (len(y) - 1)) / c)
    w_re = 1.0 / (s2 + tau2)
    mu = float(np.sum(w_re * y) / np.sum(w_re))
    weights = tau2 / (tau2 + s2)
    post = weights * y + (1 - weights) * mu
    return {"mu": mu, "tau2": tau2, "weights":
weights.tolist(), "posterior": post.tolist()}

def block_bootstrap_ratio(v_by_year, m_by_year, n_boot=2000,
alpha=0.05, seed=0):
    v = np.asarray(v_by_year, dtype=float)
    m = np.asarray(m_by_year, dtype=float)
    n = len(v)
    if n < 3:
        return None
    rng = np.random.default_rng(seed)
    out = []
    for b in range(n_boot):
        idx = rng.integers(0, n, n)
        msum = m[idx].sum()
        if msum > 0:
            out.append(v[idx].sum() / msum)
    return percentile_interval(out, alpha)
```

```
def garwood_ci(n, alpha=0.05):
    if n < 0:
        return None
    if n == 0:
        lo = 0.0
    else:
        lo = stats.chi2.ppf(alpha / 2, 2 * n) / 2.0
    hi = stats.chi2.ppf(1 - alpha / 2, 2 * (n + 1)) / 2.0
    return float(lo), float(hi)

def robust_z_loo(history, current, sigma_floor=0.15,
min_samples=8):
    vals = [h for h in history if h is not None and
np.isfinite(h)]
    if len(vals) < min_samples or current is None:
        return result(None, STATUS_INSUFFICIENT, "fewer than "
+ str(min_samples) + " reference years")
    med = float(np.median(vals))
    mad = float(np.median(np.abs(np.asarray(vals) - med)))
    sigma = 1.4826 * mad
    floor_used = False
    if sigma < sigma_floor:
        sigma = sigma_floor
        floor_used = True
    z = (current - med) / sigma
    note = ""
    if floor_used:
        note = "spread floor used"
    return result(z, STATUS_OK, note)

def mann_kendall_ties(x):
    x = np.asarray(x, dtype=float)
    x = x[np.isfinite(x)]
    n = len(x)
    if n < 4:
        return None
    s = 0
    for i in range(n - 1):
        s += int(np.sum(np.sign(x[i + 1:] - x[i])))
    uniq, counts = np.unique(x, return_counts=True)
    tie_term = float(np.sum(counts * (counts - 1) * (2 *
counts + 5)))
    var_s = (n * (n - 1) * (2 * n + 5) - tie_term) / 18.0
    if var_s <= 0:
        return {"S": s, "var": var_s, "z": 0.0, "p": 1.0}
    if s > 0:
        z = (s - 1) / math.sqrt(var_s)
    elif s < 0:
        z = (s + 1) / math.sqrt(var_s)
    else:
        z = 0.0
    p = 2 * (1 - stats.norm.cdf(abs(z)))
    return {"S": s, "var": var_s, "z": z, "p": p}
```

```
def diurnal_seen_union(windows, mu, sigma):
    merged = []
    for start, end in sorted(windows):
        if len(merged) > 0 and start ⩽ merged[-1][1]:
            merged[-1][1] = max(merged[-1][1], end)
        else:
            merged.append([start, end])
    total = 0.0
    for start, end in merged:
        total += stats.norm.cdf(end, mu, sigma) -
stats.norm.cdf(start, mu, sigma)
    return total

def blindspot_ratio(ba_km2, cell_days, ref_ratios):
    if ba_km2 is None:
        return result(None, "pending", "burned area not yet
available")
    if cell_days is None or cell_days ⩽ 0:
        return result(None, STATUS_UNAVAILABLE, "no detections
to compare")
    refs = [r for r in ref_ratios if r is not None and
np.isfinite(r) and r > 0]
    if len(refs) < 3:
        return result(None, STATUS_INSUFFICIENT, "fewer than 3
reference years")
    return result((ba_km2 / cell_days) /
float(np.median(refs)), STATUS_OK, "diagnostic flag, not a
correction factor")

def kalman_local_level(z, r, q, x0=None, p0=1e6):
    x = x0
    p = p0
    out = []
    for value in z:
        if x is None:
            if value is None or not np.isfinite(value):
                out.append(None)
                continue
            x = value
            p = r
            out.append(x)
            continue
        p = p + q
        if value is not None and np.isfinite(value):
            gain = p / (p + r)
            x = x + gain * (value - x)
            p = (1 - gain) * p
        out.append(x)
    return out

def calendar_cell(fire_days, valid_days, method_version,
source):
    if valid_days is None:
        return {"value": None, "coverage": None, "status":
STATUS_UNAVAILABLE, "method": method_version, "source":
source}
    if valid_days == 0:
        return {"value": None, "coverage": 0, "status":
"not_observable", "method": method_version, "source": source}
    return {"value": fire_days, "coverage": valid_days,
"status": STATUS_OK, "method": method_version, "source":
source}
```

Deliverables produced with this atlas

PATH	CONTENT
FIREWORK_CHRONOS_V2_Research_Atlas.pdf	this book
db/schema_v2.sql	Supabase schema
db/csv/*.csv (19 files)	static real data + registers
db/fetch_and_load.py	FIRMS fetch → fire_detection, cell_day
db/gee_exposure.py	GEE exposure → cell_month
formula_registry/F01–F46.yaml	registry
code/v2/chronos_core.py, audit_run.py, sim_ladder.py	reference code + audit
docs/v2/*.md	source register, conflicts, ideas, questions, error budget

AAKHRI BAAT

Yeh kitaab sach bolti hai: jo naapa woh likha, jo nahi naapa woh 'pending'. Asli FIRMS data aate hi har SIMULATED label ki jagah asli number aayega.